AMITY UNIVERSITY
UTTAR PRADESH

CSIT248

Advanced Web Technology

Submitted by

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Programme:	BCA (Hons./Hons. with Research)
Semester:	3

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S. No.	Experiment No.	Title of Experiment	Date Performed	Date Submitted	Faculty Signature
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Experiment No.: 1

Title : Demonstration of Basic HTML Tags, Fonts, Images, and Lists.

Aim : To create a web page using basic HTML tags such as Body, Horizontal Rule, Font, Image, Ordered List, and Unordered List, and to demonstrate different font styles and list types.

Source Code :

```
<!DOCTYPE html>
<html>

<head>
    <title>Web Technology Assignment</title>
</head>

<body bgcolor="yellow" text="red">


    <hr width="70%" color="blue" size="10">


    <font face="Arial" color="red" size="6">
        <center>WEB TECHNOLOGY</center>
    </font>

    <p>
        Welcome to HTML Programming. This web page demonstrates basic HTML tags
        such as Body, Horizontal Rule, Font, Image, Ordered List, and Unordered List.
    </p>

    <!-- Image -->
    <center>
        <img src="https://upload.wikimedia.org/wikipedia/commons/6/61/HTML5_logo_and_wordmark.svg"
alt="HTML Logo"
            width="200">
    </center>

    <h2>Ordered List</h2>

    <ol>
        <li>Python</li>
        <li>Java</li>
        <li>C++</li>
        <li>HTML</li>
    </ol>

    <h2>Ordered List (Alphabet)</h2>

    <ol type="A">
        <li>Python</li>
        <li>Java</li>
        <li>C++</li>
        <li>HTML</li>
    </ol>

    <h2>Ordered List (Roman Numbers)</h2>

    <ol type="I">
        <li>Python</li>
        <li>Java</li>
        <li>C++</li>
```

```
        <li>HTML</li>
    </ol>

    <h2>Unordered List (Disc)</h2>

    <ul type="disc">
        <li>Python</li>
        <li>Java</li>
        <li>C++</li>
    </ul>

    <h2>Unordered List (Circle)</h2>

    <ul type="circle">
        <li>HTML</li>
        <li>CSS</li>
        <li>JavaScript</li>
    </ul>

    <h2>Unordered List (Square)</h2>

    <ul type="square">
        <li>React</li>
        <li>Node.js</li>
        <li>MongoDB</li>
    </ul>

    <hr width="70%" color="blue" size="5">

    <center>
        <b>Created by Atul Anand</b>
    </center>

</body>

</html>
```

Output :

WEB TECHNOLOGY

Welcome to HTML Programming. This web page demonstrates basic HTML tags such as Body, Horizontal Rule, Font, Image, Ordered List, and Unordered List.



Ordered List

1. Python
2. Java
3. C++
4. HTML

Ordered List (Alphabet)

A. Python
B. Java
C. C++
D. HTML

Ordered List (Roman Numbers)

I. Python
II. Java
III. C++
IV. HTML

Experiment No.: 2

Title : Creating a Simple Burger Café Web Page Using HTML.

Aim : To design a simple restaurant web page using basic HTML elements such as headings, images, lists, paragraphs, and hyperlinks to display menu details, contact information, and social media links.

Source Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Burger Cafe's MJ</title>
</head>

<body>

    <h1>Welcome to The MJ Burger Cafe's </h1>

    <div style="text-align: center;">
        <img src="https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcQpz3ysoA5s4c9pLDCU-zuwx7n4
OXrFCjSM0rB8eZK_470bVJkFJQMcUEzC&s=10" alt="A large cheeseburger with melted cheese, lettuce, toma
to, and a toasted bun shown against a bright, appetizing background that creates a warm, inviting
cafe atmosphere" width="200" height="200">
    </div>

    <h2>Menu</h2>

    <ul>
        <li>Double Cheeseburger</li>
        <li>BBQ Burger</li>
        <li>Milkshake</li>
    </ul>

    <h2>About Us</h2>

    <p>
        The Burger Hub, established in 1990, is a cherished family-owned eatery
        renowned for its delectable burgers. We place utmost importance on utilizing
        the freshest ingredients available and ensuring an exceptional dining
        experience for our valued customers.
    </p>

    <h2>Contact Us</h2>

    <p>
        Visit us at: MJ Main Street, MJCity
        <br>
        Call us: 8102386247
        <br>
        Email: info@mjburgercafe.com
    </p>

    <p>
        Follow us on
        <a href="https://www.instagram.com/onyrdn/" target="_blank">
            Instagram.
        </a>
        <a href="https://www.facebook.com/onyrdn/" target="_blank">
            Facebook.
```

```
        </a>
        for the latest updates!
    </p>

</body>
</html>
```

Output :

Welcome to The MJ Burger Cafe's



Menu

- Double Cheeseburger
- BBQ Burger
- Milkshake

About Us

The Burger Hub, established in 1990, is a cherished family-owned eatery renowned for its delectable burgers. We place utmost importance on utilizing the freshest ingredients available and ensuring an exceptional dining experience for our valued customers.

Contact Us

Visit us at: MJ Main Street, MJCity
Call us: 8102386247
Email: info@mjburgercafe.com

Follow us on Instagram. Facebook. for the latest updates!

Experiment No.: 3

Title : Creating a Web Page with Headings, Hyperlinks, Images, and Lists.

Aim : To create a simple HTML web page that displays an image, headings, formatted text, unordered lists, and hyperlinks to demonstrate navigation using HTML.

Source Code :

```
<!DOCTYPE html>
<html>

<head>
    <title>GeeksForGeeks</title>
</head>

<body bgcolor="#f0f0eb">

    <center>
        <img src="https://media.geeksforgeeks.org/gfg-gg-logo.svg"
             alt="GFG Logo" width="120">

        <h1 style="color:green;">GeeksForGeeks</h1>
    </center>

    <h1>Table of Content</h1>

    <h2>C++</h2>

    <p>
        C++ is an object-oriented programming language that is widely used
        for competitive programming, Data Structures, Operating Systems, etc.
    </p>

    <p>
        <b><i>Some of its topic are given below:-</i></b>
    </p>

    <ul>
        <li>
            <a href="https://www.geeksforgeeks.org/cpp/basics/">
                Basics
            </a>
        </li>

        <li>
            <a href="https://www.geeksforgeeks.org/cpp/difference-c-structures-c-structures/">
                Difference Between C Structures and C++ Structures
            </a>
        </li>

        <li>
            <a href="https://www.geeksforgeeks.org/cpp/comparison-of-inheritance-in-c-and-java/">
                Comparison of Inheritance in C++ and Java
            </a>
        </li>

        <li>
            <a href="https://www.geeksforgeeks.org/cpp/comparison-of-static-keyword-in-c-and-jav
a/">
                Comparison of static keyword in C++ and Java
            </a>
        </li>
```

```
        <li>
            <a href="https://www.geeksforgeeks.org/cpp/comparison-of-exception-handling-in-c-and-j
ava/">
                Comparison of Exception Handling in C++ and Java
            </a>
        </li>

        <li>
            <a href="https://www.geeksforgeeks.org/cpp/basic-input-output-c/">
                Basic Input / Output in C++
            </a>
        </li>

        <li>
            <a href="https://www.geeksforgeeks.org/cpp/write-a-c-program-that-wont-compile-in-c/">
                Write a C program that won't compile in C++
            </a>
        </li>

        <li>
            <a href="https://www.geeksforgeeks.org/cpp/references-in-c/">
                References in C++
            </a>
        </li>
    </ul>

    <h2>
        <font color="red"><i>Java</i></font>
    </h2>

    <p>
        Java has been one amongst the foremost standard programming
        languages for several years. When compared with C++, Java doesn't
        enable several things which may cause bad or inefficient programming.
    </p>

</body>

</html>
```

Output :



GeeksForGeeks

Table of Content

C++

C++ is an object-oriented programming language that is widely used for competitive programming, Data Structures, Operating Systems, etc.

Some of its topic are given below:-

- Basics
- Difference Between C Structures and C++ Structures
- Comparison of Inheritance in C++ and Java
- Comparison of static keyword in C++ and Java
- Comparison of Exception Handling in C++ and Java
- Basic Input / Output in C++
- Write a C program that won't compile in C++
- References in C++

Java

Java has been one amongst the foremost standard programming languages for several years. When compared with C++, Java doesn't enable several things which may cause bad or inefficient programming.

Experiment No.: 4

Title : Student Registration Form with Rich Form Controls.

Aim : To design a comprehensive student registration web form utilizing diverse input controls (text, radio, checkboxes, date, select dropdowns, file uploads, textareas) and structured fieldsets.

Source Code :

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description"
    content="Experiment 1: Design a web page to develop a student Registration Form using HTML - C
SIT248 Advanced Web Technology">
  <title>Experiment 01 - Student Registration Form (Inline CSS)</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&display=swa
p" rel="stylesheet">
</head>

<body
  style="margin: 0; padding: 0; font-family: 'Inter', system-ui, -apple-system, sans-serif; backgr
ound-color: #0f172a; color: #f8fafc; line-height: 1.6; min-height: 100vh;">

  <!-- Navigation (Inline CSS) -->
  <nav
    style="position: sticky; top: 0; z-index: 100; background: rgba(15, 23, 42, 0.92); backdrop-fi
lter: blur(12px); border-bottom: 1px solid #334155; padding: 1rem 2rem; display: flex; justify-con
tent: space-between; align-items: center;">
    <a href="../index.html"
      style="font-weight: 800; font-size: 1.2rem; color: #818cf8; text-decoration: none; display:
flex; align-items: center; gap: 0.5rem;">
      <span
        style="background: rgba(99, 102, 241, 0.15); border: 1px solid rgba(99, 102, 241, 0.3); pa
dding: 0.2rem 0.5rem; border-radius: 6px;">&larr;</span>
      CSIT248 Lab Work
    </a>
    <div style="display: flex; gap: 1rem; align-items: center;">
      <span
        style="font-size: 0.8rem; background: rgba(99, 102, 241, 0.12); color: #a5b4fc; border: 1p
x solid rgba(99, 102, 241, 0.25); padding: 0.3rem 0.75rem; border-radius: 9999px; font-weight: 60
0;">
        Formatted Exclusively with Inline CSS
      </span>
      <a href="../index.html"
        style="background: #6366f1; color: #ffffff; padding: 0.4rem 0.9rem; border-radius: 6px; te
xt-decoration: none; font-size: 0.85rem; font-weight: 600;">
        All Experiments
      </a>
    </div>
  </nav>

  <main style="max-width: 900px; margin: 0 auto; padding: 2.5rem 1.5rem;">

    <!-- Header (Inline CSS) -->
    <header style="text-align: center; margin-bottom: 2rem;">
      <div
```

```
        style="display: inline-block; background: rgba(99, 102, 241, 0.15); color: #a5b4fc; borde
r: 1px solid rgba(99, 102, 241, 0.3); padding: 0.25rem 0.75rem; border-radius: 9999px; font-size:
0.85rem; font-weight: 600; margin-bottom: 0.75rem;">
        Experiment 01 &bull; Module II (Forms &amp;amp; Form Elements)
      </div>
      <h1 style="font-size: 2.25rem; font-weight: 800; color: #ffffff; margin: 0 0 0.5rem 0;">Stud
ent Registration Form
      </h1>
      <p style="color: #94a3b8; font-size: 1.05rem; margin: 0;">Amity University Noida &bull; Depa
rtment of Computer
        Applications</p>
    </header>

    <!-- Registration Form (Inline CSS) -->
    <form action="#" method="post"
      onsubmit="event.preventDefault(); alert('✅ Registration Form Submitted Successfully!\n\nTh
ank you, your application has been recorded.');"
      style="background: #1e293b; border: 1px solid #334155; border-radius: 16px; padding: 2.25re
m; box-shadow: 0 10px 25px rgba(0, 0, 0, 0.3);">

      <!-- Fieldset 1: Personal Details -->
      <fieldset
        style="border: 1px solid #475569; border-radius: 10px; padding: 1.5rem; margin-bottom: 1.7
5rem; background: rgba(15, 23, 42, 0.6);">
        <legend style="color: #38bdf8; font-weight: 700; font-size: 1.1rem; padding: 0 0.5rem;">
          👤 Personal Information
        </legend>

        <div
          style="display: grid; grid-template-columns: repeat(auto-fit, minmax(240px, 1fr)); gap:
1.25rem; margin-top: 0.5rem;">
          <!-- First Name -->
          <div>
            <label for="fname"
              style="display: block; font-size: 0.85rem; font-weight: 600; color: #cbd5e1; margin-
bottom: 0.35rem;">First
              Name <span style="color: #ef4444;">*</span></label>
            <input type="text" id="fname" name="firstname" required placeholder="e.g. Atul"
              style="width: 100%; box-sizing: border-box; background: #0f172a; border: 1px solid #
475569; color: #ffffff; padding: 0.65rem 0.85rem; border-radius: 6px; font-size: 0.9rem; outline:
none;">
          </div>

          <!-- Last Name -->
          <div>
            <label for="lname"
              style="display: block; font-size: 0.85rem; font-weight: 600; color: #cbd5e1; margin-
bottom: 0.35rem;">Last
              Name <span style="color: #ef4444;">*</span></label>
            <input type="text" id="lname" name="lastname" required placeholder="e.g. Anand"
              style="width: 100%; box-sizing: border-box; background: #0f172a; border: 1px solid #
475569; color: #ffffff; padding: 0.65rem 0.85rem; border-radius: 6px; font-size: 0.9rem; outline:
none;">
          </div>

          <!-- Father's Name -->
          <div>
            <label for="fathername"
              style="display: block; font-size: 0.85rem; font-weight: 600; color: #cbd5e1; margin-
bottom: 0.35rem;">Father's
              / Guardian's Name</label>
            <input type="text" id="fathername" name="fathername" placeholder="Full Name"
              style="width: 100%; box-sizing: border-box; background: #0f172a; border: 1px solid #
475569; color: #ffffff; padding: 0.65rem 0.85rem; border-radius: 6px; font-size: 0.9rem; outline:
none;">
          </div>
```

```
          <!-- Date of Birth -->
          <div>
            <label for="dob"
              style="display: block; font-size: 0.85rem; font-weight: 600; color: #cbd5e1; margin-
bottom: 0.35rem;">Date
              of Birth <span style="color: #ef4444;">*</span></label>
            <input type="date" id="dob" name="dob" required
              style="width: 100%; box-sizing: border-box; background: #0f172a; border: 1px solid #
475569; color: #ffffff; padding: 0.65rem 0.85rem; border-radius: 6px; font-size: 0.9rem; outline:
none;">
          </div>
        </div>

        <!-- Gender Radio Buttons -->
        <div style="margin-top: 1.25rem;">
          <label
            style="display: block; font-size: 0.85rem; font-weight: 600; color: #cbd5e1; margin-bo
ttom: 0.5rem;">Gender
            <span style="color: #ef4444;">*</span></label>
          <div style="display: flex; gap: 1.5rem; align-items: center; font-size: 0.9rem; color: #
e2e8f0;">
            <label style="display: flex; align-items: center; gap: 0.4rem; cursor: pointer;">
              <input type="radio" name="gender" value="Male" checked style="accent-color: #6366f
1;"> Male
            </label>
            <label style="display: flex; align-items: center; gap: 0.4rem; cursor: pointer;">
              <input type="radio" name="gender" value="Female" style="accent-color: #6366f1;"> Fem
ale
            </label>
            <label style="display: flex; align-items: center; gap: 0.4rem; cursor: pointer;">
              <input type="radio" name="gender" value="Other" style="accent-color: #6366f1;"> Othe
r
            </label>
          </div>
        </div>
      </fieldset>

      <!-- Fieldset 2: Academic & Program Details -->
      <fieldset
        style="border: 1px solid #475569; border-radius: 10px; padding: 1.5rem; margin-bottom: 1.7
5rem; background: rgba(15, 23, 42, 0.6);">
        <legend style="color: #34d399; font-weight: 700; font-size: 1.1rem; padding: 0 0.5rem;">
          🎓 Academic Enrollment Details
        </legend>

        <div
          style="display: grid; grid-template-columns: repeat(auto-fit, minmax(240px, 1fr)); gap:
1.25rem; margin-top: 0.5rem;">
          <!-- Program Dropdown -->
          <div>
            <label for="program"
              style="display: block; font-size: 0.85rem; font-weight: 600; color: #cbd5e1; margin-
bottom: 0.35rem;">Enrolled
              Program <span style="color: #ef4444;">*</span></label>
            <select id="program" name="program" required
              style="width: 100%; box-sizing: border-box; background: #0f172a; border: 1px solid #
475569; color: #ffffff; padding: 0.65rem 0.85rem; border-radius: 6px; font-size: 0.9rem; outline:
none;">
              <option value="BCA-Hons">BCA (Hons.) with Research</option>
              <option value="BCA-Gen">BCA (General)</option>
              <option value="BTech-CSE">B.Tech Computer Science &amp;amp; Engineering</option>
              <option value="MCA">MCA (Master of Computer Applications)</option>
              <option value="BSc-IT">B.Sc. Information Technology</option>
            </select>
          </div>
```

```
          <!-- Current Semester Dropdown -->
          <div>
            <label for="semester"
              style="display: block; font-size: 0.85rem; font-weight: 600; color: #cbd5e1; margin-
bottom: 0.35rem;">Current
              Semester <span style="color: #ef4444;">*</span></label>
            <select id="semester" name="semester" required
              style="width: 100%; box-sizing: border-box; background: #0f172a; border: 1px solid #
475569; color: #ffffff; padding: 0.65rem 0.85rem; border-radius: 6px; font-size: 0.9rem; outline:
none;">
              <option value="Sem-1">Semester I</option>
              <option value="Sem-2">Semester II</option>
              <option value="Sem-3" selected>Semester III</option>
              <option value="Sem-4">Semester IV</option>
              <option value="Sem-5">Semester V</option>
              <option value="Sem-6">Semester VI</option>
            </select>
          </div>

          <!-- Enrollment / Roll Number -->
          <div>
            <label for="enrollment"
              style="display: block; font-size: 0.85rem; font-weight: 600; color: #cbd5e1; margin-
bottom: 0.35rem;">Enrollment
              No. <span style="color: #ef4444;">*</span></label>
            <input type="text" id="enrollment" name="enrollment" required placeholder="e.g. A23052
21001"
              style="width: 100%; box-sizing: border-box; background: #0f172a; border: 1px solid #
475569; color: #ffffff; padding: 0.65rem 0.85rem; border-radius: 6px; font-size: 0.9rem; outline:
none;">
          </div>
        </div>

        <!-- Elective Checkboxes -->
        <div style="margin-top: 1.25rem;">
          <label
            style="display: block; font-size: 0.85rem; font-weight: 600; color: #cbd5e1; margin-bo
ttom: 0.5rem;">Elective
            Course Interests (Select all that apply)</label>
          <div
            style="display: grid; grid-template-columns: repeat(auto-fit, minmax(200px, 1fr)); ga
p: 0.75rem; font-size: 0.875rem; color: #cbd5e1;">
            <label style="display: flex; align-items: center; gap: 0.4rem; cursor: pointer;">
              <input type="checkbox" name="electives" value="WebDev" checked style="accent-color:
#6366f1;"> Advanced
              Web Tech
            </label>
            <label style="display: flex; align-items: center; gap: 0.4rem; cursor: pointer;">
              <input type="checkbox" name="electives" value="AIML" checked style="accent-color: #6
366f1;"> AI &amp;amp;
              Machine Learning
            </label>
            <label style="display: flex; align-items: center; gap: 0.4rem; cursor: pointer;">
              <input type="checkbox" name="electives" value="CyberSec" checked style="accent-colo
r: #6366f1;">
              Cybersecurity Fundamentals
            </label>
            <label style="display: flex; align-items: center; gap: 0.4rem; cursor: pointer;">
              <input type="checkbox" name="electives" value="DataAnalytics" checked style="accent-
color: #6366f1;"> Data
              Analytics &amp;amp; Mining
            </label>
          </div>
        </div>
      </fieldset>
```

```
      <!-- Fieldset 3: Contact Details & Address -->
      <fieldset
        style="border: 1px solid #475569; border-radius: 10px; padding: 1.5rem; margin-bottom: 1.7
5rem; background: rgba(15, 23, 42, 0.6);">
        <legend style="color: #fbbf24; font-weight: 700; font-size: 1.1rem; padding: 0 0.5rem;">
          📞 Contact &amp;amp; Communication Details
        </legend>

        <div
          style="display: grid; grid-template-columns: repeat(auto-fit, minmax(240px, 1fr)); gap:
1.25rem; margin-top: 0.5rem;">
          <div>
            <label for="email"
              style="display: block; font-size: 0.85rem; font-weight: 600; color: #cbd5e1; margin-
bottom: 0.35rem;">Email
              Address <span style="color: #ef4444;">*</span></label>
            <input type="email" id="email" name="email" required placeholder="e.g. atulll0923@gmai
l.com"
              style="width: 100%; box-sizing: border-box; background: #0f172a; border: 1px solid #
475569; color: #ffffff; padding: 0.65rem 0.85rem; border-radius: 6px; font-size: 0.9rem; outline:
none;">
          </div>

          <div>
            <label for="mobile"
              style="display: block; font-size: 0.85rem; font-weight: 600; color: #cbd5e1; margin-
bottom: 0.35rem;">Mobile
              Number <span style="color: #ef4444;">*</span></label>
            <input type="tel" id="mobile" name="mobile" required pattern="[0-9]{10}" placeholder
="10-digit Mobile No."
              style="width: 100%; box-sizing: border-box; background: #0f172a; border: 1px solid #
475569; color: #ffffff; padding: 0.65rem 0.85rem; border-radius: 6px; font-size: 0.9rem; outline:
none;">
          </div>
        </div>

        <div style="margin-top: 1.25rem;">
          <label for="address"
            style="display: block; font-size: 0.85rem; font-weight: 600; color: #cbd5e1; margin-bo
ttom: 0.35rem;">Residential
            / Permanent Address</label>
          <textarea id="address" name="address" rows="3" placeholder="House/Flat No., Street, Land
mark, City..."
            style="width: 100%; box-sizing: border-box; background: #0f172a; border: 1px solid #47
5569; color: #ffffff; padding: 0.65rem 0.85rem; border-radius: 6px; font-size: 0.9rem; outline: no
ne; font-family: inherit; resize: vertical;"></textarea>
        </div>
      </fieldset>

      <!-- Fieldset 4: Documents Upload & Declaration -->
      <fieldset
        style="border: 1px solid #475569; border-radius: 10px; padding: 1.5rem; margin-bottom: 2re
m; background: rgba(15, 23, 42, 0.6);">
        <legend style="color: #a78bfa; font-weight: 700; font-size: 1.1rem; padding: 0 0.5rem;">
          📎 Attachments &amp;amp; Declaration
        </legend>

        <div style="margin-top: 0.5rem;">
          <label for="photo"
            style="display: block; font-size: 0.85rem; font-weight: 600; color: #cbd5e1; margin-bo
ttom: 0.35rem;">Upload
            Student Photograph (JPG, PNG)</label>
          <input type="file" id="photo" name="photo" accept="image/*"
            style="width: 100%; box-sizing: border-box; background: #0f172a; border: 1px solid #47
5569; color: #94a3b8; padding: 0.5rem; border-radius: 6px; font-size: 0.85rem;">
```

```
        </div>

        <div style="margin-top: 1.25rem;">
          <label
            style="display: flex; align-items: flex-start; gap: 0.5rem; font-size: 0.85rem; color:
#cbd5e1; cursor: pointer;">
            <input type="checkbox" required style="accent-color: #6366f1; margin-top: 0.2rem;">
            <span>I hereby declare that all information provided in this registration form is auth
entic and accurate to
              the best of my knowledge.</span>
          </label>
        </div>
      </fieldset>

      <!-- Action Buttons (Inline CSS) -->
      <div style="display: flex; gap: 1rem; justify-content: flex-end; flex-wrap: wrap;">
        <button type="reset"
          style="background: rgba(239, 68, 68, 0.15); color: #f87171; border: 1px solid rgba(239,
68, 68, 0.3); padding: 0.75rem 1.5rem; border-radius: 8px; font-size: 0.95rem; font-weight: 600; c
ursor: pointer;">
          ↺ Reset Form
        </button>
        <button type="submit"
          style="background: #6366f1; color: #ffffff; border: 1px solid #6366f1; padding: 0.75rem
2rem; border-radius: 8px; font-size: 0.95rem; font-weight: 700; cursor: pointer; box-shadow: 0 4px
14px rgba(99, 102, 241, 0.4);">
          ✓ Submit Registration
        </button>
      </div>

    </form>

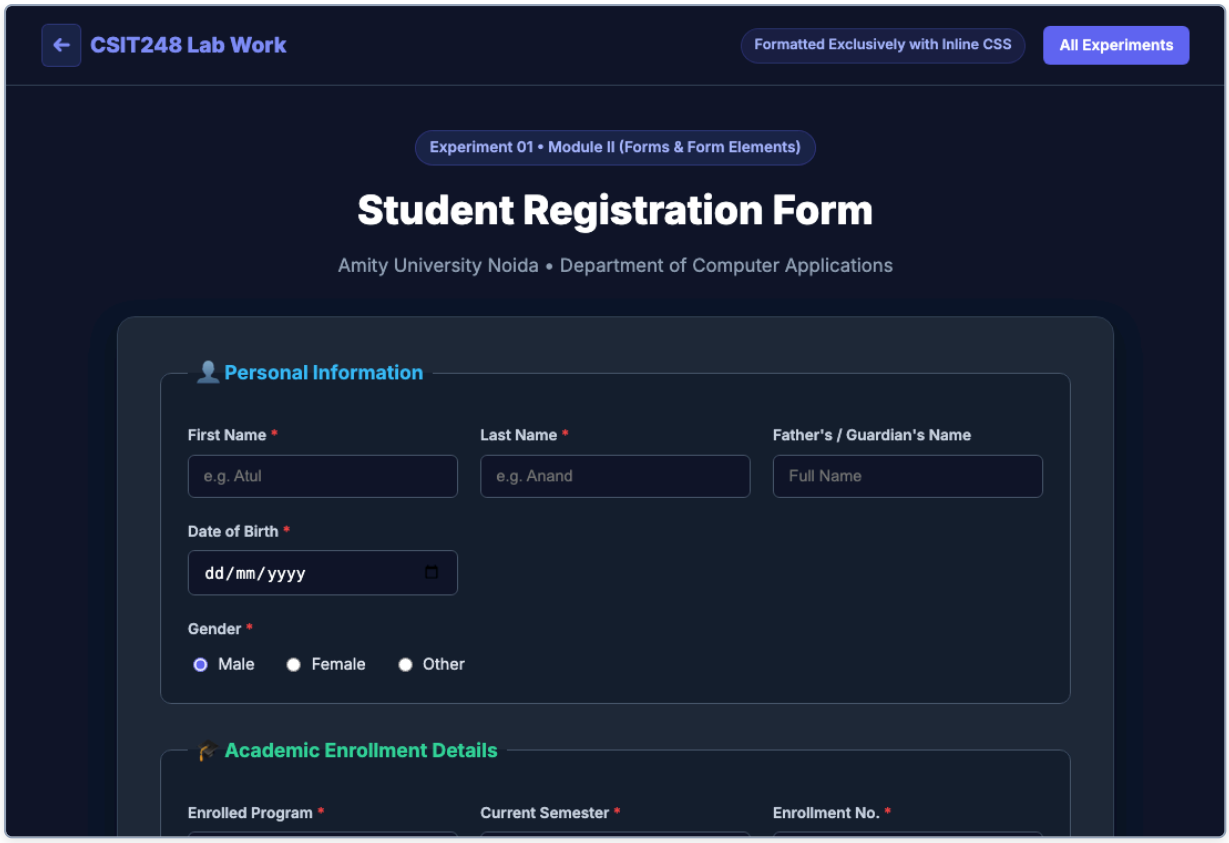
  </main>

  <!-- Footer (Inline CSS) -->
  <footer
    style="background: #090d16; border-top: 1px solid #1e293b; padding: 2rem 1.5rem; text-align: c
enter; color: #94a3b8; font-size: 0.875rem; margin-top: 3rem;">
    <p style="margin: 0 0 0.5rem 0;">Experiment 01 &bull; CSIT248 Advanced Web Technology &bull; S
tudent Registration
      Form</p>
    <p style="margin: 0 0 0.75rem 0; color: #64748b; font-size: 0.8rem;">Amity University Noida &b
ull; Formatted using
      100% Inline CSS</p>
    <p style="margin: 0;"><a href="../index.html"
        style="color: #818cf8; text-decoration: none; font-weight: 600;">&uarr; Back to Lab Work P
ortal</a></p>
  </footer>

</body>

</html>
```

Output :


CSIT248 Lab Work
Formatted Exclusively with Inline CSS
All Experiments
Experiment 01 • Module II (Forms & Form Elements)
Student Registration Form
Amity University Noida • Department of Computer Applications
Personal Information
First Name *
e.g. Atul
Last Name *
e.g. Anand
Father's / Guardian's Name
Full Name
Date of Birth *
dd/mm/yyyy
Gender *
Male
Female
Other
Academic Enrollment Details
Enrolled Program *
Current Semester *
Enrollment No. *

Experiment No.: 5

Title : Personal Introduction Web Page Showcasing Bio, Timeline, and Skills.

Aim : To design a web page to introduce yourself, showcasing personal bio, academic timeline, technical competencies, definition lists, marquee banner, and contact details.

Source Code :

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description"
    content="Experiment 2: Design a web page to introduce yourself - CSIT248 Advanced Web Technolo
gy">
  <title>Experiment 02 - Introduce Yourself (Inline CSS)</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&display=swa
p" rel="stylesheet">
</head>

<body
  style="margin: 0; padding: 0; font-family: 'Inter', system-ui, -apple-system, sans-serif; backgr
ound-color: #0f172a; color: #f8fafc; line-height: 1.7; min-height: 100vh;">

  <!-- ================================================
       NAVIGATION BAR (Formatted via Inline CSS)
       ================================================ -->
  <nav
    style="position: sticky; top: 0; z-index: 100; background: rgba(15, 23, 42, 0.92); backdrop-fi
lter: blur(12px); border-bottom: 1px solid #334155; padding: 1rem 2rem; display: flex; justify-con
tent: space-between; align-items: center; flex-wrap: wrap; gap: 1rem;">
    <a href="../index.html"
      style="font-weight: 800; font-size: 1.2rem; color: #818cf8; text-decoration: none; display:
flex; align-items: center; gap: 0.5rem;">
      <span
        style="background: rgba(99, 102, 241, 0.15); border: 1px solid rgba(99, 102, 241, 0.3); pa
dding: 0.2rem 0.5rem; border-radius: 6px;">&larr;</span>
      CSIT248 Lab Work
    </a>
    <ul
      style="display: flex; list-style: none; gap: 1.25rem; margin: 0; padding: 0; align-items: ce
nter; flex-wrap: wrap;">
      <li><a href="#about" style="color: #94a3b8; text-decoration: none; font-size: 0.9rem; font-w
eight: 500;">About</a>
      </li>
      <li><a href="#education"
          style="color: #94a3b8; text-decoration: none; font-size: 0.9rem; font-weight: 500;">Educ
ation</a></li>
      <li><a href="#skills" style="color: #94a3b8; text-decoration: none; font-size: 0.9rem; font-
weight: 500;">Skills
          &amp; Domains</a></li>
      <li><a href="#projects"
          style="color: #94a3b8; text-decoration: none; font-size: 0.9rem; font-weight: 500;">Proj
ects</a></li>
      <li><a href="#hobbies"
          style="color: #94a3b8; text-decoration: none; font-size: 0.9rem; font-weight: 500;">Hobb
ies</a></li>
      <li><a href="#contact"
```

```
          style="color: #94a3b8; text-decoration: none; font-size: 0.9rem; font-weight: 500;">Cont
act</a></li>
      <li>
        <a href="../index.html"
          style="background: #6366f1; color: #ffffff; padding: 0.35rem 0.85rem; border-radius: 6p
x; text-decoration: none; font-size: 0.85rem; font-weight: 600;">
          All Experiments
        </a>
      </li>
    </ul>
  </nav>

  <!-- ============================================
       MARQUEE TAG (Formatted via Inline CSS)
       ============================================ -->
  <div
    style="background: linear-gradient(90deg, #4338ca, #6366f1, #06b6d4); color: #ffffff; padding:
0.45rem 1rem; font-size: 0.85rem; font-weight: 600;">
    <marquee behavior="scroll" direction="left" scrollamount="6">
      🚀 Welcome to Atul Anand's Portfolio | BCA (Hons.) with Research Student at Amity Universit
y Noida | Exploring Web
      Development, AI/ML, Cybersecurity &amp;amp; Data Analytics!
    </marquee>
  </div>

  <!-- ============================================
       HERO HEADER (Formatted via Inline CSS)
       ============================================ -->
  <header id="about"
    style="padding: 4rem 1.5rem 3rem 1.5rem; text-align: center; background: radial-gradient(circl
e at 50% 25%, rgba(99, 102, 241, 0.18) 0%, rgba(15, 23, 42, 0) 70%); border-bottom: 1px solid #1e2
93b;">
    <div
      style="width: 140px; height: 140px; margin: 0 auto 1.5rem auto; border-radius: 50%; padding:
4px; background: linear-gradient(135deg, #6366f1, #ec4899); box-shadow: 0 10px 25px rgba(99, 102,
241, 0.35);">
      <!-- Real Portrait Photograph -->
      <img src="atul_anand.jpg" alt="Atul Anand"
        style="width: 100%; height: 100%; object-fit: cover; border-radius: 50%; display: block; b
order: 2px solid #0f172a;">
    </div>
    <h1 style="font-size: 2.75rem; font-weight: 800; color: #ffffff; margin: 0 0 0.5rem 0; letter-
spacing: -0.025em;">
      Atul Anand</h1>
    <p style="color: #38bdf8; font-size: 1.25rem; font-weight: 600; margin: 0 0 0.5rem 0;">BCA Stu
dent / Aspiring
      Software Developer</p>
    <p style="color: #cbd5e1; font-size: 1rem; font-weight: 500; margin: 0 0 1rem 0;">BCA (Hons.)
with Research &bull;
      Semester III &bull; Amity University Noida</p>

    <p style="max-width: 720px; margin: 0 auto 1.25rem auto; color: #94a3b8; font-size: 1.05rem; l
ine-height: 1.6;">
      Passionate about building intuitive, accessible, and high-performance digital experiences wh
ile exploring Web
      Development, AI/ML, Cybersecurity, and Data Analytics.
    </p>

    <!-- Focus Badges (Inline CSS) -->
    <div style="display: flex; justify-content: center; gap: 0.6rem; flex-wrap: wrap;">
      <span
        style="font-size: 0.8rem; background: rgba(99, 102, 241, 0.15); color: #a5b4fc; border: 1p
x solid rgba(99, 102, 241, 0.3); padding: 0.25rem 0.75rem; border-radius: 9999px; font-weight: 60
0;">
        Web Development</span>
      <span
```

```
        style="font-size: 0.8rem; background: rgba(6, 182, 212, 0.15); color: #67e8f9; border: 1px
solid rgba(6, 182, 212, 0.3); padding: 0.25rem 0.75rem; border-radius: 9999px; font-weight: 600;">
🤖
        AI &amp;amp; Machine Learning</span>
      <span
        style="font-size: 0.8rem; background: rgba(245, 158, 11, 0.15); color: #fcd34d; border: 1p
x solid rgba(245, 158, 11, 0.3); padding: 0.25rem 0.75rem; border-radius: 9999px; font-weight: 60
0;">🛡️
        Cybersecurity</span>
      <span
        style="font-size: 0.8rem; background: rgba(16, 185, 129, 0.15); color: #6ee7b7; border: 1p
x solid rgba(16, 185, 129, 0.3); padding: 0.25rem 0.75rem; border-radius: 9999px; font-weight: 60
0;">📊
        Data Analytics</span>
    </div>
  </header>

  <!-- ================================================
       MAIN CONTAINER (Formatted via Inline CSS)
       ================================================ -->
  <main style="max-width: 1000px; margin: 0 auto; padding: 2.5rem 1.5rem;">

    <!-- Section 1: Detailed Bio with Text Formatting Tags -->
    <section
      style="background: #1e293b; border: 1px solid #334155; border-radius: 14px; padding: 2rem; m
argin-bottom: 2.5rem; box-shadow: 0 8px 20px rgba(0, 0, 0, 0.2);">
      <h2
        style="font-size: 1.5rem; font-weight: 700; color: #f8fafc; margin: 0 0 1rem 0; border-bot
tom: 2px solid #334155; padding-bottom: 0.5rem;">
        <span style="color: #6366f1;">01.</span> About Me &amp;amp; Profile Focus
      </h2>
      <p style="color: #e2e8f0; font-size: 1rem; margin: 0 0 1rem 0;">
        Hello! I am a dedicated <strong>BCA (Hons.) with Research</strong> student at <em>Amity Un
iversity Noida</em>,
        specializing in <strong style="color: #38bdf8;">Computer Applications</strong>. My profile
focus centers around
        <u style="text-decoration-color: #6366f1;">building practical, user-friendly, and innovati
ve digital
          solutions</u> to solve real-world problems.
      </p>
      <p style="color: #e2e8f0; font-size: 1rem; margin: 0 0 1.25rem 0;">
        Currently in <mark
          style="background: rgba(99, 102, 241, 0.25); color: #c7d2fe; padding: 0.15rem 0.4rem; bo
rder-radius: 4px;">Semester
          III</mark>, actively strengthening my skills in full stack web technologies, intelligent
algorithms, secure
        system design, and data-driven analytical insights.
      </p>
      <hr style="margin: 1.5rem 0; border: 0; border-top: 1px solid #334155;">
      <p style="color: #94a3b8; font-size: 0.9rem; margin: 0;">
        <small><strong style="color: #cbd5e1;">Core Philosophy:</strong> Innovation &amp;times; Practi
cality<sup>2</sup> =
          User Impact</small> &amp;bull;
        <small><strong style="color: #cbd5e1;">Lab Course:</strong> CSIT248 Advanced Web Technolog
y</small>
      </p>
    </section>

    <!-- Section 2: Educational Background (HTML Table with Inline CSS) -->
    <section id="education" style="margin-bottom: 2.5rem;">
      <h2
        style="font-size: 1.5rem; font-weight: 700; color: #f8fafc; margin: 0 0 1rem 0; border-bot
tom: 2px solid #334155; padding-bottom: 0.5rem;">
        <span style="color: #6366f1;">02.</span> Educational Background
      </h2>
      <div
```

```
        style="background: #1e293b; border: 1px solid #334155; border-radius: 14px; overflow-x: au
to; box-shadow: 0 8px 20px rgba(0, 0, 0, 0.2);">
        <table style="width: 100%; border-collapse: collapse; text-align: left; font-size: 0.95re
m;">
          <thead>
            <tr style="background: #1e1b4b; color: #a5b4fc; border-bottom: 1px solid #334155;">
              <th style="padding: 1rem 1.25rem; font-weight: 600;">Degree / Program</th>
              <th style="padding: 1rem 1.25rem; font-weight: 600;">Institution / University</th>
              <th style="padding: 1rem 1.25rem; font-weight: 600;">Field / Discipline</th>
              <th style="padding: 1rem 1.25rem; font-weight: 600;">Academic Timeline</th>
              <th style="padding: 1rem 1.25rem; font-weight: 600;">Current Status</th>
            </tr>
          </thead>
          <tbody>
            <tr style="border-bottom: 1px solid #334155;">
              <td style="padding: 1rem 1.25rem; color: #ffffff; font-weight: 600;">BCA (Hons.) wit
h Research</td>
              <td style="padding: 1rem 1.25rem; color: #cbd5e1;">Amity University Noida</td>
              <td style="padding: 1rem 1.25rem; color: #cbd5e1;">Computer Applications</td>
              <td style="padding: 1rem 1.25rem; color: #cbd5e1;">2025 &ndash; 2029</td>
              <td style="padding: 1rem 1.25rem; color: #34d399; font-weight: 600;">Pursuing (Semes
ter III)</td>
            </tr>
            <tr style="border-bottom: 1px solid #334155; background: rgba(255, 255, 255, 0.015);">
              <td style="padding: 1rem 1.25rem; color: #ffffff; font-weight: 600;">Senior Secondar
y (Class XII)</td>
              <td style="padding: 1rem 1.25rem; color: #cbd5e1;">Shivam School</td>
              <td style="padding: 1rem 1.25rem; color: #cbd5e1;">PCM &amp;amp; Fine-Art</td>
              <td style="padding: 1rem 1.25rem; color: #cbd5e1;">2020 &ndash; 2022</td>
              <td style="padding: 1rem 1.25rem; color: #38bdf8; font-weight: 600;">Completed (Dist
inction)</td>
            </tr>
            <tr>
              <td style="padding: 1rem 1.25rem; color: #ffffff; font-weight: 600;">Secondary Schoo
l (Class X)</td>
              <td style="padding: 1rem 1.25rem; color: #cbd5e1;">DAV Public School</td>
              <td style="padding: 1rem 1.25rem; color: #cbd5e1;">General Sciences &amp;amp; Math</td>
              <td style="padding: 1rem 1.25rem; color: #cbd5e1;">2018 &ndash; 2020</td>
              <td style="padding: 1rem 1.25rem; color: #38bdf8; font-weight: 600;">Completed (Dist
inction)</td>
            </tr>
          </tbody>
        </table>
      </div>
    </section>

    <!-- Section 3: Technical Competencies & Interest Domains (Inline CSS) -->
    <section id="skills" style="margin-bottom: 2.5rem;">
      <h2
        style="font-size: 1.5rem; font-weight: 700; color: #f8fafc; margin: 0 0 1.25rem 0; border-
bottom: 2px solid #334155; padding-bottom: 0.5rem;">
        <span style="color: #6366f1;">03.</span> Technical Competencies &amp;amp; Domains of Interest
      </h2>
      <div style="display: grid; grid-template-columns: repeat(auto-fit, minmax(280px, 1fr)); gap:
1.5rem;">

        <!-- Domain 1: Web Development -->
        <div
          style="background: #1e293b; border: 1px solid #334155; border-radius: 12px; padding: 1.5
rem; box-shadow: 0 6px 16px rgba(0, 0, 0, 0.2);">
          <h3 style="color: #38bdf8; font-size: 1.15rem; margin: 0 0 1rem 0; font-weight: 700;">
🌐 Web Development</h3>
          <ul style="margin: 0; padding-left: 1.25rem; color: #cbd5e1; font-size: 0.9rem; line-hei
ght: 1.8;">
            <li>Semantic HTML5, Image Mapping &amp;amp; Canvas</li>
            <li>CSS3 (Flexbox, Grid, Inline Styling, Animations)</li>
```

```
            <li>JavaScript (ES6+, DOM Manipulation &amp; Validation)</li>
            <li>PHP &amp; MySQL Relational Database Integration</li>
          </ul>
        </div>

        <!-- Domain 2: AI / ML & Data Analytics -->
        <div
          style="background: #1e293b; border: 1px solid #334155; border-radius: 12px; padding: 1.5
rem; box-shadow: 0 6px 16px rgba(0, 0, 0, 0.2);">
          <h3 style="color: #34d399; font-size: 1.15rem; margin: 0 0 1rem 0; font-weight: 700;">
🤖 AI/ML &amp; Data
            Analytics</h3>
          <ul style="margin: 0; padding-left: 1.25rem; color: #cbd5e1; font-size: 0.9rem; line-hei
ght: 1.8;">
            <li>Python Programming &amp; Algorithmic Foundations</li>
            <li>Machine Learning Core Concepts &amp; Modeling</li>
            <li>Data Analysis (Pandas, NumPy, Data Cleaning)</li>
            <li>Data Visualization &amp; Exploratory Data Analysis</li>
          </ul>
        </div>

        <!-- Domain 3: Cybersecurity & Tools -->
        <div
          style="background: #1e293b; border: 1px solid #334155; border-radius: 12px; padding: 1.5
rem; box-shadow: 0 6px 16px rgba(0, 0, 0, 0.2);">
          <h3 style="color: #fbbf24; font-size: 1.15rem; margin: 0 0 1rem 0; font-weight: 700;">
🛡️ Cybersecurity &amp;
            Tools</h3>
          <ol style="margin: 0; padding-left: 1.25rem; color: #cbd5e1; font-size: 0.9rem; line-hei
ght: 1.8;">
            <li>Web Security Principles &amp; Input Sanitization</li>
            <li>Network Protocols &amp; Cryptography Basics</li>
            <li>Git &amp; GitHub Version Control Workflows</li>
            <li>Linux Terminal &amp; VS Code Development</li>
          </ol>
        </div>

      </div>
    </section>

    <!-- Section 4: Definition List (Inline CSS) -->
    <section style="margin-bottom: 2.5rem;">
      <h2
        style="font-size: 1.5rem; font-weight: 700; color: #f8fafc; margin: 0 0 1.25rem 0; border-
bottom: 2px solid #334155; padding-bottom: 0.5rem;">
        <span style="color: #6366f1;">04.</span> Personal Attributes &amp; Strengths
      </h2>
      <dl
        style="display: grid; grid-template-columns: 180px 1fr; gap: 1rem 1.5rem; background: #1e2
93b; padding: 1.75rem; border-radius: 14px; border: 1px solid #334155; margin: 0; box-shadow: 0 8p
x 20px rgba(0, 0, 0, 0.2);">
        <dt style="font-weight: 700; color: #38bdf8; font-size: 0.95rem;">Solution Builder</dt>
        <dd style="margin: 0; color: #cbd5e1; font-size: 0.95rem;">Focus on engineering practical,
user-friendly, and
          accessible software solutions that address tangible challenges.</dd>

        <dt style="font-weight: 700; color: #38bdf8; font-size: 0.95rem;">Research-Oriented</dt>
        <dd style="margin: 0; color: #cbd5e1; font-size: 0.95rem;">Eager to explore emerging techn
ologies in Artificial
          Intelligence, predictive data modeling, and cybersecurity.</dd>

        <dt style="font-weight: 700; color: #38bdf8; font-size: 0.95rem;">Collaborative</dt>
        <dd style="margin: 0; color: #cbd5e1; font-size: 0.95rem;">Strong communicator in peer pro
gramming, technical
          writing, code reviews, and structured academic coursework.</dd>
```

```
        <dt style="font-weight: 700; color: #38bdf8; font-size: 0.95rem;">Design Mindset</dt>
        <dd style="margin: 0; color: #cbd5e1; font-size: 0.95rem;">Prioritizes high visual aesthet
ics, responsive
          layouts, intuitive UX flows, and robust error handling.</dd>
      </dl>
    </section>

    <!-- Section 5: Projects Showcase (Inline CSS) -->
    <section id="projects" style="margin-bottom: 2.5rem;">
      <h2
        style="font-size: 1.5rem; font-weight: 700; color: #f8fafc; margin: 0 0 1.25rem 0; border-
bottom: 2px solid #334155; padding-bottom: 0.5rem;">
        <span style="color: #6366f1;">05.</span> Featured Academic Projects
      </h2>
      <div style="display: grid; grid-template-columns: repeat(auto-fit, minmax(280px, 1fr)); gap:
1.5rem;">

        <!-- Project 1 -->
        <article
          style="background: #1e293b; border: 1px solid #334155; border-radius: 12px; overflow: hi
dden; box-shadow: 0 6px 16px rgba(0, 0, 0, 0.2);">
          <div
            style="height: 120px; background: linear-gradient(135deg, #1e1b4b, #312e81); display:
flex; align-items: center; justify-content: center; color: #a5b4fc;">
            <svg width="44" height="44" viewBox="0 0 24 24" fill="none" stroke="currentColor" stro
ke-width="1.5">
              <rect x="2" y="3" width="20" height="14" rx="2" ry="2"></rect>
              <line x1="8" y1="21" x2="16" y2="21"></line>
              <line x1="12" y1="17" x2="12" y2="21"></line>
            </svg>
          </div>
          <div style="padding: 1.25rem;">
            <h4 style="font-size: 1.1rem; color: #ffffff; margin: 0 0 0.5rem 0;">Interactive Web P
ortal</h4>
            <p style="color: #94a3b8; font-size: 0.875rem; margin: 0 0 1rem 0;">A multi-page respo
nsive web platform
              demonstrating semantic HTML5, CSS animations, and front-level validation.</p>
            <a href="../index.html"
              style="color: #38bdf8; font-size: 0.85rem; font-weight: 600; text-decoration: non
e;">View Portfolio Hub
              &rarr;</a>
          </div>
        </article>

        <!-- Project 2 -->
        <article
          style="background: #1e293b; border: 1px solid #334155; border-radius: 12px; overflow: hi
dden; box-shadow: 0 6px 16px rgba(0, 0, 0, 0.2);">
          <div
            style="height: 120px; background: linear-gradient(135deg, #064e3b, #047857); display:
flex; align-items: center; justify-content: center; color: #6ee7b7;">
            <svg width="44" height="44" viewBox="0 0 24 24" fill="none" stroke="currentColor" stro
ke-width="1.5">
              <path d="M6 2L3 6v14a2 2 0 0 0 2 2h14a2 2 0 0 0 2-2V6l-3-4z"></path>
              <line x1="3" y1="6" x2="21" y2="6"></line>
              <path d="M16 10a4 4 0 0 1-8 0"></path>
            </svg>
          </div>
          <div style="padding: 1.25rem;">
            <h4 style="font-size: 1.1rem; color: #ffffff; margin: 0 0 0.5rem 0;">E-Commerce Catalo
g System</h4>
            <p style="color: #94a3b8; font-size: 0.875rem; margin: 0 0 1rem 0;">Tabular inventory
catalog for computer
              peripherals with specifications, stock indicators, and price calculation logic.</p>
            <a href="../experiment-11/index.html"
              style="color: #34d399; font-size: 0.85rem; font-weight: 600; text-decoration: non
```

```
e;">Open Catalog
              &rarr;</a>
          </div>
        </article>

        <!-- Project 3 -->
        <article
          style="background: #1e293b; border: 1px solid #334155; border-radius: 12px; overflow: hi
dden; box-shadow: 0 6px 16px rgba(0, 0, 0, 0.2);">
          <div
            style="height: 120px; background: linear-gradient(135deg, #701a75, #a21caf); display:
flex; align-items: center; justify-content: center; color: #f472b6;">
            <svg width="44" height="44" viewBox="0 0 24 24" fill="none" stroke="currentColor" stro
ke-width="1.5">
              <polygon points="1 6 1 22 8 18 16 22 23 18 23 2 16 6 8 2 1 6"></polygon>
              <line x1="8" y1="2" x2="8" y2="18"></line>
              <line x1="16" y1="6" x2="16" y2="22"></line>
            </svg>
          </div>
          <div style="padding: 1.25rem;">
            <h4 style="font-size: 1.1rem; color: #ffffff; margin: 0 0 0.5rem 0;">Client Image Mapp
ing Demo</h4>
            <p style="color: #94a3b8; font-size: 0.875rem; margin: 0 0 1rem 0;">Hotspot coordinate
mapping interface for
              interactive graphic exploration with rectangular, circular, and polygonal regions.</
p>
            <a href="../experiment-4/index.html"
              style="color: #f472b6; font-size: 0.85rem; font-weight: 600; text-decoration: non
e;">Explore Mapping
              &rarr;</a>
          </div>
        </article>

      </div>
    </section>

    <!-- Section 6: Hobbies (Inline CSS) -->
    <section id="hobbies"
      style="background: #1e293b; border: 1px solid #334155; border-radius: 14px; padding: 2rem; m
argin-bottom: 2.5rem; box-shadow: 0 8px 20px rgba(0, 0, 0, 0.2);">
      <h2
        style="font-size: 1.5rem; font-weight: 700; color: #f8fafc; margin: 0 0 1rem 0; border-bot
tom: 2px solid #334155; padding-bottom: 0.5rem;">
        <span style="color: #6366f1;">06.</span> Hobbies &amp;amp; Extracurricular Interests
      </h2>
      <ul style="margin: 0; padding-left: 1.25rem; color: #cbd5e1; font-size: 0.95rem; line-heigh
t: 2;">
        <li>💻 <strong style="color: #ffffff;">Tech Exploration:</strong> Experimenting with new
web tools, AI
          libraries, and open-source software.</li>
        <li>♟️ <strong style="color: #ffffff;">Competitive Chess:</strong> Enjoy rapid and tactic
al chess games to
          sharpen analytical problem-solving.</li>
        <li>📷 <strong style="color: #ffffff;">Photography:</strong> Capturing urban landscape sh
ots, architectures, and
          natural lighting.</li>
        <li>🏃 <strong style="color: #ffffff;">Fitness &amp;amp; Running:</strong> Regular workout ro
utines and distance
          running for physical wellness.</li>
      </ul>
    </section>

    <!-- Section 7: Contact & Address Tag (Inline CSS) -->
    <section id="contact"
      style="background: #1e293b; border: 1px solid #334155; border-radius: 14px; padding: 2rem; m
argin-bottom: 2rem; box-shadow: 0 8px 20px rgba(0, 0, 0, 0.2);">
```

```
      <h2
        style="font-size: 1.5rem; font-weight: 700; color: #f8fafc; margin: 0 0 1rem 0; border-bot
tom: 2px solid #334155; padding-bottom: 0.5rem;">
        <span style="color: #6366f1;">07.</span> Contact &amp;amp; Connect
      </h2>
      <p style="color: #94a3b8; font-size: 0.95rem; margin: 0 0 1.25rem 0;">Feel free to connect f
or academic
        collaboration, project discussions, or tech innovations:</p>

      <!-- Demonstrating <address> tag with Inline CSS -->
      <address style="font-style: normal; line-height: 1.9; color: #cbd5e1; font-size: 0.95rem; ma
rgin-bottom: 1.5rem;">
        <strong style="color: #ffffff; font-size: 1.1rem;">Atul Anand</strong><br>
        BCA (Hons.) with Research &bull; Semester III<br>
        Department of Computer Applications<br>
        Amity University Noida, Uttar Pradesh, India
      </address>

      <!-- Connect Action Buttons (Inline CSS) -->
      <div style="display: flex; gap: 0.85rem; flex-wrap: wrap;">

        <!-- LinkedIn -->
        <a href="https://www.linkedin.com" target="_blank" rel="noopener noreferrer"
          style="display: flex; align-items: center; gap: 0.5rem; background: #0077b5; color: #fff
fff; text-decoration: none; padding: 0.65rem 1.1rem; border-radius: 8px; font-weight: 600; font-si
ze: 0.875rem; box-shadow: 0 4px 12px rgba(0, 119, 181, 0.3);">
          <svg width="18" height="18" fill="currentColor" viewBox="0 0 24 24">
            <path
              d="M19 3a2 2 0 0 1 2 2v14a2 2 0 0 1-2 2H5a2 2 0 0 1-2-2V5a2 2 0 0 1 2-2h14m-.5 15.5v
-5.3a3.26 3.26 0 0 0-3.26-3.26c-.85 0-1.84.52-2.28 1.3v-1.11h-2.79v8.37h2.79v-4.93c0-.77.62-1.4 1.
39-1.4a1.4 1.4 0 0 1 1.4 1.4v4.93h2.75M6.46 10.9v8.37H9.2V10.9H6.46M7.83 6.2a1.66 1.66 0 0 0-1.67
1.66c0 .92.75 1.67 1.67 1.67s1.66-.75 1.66-1.67A1.66 1.66 0 0 0 7.83 6.2z" />
          </svg>
          LinkedIn
        </a>

        <!-- GitHub -->
        <a href="https://github.com" target="_blank" rel="noopener noreferrer"
          style="display: flex; align-items: center; gap: 0.5rem; background: #0f172a; color: #fff
fff; border: 1px solid #475569; text-decoration: none; padding: 0.65rem 1.1rem; border-radius: 8p
x; font-weight: 600; font-size: 0.875rem; box-shadow: 0 4px 12px rgba(0, 0, 0, 0.3);">
          <svg width="18" height="18" fill="currentColor" viewBox="0 0 24 24">
            <path fill-rule="evenodd" clip-rule="evenodd"
              d="M12 2C6.477 2 2 6.484 2 12.017c0 4.425 2.865 8.18 6.839 9.504.5.092.682-.217.682
-.483 0-.237-.008-.868-.013-1.703-2.782.605-3.369-1.343-3.369-1.343-.454-1.158-1.11-1.466-1.11-1.4
66-.908-.62.069-.608.069-.608 1.003.07 1.53 1.032 1.53 1.032.892 1.53 2.341 1.088 2.91.832.092-.64
7.35-1.088.636-1.338-2.22-.253-4.555-1.113-4.555-4.951 0-1.093.39-1.988 1.029-2.688-.103-.253-.446
-1.272.098-2.65 0 0 .84-.27 2.75 1.026A9.564 9.564 0 0112 6.844c.85.004 1.705.115 2.504.337 1.909-
1.296 2.747-1.027 2.747-1.027.546 1.379.202 2.398.1 2.651.64.7 1.028 1.595 1.028 2.688 0 3.848-2.3
39 4.695-4.566 4.943.359.309.678.92.678 1.855 0 1.338-.012 2.419-.012 2.747 0 .268.18.58.688.482A1
0.019 10.019 0 0022 12.017C22 6.484 17.522 2 12 2z" />
          </svg>
          GitHub
        </a>

        <!-- Open Gmail Directly -->
        <a href="https://mail.google.com/mail/?view=cm&fs=1&to=atulll0923@gmail.com" target="_blan
k"
          rel="noopener noreferrer"
          style="display: flex; align-items: center; gap: 0.5rem; background: linear-gradient(135d
eg, #ea4335, #c5221f); color: #ffffff; text-decoration: none; padding: 0.65rem 1.1rem; border-radi
us: 8px; font-weight: 600; font-size: 0.875rem; box-shadow: 0 4px 14px rgba(234, 67, 53, 0.35);">
          <svg width="18" height="18" fill="none" stroke="currentColor" stroke-width="2" viewBox
="0 0 24 24">
            <path d="M4 4h16c1.1 0 2 .9 2 2v12c0 1.1-.9 2-2 2H4c-1.1 0-2-.9-2-2V6c0-1.1.9-2 2-2z">
</path>
```

```
            <polyline points="22,6 12,13 2,6"></polyline>
          </svg>
          Open Gmail
        </a>

        <!-- Copy Email Button -->
        <button onclick="copyEmail(this)" type="button"
          style="display: flex; align-items: center; gap: 0.5rem; background: #334155; color: #f8f
afc; border: 1px solid #64748b; padding: 0.65rem 1.1rem; border-radius: 8px; font-weight: 600; fon
t-size: 0.875rem; cursor: pointer;">
          📋 <span id="copy-btn-text">Copy Email</span>
        </button>

      </div>
    </section>

  </main>

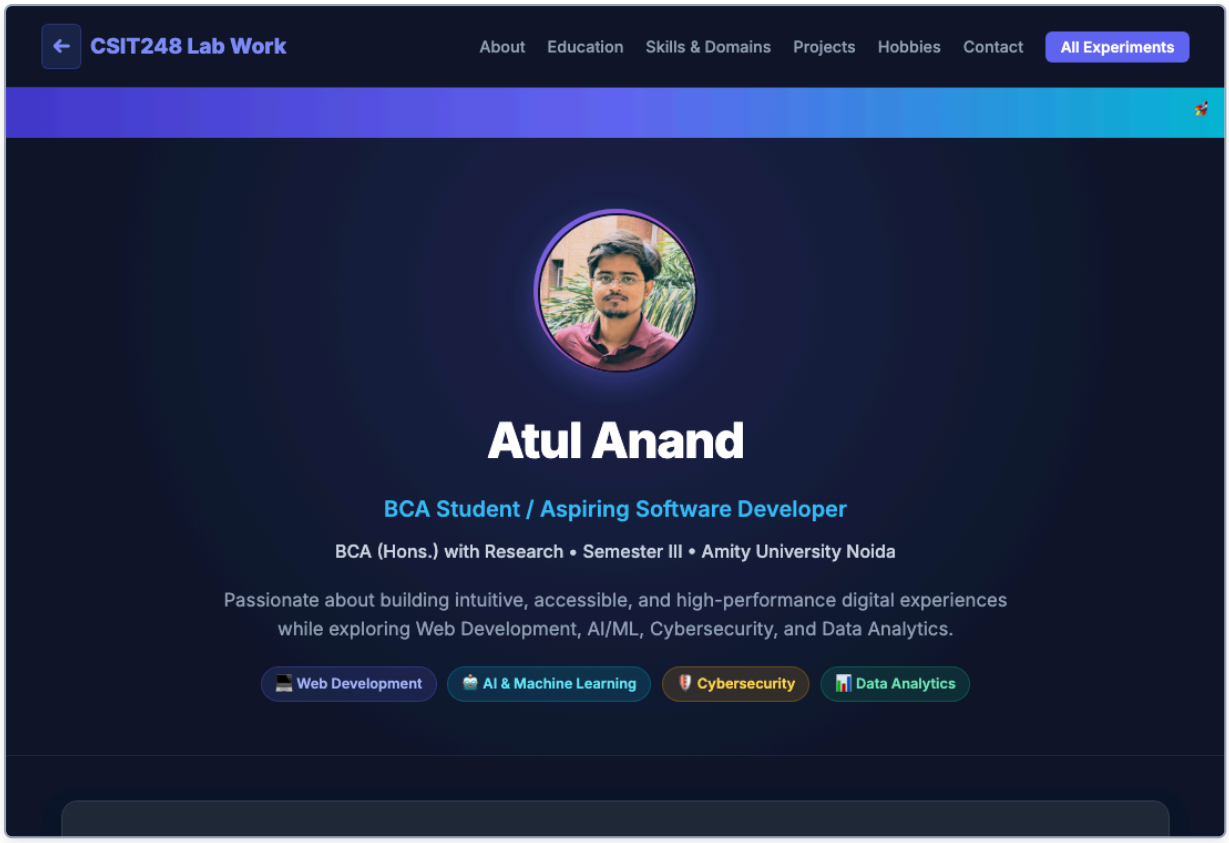
  <!-- JavaScript to Copy Email -->
  <script>
    function copyEmail(btn) {
      const email = 'atulll0923@gmail.com';
      navigator.clipboard.writeText(email).then(() => {
        const textSpan = btn.querySelector('#copy-btn-text');
        textSpan.textContent = 'Copied! ✅';
        btn.style.background = '#10b981';
        btn.style.borderColor = '#34d399';
        setTimeout(() => {
          textSpan.textContent = 'Copy Email';
          btn.style.background = '#334155';
          btn.style.borderColor = '#64748b';
        }, 2500);
      }).catch(err => {
        prompt('Copy email manually:', email);
      });
    }
  </script>

  <!-- ============================================
       FOOTER (Formatted via Inline CSS)
       ============================================ -->
  <footer
    style="background: #090d16; border-top: 1px solid #1e293b; padding: 2rem 1.5rem; text-align: c
enter; color: #94a3b8; font-size: 0.875rem;">
    <p style="margin: 0 0 0.5rem 0;">Experiment 02 &bull; CSIT248 Advanced Web Technology &bull; D
esigned by Atul Anand
      (BCA Hons. with Research)</p>
    <p style="margin: 0 0 0.75rem 0; color: #64748b; font-size: 0.8rem;">Amity University Noida &b
ull; Formatted using
      100% Inline CSS</p>
    <p style="margin: 0;"><a href="../index.html"
        style="color: #818cf8; text-decoration: none; font-weight: 600;">&uarr; Back to Lab Work P
ortal</a></p>
  </footer>

</body>

</html>
```

Output :


CSIT248 Lab Work
About
Education
Skills & Domains
Projects
Hobbies
Contact
All Experiments
Atul Anand
BCA Student / Aspiring Software Developer
BCA (Hons.) with Research • Semester III • Amity University Noida
Passionate about building intuitive, accessible, and high-performance digital experiences while exploring Web Development, AI/ML, Cybersecurity, and Data Analytics.
Web Development
AI & Machine Learning
Cybersecurity
Data Analytics

Experiment No.: 6

Title : Weekly Academic Timetable Matrix with Colspan and Rowspan.

Aim : To construct a weekly semester timetable matrix using HTML table elements, cell padding, cell spacing, and column/row spans to represent overlapping lecture slots and laboratory practicals.

Source Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Experiment 3: Design a web page to display timetable of curren
t semester - CSIT248 Advanced Web Technology">
  <title>Experiment 03 - Current Semester Timetable (Inline CSS)</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&display=swa
p" rel="stylesheet">
</head>
<body style="margin: 0; padding: 0; font-family: 'Inter', system-ui, -apple-system, sans-serif; ba
ckground-color: #0f172a; color: #f8fafc; line-height: 1.6; min-height: 100vh;">

  <!-- Navigation (Inline CSS) -->
  <nav style="position: sticky; top: 0; z-index: 100; background: rgba(15, 23, 42, 0.92); backdrop
-filter: blur(12px); border-bottom: 1px solid #334155; padding: 1rem 2rem; display: flex; justify-
content: space-between; align-items: center;">
    <a href="../index.html" style="font-weight: 800; font-size: 1.2rem; color: #818cf8; text-decor
ation: none; display: flex; align-items: center; gap: 0.5rem;">
      <span style="background: rgba(99, 102, 241, 0.15); border: 1px solid rgba(99, 102, 241, 0.
3); padding: 0.2rem 0.5rem; border-radius: 6px;">&larr;</span>
      CSIT248 Lab Work
    </a>
    <div style="display: flex; gap: 1rem; align-items: center;">
      <span style="font-size: 0.8rem; background: rgba(99, 102, 241, 0.12); color: #a5b4fc; borde
r: 1px solid rgba(99, 102, 241, 0.25); padding: 0.3rem 0.75rem; border-radius: 9999px; font-weigh
t: 600;">
        Formatted Exclusively with Inline CSS
      </span>
      <a href="../index.html" style="background: #6366f1; color: #ffffff; padding: 0.4rem 0.9rem;
border-radius: 6px; text-decoration: none; font-size: 0.85rem; font-weight: 600;">
        All Experiments
      </a>
    </div>
  </nav>

  <main style="max-width: 1300px; margin: 0 auto; padding: 2.5rem 1.5rem;">

    <!-- Header (Inline CSS) -->
    <header style="text-align: center; margin-bottom: 2rem;">
      <h1 style="font-size: 2.25rem; font-weight: 800; color: #ffffff; margin: 0 0 0.5rem 0;">Depa
rtment of Computer Applications</h1>
      <p style="color: #94a3b8; font-size: 1.05rem; margin: 0 0 1rem 0;">Amity University Noida &b
ull; Academic Year 2025&ndash;2026</p>

      <div style="display: flex; justify-content: center; gap: 0.75rem; flex-wrap: wrap;">
        <span style="background: rgba(99, 102, 241, 0.15); color: #a5b4fc; border: 1px solid rgba
(99, 102, 241, 0.3); padding: 0.25rem 0.75rem; border-radius: 9999px; font-size: 0.85rem; font-wei
ght: 600;">Program: BCA (Hons.) with Research</span>
        <span style="background: rgba(99, 102, 241, 0.15); color: #a5b4fc; border: 1px solid rgba
```

```
(99, 102, 241, 0.3); padding: 0.25rem 0.75rem; border-radius: 9999px; font-size: 0.85rem; font-wei
ght: 600;">Semester: III</span>
        <span style="background: rgba(99, 102, 241, 0.15); color: #a5b4fc; border: 1px solid rgba
(99, 102, 241, 0.3); padding: 0.25rem 0.75rem; border-radius: 9999px; font-size: 0.85rem; font-wei
ght: 600;">Classroom: Room 408 (Academic Block)</span>
        <span style="background: rgba(99, 102, 241, 0.15); color: #a5b4fc; border: 1px solid rgba
(99, 102, 241, 0.3); padding: 0.25rem 0.75rem; border-radius: 9999px; font-size: 0.85rem; font-wei
ght: 600;">Total Credits: 24</span>
      </div>
    </header>

    <!-- HTML Timetable Table (Inline CSS) -->
    <div style="background: #1e293b; border: 1px solid #334155; border-radius: 14px; padding: 1.5r
em; box-shadow: 0 10px 25px rgba(0, 0, 0, 0.3); overflow-x: auto; margin-bottom: 2.5rem;">
      <table border="1" cellpadding="8" cellspacing="0" style="width: 100%; border-collapse: colla
pse; text-align: center; min-width: 950px; border-color: #334155;">
        <caption style="font-size: 1.25rem; font-weight: 700; color: #f8fafc; padding-bottom: 1re
m; text-align: left;">
          Weekly Academic Schedule &amp;amp; Lab Allocations (Table Structure)
        </caption>
        <thead>
          <tr style="background: #1e1b4b; color: #e0e7ff;">
            <th rowspan="2" style="width: 110px; padding: 0.75rem 0.5rem; border: 1px solid #47556
9; font-weight: 700;">Day / Time</th>
            <th style="padding: 0.75rem 0.5rem; border: 1px solid #475569; font-weight: 700;">
              Period I
              <span style="font-size: 0.75rem; font-weight: 500; color: #a5b4fc; display: block; m
argin-top: 0.2rem;">09:00 - 10:00</span>
            </th>
            <th style="padding: 0.75rem 0.5rem; border: 1px solid #475569; font-weight: 700;">
              Period II
              <span style="font-size: 0.75rem; font-weight: 500; color: #a5b4fc; display: block; m
argin-top: 0.2rem;">10:00 - 11:00</span>
            </th>
            <th style="padding: 0.75rem 0.5rem; border: 1px solid #475569; font-weight: 700;">
              Period III
              <span style="font-size: 0.75rem; font-weight: 500; color: #a5b4fc; display: block; m
argin-top: 0.2rem;">11:00 - 12:00</span>
            </th>
            <th style="padding: 0.75rem 0.5rem; border: 1px solid #475569; font-weight: 700;">
              Period IV
              <span style="font-size: 0.75rem; font-weight: 500; color: #a5b4fc; display: block; m
argin-top: 0.2rem;">12:00 - 01:00</span>
            </th>
            <th rowspan="2" style="width: 60px; padding: 0.75rem 0.5rem; border: 1px solid #47556
9; font-weight: 700;">
              Lunch
              <span style="font-size: 0.75rem; font-weight: 500; color: #a5b4fc; display: block; m
argin-top: 0.2rem;">01:00 - 02:00</span>
            </th>
            <th style="padding: 0.75rem 0.5rem; border: 1px solid #475569; font-weight: 700;">
              Period V
              <span style="font-size: 0.75rem; font-weight: 500; color: #a5b4fc; display: block; m
argin-top: 0.2rem;">02:00 - 03:00</span>
            </th>
            <th style="padding: 0.75rem 0.5rem; border: 1px solid #475569; font-weight: 700;">
              Period VI
              <span style="font-size: 0.75rem; font-weight: 500; color: #a5b4fc; display: block; m
argin-top: 0.2rem;">03:00 - 04:00</span>
            </th>
            <th style="padding: 0.75rem 0.5rem; border: 1px solid #475569; font-weight: 700;">
              Period VII
              <span style="font-size: 0.75rem; font-weight: 500; color: #a5b4fc; display: block; m
argin-top: 0.2rem;">04:00 - 05:00</span>
            </th>
          </tr>
```

```
          <tr style="background: #1e1b4b; color: #cbd5e1; font-size: 0.8rem;">
            <th style="border: 1px solid #475569; padding: 0.4rem;">Slot 1</th>
            <th style="border: 1px solid #475569; padding: 0.4rem;">Slot 2</th>
            <th style="border: 1px solid #475569; padding: 0.4rem;">Slot 3</th>
            <th style="border: 1px solid #475569; padding: 0.4rem;">Slot 4</th>
            <th style="border: 1px solid #475569; padding: 0.4rem;">Slot 5</th>
            <th style="border: 1px solid #475569; padding: 0.4rem;">Slot 6</th>
            <th style="border: 1px solid #475569; padding: 0.4rem;">Slot 7</th>
          </tr>
        </thead>
        <tbody>
          <!-- MONDAY -->
          <tr>
            <th style="background: #1e293b; color: #38bdf8; font-weight: 700; border: 1px solid #4
75569; padding: 0.75rem 0.5rem;">MONDAY</th>
            <td colspan="2" style="border: 1px solid #475569; padding: 0.5rem;">
              <div style="background: rgba(6, 182, 212, 0.25); color: #67e8f9; border: 1px solid r
gba(6, 182, 212, 0.5); padding: 0.4rem 0.3rem; border-radius: 6px; font-weight: 600;">
                <span style="font-size: 0.85rem; font-weight: 800; display: block;">CSIT244 (Lab)
</span>
                <span style="font-size: 0.75rem; opacity: 0.85; display: block;">Database Systems
Lab &bull; Lab 204</span>
                <span style="font-size: 0.7rem; text-transform: uppercase; letter-spacing: 0.05em;
display: block;">Practical (2 hrs)</span>
              </div>
            </td>
            <td style="border: 1px solid #475569; padding: 0.5rem;">
              <div style="background: rgba(59, 130, 246, 0.2); color: #93c5fd; border: 1px solid r
gba(59, 130, 246, 0.4); padding: 0.4rem 0.3rem; border-radius: 6px; font-weight: 600;">
                <span style="font-size: 0.85rem; font-weight: 800; display: block;">CSIT248</span>
                <span style="font-size: 0.75rem; opacity: 0.85; display: block;">Adv. Web Tech &bu
ll; R-408</span>
                <span style="font-size: 0.7rem; text-transform: uppercase; display: block;">Lectur
e</span>
              </div>
            </td>
            <td style="border: 1px solid #475569; padding: 0.5rem;">
              <div style="background: rgba(139, 92, 246, 0.2); color: #c4b5fd; border: 1px solid r
gba(139, 92, 246, 0.4); padding: 0.4rem 0.3rem; border-radius: 6px; font-weight: 600;">
                <span style="font-size: 0.85rem; font-weight: 800; display: block;">MATH302</span>
                <span style="font-size: 0.75rem; opacity: 0.85; display: block;">Discrete Math &bu
ll; R-408</span>
                <span style="font-size: 0.7rem; text-transform: uppercase; display: block;">Lectur
e</span>
              </div>
            </td>
            <!-- LUNCH BREAK ROWSPAN: Spanning across all 5 Days (Mon-Fri) -->
            <td rowspan="5" style="background: #0f172a; color: #94a3b8; font-weight: 800; letter-s
pacing: 0.25em; writing-mode: vertical-lr; text-transform: uppercase; font-size: 0.85rem; padding:
1rem 0.5rem; border: 1px solid #475569;">
              LUNCH BREAK (1:00 PM - 2:00 PM)
            </td>
            <td style="border: 1px solid #475569; padding: 0.5rem;">
              <div style="background: rgba(236, 72, 153, 0.2); color: #f9a8d4; border: 1px solid r
gba(236, 72, 153, 0.4); padding: 0.4rem 0.3rem; border-radius: 6px; font-weight: 600;">
                <span style="font-size: 0.85rem; font-weight: 800; display: block;">CSIT242</span>
                <span style="font-size: 0.75rem; opacity: 0.85; display: block;">Computer Networks
&bull; R-408</span>
                <span style="font-size: 0.7rem; text-transform: uppercase; display: block;">Lectur
e</span>
              </div>
            </td>
            <td style="border: 1px solid #475569; padding: 0.5rem;">
              <div style="background: rgba(245, 158, 11, 0.2); color: #fcd34d; border: 1px solid r
gba(245, 158, 11, 0.4); padding: 0.4rem 0.3rem; border-radius: 6px; font-weight: 600;">
                <span style="font-size: 0.85rem; font-weight: 800; display: block;">CSIT240</span>
```

```
                <span style="font-size: 0.75rem; opacity: 0.85; display: block;">Operating Systems
&bull; R-408</span>
                <span style="font-size: 0.7rem; text-transform: uppercase; display: block;">Lectur
e</span>
              </div>
            </td>
            <td style="border: 1px solid #475569; padding: 0.5rem;">
              <div style="background: rgba(255, 255, 255, 0.04); color: #94a3b8; border: 1px solid
#334155; padding: 0.4rem 0.3rem; border-radius: 6px; font-weight: 600;">
                <span style="font-size: 0.85rem; font-weight: 800; display: block;">Library / Ment
or</span>
                <span style="font-size: 0.75rem; opacity: 0.85; display: block;">Central Library</
span>
                <span style="font-size: 0.7rem; text-transform: uppercase; display: block;">Self S
tudy</span>
              </div>
            </td>
          </tr>

          <!-- TUESDAY -->
          <tr>
            <th style="background: #1e293b; color: #38bdf8; font-weight: 700; border: 1px solid #4
75569; padding: 0.75rem 0.5rem;">TUESDAY</th>
            <td style="border: 1px solid #475569; padding: 0.5rem;">
              <div style="background: rgba(245, 158, 11, 0.2); color: #fcd34d; border: 1px solid r
gba(245, 158, 11, 0.4); padding: 0.4rem 0.3rem; border-radius: 6px; font-weight: 600;">
                <span style="font-size: 0.85rem; font-weight: 800; display: block;">CSIT240</span>
                <span style="font-size: 0.75rem; opacity: 0.85; display: block;">Operating Systems
&bull; R-408</span>
                <span style="font-size: 0.7rem; text-transform: uppercase; display: block;">Lectur
e</span>
              </div>
            </td>
            <td style="border: 1px solid #475569; padding: 0.5rem;">
              <div style="background: rgba(59, 130, 246, 0.2); color: #93c5fd; border: 1px solid r
gba(59, 130, 246, 0.4); padding: 0.4rem 0.3rem; border-radius: 6px; font-weight: 600;">
                <span style="font-size: 0.85rem; font-weight: 800; display: block;">CSIT248</span>
                <span style="font-size: 0.75rem; opacity: 0.85; display: block;">Adv. Web Tech &bu
ll; R-408</span>
                <span style="font-size: 0.7rem; text-transform: uppercase; display: block;">Lectur
e</span>
              </div>
            </td>
            <td style="border: 1px solid #475569; padding: 0.5rem;">
              <div style="background: rgba(236, 72, 153, 0.2); color: #f9a8d4; border: 1px solid r
gba(236, 72, 153, 0.4); padding: 0.4rem 0.3rem; border-radius: 6px; font-weight: 600;">
                <span style="font-size: 0.85rem; font-weight: 800; display: block;">CSIT242</span>
                <span style="font-size: 0.75rem; opacity: 0.85; display: block;">Computer Networks
&bull; R-408</span>
                <span style="font-size: 0.7rem; text-transform: uppercase; display: block;">Lectur
e</span>
              </div>
            </td>
            <td style="border: 1px solid #475569; padding: 0.5rem;">
              <div style="background: rgba(16, 185, 129, 0.2); color: #6ee7b7; border: 1px solid r
gba(16, 185, 129, 0.4); padding: 0.4rem 0.3rem; border-radius: 6px; font-weight: 600;">
                <span style="font-size: 0.85rem; font-weight: 800; display: block;">CSIT244</span>
                <span style="font-size: 0.75rem; opacity: 0.85; display: block;">DBMS Theory &bul
l; R-408</span>
                <span style="font-size: 0.7rem; text-transform: uppercase; display: block;">Lectur
e</span>
              </div>
            </td>
            <td colspan="2" style="border: 1px solid #475569; padding: 0.5rem;">
              <div style="background: rgba(6, 182, 212, 0.25); color: #67e8f9; border: 1px solid r
gba(6, 182, 212, 0.5); padding: 0.4rem 0.3rem; border-radius: 6px; font-weight: 600;">
```

```
                <span style="font-size: 0.85rem; font-weight: 800; display: block;">CSIT242 (Lab)
</span>
                <span style="font-size: 0.75rem; opacity: 0.85; display: block;">Network Simulatio
n Lab &bull; Lab 105</span>
                <span style="font-size: 0.7rem; text-transform: uppercase; display: block;">Practi
cal (2 hrs)</span>
              </div>
            </td>
            <td style="border: 1px solid #475569; padding: 0.5rem;">
              <div style="background: rgba(139, 92, 246, 0.2); color: #c4b5fd; border: 1px solid r
gba(139, 92, 246, 0.4); padding: 0.4rem 0.3rem; border-radius: 6px; font-weight: 600;">
                <span style="font-size: 0.85rem; font-weight: 800; display: block;">MATH302</span>
                <span style="font-size: 0.75rem; opacity: 0.85; display: block;">Tutorial &bull; R
-408</span>
                <span style="font-size: 0.7rem; text-transform: uppercase; display: block;">Tutori
al</span>
              </div>
            </td>
          </tr>

          <!-- WEDNESDAY -->
          <tr>
            <th style="background: #1e293b; color: #38bdf8; font-weight: 700; border: 1px solid #4
75569; padding: 0.75rem 0.5rem;">WEDNESDAY</th>
            <td style="border: 1px solid #475569; padding: 0.5rem;">
              <div style="background: rgba(16, 185, 129, 0.2); color: #6ee7b7; border: 1px solid r
gba(16, 185, 129, 0.4); padding: 0.4rem 0.3rem; border-radius: 6px; font-weight: 600;">
                <span style="font-size: 0.85rem; font-weight: 800; display: block;">CSIT244</span>
                <span style="font-size: 0.75rem; opacity: 0.85; display: block;">DBMS Theory &bul
l; R-408</span>
                <span style="font-size: 0.7rem; text-transform: uppercase; display: block;">Lectur
e</span>
              </div>
            </td>
            <td style="border: 1px solid #475569; padding: 0.5rem;">
              <div style="background: rgba(236, 72, 153, 0.2); color: #f9a8d4; border: 1px solid r
gba(236, 72, 153, 0.4); padding: 0.4rem 0.3rem; border-radius: 6px; font-weight: 600;">
                <span style="font-size: 0.85rem; font-weight: 800; display: block;">CSIT242</span>
                <span style="font-size: 0.75rem; opacity: 0.85; display: block;">Computer Networks
&bull; R-408</span>
                <span style="font-size: 0.7rem; text-transform: uppercase; display: block;">Lectur
e</span>
              </div>
            </td>
            <td style="border: 1px solid #475569; padding: 0.5rem;">
              <div style="background: rgba(139, 92, 246, 0.2); color: #c4b5fd; border: 1px solid r
gba(139, 92, 246, 0.4); padding: 0.4rem 0.3rem; border-radius: 6px; font-weight: 600;">
                <span style="font-size: 0.85rem; font-weight: 800; display: block;">MATH302</span>
                <span style="font-size: 0.75rem; opacity: 0.85; display: block;">Discrete Math &bu
ll; R-408</span>
                <span style="font-size: 0.7rem; text-transform: uppercase; display: block;">Lectur
e</span>
              </div>
            </td>
            <td style="border: 1px solid #475569; padding: 0.5rem;">
              <div style="background: rgba(245, 158, 11, 0.2); color: #fcd34d; border: 1px solid r
gba(245, 158, 11, 0.4); padding: 0.4rem 0.3rem; border-radius: 6px; font-weight: 600;">
                <span style="font-size: 0.85rem; font-weight: 800; display: block;">CSIT240</span>
                <span style="font-size: 0.75rem; opacity: 0.85; display: block;">Operating Systems
&bull; R-408</span>
                <span style="font-size: 0.7rem; text-transform: uppercase; display: block;">Lectur
e</span>
              </div>
            </td>
            <td colspan="2" style="border: 1px solid #475569; padding: 0.5rem;">
              <div style="background: rgba(6, 182, 212, 0.25); color: #67e8f9; border: 1px solid r
```

```
gba(6, 182, 212, 0.5); padding: 0.4rem 0.3rem; border-radius: 6px; font-weight: 600;">
                <span style="font-size: 0.85rem; font-weight: 800; display: block;">CSIT248 (Lab)
</span>
                <span style="font-size: 0.75rem; opacity: 0.85; display: block;">Adv. Web Tech Lab
&bull; Lab 302</span>
                <span style="font-size: 0.7rem; text-transform: uppercase; display: block;">Practi
cal (2 hrs)</span>
              </div>
            </td>
            <td style="border: 1px solid #475569; padding: 0.5rem;">
              <div style="background: rgba(255, 255, 255, 0.04); color: #94a3b8; border: 1px solid
#334155; padding: 0.4rem 0.3rem; border-radius: 6px; font-weight: 600;">
                <span style="font-size: 0.85rem; font-weight: 800; display: block;">Club Activitie
s</span>
                <span style="font-size: 0.75rem; opacity: 0.85; display: block;">Student Center</s
pan>
                <span style="font-size: 0.7rem; text-transform: uppercase; display: block;">Co-Cur
ricular</span>
              </div>
            </td>
          </tr>

          <!-- THURSDAY -->
          <tr>
            <th style="background: #1e293b; color: #38bdf8; font-weight: 700; border: 1px solid #4
75569; padding: 0.75rem 0.5rem;">THURSDAY</th>
            <td style="border: 1px solid #475569; padding: 0.5rem;">
              <div style="background: rgba(59, 130, 246, 0.2); color: #93c5fd; border: 1px solid r
gba(59, 130, 246, 0.4); padding: 0.4rem 0.3rem; border-radius: 6px; font-weight: 600;">
                <span style="font-size: 0.85rem; font-weight: 800; display: block;">CSIT248</span>
                <span style="font-size: 0.75rem; opacity: 0.85; display: block;">Adv. Web Tech &bu
ll; R-408</span>
                <span style="font-size: 0.7rem; text-transform: uppercase; display: block;">Lectur
e</span>
              </div>
            </td>
            <td style="border: 1px solid #475569; padding: 0.5rem;">
              <div style="background: rgba(16, 185, 129, 0.2); color: #6ee7b7; border: 1px solid r
gba(16, 185, 129, 0.4); padding: 0.4rem 0.3rem; border-radius: 6px; font-weight: 600;">
                <span style="font-size: 0.85rem; font-weight: 800; display: block;">CSIT244</span>
                <span style="font-size: 0.75rem; opacity: 0.85; display: block;">DBMS Theory &bul
l; R-408</span>
                <span style="font-size: 0.7rem; text-transform: uppercase; display: block;">Lectur
e</span>
              </div>
            </td>
            <td colspan="2" style="border: 1px solid #475569; padding: 0.5rem;">
              <div style="background: rgba(6, 182, 212, 0.25); color: #67e8f9; border: 1px solid r
gba(6, 182, 212, 0.5); padding: 0.4rem 0.3rem; border-radius: 6px; font-weight: 600;">
                <span style="font-size: 0.85rem; font-weight: 800; display: block;">CSIT240 (Lab)
</span>
                <span style="font-size: 0.75rem; opacity: 0.85; display: block;">Linux / OS Lab &b
ull; Lab 208</span>
                <span style="font-size: 0.7rem; text-transform: uppercase; display: block;">Practi
cal (2 hrs)</span>
              </div>
            </td>
            <td style="border: 1px solid #475569; padding: 0.5rem;">
              <div style="background: rgba(139, 92, 246, 0.2); color: #c4b5fd; border: 1px solid r
gba(139, 92, 246, 0.4); padding: 0.4rem 0.3rem; border-radius: 6px; font-weight: 600;">
                <span style="font-size: 0.85rem; font-weight: 800; display: block;">MATH302</span>
                <span style="font-size: 0.75rem; opacity: 0.85; display: block;">Discrete Math &bu
ll; R-408</span>
                <span style="font-size: 0.7rem; text-transform: uppercase; display: block;">Lectur
e</span>
              </div>
```

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            </td>
            <td style="border: 1px solid #475569; padding: 0.5rem;">
              <div style="background: rgba(236, 72, 153, 0.2); color: #f9a8d4; border: 1px solid r
gba(236, 72, 153, 0.4); padding: 0.4rem 0.3rem; border-radius: 6px; font-weight: 600;">
                <span style="font-size: 0.85rem; font-weight: 800; display: block;">CSIT242</span>
                <span style="font-size: 0.75rem; opacity: 0.85; display: block;">Computer Networks
&bull; R-408</span>
                <span style="font-size: 0.7rem; text-transform: uppercase; display: block;">Lectur
e</span>
              </div>
            </td>
            <td style="border: 1px solid #475569; padding: 0.5rem;">
              <div style="background: rgba(99, 102, 241, 0.15); color: #c7d2fe; border: 1px solid
rgba(99, 102, 241, 0.3); padding: 0.4rem 0.3rem; border-radius: 6px; font-weight: 600;">
                <span style="font-size: 0.85rem; font-weight: 800; display: block;">PSDA Seminar</
span>
                <span style="font-size: 0.75rem; opacity: 0.85; display: block;">Auditorium 2</spa
n>
                <span style="font-size: 0.7rem; text-transform: uppercase; display: block;">Presen
tation</span>
              </div>
            </td>
          </tr>

          <!-- FRIDAY -->
          <tr>
            <th style="background: #1e293b; color: #38bdf8; font-weight: 700; border: 1px solid #4
75569; padding: 0.75rem 0.5rem;">FRIDAY</th>
            <td style="border: 1px solid #475569; padding: 0.5rem;">
              <div style="background: rgba(245, 158, 11, 0.2); color: #fcd34d; border: 1px solid r
gba(245, 158, 11, 0.4); padding: 0.4rem 0.3rem; border-radius: 6px; font-weight: 600;">
                <span style="font-size: 0.85rem; font-weight: 800; display: block;">CSIT240</span>
                <span style="font-size: 0.75rem; opacity: 0.85; display: block;">Operating Systems
&bull; R-408</span>
                <span style="font-size: 0.7rem; text-transform: uppercase; display: block;">Lectur
e</span>
              </div>
            </td>
            <td style="border: 1px solid #475569; padding: 0.5rem;">
              <div style="background: rgba(139, 92, 246, 0.2); color: #c4b5fd; border: 1px solid r
gba(139, 92, 246, 0.4); padding: 0.4rem 0.3rem; border-radius: 6px; font-weight: 600;">
                <span style="font-size: 0.85rem; font-weight: 800; display: block;">MATH302</span>
                <span style="font-size: 0.75rem; opacity: 0.85; display: block;">Discrete Math &bu
ll; R-408</span>
                <span style="font-size: 0.7rem; text-transform: uppercase; display: block;">Lectur
e</span>
              </div>
            </td>
            <td style="border: 1px solid #475569; padding: 0.5rem;">
              <div style="background: rgba(59, 130, 246, 0.2); color: #93c5fd; border: 1px solid r
gba(59, 130, 246, 0.4); padding: 0.4rem 0.3rem; border-radius: 6px; font-weight: 600;">
                <span style="font-size: 0.85rem; font-weight: 800; display: block;">CSIT248</span>
                <span style="font-size: 0.75rem; opacity: 0.85; display: block;">Adv. Web Tech &bu
ll; R-408</span>
                <span style="font-size: 0.7rem; text-transform: uppercase; display: block;">Lectur
e</span>
              </div>
            </td>
            <td style="border: 1px solid #475569; padding: 0.5rem;">
              <div style="background: rgba(16, 185, 129, 0.2); color: #6ee7b7; border: 1px solid r
gba(16, 185, 129, 0.4); padding: 0.4rem 0.3rem; border-radius: 6px; font-weight: 600;">
                <span style="font-size: 0.85rem; font-weight: 800; display: block;">CSIT244</span>
                <span style="font-size: 0.75rem; opacity: 0.85; display: block;">DBMS Theory &bul
l; R-408</span>
                <span style="font-size: 0.7rem; text-transform: uppercase; display: block;">Lectur
e</span>
```

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              </div>
            </td>
            <td colspan="2" style="border: 1px solid #475569; padding: 0.5rem;">
              <div style="background: rgba(245, 158, 11, 0.15); color: #fde68a; border: 1px solid
rgba(245, 158, 11, 0.3); padding: 0.4rem 0.3rem; border-radius: 6px; font-weight: 600;">
                <span style="font-size: 0.85rem; font-weight: 800; display: block;">Capstone Proje
ct</span>
                <span style="font-size: 0.75rem; opacity: 0.85; display: block;">Incubation Lab &b
ull; Lab 401</span>
                <span style="font-size: 0.7rem; text-transform: uppercase; display: block;">Projec
t Work (2 hrs)</span>
              </div>
            </td>
            <td style="border: 1px solid #475569; padding: 0.5rem;">
              <div style="background: rgba(16, 185, 129, 0.15); color: #a7f3d0; border: 1px solid
rgba(16, 185, 129, 0.3); padding: 0.4rem 0.3rem; border-radius: 6px; font-weight: 600;">
                <span style="font-size: 0.85rem; font-weight: 800; display: block;">Sports &amp; F
itness</span>
                <span style="font-size: 0.75rem; opacity: 0.85; display: block;">Sports Complex</s
pan>
                <span style="font-size: 0.7rem; text-transform: uppercase; display: block;">Extrac
urricular</span>
              </div>
            </td>
          </tr>
        </tbody>
        <tfoot>
          <tr>
            <td colspan="9" style="background: #111827; color: #94a3b8; font-size: 0.8rem; paddin
g: 0.75rem 1rem; text-align: left; border: 1px solid #475569;">
              &bull; <strong>Note:</strong> Practical lab sessions are of 2 consecutive hours. Att
endance is mandatory (minimum 75% as per university criteria).
            </td>
          </tr>
        </tfoot>
      </table>
    </div>

    <!-- Course & Faculty Legend Table (Inline CSS) -->
    <section style="background: #1e293b; border: 1px solid #334155; border-radius: 14px; padding:
1.75rem; box-shadow: 0 10px 25px rgba(0, 0, 0, 0.3);">
      <h2 style="font-size: 1.25rem; font-weight: 700; color: #ffffff; margin: 0 0 1rem 0; border-
bottom: 2px solid #334155; padding-bottom: 0.5rem;">
        Subject &amp; Faculty Legend
      </h2>
      <div style="overflow-x: auto;">
        <table style="width: 100%; border-collapse: collapse; text-align: left; font-size: 0.9re
m;">
          <thead>
            <tr style="background: rgba(255, 255, 255, 0.04); color: #a5b4fc; border-bottom: 1px s
olid #334155;">
              <th style="padding: 0.75rem 1rem;">Color</th>
              <th style="padding: 0.75rem 1rem;">Course Code</th>
              <th style="padding: 0.75rem 1rem;">Course Title</th>
              <th style="padding: 0.75rem 1rem;">Faculty Name</th>
              <th style="padding: 0.75rem 1rem;">Credits (L-T-P)</th>
              <th style="padding: 0.75rem 1rem;">Venue</th>
            </tr>
          </thead>
          <tbody>
            <tr style="border-bottom: 1px solid #334155;">
              <td style="padding: 0.75rem 1rem;"><span style="display: inline-block; width: 12px;
height: 12px; border-radius: 3px; background: #3b82f6;"></span></td>
              <td style="padding: 0.75rem 1rem; color: #ffffff; font-weight: 700;">CSIT248</td>
              <td style="padding: 0.75rem 1rem; color: #cbd5e1;">Advanced Web Technology</td>
              <td style="padding: 0.75rem 1rem; color: #cbd5e1;">Dr. V. Sharma / Prof. R. Verma</t
```

```
d>
              <td style="padding: 0.75rem 1rem; color: #cbd5e1;">2-0-2 (6 Units)</td>
              <td style="padding: 0.75rem 1rem; color: #38bdf8;">Room 408 / Lab 302</td>
            </tr>
            <tr style="border-bottom: 1px solid #334155; background: rgba(255, 255, 255, 0.015);">
              <td style="padding: 0.75rem 1rem;"><span style="display: inline-block; width: 12px;
height: 12px; border-radius: 3px; background: #10b981;"></span></td>
              <td style="padding: 0.75rem 1rem; color: #ffffff; font-weight: 700;">CSIT244</td>
              <td style="padding: 0.75rem 1rem; color: #cbd5e1;">Database Management Systems</td>
              <td style="padding: 0.75rem 1rem; color: #cbd5e1;">Dr. P. Gupta</td>
              <td style="padding: 0.75rem 1rem; color: #cbd5e1;">3-0-2 (5 Units)</td>
              <td style="padding: 0.75rem 1rem; color: #38bdf8;">Room 408 / Lab 204</td>
            </tr>
            <tr style="border-bottom: 1px solid #334155;">
              <td style="padding: 0.75rem 1rem;"><span style="display: inline-block; width: 12px;
height: 12px; border-radius: 3px; background: #f59e0b;"></span></td>
              <td style="padding: 0.75rem 1rem; color: #ffffff; font-weight: 700;">CSIT240</td>
              <td style="padding: 0.75rem 1rem; color: #cbd5e1;">Operating Systems</td>
              <td style="padding: 0.75rem 1rem; color: #cbd5e1;">Prof. A. K. Singh</td>
              <td style="padding: 0.75rem 1rem; color: #cbd5e1;">3-0-2 (5 Units)</td>
              <td style="padding: 0.75rem 1rem; color: #38bdf8;">Room 408 / Lab 208</td>
            </tr>
            <tr style="border-bottom: 1px solid #334155; background: rgba(255, 255, 255, 0.015);">
              <td style="padding: 0.75rem 1rem;"><span style="display: inline-block; width: 12px;
height: 12px; border-radius: 3px; background: #ec4899;"></span></td>
              <td style="padding: 0.75rem 1rem; color: #ffffff; font-weight: 700;">CSIT242</td>
              <td style="padding: 0.75rem 1rem; color: #cbd5e1;">Computer Networks</td>
              <td style="padding: 0.75rem 1rem; color: #cbd5e1;">Dr. S. Nair</td>
              <td style="padding: 0.75rem 1rem; color: #cbd5e1;">3-0-2 (5 Units)</td>
              <td style="padding: 0.75rem 1rem; color: #38bdf8;">Room 408 / Lab 105</td>
            </tr>
            <tr>
              <td style="padding: 0.75rem 1rem;"><span style="display: inline-block; width: 12px;
height: 12px; border-radius: 3px; background: #8b5cf6;"></span></td>
              <td style="padding: 0.75rem 1rem; color: #ffffff; font-weight: 700;">MATH302</td>
              <td style="padding: 0.75rem 1rem; color: #cbd5e1;">Discrete Mathematical Structures
</td>
              <td style="padding: 0.75rem 1rem; color: #cbd5e1;">Prof. K. Raman</td>
              <td style="padding: 0.75rem 1rem; color: #cbd5e1;">3-1-0 (4 Units)</td>
              <td style="padding: 0.75rem 1rem; color: #38bdf8;">Room 408</td>
            </tr>
          </tbody>
        </table>
      </div>
    </section>

  </main>

  <!-- Footer (Inline CSS) -->
  <footer style="background: #090d16; border-top: 1px solid #1e293b; padding: 2rem 1.5rem; text-al
ign: center; color: #94a3b8; font-size: 0.875rem; margin-top: 3rem;">
    <p style="margin: 0 0 0.5rem 0;">Experiment 03 &bull; CSIT248 Advanced Web Technology &bull; T
able &amp; Layout Demonstration</p>
    <p style="margin: 0 0 0.75rem 0; color: #64748b; font-size: 0.8rem;">Formatted using 100% Inli
ne CSS properties.</p>
    <p style="margin: 0;"><a href="../index.html" style="color: #818cf8; text-decoration: none; fo
nt-weight: 600;">&uarr; Back to Lab Work Portal</a></p>
  </footer>

</body>
</html>
```

Output :

CSIT248 Lab Work

Formatted Exclusively with Inline CSS

All Experiments

Department of Computer Applications

Amity University Noida • Academic Year 2025–2026

Program: BCA (Hons.) with Research | Semester: III | Classroom: Room 408 (Academic Block) | Total Credits: 24

Weekly Academic Schedule & Lab Allocations (Table Structure)

Day / Time	Period I 09:00 - 10:00 Slot 1	Period II 10:00 - 11:00 Slot 2	Period III 11:00 - 12:00 Slot 3	Period IV 12:00 - 01:00 Slot 4	Lunch 01:00 - 02:00	Period V 02:00 - 03:00 Slot 5	Period VI 03:00 - 04:00 Slot 6	Period VII 04:00 - 05:00 Slot 7
MONDAY	CSIT244 (Lab) Database Systems Lab • Lab 204 PRACTICAL (2 HRS)		CSIT248 Adv. Web Tech • R-408 LECTURE	MATH302 Discrete Math • R-408 LECTURE	LUNCH BREAK	CSIT242 Computer Networks • R-408 LECTURE	CSIT240 Operating Systems • R-408 LECTURE	Library / Mentor Central Library SELF STUDY
TUESDAY	CSIT240 Operating Systems • R-408 LECTURE	CSIT248 Adv. Web Tech • R-408 LECTURE	CSIT242 Computer Networks • R-408 LECTURE	CSIT244 DBMS Theory • R-408 LECTURE		CSIT242 (Lab) Network Simulation Lab • Lab 105 PRACTICAL (2 HRS)		MATH302 Tutorial • R-408 TUTORIAL

Experiment No.: 7

Title : Client-Side Image Mapping with Poly and Rect Hotspots.

Aim : To implement interactive client-side image mapping using HTML <map> and <area> tags with circular, polygonal, and rectangular hotspot coordinates linked to target resources.

Source Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Experiment 4: Design a web page to demonstrate Image Mapping - CSIT248 Advanced Web Technology">
  <title>Experiment 04 - Client-Side Image Mapping (Inline CSS)</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&display=swap" rel="stylesheet">
</head>
<body style="margin: 0; padding: 0; font-family: 'Inter', system-ui, -apple-system, sans-serif; background-color: #0f172a; color: #f8fafc; line-height: 1.6; min-height: 100vh;">

  <!-- Navigation (Inline CSS) -->
  <nav style="position: sticky; top: 0; z-index: 100; background: rgba(15, 23, 42, 0.92); backdrop-filter: blur(12px); border-bottom: 1px solid #334155; padding: 1rem 2rem; display: flex; justify-content: space-between; align-items: center;">
    <a href="../index.html" style="font-weight: 800; font-size: 1.2rem; color: #818cf8; text-decoration: none; display: flex; align-items: center; gap: 0.5rem;">
      <span style="background: rgba(99, 102, 241, 0.15); border: 1px solid rgba(99, 102, 241, 0.3); padding: 0.2rem 0.5rem; border-radius: 6px;">&larr;</span>
      CSIT248 Lab Work
    </a>
    <div style="display: flex; gap: 1rem; align-items: center;">
      <span style="font-size: 0.8rem; background: rgba(99, 102, 241, 0.12); color: #a5b4fc; border: 1px solid rgba(99, 102, 241, 0.25); padding: 0.3rem 0.75rem; border-radius: 9999px; font-weight: 600;">
        Formatted Exclusively with Inline CSS
      </span>
      <a href="../index.html" style="background: #6366f1; color: #ffffff; padding: 0.4rem 0.9rem; border-radius: 6px; text-decoration: none; font-size: 0.85rem; font-weight: 600;">
        All Experiments
      </a>
    </div>
  </nav>

  <main style="max-width: 1200px; margin: 0 auto; padding: 2.5rem 1.5rem;">

    <!-- Header (Inline CSS) -->
    <header style="text-align: center; margin-bottom: 2rem;">
      <h1 style="font-size: 2.25rem; font-weight: 800; color: #ffffff; margin: 0 0 0.5rem 0;">Demonstration of HTML5 Image Mapping</h1>
      <p style="color: #94a3b8; font-size: 1.05rem; max-width: 750px; margin: 0 auto;">
        Interactive demonstration of the HTML <code>&lt;map&gt;</code> and <code>&lt;area&gt;</code> tags using rectangular (<code>rect</code>), circular (<code>circle</code>), and polygonal (<code>poly</code>) clickable hotspot regions.
      </p>
    </header>

    <!-- Interactive Demo Layout (Inline CSS) -->
```

```
    <div style="display: grid; grid-template-columns: repeat(auto-fit, minmax(340px, 1fr)); gap:
1.5rem; margin-bottom: 2.5rem;">

      <!-- Main Visual Map Box (Inline CSS) -->
      <section style="background: #1e293b; border: 1px solid #334155; border-radius: 14px; paddin
g: 1.25rem; box-shadow: 0 10px 25px rgba(0, 0, 0, 0.3); display: flex; flex-direction: column; ali
gn-items: center;">
        <div style="position: relative; width: 100%; max-width: 800px; border-radius: 8px; overflo
w: hidden; border: 1px solid #475569; background: #000;">
          <!-- The mapped image with usemap attribute -->
          <img
            src="workstation.svg"
            alt="Workstation Setup Map"
            width="800"
            height="500"
            usemap="#workstation-map"
            id="workstation-img"
            style="display: block; width: 100%; height: auto; user-select: none;"
          >

          <!-- Standard HTML5 Client-Side Image Map -->
          <map name="workstation-map" id="workstation-map">
            <!-- 1. Monitor (RECT) -->
            <area
              shape="rect"
              coords="220,60,580,280"
              href="#monitor"
              alt="4K UltraWide Monitor"
              title="Click to inspect Monitor [RECT: 220,60,580,280]"
              data-name="4K UltraWide Curved Monitor"
              data-type="Output Display Device"
              data-shape="rect: 220,60,580,280"
              data-desc="Primary visual output device. 34-inch IPS panel with 3840x1440 resolutio
n, 144Hz refresh rate, and 99% sRGB color gamut for programming and multimedia workflows."
            >

            <!-- 2. Mechanical Keyboard (RECT) -->
            <area
              shape="rect"
              coords="240,330,560,420"
              href="#keyboard"
              alt="Mechanical Keyboard"
              title="Click to inspect Keyboard [RECT: 240,330,560,420]"
              data-name="Custom Mechanical Keyboard"
              data-type="Alphanumeric Input Device"
              data-shape="rect: 240,330,560,420"
              data-desc="Tactile 75% ANSI layout keyboard with hot-swappable lubricated switches,
double-shot PBT keycaps, and RGB per-key backlighting."
            >

            <!-- 3. Ergonomic Mouse (CIRCLE) -->
            <area
              shape="circle"
              coords="630,375,35"
              href="#mouse"
              alt="Ergonomic Optical Mouse"
              title="Click to inspect Mouse [CIRCLE: 630,375,35]"
              data-name="Wireless Ergonomic Mouse"
              data-type="Pointing Input Device"
              data-shape="circle: (630, 375), r=35"
              data-desc="Precision optical sensor with up to 26,000 DPI, programmable macro button
s, and hyper-fast magnetic scroll wheel."
            >

            <!-- 4. Coffee Mug (CIRCLE) -->
            <area
```

```
              shape="circle"
              coords="150,360,35"
              href="#coffee"
              alt="Programmer Coffee Mug"
              title="Click to inspect Coffee Mug [CIRCLE: 150,360,35]"
              data-name="Developer Coffee Mug"
              data-type="Productivity Fuel"
              data-shape="circle: (150, 360), r=35"
              data-desc="Ceramic insulated mug holding freshly brewed dark roast espresso — essent
ial fuel for debugging code and writing clean HTML5!"
            >

            <!-- 5. PC Tower (POLY) -->
            <area
              shape="poly"
              coords="640,70,760,70,760,300,640,300"
              href="#tower"
              alt="High-Performance Desktop Tower"
              title="Click to inspect PC Tower [POLY: 640,70,760,70,760,300,640,300]"
              data-name="High-Performance Desktop Tower"
              data-type="Central Processing Unit (CPU)"
              data-shape="poly: 640,70, 760,70, 760,300, 640,300"
              data-desc="Custom liquid-cooled workstation containing an AMD Ryzen 9 16-Core proces
sor, 64GB DDR5 RAM, and NVIDIA RTX GPU."
            >

            <!-- 6. Studio Headphones (POLY) -->
            <area
              shape="poly"
              coords="60,110,120,70,170,110,170,220,120,250,60,220"
              href="#headphones"
              alt="Studio Monitoring Headphones"
              title="Click to inspect Headphones [POLY: 60,110,120,70,170,110,170,220,120,250,60,2
20]"
              data-name="Studio Monitoring Headphones"
              data-type="Audio Output Peripheral"
              data-shape="poly: 60,110, 120,70, 170,110, 170,220, 120,250, 60,220"
              data-desc="Open-back planar magnetic headphones for deep focus music, spatial audio
rendering, and crisp voice communications."
            >
          </map>
        </div>

        <div style="width: 100%; margin-top: 0.75rem; display: flex; justify-content: space-betwee
n; background: #090d16; padding: 0.5rem 1rem; border-radius: 6px; font-family: monospace; font-siz
e: 0.85rem; color: #a5b4fc; box-sizing: border-box;">
          <span id="coord-display">Pointer Coordinates: (X: --, Y: --)</span>
          <span style="color: #6ee7b7;">Hover or Click any hotspot area!</span>
        </div>
      </section>

      <!-- Side Inspector Panel (Inline CSS) -->
      <aside style="background: #1e293b; border: 1px solid #334155; border-radius: 14px; padding:
1.5rem; display: flex; flex-direction: column; gap: 1rem; box-shadow: 0 10px 25px rgba(0, 0, 0, 0.
3);">
        <h3 style="font-size: 1.25rem; color: #ffffff; border-bottom: 1px solid #334155; padding-b
ottom: 0.75rem; margin: 0; display: flex; align-items: center; gap: 0.5rem;">
          <svg width="20" height="20" viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke
-width="2">
            <circle cx="12" cy="12" r="10"></circle>
            <line x1="12" y1="16" x2="12" y2="12"></line>
            <line x1="12" y1="8" x2="12.01" y2="8"></line>
          </svg>
          Hotspot Inspector
        </h3>
```

```
        <div id="device-info-box" style="background: rgba(99, 102, 241, 0.1); border: 1px solid rg
ba(99, 102, 241, 0.3); border-radius: 10px; padding: 1.25rem; min-height: 160px;">
          <div id="device-name" style="font-size: 1.2rem; font-weight: 700; color: #38bdf8; margin
-bottom: 0.35rem;">Select an Element</div>
          <div id="device-shape" style="font-size: 0.8rem; background: #1e1b4b; color: #a5b4fc; pa
dding: 0.15rem 0.5rem; border-radius: 4px; display: inline-block; margin-bottom: 0.75rem; font-fam
ily: monospace;">Shape: Hover over image</div>
          <p id="device-desc" style="font-size: 0.9rem; color: #e2e8f0; line-height: 1.5; margin:
0;">
            Move your mouse cursor over the workstation image or click on the hotspots (Monitor, K
eyboard, Mouse, Mug, PC Tower, Headphones) to test the HTML Image Map.
          </p>
        </div>

        <h4 style="font-size: 0.95rem; color: #94a3b8; margin: 0.5rem 0 0 0;">Quick Hotspot Links:
</h4>
        <ul style="list-style: none; display: flex; flex-direction: column; gap: 0.5rem; margin:
0; padding: 0;">
          <li onclick="triggerArea('Monitor')" style="padding: 0.6rem 0.85rem; background: rgba(25
5, 255, 255, 0.04); border: 1px solid #334155; border-radius: 6px; font-size: 0.85rem; cursor: poi
nter; display: flex; justify-content: space-between; align-items: center;">
            <span>🖥️ 4K Monitor</span>
            <code style="font-size: 0.75rem; color: #38bdf8; font-family: monospace;">shape="rect"
</code>
          </li>
          <li onclick="triggerArea('Keyboard')" style="padding: 0.6rem 0.85rem; background: rgba(2
55, 255, 255, 0.04); border: 1px solid #334155; border-radius: 6px; font-size: 0.85rem; cursor: po
inter; display: flex; justify-content: space-between; align-items: center;">
            <span>⌨️ Keyboard</span>
            <code style="font-size: 0.75rem; color: #38bdf8; font-family: monospace;">shape="rect"
</code>
          </li>
          <li onclick="triggerArea('Mouse')" style="padding: 0.6rem 0.85rem; background: rgba(255,
255, 255, 0.04); border: 1px solid #334155; border-radius: 6px; font-size: 0.85rem; cursor: pointe
r; display: flex; justify-content: space-between; align-items: center;">
            <span>🖱️ Mouse</span>
            <code style="font-size: 0.75rem; color: #38bdf8; font-family: monospace;">shape="circl
e"</code>
          </li>
          <li onclick="triggerArea('Coffee')" style="padding: 0.6rem 0.85rem; background: rgba(25
5, 255, 255, 0.04); border: 1px solid #334155; border-radius: 6px; font-size: 0.85rem; cursor: poi
nter; display: flex; justify-content: space-between; align-items: center;">
            <span>☕ Coffee Mug</span>
            <code style="font-size: 0.75rem; color: #38bdf8; font-family: monospace;">shape="circl
e"</code>
          </li>
          <li onclick="triggerArea('PC Tower')" style="padding: 0.6rem 0.85rem; background: rgba(2
55, 255, 255, 0.04); border: 1px solid #334155; border-radius: 6px; font-size: 0.85rem; cursor: po
inter; display: flex; justify-content: space-between; align-items: center;">
            <span>🖥️ PC Tower</span>
            <code style="font-size: 0.75rem; color: #38bdf8; font-family: monospace;">shape="poly"
</code>
          </li>
          <li onclick="triggerArea('Headphones')" style="padding: 0.6rem 0.85rem; background: rgba
(255, 255, 255, 0.04); border: 1px solid #334155; border-radius: 6px; font-size: 0.85rem; cursor:
pointer; display: flex; justify-content: space-between; align-items: center;">
            <span>🎧 Headphones</span>
            <code style="font-size: 0.75rem; color: #38bdf8; font-family: monospace;">shape="poly"
</code>
          </li>
        </ul>
      </aside>

    </div>

    <!-- Technical Code Reference (Inline CSS) -->
```

```
    <section style="background: #1e293b; border: 1px solid #334155; border-radius: 14px; padding:
1.5rem; box-shadow: 0 10px 25px rgba(0, 0, 0, 0.3);">
      <h2 style="font-size: 1.3rem; color: #ffffff; margin: 0 0 1rem 0;">HTML5 Implementation Code
Breakdown</h2>
      <pre style="background: #090d16; border: 1px solid #334155; border-radius: 8px; padding: 1.2
5rem; overflow-x: auto; font-family: 'Fira Code', monospace; font-size: 0.85rem; color: #e2e8f0; l
ine-height: 1.5; margin: 0;"><code>&lt;!-- Associating Image with Map using usemap --&gt;
&lt;img src="workstation.svg" width="800" height="500" alt="Workstation" usemap="#workstation-ma
p"&gt;

&lt;!-- Client-Side Image Map Definition --&gt;
&lt;map name="workstation-map"&gt;
  &lt;!-- 1. RECTANGLE: coords="x1,y1, x2,y2" --&gt;
  &lt;area shape="rect" coords="220,60,580,280" href="#monitor" alt="Monitor"&gt;
  &lt;area shape="rect" coords="240,330,560,420" href="#keyboard" alt="Keyboard"&gt;

  &lt;!-- 2. CIRCLE: coords="centerX, centerY, radius" --&gt;
  &lt;area shape="circle" coords="630,375,35" href="#mouse" alt="Mouse"&gt;
  &lt;area shape="circle" coords="150,360,35" href="#coffee" alt="Coffee Mug"&gt;

  &lt;!-- 3. POLYGON: coords="x1,y1, x2,y2, x3,y3, ... xn,yn" --&gt;
  &lt;area shape="poly" coords="640,70,760,70,760,300,640,300" href="#tower" alt="PC Tower"&gt;
  &lt;area shape="poly" coords="60,110,120,70,170,110,170,220,120,250,60,220" href="#headphones" a
lt="Headphones"&gt;
&lt;/map&gt;</code></pre>
    </section>

  </main>

  <!-- Interactive JavaScript Helper for coordinate tracker and inspector -->
  <script>
    const img = document.getElementById('workstation-img');
    const coordDisplay = document.getElementById('coord-display');
    const devName = document.getElementById('device-name');
    const devShape = document.getElementById('device-shape');
    const devDesc = document.getElementById('device-desc');
    const areas = document.querySelectorAll('map area');

    // Real-time Coordinate tracking on image
    img.addEventListener('mousemove', (e) => {
      const rect = img.getBoundingClientRect();
      const scaleX = 800 / rect.width;
      const scaleY = 500 / rect.height;
      const x = Math.round((e.clientX - rect.left) * scaleX);
      const y = Math.round((e.clientY - rect.top) * scaleY);
      coordDisplay.textContent = `Pointer Coordinates: (X: ${x}, Y: ${y})`;
    });

    // Area click and hover handlers
    areas.forEach(area => {
      area.addEventListener('click', (e) => {
        e.preventDefault();
        displayDetails(area);
      });
      area.addEventListener('mouseenter', () => {
        displayDetails(area);
      });
    });

    function displayDetails(area) {
      devName.textContent = area.dataset.name;
      devShape.textContent = `Type: ${area.dataset.type} | Shape: ${area.dataset.shape}`;
      devDesc.textContent = area.dataset.desc;
    }

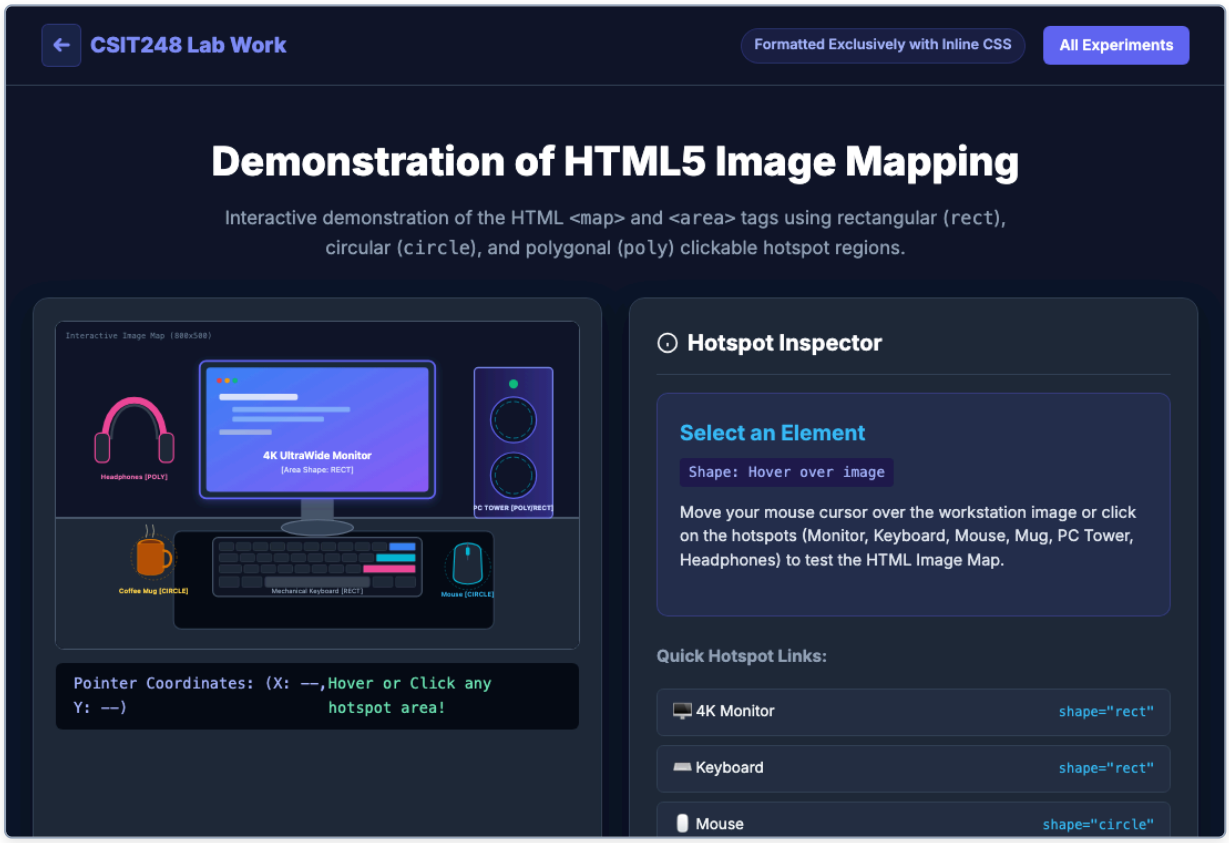
    function triggerArea(keyword) {
```

```
      const targetArea = Array.from(areas).find(a => a.dataset.name.toLowerCase().includes(keywor
d.toLowerCase()));
      if (targetArea) {
        displayDetails(targetArea);
      }
    }
  </script>

  <!-- Footer (Inline CSS) -->
  <footer style="background: #090d16; border-top: 1px solid #1e293b; padding: 2rem 1.5rem; text-al
ign: center; color: #94a3b8; font-size: 0.875rem; margin-top: 3rem;">
    <p style="margin: 0 0 0.5rem 0;">Experiment 04 &bull; CSIT248 Advanced Web Technology &bull; I
mage Mapping Demonstration</p>
    <p style="margin: 0 0 0.75rem 0; color: #64748b; font-size: 0.8rem;">Formatted using 100% Inli
ne CSS properties.</p>
    <p style="margin: 0;"><a href="../index.html" style="color: #818cf8; text-decoration: none; fo
nt-weight: 600;">&uarr; Back to Lab Work Portal</a></p>
  </footer>

</body>
</html>
```

Output :



Experiment No.: 8

Title : Nested Ordered and Unordered Categorized Lists.

Aim : To design an HTML web page demonstrating complex hierarchical nested ordered lists, unordered lists with varied bullet types (disc, circle, square), and styled form button controls.

Source Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Experiment 5: Write a program A) To add a button on a form. B) To add ordered list in page - CSIT248 Advanced Web Technology">
  <title>Experiment 05 - Form Buttons &amp;amp; Ordered Lists (Inline CSS)</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&display=swap" rel="stylesheet">
</head>
<body style="margin: 0; padding: 0; font-family: 'Inter', system-ui, -apple-system, sans-serif; background-color: #0f172a; color: #f8fafc; line-height: 1.6; min-height: 100vh;">

  <!-- Navigation (Inline CSS) -->
  <nav style="position: sticky; top: 0; z-index: 100; background: rgba(15, 23, 42, 0.92); backdrop-filter: blur(12px); border-bottom: 1px solid #334155; padding: 1rem 2rem; display: flex; justify-content: space-between; align-items: center;">
    <a href="../index.html" style="font-weight: 800; font-size: 1.2rem; color: #818cf8; text-decoration: none; display: flex; align-items: center; gap: 0.5rem;">
      <span style="background: rgba(99, 102, 241, 0.15); border: 1px solid rgba(99, 102, 241, 0.3); padding: 0.2rem 0.5rem; border-radius: 6px;">&larr;</span>
      CSIT248 Lab Work
    </a>
    <div style="display: flex; gap: 1rem; align-items: center;">
      <span style="font-size: 0.8rem; background: rgba(99, 102, 241, 0.12); color: #a5b4fc; border: 1px solid rgba(99, 102, 241, 0.25); padding: 0.3rem 0.75rem; border-radius: 9999px; font-weight: 600;">
        Formatted Exclusively with Inline CSS
      </span>
      <a href="../index.html" style="background: #6366f1; color: #ffffff; padding: 0.4rem 0.9rem; border-radius: 6px; text-decoration: none; font-size: 0.85rem; font-weight: 600;">
        All Experiments
      </a>
    </div>
  </nav>

  <main style="max-width: 1100px; margin: 0 auto; padding: 2.5rem 1.5rem;">

    <!-- Header (Inline CSS) -->
    <header style="text-align: center; margin-bottom: 2.5rem;">
      <div style="display: inline-block; background: rgba(99, 102, 241, 0.15); color: #a5b4fc; border: 1px solid rgba(99, 102, 241, 0.3); padding: 0.25rem 0.75rem; border-radius: 9999px; font-size: 0.85rem; font-weight: 600; margin-bottom: 0.75rem;">
        Experiment 05 &bull; Modules I &amp; II
      </div>
      <h1 style="font-size: 2.25rem; font-weight: 800; color: #ffffff; margin: 0 0 0.5rem 0;">Form Buttons &amp; Ordered Lists</h1>
      <p style="color: #94a3b8; font-size: 1.05rem; margin: 0;">Comprehensive practical implementation of HTML button types and ordered list variations</p>
    </header>
```

```
    <!-- ================================================
         PART A: Form Buttons Demonstration
         ================================================ -->
    <section style="background: #1e293b; border: 1px solid #334155; border-radius: 16px; padding: 2rem; margin-bottom: 2.5rem; box-shadow: 0 10px 25px rgba(0, 0, 0, 0.25);">
      <h2 style="font-size: 1.5rem; font-weight: 700; color: #38bdf8; margin: 0 0 0.5rem 0; display: flex; align-items: center; gap: 0.5rem; border-bottom: 2px solid #334155; padding-bottom: 0.5rem;">
        <span>🔘</span> Part A: Adding Different Buttons to a Web Form
      </h2>
      <p style="color: #cbd5e1; font-size: 0.95rem; margin: 0.5rem 0 1.5rem 0;">
        HTML offers multiple approaches to add functional buttons on forms using both the <code>&lt;button&gt;</code> and <code>&lt;input&gt;</code> tags with different <code>type</code> attributes:
      </p>

      <form action="#" method="post" onsubmit="event.preventDefault(); alert('Form Submitted!');" style="background: rgba(15, 23, 42, 0.6); border: 1px solid #475569; border-radius: 12px; padding: 1.5rem;">
        <div style="margin-bottom: 1.25rem;">
          <label for="sample-input" style="display: block; font-size: 0.85rem; font-weight: 600; color: #cbd5e1; margin-bottom: 0.35rem;">Sample Text Input:</label>
          <input type="text" id="sample-input" placeholder="Type text here to test reset/submit..." style="width: 100%; box-sizing: border-box; background: #0f172a; border: 1px solid #475569; color: #ffffff; padding: 0.65rem 0.85rem; border-radius: 6px; font-size: 0.9rem; outline: none;">
        </div>

        <h3 style="font-size: 1rem; color: #a5b4fc; margin: 1.25rem 0 0.75rem 0;">Interactive Buttons Showcase:</h3>

        <div style="display: grid; grid-template-columns: repeat(auto-fit, minmax(220px, 1fr)); gap: 1rem;">

          <!-- 1. Submit Button -->
          <div style="background: #1e293b; border: 1px solid #334155; padding: 1rem; border-radius: 8px;">
            <span style="font-size: 0.75rem; color: #94a3b8; font-family: monospace; display: block; margin-bottom: 0.4rem;">type="submit"</span>
            <button type="submit" style="width: 100%; background: #6366f1; color: white; border: none; padding: 0.6rem; border-radius: 6px; font-weight: 700; cursor: pointer;">
              🚀 Submit Form
            </button>
          </div>

          <!-- 2. Reset Button -->
          <div style="background: #1e293b; border: 1px solid #334155; padding: 1rem; border-radius: 8px;">
            <span style="font-size: 0.75rem; color: #94a3b8; font-family: monospace; display: block; margin-bottom: 0.4rem;">type="reset"</span>
            <button type="reset" style="width: 100%; background: rgba(239, 68, 68, 0.15); color: #f87171; border: 1px solid rgba(239, 68, 68, 0.3); padding: 0.6rem; border-radius: 6px; font-weight: 600; cursor: pointer;">
              ↺ Reset Form
            </button>
          </div>

          <!-- 3. Push Button with JS -->
          <div style="background: #1e293b; border: 1px solid #334155; padding: 1rem; border-radius: 8px;">
            <span style="font-size: 0.75rem; color: #94a3b8; font-family: monospace; display: block; margin-bottom: 0.4rem;">type="button" + onclick</span>
            <button type="button" onclick="alert('💡 Push Button Clicked! JavaScript Event Triggered.');" style="width: 100%; background: rgba(16, 185, 129, 0.15); color: #34d399; border: 1px solid rgba(16, 185, 129, 0.3); padding: 0.6rem; border-radius: 6px; font-weight: 600; cursor: pointer;">
```

```
              ⚡ Trigger JS Alert
            </button>
          </div>

          <!-- 4. Input Button Tag -->
          <div style="background: #1e293b; border: 1px solid #334155; padding: 1rem; border-radius: 8px;">
            <span style="font-size: 0.75rem; color: #94a3b8; font-family: monospace; display: block; margin-bottom: 0.4rem;">&lt;input type="button"&gt;</span>
            <input type="button" value="📩 Input Button" onclick="alert('Standard HTML <input type=button> executed!');" style="width: 100%; background: rgba(245, 158, 11, 0.15); color: #fbbf24; border: 1px solid rgba(245, 158, 11, 0.3); padding: 0.6rem; border-radius: 6px; font-weight: 600; cursor: pointer;">
          </div>

        </div>
      </form>
    </section>

    <!-- ================================================
         PART B: Ordered Lists Demonstration
         ================================================ -->
    <section style="background: #1e293b; border: 1px solid #334155; border-radius: 16px; padding: 2rem; box-shadow: 0 10px 25px rgba(0, 0, 0, 0.25);">
      <h2 style="font-size: 1.5rem; font-weight: 700; color: #34d399; margin: 0 0 0.5rem 0; display: flex; align-items: center; gap: 0.5rem; border-bottom: 2px solid #334155; padding-bottom: 0.5rem;">
        <span>🔢</span> Part B: Adding Ordered Lists to a Web Page
      </h2>
      <p style="color: #cbd5e1; font-size: 0.95rem; margin: 0.5rem 0 1.5rem 0;">
        HTML ordered lists (<code>&lt;ol&gt;</code>) support various numbering systems via the <code>type</code>, <code>start</code>, and <code>reversed</code> attributes:
      </p>

      <div style="display: grid; grid-template-columns: repeat(auto-fit, minmax(300px, 1fr)); gap: 1.5rem;">

        <!-- List 1: Decimal (1, 2, 3) -->
        <div style="background: rgba(15, 23, 42, 0.6); border: 1px solid #475569; border-radius: 12px; padding: 1.25rem;">
          <h3 style="font-size: 1rem; color: #38bdf8; margin: 0 0 0.5rem 0;">1. Default Decimal (<code>type="1"</code>)</h3>
          <p style="font-size: 0.8rem; color: #94a3b8; margin: 0 0 0.75rem 0;">Web Development Lifecycle Steps:</p>
          <ol type="1" style="color: #e2e8f0; font-size: 0.9rem; padding-left: 1.5rem; margin: 0; line-height: 1.8;">
            <li>Requirement Analysis &amp; Wireframing</li>
            <li>UI/UX Design in Figma</li>
            <li>HTML5 Semantic Coding</li>
            <li>CSS3 Layout Styling &amp; Responsiveness</li>
            <li>JavaScript Interactivity &amp; Validation</li>
          </ol>
        </div>

        <!-- List 2: Uppercase Letters (A, B, C) -->
        <div style="background: rgba(15, 23, 42, 0.6); border: 1px solid #475569; border-radius: 12px; padding: 1.25rem;">
          <h3 style="font-size: 1rem; color: #34d399; margin: 0 0 0.5rem 0;">2. Uppercase Letters (<code>type="A"</code>)</h3>
          <p style="font-size: 0.8rem; color: #94a3b8; margin: 0 0 0.75rem 0;">Core Technical Domains:</p>
          <ol type="A" style="color: #e2e8f0; font-size: 0.9rem; padding-left: 1.5rem; margin: 0; line-height: 1.8;">
            <li>Full Stack Web Development</li>
            <li>Artificial Intelligence &amp; Machine Learning</li>
            <li>Cybersecurity &amp; Information Assurance</li>
```

```
            <li>Cloud Architecture &amp;amp; DevOps</li>
          </ol>
        </div>

        <!-- List 3: Uppercase Roman Numerals (I, II, III) -->
        <div style="background: rgba(15, 23, 42, 0.6); border: 1px solid #475569; border-radius: 1
2px; padding: 1.25rem;">
          <h3 style="font-size: 1rem; color: #fbbf24; margin: 0 0 0.5rem 0;">3. Roman Numerals (<c
ode>type="I"</code>)</h3>
          <p style="font-size: 0.8rem; color: #94a3b8; margin: 0 0 0.75rem 0;">Academic Semesters
Overview:</p>
          <ol type="I" style="color: #e2e8f0; font-size: 0.9rem; padding-left: 1.5rem; margin: 0;
line-height: 1.8;">
            <li>Foundation in Computer Science</li>
            <li>Data Structures &amp;amp; OOPs with C++</li>
            <li>Advanced Web Technology &amp;amp; DBMS</li>
            <li>Operating Systems &amp;amp; Networks</li>
          </ol>
        </div>

        <!-- List 4: Custom Start & Reversed -->
        <div style="background: rgba(15, 23, 42, 0.6); border: 1px solid #475569; border-radius: 1
2px; padding: 1.25rem;">
          <h3 style="font-size: 1rem; color: #f472b6; margin: 0 0 0.5rem 0;">4. Custom Start &amp;
Reversed (<code>start="10" reversed</code>)</h3>
          <p style="font-size: 0.8rem; color: #94a3b8; margin: 0 0 0.75rem 0;">Countdown Launch Se
quence:</p>
          <ol start="10" reversed style="color: #e2e8f0; font-size: 0.9rem; padding-left: 1.5rem;
margin: 0; line-height: 1.8;">
            <li>System Diagnostics Complete</li>
            <li>Database Connection Verified</li>
            <li>Deployment Sequence Initialized</li>
            <li>Application Online</li>
          </ol>
        </div>

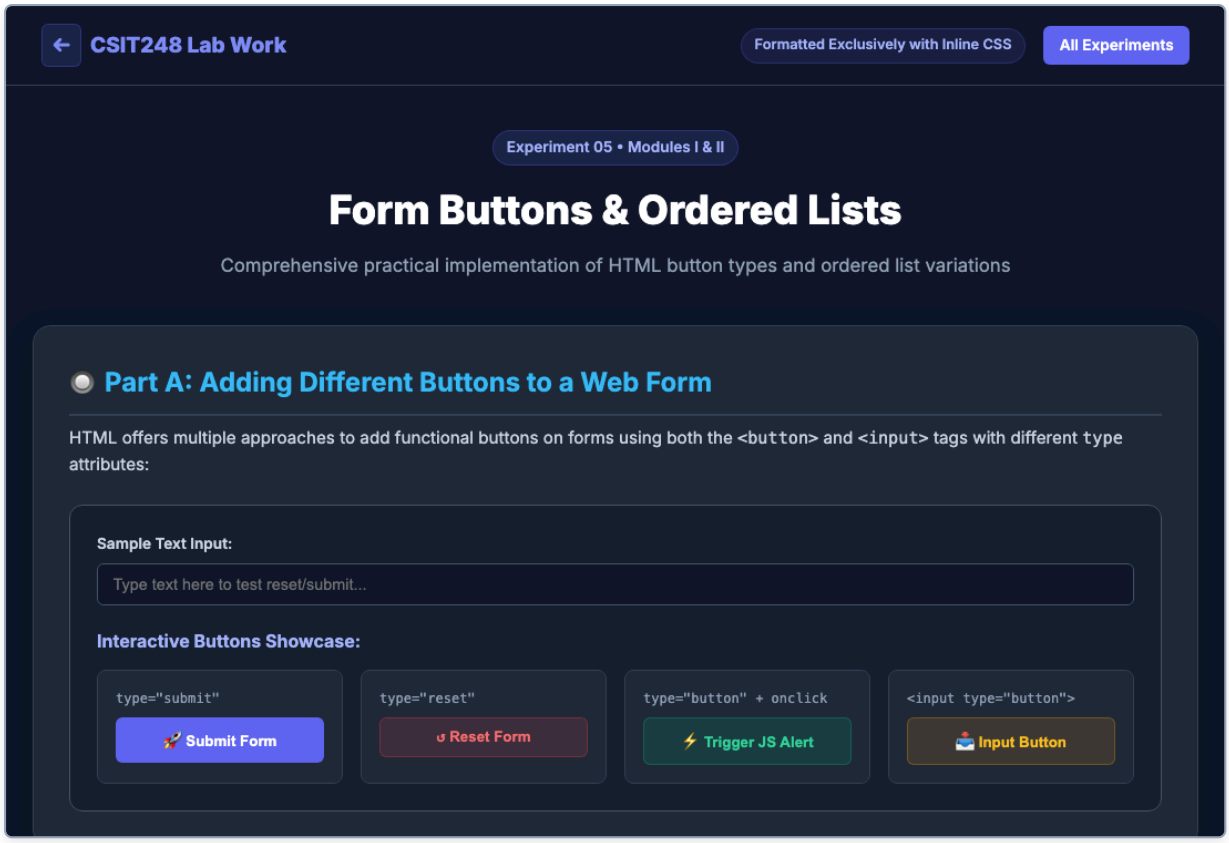
      </div>
    </section>

  </main>

  <!-- Footer (Inline CSS) -->
  <footer style="background: #090d16; border-top: 1px solid #1e293b; padding: 2rem 1.5rem; text-al
ign: center; color: #94a3b8; font-size: 0.875rem; margin-top: 3rem;">
    <p style="margin: 0 0 0.5rem 0;">Experiment 05 &bull; CSIT248 Advanced Web Technology &bull; F
orm Buttons &amp;amp; Ordered Lists</p>
    <p style="margin: 0 0 0.75rem 0; color: #64748b; font-size: 0.8rem;">Amity University Noida &b
ull; Formatted using 100% Inline CSS</p>
    <p style="margin: 0;"><a href="../index.html" style="color: #818cf8; text-decoration: none; fo
nt-weight: 600;">&amp;uarr; Back to Lab Work Portal</a></p>
  </footer>

</body>
</html>
```

Output :


CSIT248 Lab Work
Formatted Exclusively with Inline CSS
All Experiments
Experiment 05 • Modules I & II
Form Buttons & Ordered Lists
Comprehensive practical implementation of HTML button types and ordered list variations
Part A: Adding Different Buttons to a Web Form
HTML offers multiple approaches to add functional buttons on forms using both the <button> and <input> tags with different type attributes:
Sample Text Input:
Type text here to test reset/submit...
Interactive Buttons Showcase:
type="submit"
Submit Form
type="reset"
Reset Form
type="button" + onclick
Trigger JS Alert
<input type="button">
Input Button

Experiment No.: 9

Title : Computer Hardware Table Gallery with Borders and Cellpadding.

Aim : To construct a computer hardware image gallery using HTML table structure, specifying borders, cell padding, cell spacing, captions, and aligned thumbnail descriptions.

Source Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Experiment 6: Design a webpage to display Image gallery using
table - CSIT248 Advanced Web Technology">
  <title>Experiment 06 - Image Gallery using Table (Inline CSS)</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&display=swa
p" rel="stylesheet">
</head>
<body style="margin: 0; padding: 0; font-family: 'Inter', system-ui, -apple-system, sans-serif; ba
ckground-color: #0f172a; color: #f8fafc; line-height: 1.6; min-height: 100vh;">

  <!-- Navigation (Inline CSS) -->
  <nav style="position: sticky; top: 0; z-index: 100; background: rgba(15, 23, 42, 0.92); backdrop
-filter: blur(12px); border-bottom: 1px solid #334155; padding: 1rem 2rem; display: flex; justify-
content: space-between; align-items: center;">
    <a href="../index.html" style="font-weight: 800; font-size: 1.2rem; color: #818cf8; text-decor
ation: none; display: flex; align-items: center; gap: 0.5rem;">
      <span style="background: rgba(99, 102, 241, 0.15); border: 1px solid rgba(99, 102, 241, 0.
3); padding: 0.2rem 0.5rem; border-radius: 6px;">&larr;</span>
      CSIT248 Lab Work
    </a>
    <div style="display: flex; gap: 1rem; align-items: center;">
      <span style="font-size: 0.8rem; background: rgba(99, 102, 241, 0.12); color: #a5b4fc; borde
r: 1px solid rgba(99, 102, 241, 0.25); padding: 0.3rem 0.75rem; border-radius: 9999px; font-weigh
t: 600;">
        Formatted Exclusively with Inline CSS
      </span>
      <a href="../index.html" style="background: #6366f1; color: #ffffff; padding: 0.4rem 0.9rem;
border-radius: 6px; text-decoration: none; font-size: 0.85rem; font-weight: 600;">
        All Experiments
      </a>
    </div>
  </nav>

  <main style="max-width: 1200px; margin: 0 auto; padding: 2.5rem 1.5rem;">

    <!-- Header (Inline CSS) -->
    <header style="text-align: center; margin-bottom: 2rem;">
      <div style="display: inline-block; background: rgba(99, 102, 241, 0.15); color: #a5b4fc; bor
der: 1px solid rgba(99, 102, 241, 0.3); padding: 0.25rem 0.75rem; border-radius: 9999px; font-siz
e: 0.85rem; font-weight: 600; margin-bottom: 0.75rem;">
        Experiment 06 &bull; Module II (Tables &amp; Multimedia)
      </div>
      <h1 style="font-size: 2.25rem; font-weight: 800; color: #ffffff; margin: 0 0 0.5rem 0;">HTML
Table Image Gallery</h1>
      <p style="color: #94a3b8; font-size: 1.05rem; margin: 0;">Multi-column structured photograph
ic showcase organized using semantic HTML table rows and cells</p>
    </header>
```

```
    <!-- Table Gallery Container (Inline CSS) -->
    <div style="background: #1e293b; border: 1px solid #334155; border-radius: 16px; overflow-x: a
uto; box-shadow: 0 10px 25px rgba(0, 0, 0, 0.3); padding: 1.5rem;">
      <table border="1" cellpadding="12" cellspacing="0" style="width: 100%; border-collapse: coll
apse; text-align: center; border-color: #334155;">
        <caption style="font-size: 1.25rem; font-weight: 700; color: #f8fafc; padding: 1rem; text-
align: left; background: #1e1b4b; border-radius: 8px 8px 0 0; border: 1px solid #334155; border-bo
ttom: none;">
          📸 Curated Photography &amp;amp; Art Gallery Grid
        </caption>

        <!-- Table Header -->
        <thead>
          <tr style="background: rgba(15, 23, 42, 0.9); color: #a5b4fc; font-size: 0.9rem;">
            <th style="padding: 1rem; border: 1px solid #475569; width: 33.33%;">Column 1: Nature
&amp;amp; Serenity</th>
            <th style="padding: 1rem; border: 1px solid #475569; width: 33.33%;">Column 2: Archite
cture &amp;amp; Urban</th>
            <th style="padding: 1rem; border: 1px solid #475569; width: 33.33%;">Column 3: Computi
ng &amp;amp; Cosmos</th>
          </tr>
        </thead>

        <tbody>
          <!-- ROW 1 -->
          <tr>
            <!-- Image 1 -->
            <td style="border: 1px solid #475569; padding: 1.25rem; vertical-align: top; backgroun
d: rgba(15, 23, 42, 0.5);">
              <div style="background: #0f172a; border: 1px solid #334155; border-radius: 10px; ove
rflow: hidden; padding: 0.75rem;">
                <svg width="100%" height="160" viewBox="0 0 300 160" fill="none" style="border-rad
ius: 6px;">
                  <rect width="300" height="160" fill="#064e3b" />
                  <circle cx="150" cy="140" r="100" fill="#047857" />
                  <circle cx="230" cy="50" r="30" fill="#fef08a" opacity="0.9" />
                  <polygon points="60,160 110,90 160,160" fill="#065f46" />
                  <polygon points="130,160 190,70 250,160" fill="#022c22" />
                  <polygon points="10,160 70,110 130,160" fill="#047857" />
                </svg>
                <h4 style="color: #34d399; font-size: 1rem; margin: 0.75rem 0 0.25rem 0;">Alpine F
orest Sunset</h4>
                <p style="color: #94a3b8; font-size: 0.8rem; margin: 0;">Pine valleys nestled in m
ountainous slopes during golden hour.</p>
              </div>
            </td>

            <!-- Image 2 -->
            <td style="border: 1px solid #475569; padding: 1.25rem; vertical-align: top; backgroun
d: rgba(15, 23, 42, 0.5);">
              <div style="background: #0f172a; border: 1px solid #334155; border-radius: 10px; ove
rflow: hidden; padding: 0.75rem;">
                <svg width="100%" height="160" viewBox="0 0 300 160" fill="none" style="border-rad
ius: 6px;">
                  <rect width="300" height="160" fill="#1e1b4b" />
                  <circle cx="150" cy="80" r="50" fill="#4338ca" opacity="0.4" />
                  <!-- Skyscrapers -->
                  <rect x="40" y="50" width="40" height="110" fill="#312e81" stroke="#4f46e5" stro
ke-width="1" />
                  <rect x="90" y="30" width="50" height="130" fill="#1e1b4b" stroke="#6366f1" stro
ke-width="1.5" />
                  <rect x="150" y="60" width="45" height="100" fill="#312e81" stroke="#4f46e5" str
oke-width="1" />
                  <rect x="205" y="40" width="55" height="120" fill="#1e1b4b" stroke="#818cf8" str
oke-width="1" />
```

```
                <!-- Windows dots -->
                <circle cx="115" cy="50" r="3" fill="#fcd34d" />
                <circle cx="115" cy="70" r="3" fill="#fcd34d" />
                <circle cx="230" cy="60" r="3" fill="#38bdf8" />
              </svg>
              <h4 style="color: #818cf8; font-size: 1rem; margin: 0.75rem 0 0.25rem 0;">Metropol
is Skyline</h4>
              <p style="color: #94a3b8; font-size: 0.8rem; margin: 0;">Towering urban skyscraper
s glowing under evening twilight.</p>
            </div>
          </td>

          <!-- Image 3 -->
          <td style="border: 1px solid #475569; padding: 1.25rem; vertical-align: top; backgroun
d: rgba(15, 23, 42, 0.5);">
            <div style="background: #0f172a; border: 1px solid #334155; border-radius: 10px; ove
rflow: hidden; padding: 0.75rem;">
              <svg width="100%" height="160" viewBox="0 0 300 160" fill="none" style="border-rad
ius: 6px;">
                <rect width="300" height="160" fill="#090d16" />
                <circle cx="150" cy="80" r="45" fill="#3b82f6" opacity="0.3" />
                <circle cx="150" cy="80" r="30" fill="#60a5fa" />
                <!-- Orbit rings -->
                <ellipse cx="150" cy="80" rx="90" ry="25" stroke="#06b6d4" stroke-width="1.5" st
roke-dasharray="4 2" fill="none" />
                <circle cx="210" cy="70" r="8" fill="#ec4899" />
                <circle cx="80" cy="90" r="5" fill="#f59e0b" />
              </svg>
              <h4 style="color: #38bdf8; font-size: 1rem; margin: 0.75rem 0 0.25rem 0;">Deep Spa
ce Exoplanet</h4>
              <p style="color: #94a3b8; font-size: 0.8rem; margin: 0;">Planetary orbits floating
across astronomical stellar fields.</p>
            </div>
          </td>
        </tr>

        <!-- ROW 2 -->
        <tr>
          <!-- Image 4 -->
          <td style="border: 1px solid #475569; padding: 1.25rem; vertical-align: top; backgroun
d: rgba(15, 23, 42, 0.5);">
            <div style="background: #0f172a; border: 1px solid #334155; border-radius: 10px; ove
rflow: hidden; padding: 0.75rem;">
              <svg width="100%" height="160" viewBox="0 0 300 160" fill="none" style="border-rad
ius: 6px;">
                <rect width="300" height="160" fill="#0c4a6e" />
                <path d="M0 120 Q 75 90, 150 120 T 300 120 L 300 160 L 0 160 Z" fill="#0369a1" /
>
                <path d="M0 135 Q 75 110, 150 135 T 300 135 L 300 160 L 0 160 Z" fill="#0284c7"
/>
                <circle cx="80" cy="50" r="22" fill="#fbbf24" />
              </svg>
              <h4 style="color: #38bdf8; font-size: 1rem; margin: 0.75rem 0 0.25rem 0;">Ocean Wa
ves Horizon</h4>
              <p style="color: #94a3b8; font-size: 0.8rem; margin: 0;">Rhythmic ocean surf rolli
ng across a tropical sandy shoreline.</p>
            </div>
          </td>

          <!-- Image 5 -->
          <td style="border: 1px solid #475569; padding: 1.25rem; vertical-align: top; backgroun
d: rgba(15, 23, 42, 0.5);">
            <div style="background: #0f172a; border: 1px solid #334155; border-radius: 10px; ove
rflow: hidden; padding: 0.75rem;">
              <svg width="100%" height="160" viewBox="0 0 300 160" fill="none" style="border-rad
ius: 6px;">
```

```
                  <rect width="300" height="160" fill="#451a03" />
                  <polygon points="150,20 230,140 70,140" fill="#b45309" stroke="#f59e0b" stroke-w
idth="2" />
                  <polygon points="150,20 190,140 70,140" fill="#78350f" />
                  <circle cx="240" cy="40" r="15" fill="#fde68a" />
                </svg>
                <h4 style="color: #fbbf24; font-size: 1rem; margin: 0.75rem 0 0.25rem 0;">Desert P
yramid Heritage</h4>
                <p style="color: #94a3b8; font-size: 0.8rem; margin: 0;">Monumental ancient stone
architecture under starry desert skies.</p>
              </div>
            </td>

            <!-- Image 6 -->
            <td style="border: 1px solid #475569; padding: 1.25rem; vertical-align: top; backgroun
d: rgba(15, 23, 42, 0.5);">
              <div style="background: #0f172a; border: 1px solid #334155; border-radius: 10px; ove
rflow: hidden; padding: 0.75rem;">
                <svg width="100%" height="160" viewBox="0 0 300 160" fill="none" style="border-rad
ius: 6px;">
                  <rect width="300" height="160" fill="#18181b" />
                  <rect x="80" y="30" width="140" height="100" rx="10" fill="#27272a" stroke="#10b
981" stroke-width="2" />
                  <!-- Circuit lines -->
                  <path d="M100 80 H 130 V 50 H 170 V 80 H 200" stroke="#10b981" stroke-width="2"
fill="none" />
                  <circle cx="150" cy="80" r="12" fill="#10b981" />
                </svg>
                <h4 style="color: #34d399; font-size: 1rem; margin: 0.75rem 0 0.25rem 0;">Silicon
Microchip Circuit</h4>
                <p style="color: #94a3b8; font-size: 0.8rem; margin: 0;">High-density semiconducto
r nanometer logic microprocessor.</p>
              </div>
            </td>
          </tr>
        </tbody>

        <!-- Table Footer -->
        <tfoot>
          <tr style="background: #0f172a; color: #94a3b8; font-size: 0.85rem;">
            <td colspan="3" style="padding: 1rem; border: 1px solid #475569; text-align: center;">
              Displaying <strong>6 Curated Images</strong> across 2 Rows &times; 3 Columns structu
red via HTML <code>&lt;table&gt;</code>.
            </td>
          </tr>
        </tfoot>
      </table>
    </div>

  </main>

  <!-- Footer (Inline CSS) -->
  <footer style="background: #090d16; border-top: 1px solid #1e293b; padding: 2rem 1.5rem; text-al
ign: center; color: #94a3b8; font-size: 0.875rem; margin-top: 3rem;">
    <p style="margin: 0 0 0.5rem 0;">Experiment 06 &bull; CSIT248 Advanced Web Technology &bull; T
able Image Gallery</p>
    <p style="margin: 0 0 0.75rem 0; color: #64748b; font-size: 0.8rem;">Amity University Noida &b
ull; Formatted using 100% Inline CSS</p>
    <p style="margin: 0;"><a href="../index.html" style="color: #818cf8; text-decoration: none; fo
nt-weight: 600;">&uarr; Back to Lab Work Portal</a></p>
  </footer>

</body>
</html>
```

Output :

CSIT248 Lab Work

Formatted Exclusively with Inline CSS

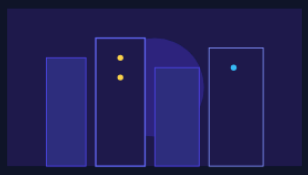
All Experiments

Experiment 06 • Module II (Tables & Multimedia)

HTML Table Image Gallery

Multi-column structured photographic showcase organized using semantic HTML table rows and cells

📸 Curated Photography & Art Gallery Grid

Column 1: Nature & Serenity	Column 2: Architecture & Urban	Column 3: Computing & Cosmos
 **Alpine Forest Sunset** Pine valleys nestled in mountainous slopes during golden hour.	 **Metropolis Skyline** Towering urban skyscrapers glowing under evening twilight.	 **Deep Space Exoplanet** Planetary orbits floating across astronomical stellar fields.

Experiment No.: 10

Title : Client-Side History Navigation using JavaScript window.history.

Aim : To demonstrate client-side web browser history manipulation utilizing JavaScript window.history methods (back(), forward(), go(), and history.length).

Source Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Experiment 7: Design a web script to show the functionality of FORWARD and BACK button of History class using JavaScript - CSIT248 Advanced Web Technology">
  <title>Experiment 07 - JavaScript History Object (Inline CSS)</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&display=swap" rel="stylesheet">
</head>
<body style="margin: 0; padding: 0; font-family: 'Inter', system-ui, -apple-system, sans-serif; background-color: #0f172a; color: #f8fafc; line-height: 1.6; min-height: 100vh;">

  <!-- Navigation (Inline CSS) -->
  <nav style="position: sticky; top: 0; z-index: 100; background: rgba(15, 23, 42, 0.92); backdrop-filter: blur(12px); border-bottom: 1px solid #334155; padding: 1rem 2rem; display: flex; justify-content: space-between; align-items: center;">
    <a href="../index.html" style="font-weight: 800; font-size: 1.2rem; color: #818cf8; text-decoration: none; display: flex; align-items: center; gap: 0.5rem;">
      <span style="background: rgba(99, 102, 241, 0.15); border: 1px solid rgba(99, 102, 241, 0.3); padding: 0.2rem 0.5rem; border-radius: 6px;">&larr;</span>
      CSIT248 Lab Work
    </a>
    <div style="display: flex; gap: 1rem; align-items: center;">
      <span style="font-size: 0.8rem; background: rgba(99, 102, 241, 0.12); color: #a5b4fc; border: 1px solid rgba(99, 102, 241, 0.25); padding: 0.3rem 0.75rem; border-radius: 9999px; font-weight: 600;">
        Formatted Exclusively with Inline CSS
      </span>
      <a href="../index.html" style="background: #6366f1; color: #ffffff; padding: 0.4rem 0.9rem; border-radius: 6px; text-decoration: none; font-size: 0.85rem; font-weight: 600;">
        All Experiments
      </a>
    </div>
  </nav>

  <main style="max-width: 1000px; margin: 0 auto; padding: 2.5rem 1.5rem;">

    <!-- Header (Inline CSS) -->
    <header style="text-align: center; margin-bottom: 2rem;">
      <div style="display: inline-block; background: rgba(99, 102, 241, 0.15); color: #a5b4fc; border: 1px solid rgba(99, 102, 241, 0.3); padding: 0.25rem 0.75rem; border-radius: 9999px; font-size: 0.85rem; font-weight: 600; margin-bottom: 0.75rem;">
        Experiment 07 &bull; Module IV (JavaScript Event Handling &amp; BOM)
      </div>
      <h1 style="font-size: 2.25rem; font-weight: 800; color: #ffffff; margin: 0 0 0.5rem 0;">JavaScript History Object Navigator</h1>
      <p style="color: #94a3b8; font-size: 1.05rem; margin: 0;">Demonstration of <code>history.back()</code>, <code>history.forward()</code>, and <code>history.go()</code> functionality</p>
    </header>
```

```
    <!-- Interactive History Controller Box (Inline CSS) -->
    <section style="background: #1e293b; border: 1px solid #334155; border-radius: 16px; padding:
2rem; margin-bottom: 2rem; box-shadow: 0 10px 25px rgba(0, 0, 0, 0.3);">

      <!-- Current Virtual Page Indicator -->
      <div style="background: rgba(15, 23, 42, 0.8); border: 1px solid #475569; border-radius: 12p
x; padding: 1.75rem; text-align: center; margin-bottom: 1.75rem;">
        <span style="font-size: 0.85rem; text-transform: uppercase; letter-spacing: 0.1em; color:
#38bdf8; font-weight: 700; display: block; margin-bottom: 0.5rem;">
          Simulated Browsing Session (HTML5 History API)
        </span>
        <h2 id="current-step-title" style="font-size: 1.85rem; font-weight: 800; color: #ffffff; m
argin: 0 0 0.5rem 0;">
          📍 Step 1: Course Overview (CSIT248)
        </h2>
        <p id="current-step-desc" style="color: #cbd5e1; font-size: 0.95rem; max-width: 600px; mar
gin: 0 auto;">
          Welcome to the initial entry point of our lab browsing history stack. Click "Push Next P
age" to simulate browser navigation.
        </p>
      </div>

      <!-- Control Buttons (Inline CSS) -->
      <div style="display: flex; justify-content: center; gap: 1rem; flex-wrap: wrap; margin-botto
m: 1.5rem;">
        <button onclick="goBack()" style="background: rgba(239, 68, 68, 0.15); color: #f87171; bor
der: 1px solid rgba(239, 68, 68, 0.3); padding: 0.75rem 1.5rem; border-radius: 8px; font-weight: 7
00; font-size: 0.95rem; cursor: pointer; display: flex; align-items: center; gap: 0.5rem;">
          &larr; history.back()
        </button>

        <button onclick="pushNextState()" style="background: #6366f1; color: #ffffff; border: 1px
solid #6366f1; padding: 0.75rem 1.75rem; border-radius: 8px; font-weight: 700; font-size: 0.95rem;
cursor: pointer; display: flex; align-items: center; gap: 0.5rem; box-shadow: 0 4px 14px rgba(99,
102, 241, 0.4);">
          ✚ Push Next Page (history.pushState)
        </button>

        <button onclick="goForward()" style="background: rgba(16, 185, 129, 0.15); color: #34d399;
border: 1px solid rgba(16, 185, 129, 0.3); padding: 0.75rem 1.5rem; border-radius: 8px; font-weigh
t: 700; font-size: 0.95rem; cursor: pointer; display: flex; align-items: center; gap: 0.5rem;">
          history.forward() &rarr;
        </button>
      </div>

      <!-- History State Metrics (Inline CSS) -->
      <div style="display: grid; grid-template-columns: repeat(auto-fit, minmax(200px, 1fr)); gap:
1rem; background: #0f172a; padding: 1.25rem; border-radius: 10px; border: 1px solid #334155;">
        <div>
          <span style="font-size: 0.8rem; color: #94a3b8; display: block;">window.history.length:
</span>
          <strong id="history-length" style="font-size: 1.3rem; color: #38bdf8;">--</strong>
        </div>
        <div>
          <span style="font-size: 0.8rem; color: #94a3b8; display: block;">Current Step Index:</sp
an>
          <strong id="current-index" style="font-size: 1.3rem; color: #34d399;">1 / 4</strong>
        </div>
        <div>
          <span style="font-size: 0.8rem; color: #94a3b8; display: block;">Last Triggered Action:
</span>
          <strong id="last-action" style="font-size: 0.95rem; color: #fbbf24;">Initial Load</stron
g>
        </div>
      </div>
```

```
    </section>

    <!-- Technical Code Explanation (Inline CSS) -->
    <section style="background: #1e293b; border: 1px solid #334155; border-radius: 16px; padding:
2rem; box-shadow: 0 10px 25px rgba(0, 0, 0, 0.25);">
      <h3 style="font-size: 1.25rem; font-weight: 700; color: #a5b4fc; margin: 0 0 1rem 0;">
        📖 JavaScript History Class Implementation Reference
      </h3>
      <pre style="background: #090d16; border: 1px solid #334155; border-radius: 8px; padding: 1.2
5rem; overflow-x: auto; font-family: 'Fira Code', monospace; font-size: 0.85rem; color: #e2e8f0; l
ine-height: 1.6; margin: 0;"><code>// 1. Move one page backward in browser history
function goBack() {
  window.history.back(); // Equivalent to window.history.go(-1)
}

// 2. Move one page forward in browser history
function goForward() {
  window.history.forward(); // Equivalent to window.history.go(1)
}

// 3. Jump specific relative offset
function jumpHistory(offset) {
  window.history.go(offset);
}

// 4. Inspect session history length
console.log("Total entries in session:", window.history.length);</code></pre>
    </section>

  </main>

  <!-- Interactive JavaScript -->
  <script>
    const pages = [
      { id: 1, title: "📍 Step 1: Course Overview (CSIT248)", desc: "Welcome to the initial entry
point of our lab browsing history stack. Click 'Push Next Page' to simulate browser navigation."
},
      { id: 2, title: "📄 Step 2: HTML5 & DOM Architecture", desc: "You have advanced to Step 2!
The browser history stack now has previous entries available for history.back()." },
      { id: 3, title: "🎨 Step 3: Cascading Style Sheets (CSS3)", desc: "Step 3 reached: Testing
deep history chain traversal. You can now step backward twice or continue forward." },
      { id: 4, title: "⚡ Step 4: JavaScript Event Loop & History API", desc: "Step 4 reached: Term
inal point of this simulated history session. Try using history.back() to navigate back!" }
    ];

    let currentStep = 1;

    function updateDisplay(step, actionText) {
      const page = pages.find(p => p.id === step) || pages[0];
      document.getElementById('current-step-title').textContent = page.title;
      document.getElementById('current-step-desc').textContent = page.desc;
      document.getElementById('history-length').textContent = window.history.length;
      document.getElementById('current-index').textContent = `${step} / ${pages.length}`;
      if (actionText) {
        document.getElementById('last-action').textContent = actionText;
      }
    }

    function pushNextState() {
      if (currentStep < pages.length) {
        currentStep++;
        history.pushState({ step: currentStep }, `Step ${currentStep}`, `#step-${currentStep}`);
        updateDisplay(currentStep, `history.pushState(#step-${currentStep})`);
      } else {
        alert("Maximum simulated steps reached! Use history.back() to move backward.");
      }
```

```
    }

    function goBack() {
      if (currentStep > 1) {
        currentStep--;
        history.back();
        updateDisplay(currentStep, "window.history.back()");
      } else {
        alert("⚠️ You are already at the earliest page in this session history stack!");
      }
    }

    function goForward() {
      if (currentStep < pages.length) {
        currentStep++;
        history.forward();
        updateDisplay(currentStep, "window.history.forward()");
      } else {
        alert("⚠️ No forward history entry exists ahead in this session stack!");
      }
    }

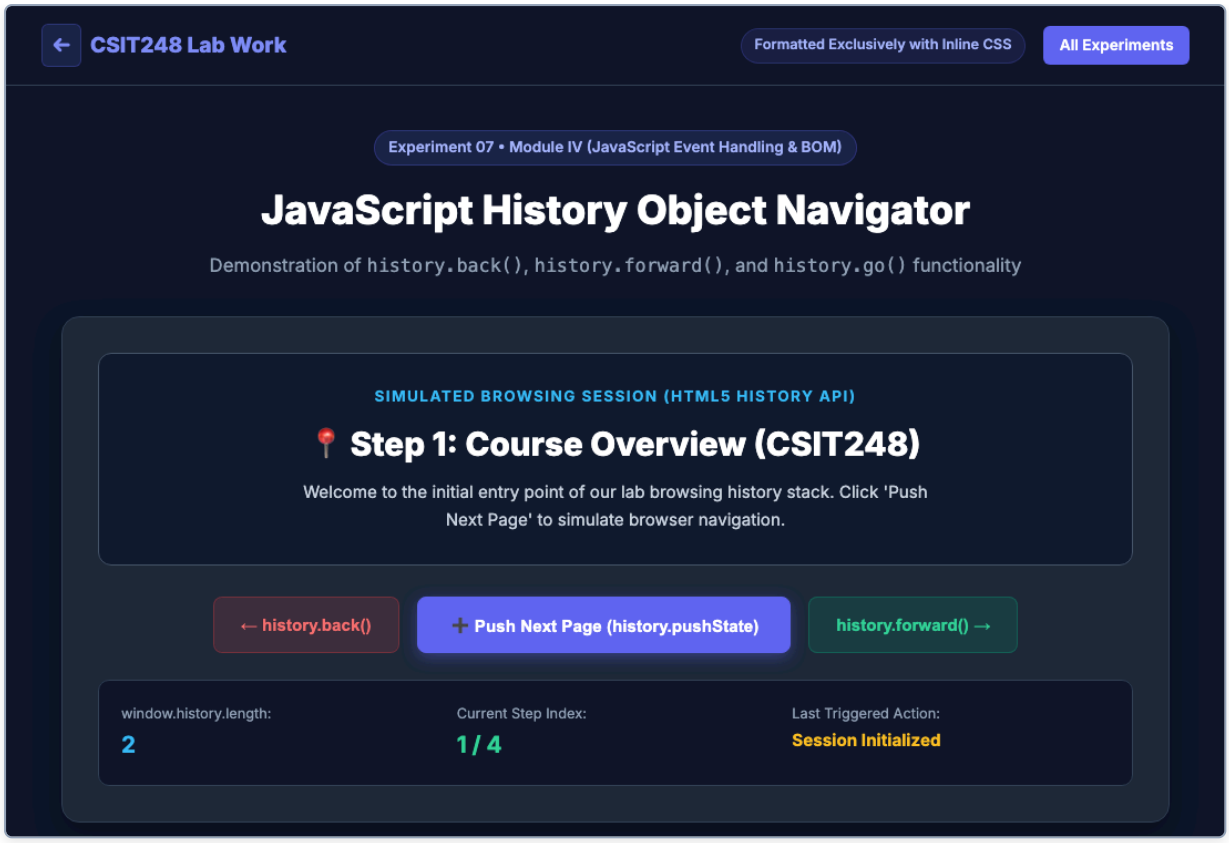
    window.addEventListener('popstate', (event) => {
      if (event.state && event.state.step) {
        currentStep = event.state.step;
        updateDisplay(currentStep, "popstate Event Triggered");
      }
    });

    // Initialize display on load
    updateDisplay(1, "Session Initialized");
  </script>

  <!-- Footer (Inline CSS) -->
  <footer style="background: #090d16; border-top: 1px solid #1e293b; padding: 2rem 1.5rem; text-al
ign: center; color: #94a3b8; font-size: 0.875rem; margin-top: 3rem;">
    <p style="margin: 0 0 0.5rem 0;">Experiment 07 &bull; CSIT248 Advanced Web Technology &bull; J
avaScript History Class</p>
    <p style="margin: 0 0 0.75rem 0; color: #64748b; font-size: 0.8rem;">Amity University Noida &b
ull; Formatted using 100% Inline CSS</p>
    <p style="margin: 0;"><a href="../index.html" style="color: #818cf8; text-decoration: none; fo
nt-weight: 600;">&uarr; Back to Lab Work Portal</a></p>
  </footer>

</body>
</html>
```

Output :


CSIT248 Lab Work
Formatted Exclusively with Inline CSS
All Experiments
Experiment 07 • Module IV (JavaScript Event Handling & BOM)
JavaScript History Object Navigator
Demonstration of history.back(), history.forward(), and history.go() functionality
SIMULATED BROWSING SESSION (HTML5 HISTORY API)
📍 Step 1: Course Overview (CSIT248)
Welcome to the initial entry point of our lab browsing history stack. Click 'Push Next Page' to simulate browser navigation.
← history.back()
➕ Push Next Page (history.pushState)
history.forward() →
window.history.length:
2
Current Step Index:
1 / 4
Last Triggered Action:
Session Initialized

Experiment No.: 11

Title : Comprehensive Front-Level Form Validation with Regular Expressions.

Aim : To implement robust client-side front-level form validation in JavaScript using regular expressions for student registration details, validating email, phone, passwords, and required fields before submission.

Source Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Experiment 8: Write a program to show front level validation i
n a Registration Form - CSIT248 Advanced Web Technology">
  <title>Experiment 08 - Front-Level Form Validation (Inline CSS)</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&display=swa
p" rel="stylesheet">
</head>
<body style="margin: 0; padding: 0; font-family: 'Inter', system-ui, -apple-system, sans-serif; ba
ckground-color: #0f172a; color: #f8fafc; line-height: 1.6; min-height: 100vh;">

  <!-- Navigation (Inline CSS) -->
  <nav style="position: sticky; top: 0; z-index: 100; background: rgba(15, 23, 42, 0.92); backdrop
-filter: blur(12px); border-bottom: 1px solid #334155; padding: 1rem 2rem; display: flex; justify-
content: space-between; align-items: center;">
    <a href="../index.html" style="font-weight: 800; font-size: 1.2rem; color: #818cf8; text-decor
ation: none; display: flex; align-items: center; gap: 0.5rem;">
      <span style="background: rgba(99, 102, 241, 0.15); border: 1px solid rgba(99, 102, 241, 0.
3); padding: 0.2rem 0.5rem; border-radius: 6px;">&larr;</span>
      CSIT248 Lab Work
    </a>
    <div style="display: flex; gap: 1rem; align-items: center;">
      <span style="font-size: 0.8rem; background: rgba(99, 102, 241, 0.12); color: #a5b4fc; borde
r: 1px solid rgba(99, 102, 241, 0.25); padding: 0.3rem 0.75rem; border-radius: 9999px; font-weigh
t: 600;">
        Formatted Exclusively with Inline CSS
      </span>
      <a href="../index.html" style="background: #6366f1; color: #ffffff; padding: 0.4rem 0.9rem;
border-radius: 6px; text-decoration: none; font-size: 0.85rem; font-weight: 600;">
        All Experiments
      </a>
    </div>
  </nav>

  <main style="max-width: 800px; margin: 0 auto; padding: 2.5rem 1.5rem;">

    <!-- Header (Inline CSS) -->
    <header style="text-align: center; margin-bottom: 2rem;">
      <div style="display: inline-block; background: rgba(99, 102, 241, 0.15); color: #a5b4fc; bor
der: 1px solid rgba(99, 102, 241, 0.3); padding: 0.25rem 0.75rem; border-radius: 9999px; font-siz
e: 0.85rem; font-weight: 600; margin-bottom: 0.75rem;">
        Experiment 08 &bull; Module IV (Front-Level JavaScript Form Validation)
      </div>
      <h1 style="font-size: 2.25rem; font-weight: 800; color: #ffffff; margin: 0 0 0.5rem 0;">Fron
t-Level Form Validation</h1>
      <p style="color: #94a3b8; font-size: 1.05rem; margin: 0;">Real-time regular expression &amp;
logical checks using JavaScript DOM events</p>
    </header>
```

```
    <!-- Validation Form Box (Inline CSS) -->
    <section style="background: #1e293b; border: 1px solid #334155; border-radius: 16px; padding:
2.25rem; box-shadow: 0 10px 25px rgba(0, 0, 0, 0.3);">

      <!-- Top Alert Banner -->
      <div id="error-summary" style="display: none; background: rgba(239, 68, 68, 0.15); border: 1
px solid rgba(239, 68, 68, 0.3); color: #fca5a5; padding: 1rem; border-radius: 8px; font-size: 0.9
rem; margin-bottom: 1.5rem;">
        ⚠️ Please resolve all highlighted errors before submitting the registration.
      </div>

      <form id="val-form" onsubmit="validateAndSubmit(event)" novalidate>

        <!-- 1. Full Name -->
        <div style="margin-bottom: 1.25rem;">
          <label for="fullname" style="display: block; font-size: 0.85rem; font-weight: 600; colo
r: #cbd5e1; margin-bottom: 0.35rem;">
            Student Full Name <span style="color: #ef4444;">*</span>
          </label>
          <input
            type="text"
            id="fullname"
            placeholder="e.g. Atul Anand"
            oninput="validateName()"
            style="width: 100%; box-sizing: border-box; background: #0f172a; border: 1px solid #47
5569; color: #ffffff; padding: 0.65rem 0.85rem; border-radius: 6px; font-size: 0.9rem; outline: no
ne;"
          >
          <span id="err-name" style="display: none; color: #f87171; font-size: 0.8rem; margin-top:
0.25rem;"></span>
        </div>

        <!-- 2. Enrollment Number -->
        <div style="margin-bottom: 1.25rem;">
          <label for="enroll-no" style="display: block; font-size: 0.85rem; font-weight: 600; colo
r: #cbd5e1; margin-bottom: 0.35rem;">
            Enrollment ID <span style="color: #ef4444;">*</span>
          </label>
          <input
            type="text"
            id="enroll-no"
            placeholder="e.g. A2305221001"
            oninput="validateEnroll()"
            style="width: 100%; box-sizing: border-box; background: #0f172a; border: 1px solid #47
5569; color: #ffffff; padding: 0.65rem 0.85rem; border-radius: 6px; font-size: 0.9rem; outline: no
ne;"
          >
          <span id="err-enroll" style="display: none; color: #f87171; font-size: 0.8rem; margin-to
p: 0.25rem;"></span>
        </div>

        <!-- 3. Email Address -->
        <div style="margin-bottom: 1.25rem;">
          <label for="user-email" style="display: block; font-size: 0.85rem; font-weight: 600; col
or: #cbd5e1; margin-bottom: 0.35rem;">
            Email Address <span style="color: #ef4444;">*</span>
          </label>
          <input
            type="email"
            id="user-email"
            placeholder="e.g. atulll0923@gmail.com"
            oninput="validateEmail()"
            style="width: 100%; box-sizing: border-box; background: #0f172a; border: 1px solid #47
5569; color: #ffffff; padding: 0.65rem 0.85rem; border-radius: 6px; font-size: 0.9rem; outline: no
ne;"
```

```
          >
          <span id="err-email" style="display: none; color: #f87171; font-size: 0.8rem; margin-to
p: 0.25rem;"></span>
        </div>

        <!-- 4. Mobile Number (10 digits) -->
        <div style="margin-bottom: 1.25rem;">
          <label for="user-mobile" style="display: block; font-size: 0.85rem; font-weight: 600; co
lor: #cbd5e1; margin-bottom: 0.35rem;">
            Mobile Number (10 Digits) <span style="color: #ef4444;">*</span>
          </label>
          <input
            type="tel"
            id="user-mobile"
            maxlength="10"
            placeholder="e.g. 9876543210"
            oninput="validateMobile()"
            style="width: 100%; box-sizing: border-box; background: #0f172a; border: 1px solid #47
5569; color: #ffffff; padding: 0.65rem 0.85rem; border-radius: 6px; font-size: 0.9rem; outline: no
ne;"
          >
          <span id="err-mobile" style="display: none; color: #f87171; font-size: 0.8rem; margin-to
p: 0.25rem;"></span>
        </div>

        <!-- 5. Password & Confirm Password Grid -->
        <div style="display: grid; grid-template-columns: repeat(auto-fit, minmax(240px, 1fr)); ga
p: 1.25rem; margin-bottom: 1.25rem;">
          <div>
            <label for="user-pwd" style="display: block; font-size: 0.85rem; font-weight: 600; col
or: #cbd5e1; margin-bottom: 0.35rem;">
              Password (Min 8 chars, 1 uppercase, 1 number) <span style="color: #ef4444;">*</span>
            </label>
            <input
              type="password"
              id="user-pwd"
              placeholder="••••••••"
              oninput="validatePwd()"
              style="width: 100%; box-sizing: border-box; background: #0f172a; border: 1px solid #
475569; color: #ffffff; padding: 0.65rem 0.85rem; border-radius: 6px; font-size: 0.9rem; outline:
none;"
            >
            <span id="err-pwd" style="display: none; color: #f87171; font-size: 0.8rem; margin-to
p: 0.25rem;"></span>
          </div>

          <div>
            <label for="user-cpwd" style="display: block; font-size: 0.85rem; font-weight: 600; co
lor: #cbd5e1; margin-bottom: 0.35rem;">
              Confirm Password <span style="color: #ef4444;">*</span>
            </label>
            <input
              type="password"
              id="user-cpwd"
              placeholder="••••••••"
              oninput="validateCPwd()"
              style="width: 100%; box-sizing: border-box; background: #0f172a; border: 1px solid #
475569; color: #ffffff; padding: 0.65rem 0.85rem; border-radius: 6px; font-size: 0.9rem; outline:
none;"
            >
            <span id="err-cpwd" style="display: none; color: #f87171; font-size: 0.8rem; margin-to
p: 0.25rem;"></span>
          </div>
        </div>

        <!-- 6. Terms Checkbox -->
```

```
        <div style="margin-bottom: 1.75rem;">
          <label style="display: flex; align-items: center; gap: 0.5rem; font-size: 0.85rem; colo
r: #cbd5e1; cursor: pointer;">
            <input type="checkbox" id="terms-chk" onchange="validateTerms()" style="accent-color:
#6366f1;">
            <span>I accept the university academic policies and front-level code validation rules.
</span>
          </label>
          <span id="err-terms" style="display: none; color: #f87171; font-size: 0.8rem; margin-to
p: 0.25rem;"></span>
        </div>

        <!-- Action Buttons -->
        <div style="display: flex; gap: 1rem; justify-content: flex-end;">
          <button type="button" onclick="resetValidationForm()" style="background: rgba(239, 68, 6
8, 0.15); color: #f87171; border: 1px solid rgba(239, 68, 68, 0.3); padding: 0.75rem 1.5rem; borde
r-radius: 8px; font-weight: 600; font-size: 0.95rem; cursor: pointer;">
            ↺ Reset
          </button>
          <button type="submit" style="background: #6366f1; color: #ffffff; border: 1px solid #636
6f1; padding: 0.75rem 2rem; border-radius: 8px; font-weight: 700; font-size: 0.95rem; cursor: poin
ter; box-shadow: 0 4px 14px rgba(99, 102, 241, 0.4);">
            ✓ Validate &amp;amp; Register
          </button>
        </div>

      </form>

    </section>

  </main>

  <!-- JavaScript Validation Logic -->
  <script>
    function setFieldState(input, errSpan, isValid, message) {
      if (isValid) {
        input.style.borderColor = '#10b981';
        errSpan.style.display = 'none';
      } else {
        input.style.borderColor = '#ef4444';
        errSpan.textContent = message;
        errSpan.style.display = 'block';
      }
      return isValid;
    }

    function validateName() {
      const input = document.getElementById('fullname');
      const err = document.getElementById('err-name');
      const val = input.value.trim();
      const isValid = /^[A-Za-z\s]{3,40}$/.test(val);
      return setFieldState(input, err, isValid, 'Name must contain only letters and be at least 3
characters.');
    }

    function validateEnroll() {
      const input = document.getElementById('enroll-no');
      const err = document.getElementById('err-enroll');
      const val = input.value.trim();
      const isValid = /^[A-Za-z0-9]{6,15}$/.test(val);
      return setFieldState(input, err, isValid, 'Enrollment number must be 6-15 alphanumeric chara
cters.');
    }

    function validateEmail() {
      const input = document.getElementById('user-email');
```

```
      const err = document.getElementById('err-email');
      const val = input.value.trim();
      const isValid = /^[^\s@]+@[^\s@]+\.[^\s@]+$/.test(val);
      return setFieldState(input, err, isValid, 'Please enter a valid email address (e.g. name@dom
ain.com).');
    }

    function validateMobile() {
      const input = document.getElementById('user-mobile');
      const err = document.getElementById('err-mobile');
      const val = input.value.trim();
      const isValid = /^[6-9]\d{9}$/.test(val);
      return setFieldState(input, err, isValid, 'Mobile number must be exactly 10 digits starting
with 6, 7, 8, or 9.');
    }

    function validatePwd() {
      const input = document.getElementById('user-pwd');
      const err = document.getElementById('err-pwd');
      const val = input.value;
      // Min 8 chars, 1 uppercase, 1 number
      const isValid = /^(?=.*[A-Z])(?=.*\d).{8,}$/.test(val);
      return setFieldState(input, err, isValid, 'Password must be at least 8 characters with 1 upp
ercase letter and 1 number.');
    }

    function validateCPwd() {
      const pwd = document.getElementById('user-pwd').value;
      const input = document.getElementById('user-cpwd');
      const err = document.getElementById('err-cpwd');
      const val = input.value;
      const isValid = (val === pwd && val.length > 0);
      return setFieldState(input, err, isValid, 'Passwords do not match.');
    }

    function validateTerms() {
      const chk = document.getElementById('terms-chk');
      const err = document.getElementById('err-terms');
      if (!chk.checked) {
        err.textContent = 'You must accept the terms before proceeding.';
        err.style.display = 'block';
        return false;
      } else {
        err.style.display = 'none';
        return true;
      }
    }

    function validateAndSubmit(e) {
      e.preventDefault();
      const isNameOk = validateName();
      const isEnrollOk = validateEnroll();
      const isEmailOk = validateEmail();
      const isMobileOk = validateMobile();
      const isPwdOk = validatePwd();
      const isCPwdOk = validateCPwd();
      const isTermsOk = validateTerms();

      const allValid = isNameOk && isEnrollOk && isEmailOk && isMobileOk && isPwdOk && isCPwdOk &&
isTermsOk;

      const summary = document.getElementById('error-summary');
      if (allValid) {
        summary.style.display = 'none';
        alert(`🎉 Validation Passed (100% Client-Side Validated!)\n\nStudent: ${document.getEleme
ntById('fullname').value}\nEnrollment: ${document.getElementById('enroll-no').value}\nEmail: ${doc
```

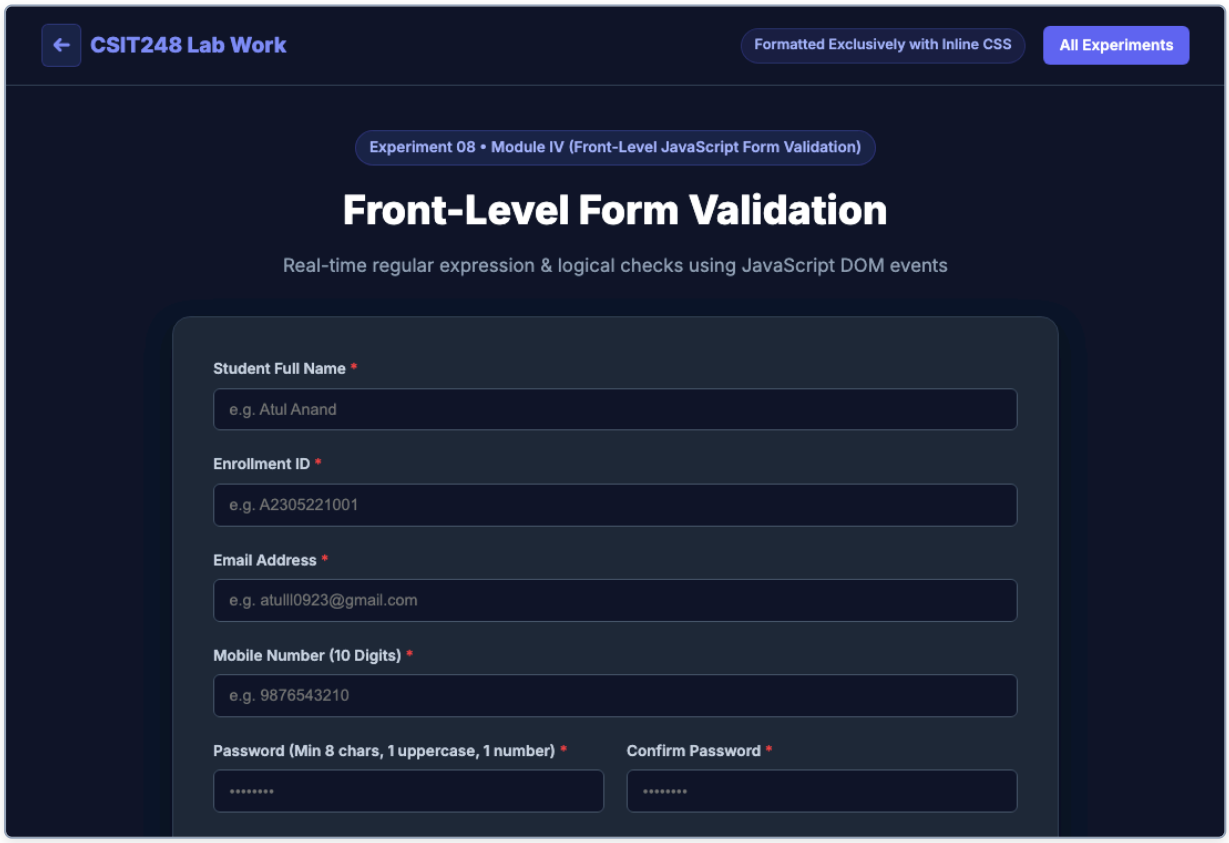
```
ument.getElementById('user-email').value}\nMobile: ${document.getElementById('user-mobile').value}
\n\nRegistration confirmed!`);
      } else {
        summary.style.display = 'block';
      }
    }

    function resetValidationForm() {
      document.getElementById('val-form').reset();
      document.querySelectorAll('#val-form input').forEach(input => {
        input.style.borderColor = '#475569';
      });
      document.querySelectorAll('#val-form span[id^="err-"]').forEach(span => {
        span.style.display = 'none';
      });
      document.getElementById('error-summary').style.display = 'none';
    }
  </script>

  <!-- Footer (Inline CSS) -->
  <footer style="background: #090d16; border-top: 1px solid #1e293b; padding: 2rem 1.5rem; text-al
ign: center; color: #94a3b8; font-size: 0.875rem; margin-top: 3rem;">
    <p style="margin: 0 0 0.5rem 0;">Experiment 08 &bull; CSIT248 Advanced Web Technology &bull; F
ront-Level Form Validation</p>
    <p style="margin: 0 0 0.75rem 0; color: #64748b; font-size: 0.8rem;">Amity University Noida &b
ull; Formatted using 100% Inline CSS</p>
    <p style="margin: 0;"><a href="../index.html" style="color: #818cf8; text-decoration: none; fo
nt-weight: 600;">&uarr; Back to Lab Work Portal</a></p>
  </footer>

</body>
</html>
```

Output :



Experiment No.: 12

Title : External and Inline CSS Design Rules Demonstration.

Aim : To demonstrate Cascading Style Sheets (CSS) priority rules, class selectors, ID selectors, inline styling, and external stylesheet integration on a course evaluation and feedback form.

Source Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Experiment 9: Create a HTML form with the use of Cascading Sty
le Sheets - CSIT248 Advanced Web Technology">
  <title>Experiment 09 - CSS Styled Feedback Form (Inline CSS)</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&display=swa
p" rel="stylesheet">
</head>
<body style="margin: 0; padding: 0; font-family: 'Inter', system-ui, -apple-system, sans-serif; ba
ckground-color: #0f172a; color: #f8fafc; line-height: 1.6; min-height: 100vh;">

  <!-- Navigation (Inline CSS) -->
  <nav style="position: sticky; top: 0; z-index: 100; background: rgba(15, 23, 42, 0.92); backdrop
-filter: blur(12px); border-bottom: 1px solid #334155; padding: 1rem 2rem; display: flex; justify-
content: space-between; align-items: center;">
    <a href="../index.html" style="font-weight: 800; font-size: 1.2rem; color: #818cf8; text-decor
ation: none; display: flex; align-items: center; gap: 0.5rem;">
      <span style="background: rgba(99, 102, 241, 0.15); border: 1px solid rgba(99, 102, 241, 0.
3); padding: 0.2rem 0.5rem; border-radius: 6px;">&larr;</span>
      CSIT248 Lab Work
    </a>
    <div style="display: flex; gap: 1rem; align-items: center;">
      <span style="font-size: 0.8rem; background: rgba(99, 102, 241, 0.12); color: #a5b4fc; borde
r: 1px solid rgba(99, 102, 241, 0.25); padding: 0.3rem 0.75rem; border-radius: 9999px; font-weigh
t: 600;">
        Formatted Exclusively with Inline CSS
      </span>
      <a href="../index.html" style="background: #6366f1; color: #ffffff; padding: 0.4rem 0.9rem;
border-radius: 6px; text-decoration: none; font-size: 0.85rem; font-weight: 600;">
        All Experiments
      </a>
    </div>
  </nav>

  <main style="max-width: 850px; margin: 0 auto; padding: 2.5rem 1.5rem;">

    <!-- Header (Inline CSS) -->
    <header style="text-align: center; margin-bottom: 2rem;">
      <div style="display: inline-block; background: rgba(99, 102, 241, 0.15); color: #a5b4fc; bor
der: 1px solid rgba(99, 102, 241, 0.3); padding: 0.25rem 0.75rem; border-radius: 9999px; font-siz
e: 0.85rem; font-weight: 600; margin-bottom: 0.75rem;">
        Experiment 09 &bull; Module III (Cascading Style Sheets &amp; UI Styling)
      </div>
      <h1 style="font-size: 2.25rem; font-weight: 800; color: #ffffff; margin: 0 0 0.5rem 0;">CSS
Styled Course Feedback Form</h1>
      <p style="color: #94a3b8; font-size: 1.05rem; margin: 0;">Demonstration of rich CSS styling
properties, color gradients, and glassmorphism on HTML forms</p>
    </header>
```

```
    <!-- Styled Feedback Form Container (Inline CSS) -->
    <form action="#" method="post" onsubmit="event.preventDefault(); alert('✨ Feedback Submitte
d! Thank you for evaluating CSIT248 Advanced Web Technology.');" style="background: linear-gradien
t(135deg, rgba(30, 41, 59, 0.9) 0%, rgba(15, 23, 42, 0.95) 100%); border: 1px solid #4338ca; borde
r-radius: 20px; padding: 2.5rem; box-shadow: 0 15px 35px rgba(0, 0, 0, 0.4), 0 0 20px rgba(99, 10
2, 241, 0.15);">

      <!-- Student & Course Details Grid -->
      <div style="display: grid; grid-template-columns: repeat(auto-fit, minmax(240px, 1fr)); gap:
1.25rem; margin-bottom: 1.5rem;">
        <div>
          <label for="student-name" style="display: block; font-size: 0.85rem; font-weight: 600; c
olor: #a5b4fc; margin-bottom: 0.4rem;">Student Name</label>
          <input type="text" id="student-name" required value="Atul Anand" style="width: 100%; box
-sizing: border-box; background: rgba(15, 23, 42, 0.8); border: 1px solid #6366f1; color: #ffffff;
padding: 0.75rem 1rem; border-radius: 8px; font-size: 0.95rem; outline: none; box-shadow: 0 0 10px
rgba(99, 102, 241, 0.2);">
        </div>

        <div>
          <label for="course-select" style="display: block; font-size: 0.85rem; font-weight: 600;
color: #a5b4fc; margin-bottom: 0.4rem;">Course Subject</label>
          <select id="course-select" style="width: 100%; box-sizing: border-box; background: rgba
(15, 23, 42, 0.8); border: 1px solid #475569; color: #ffffff; padding: 0.75rem 1rem; border-radiu
s: 8px; font-size: 0.95rem; outline: none;">
            <option value="CSIT248" selected>CSIT248: Advanced Web Technology</option>
            <option value="CSIT244">CSIT244: Database Management Systems</option>
            <option value="CSIT240">CSIT240: Operating Systems</option>
            <option value="CSIT242">CSIT242: Computer Networks</option>
          </select>
        </div>
      </div>

      <!-- Rating Sliders / Evaluation Metrics -->
      <div style="background: rgba(15, 23, 42, 0.7); border: 1px solid #334155; border-radius: 12p
x; padding: 1.5rem; margin-bottom: 1.5rem;">
        <h3 style="font-size: 1.05rem; color: #38bdf8; margin: 0 0 1rem 0; font-weight: 700;">1. C
ourse Content &amp;amp; Lab Practical Assessment</h3>

        <!-- Metric 1 -->
        <div style="margin-bottom: 1.25rem;">
          <div style="display: flex; justify-content: space-between; font-size: 0.85rem; color: #c
bd5e1; margin-bottom: 0.35rem;">
            <span>Lab Curriculum Depth &amp;amp; Practical Hands-on:</span>
            <strong style="color: #34d399;">5 / 5 (Excellent)</strong>
          </div>
          <input type="range" min="1" max="5" value="5" style="width: 100%; accent-color: #6366f1;
cursor: pointer;">
        </div>

        <!-- Metric 2 -->
        <div>
          <div style="display: flex; justify-content: space-between; font-size: 0.85rem; color: #c
bd5e1; margin-bottom: 0.35rem;">
            <span>Faculty Guidance &amp;amp; Code Review Clarity:</span>
            <strong style="color: #34d399;">5 / 5 (Excellent)</strong>
          </div>
          <input type="range" min="1" max="5" value="5" style="width: 100%; accent-color: #06b6d4;
cursor: pointer;">
        </div>
      </div>

      <!-- Preferred Web Frameworks Checkboxes -->
      <div style="background: rgba(15, 23, 42, 0.7); border: 1px solid #334155; border-radius: 12p
x; padding: 1.5rem; margin-bottom: 1.5rem;">
```

```
        <h3 style="font-size: 1.05rem; color: #fbbf24; margin: 0 0 1rem 0; font-weight: 700;">2. T
echnologies Most Engaged With:</h3>
        <div style="display: grid; grid-template-columns: repeat(auto-fit, minmax(180px, 1fr)); ga
p: 0.75rem; font-size: 0.9rem; color: #cbd5e1;">
          <label style="display: flex; align-items: center; gap: 0.5rem; cursor: pointer;">
            <input type="checkbox" checked style="accent-color: #6366f1;"> HTML5 Semantics
          </label>
          <label style="display: flex; align-items: center; gap: 0.5rem; cursor: pointer;">
            <input type="checkbox" checked style="accent-color: #6366f1;"> CSS3 Flexbox/Grid
          </label>
          <label style="display: flex; align-items: center; gap: 0.5rem; cursor: pointer;">
            <input type="checkbox" checked style="accent-color: #6366f1;"> JavaScript DOM
          </label>
          <label style="display: flex; align-items: center; gap: 0.5rem; cursor: pointer;">
            <input type="checkbox" checked style="accent-color: #6366f1;"> PHP &amp;amp; MySQL
          </label>
        </div>
      </div>

      <!-- Feedback Comments Textarea -->
      <div style="margin-bottom: 1.75rem;">
        <label for="comments" style="display: block; font-size: 0.85rem; font-weight: 600; color:
#a5b4fc; margin-bottom: 0.4rem;">Detailed Comments / Suggestions for Lab Work</label>
        <textarea id="comments" rows="4" placeholder="Share your experience regarding the practica
l assignments and web development exercises..." style="width: 100%; box-sizing: border-box; backgr
ound: rgba(15, 23, 42, 0.8); border: 1px solid #475569; color: #ffffff; padding: 0.75rem 1rem; bor
der-radius: 8px; font-size: 0.9rem; outline: none; font-family: inherit; resize: vertical;"></text
area>
      </div>

      <!-- Action Buttons (Inline CSS) -->
      <div style="display: flex; justify-content: flex-end; gap: 1rem;">
        <button type="reset" style="background: rgba(239, 68, 68, 0.15); color: #f87171; border: 1
px solid rgba(239, 68, 68, 0.3); padding: 0.75rem 1.5rem; border-radius: 8px; font-weight: 600; fo
nt-size: 0.95rem; cursor: pointer;">
          ↺ Reset Form
        </button>
        <button type="submit" style="background: linear-gradient(135deg, #6366f1 0%, #4f46e5 10
0%); color: #ffffff; border: 1px solid #818cf8; padding: 0.75rem 2rem; border-radius: 8px; font-we
ight: 700; font-size: 0.95rem; cursor: pointer; box-shadow: 0 4px 15px rgba(99, 102, 241, 0.4);">
          ⭐ Submit Evaluation
        </button>
      </div>

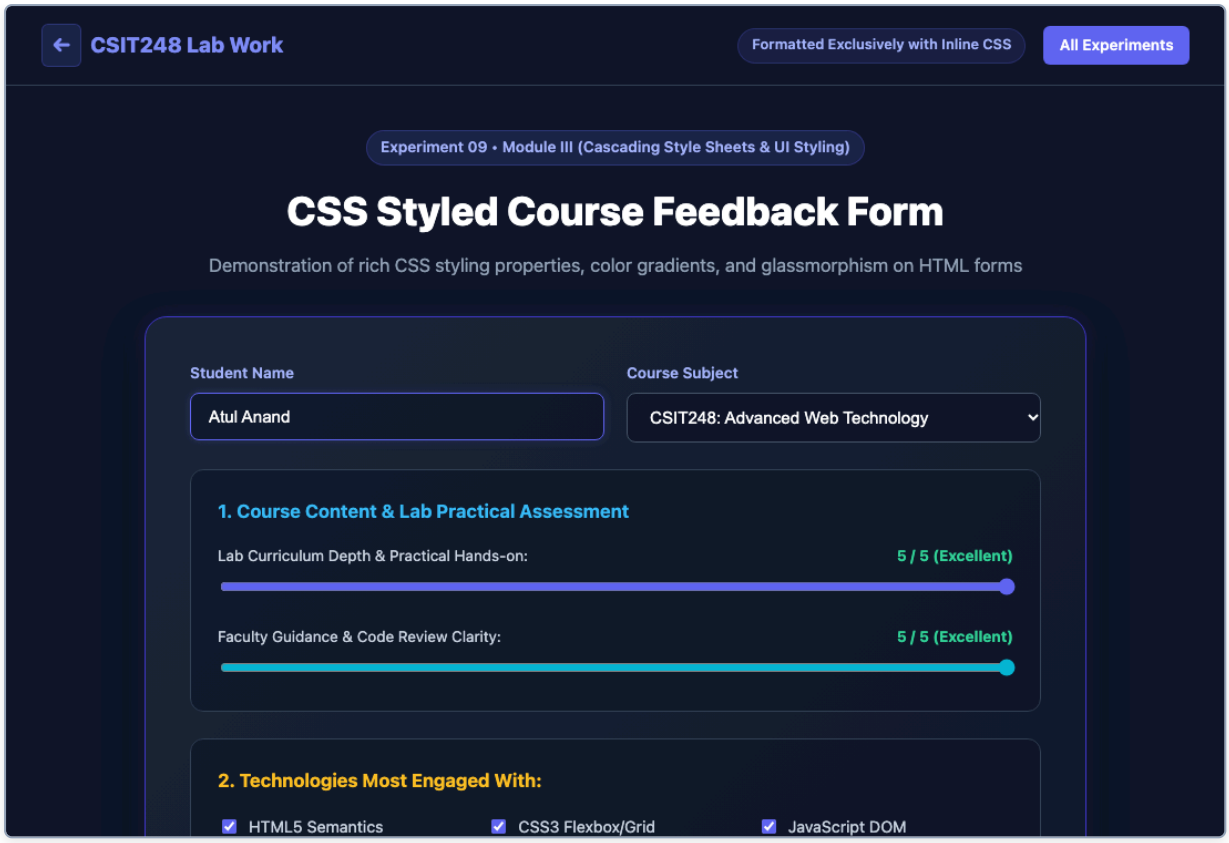
    </form>

  </main>

  <!-- Footer (Inline CSS) -->
  <footer style="background: #090d16; border-top: 1px solid #1e293b; padding: 2rem 1.5rem; text-al
ign: center; color: #94a3b8; font-size: 0.875rem; margin-top: 3rem;">
    <p style="margin: 0 0 0.5rem 0;">Experiment 09 &bull; CSIT248 Advanced Web Technology &bull; H
TML Form with CSS</p>
    <p style="margin: 0 0 0.75rem 0; color: #64748b; font-size: 0.8rem;">Amity University Noida &b
ull; Formatted using 100% Inline CSS</p>
    <p style="margin: 0;"><a href="../index.html" style="color: #818cf8; text-decoration: none; fo
nt-weight: 600;">&uarr; Back to Lab Work Portal</a></p>
  </footer>

</body>
</html>
```

Output :


CSIT248 Lab Work
Formatted Exclusively with Inline CSS
All Experiments
Experiment 09 • Module III (Cascading Style Sheets & UI Styling)
CSS Styled Course Feedback Form
Demonstration of rich CSS styling properties, color gradients, and glassmorphism on HTML forms
Student Name
Atul Anand
Course Subject
CSIT248: Advanced Web Technology
1. Course Content & Lab Practical Assessment
Lab Curriculum Depth & Practical Hands-on:
5 / 5 (Excellent)
Faculty Guidance & Code Review Clarity:
5 / 5 (Excellent)
2. Technologies Most Engaged With:
HTML5 Semantics
CSS3 Flexbox/Grid
JavaScript DOM

Experiment No.: 13

Title : DHTML Dynamic Records Management Form.

Aim : To build a dynamic HTML (DHTML) data management interface capable of adding, editing, validating, and deleting student records in the DOM dynamically without page refresh.

Source Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Experiment 10: Create a DHTML form for the dynamic use of reco
rds in a web form - CSIT248 Advanced Web Technology">
  <title>Experiment 10 - DHTML Dynamic Records Form (Inline CSS)</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&display=swa
p" rel="stylesheet">
</head>
<body style="margin: 0; padding: 0; font-family: 'Inter', system-ui, -apple-system, sans-serif; ba
ckground-color: #0f172a; color: #f8fafc; line-height: 1.6; min-height: 100vh;">

  <!-- Navigation (Inline CSS) -->
  <nav style="position: sticky; top: 0; z-index: 100; background: rgba(15, 23, 42, 0.92); backdrop
-filter: blur(12px); border-bottom: 1px solid #334155; padding: 1rem 2rem; display: flex; justify-
content: space-between; align-items: center;">
    <a href="../index.html" style="font-weight: 800; font-size: 1.2rem; color: #818cf8; text-decor
ation: none; display: flex; align-items: center; gap: 0.5rem;">
      <span style="background: rgba(99, 102, 241, 0.15); border: 1px solid rgba(99, 102, 241, 0.
3); padding: 0.2rem 0.5rem; border-radius: 6px;">&larr;</span>
      CSIT248 Lab Work
    </a>
    <div style="display: flex; gap: 1rem; align-items: center;">
      <span style="font-size: 0.8rem; background: rgba(99, 102, 241, 0.12); color: #a5b4fc; borde
r: 1px solid rgba(99, 102, 241, 0.25); padding: 0.3rem 0.75rem; border-radius: 9999px; font-weigh
t: 600;">
        Formatted Exclusively with Inline CSS
      </span>
      <a href="../index.html" style="background: #6366f1; color: #ffffff; padding: 0.4rem 0.9rem;
border-radius: 6px; text-decoration: none; font-size: 0.85rem; font-weight: 600;">
        All Experiments
      </a>
    </div>
  </nav>

  <main style="max-width: 1200px; margin: 0 auto; padding: 2.5rem 1.5rem;">

    <!-- Header (Inline CSS) -->
    <header style="text-align: center; margin-bottom: 2rem;">
      <div style="display: inline-block; background: rgba(99, 102, 241, 0.15); color: #a5b4fc; bor
der: 1px solid rgba(99, 102, 241, 0.3); padding: 0.25rem 0.75rem; border-radius: 9999px; font-siz
e: 0.85rem; font-weight: 600; margin-bottom: 0.75rem;">
        Experiment 10 &bull; Module IV &amp; V (DHTML &amp; Dynamic DOM Manipulation)
      </div>
      <h1 style="font-size: 2.25rem; font-weight: 800; color: #ffffff; margin: 0 0 0.5rem 0;">DHTM
L Dynamic Records Management Form</h1>
      <p style="color: #94a3b8; font-size: 1.05rem; margin: 0;">Add, Edit, Delete, and Calculate d
ynamic student academic records directly in the DOM</p>
    </header>
```

```
    <!-- Layout: Form + Records Grid (Inline CSS) -->
    <div style="display: grid; grid-template-columns: 360px 1fr; gap: 1.75rem; margin-bottom: 2.5r
em;">

      <!-- Left Panel: Record Entry Form (Inline CSS) -->
      <section style="background: #1e293b; border: 1px solid #334155; border-radius: 16px; paddin
g: 1.75rem; box-shadow: 0 10px 25px rgba(0, 0, 0, 0.3); height: fit-content;">
        <h2 style="font-size: 1.25rem; font-weight: 700; color: #38bdf8; margin: 0 0 1.25rem 0; bo
rder-bottom: 1px solid #334155; padding-bottom: 0.5rem;">
          ✚ Add New Record (DHTML)
        </h2>

        <form id="record-form" onsubmit="addRecord(event)">

          <div style="margin-bottom: 1rem;">
            <label for="rec-name" style="display: block; font-size: 0.85rem; font-weight: 600; col
or: #cbd5e1; margin-bottom: 0.35rem;">Student Name *</label>
            <input type="text" id="rec-name" required placeholder="e.g. Atul Anand" style="width:
100%; box-sizing: border-box; background: #0f172a; border: 1px solid #475569; color: #ffffff; padd
ing: 0.65rem 0.85rem; border-radius: 6px; font-size: 0.9rem; outline: none;">
          </div>

          <div style="margin-bottom: 1rem;">
            <label for="rec-enroll" style="display: block; font-size: 0.85rem; font-weight: 600; c
olor: #cbd5e1; margin-bottom: 0.35rem;">Enrollment ID *</label>
            <input type="text" id="rec-enroll" required placeholder="e.g. A2305221001" style="widt
h: 100%; box-sizing: border-box; background: #0f172a; border: 1px solid #475569; color: #ffffff; p
adding: 0.65rem 0.85rem; border-radius: 6px; font-size: 0.9rem; outline: none;">
          </div>

          <div style="margin-bottom: 1rem;">
            <label for="rec-subject" style="display: block; font-size: 0.85rem; font-weight: 600;
color: #cbd5e1; margin-bottom: 0.35rem;">Subject Course *</label>
            <select id="rec-subject" required style="width: 100%; box-sizing: border-box; backgrou
nd: #0f172a; border: 1px solid #475569; color: #ffffff; padding: 0.65rem 0.85rem; border-radius: 6
px; font-size: 0.9rem; outline: none;">
              <option value="CSIT248 (Adv. Web Tech)">CSIT248 (Adv. Web Tech)</option>
              <option value="CSIT244 (DBMS)">CSIT244 (DBMS)</option>
              <option value="CSIT240 (OS)">CSIT240 (OS)</option>
              <option value="CSIT242 (CN)">CSIT242 (CN)</option>
              <option value="MATH302 (Math)">MATH302 (Math)</option>
            </select>
          </div>

          <div style="display: grid; grid-template-columns: 1fr 1fr; gap: 1rem; margin-bottom: 1.5
rem;">
            <div>
              <label for="rec-marks" style="display: block; font-size: 0.85rem; font-weight: 600;
color: #cbd5e1; margin-bottom: 0.35rem;">Marks (0-100) *</label>
              <input type="number" id="rec-marks" min="0" max="100" required placeholder="95" styl
e="width: 100%; box-sizing: border-box; background: #0f172a; border: 1px solid #475569; color: #ff
ffff; padding: 0.65rem 0.85rem; border-radius: 6px; font-size: 0.9rem; outline: none;">
            </div>

            <div>
              <label for="rec-attendance" style="display: block; font-size: 0.85rem; font-weight:
600; color: #cbd5e1; margin-bottom: 0.35rem;">Attendance % *</label>
              <input type="number" id="rec-attendance" min="0" max="100" required placeholder="92"
style="width: 100%; box-sizing: border-box; background: #0f172a; border: 1px solid #475569; color:
#ffffff; padding: 0.65rem 0.85rem; border-radius: 6px; font-size: 0.9rem; outline: none;">
            </div>
          </div>

          <button type="submit" style="width: 100%; background: #6366f1; color: #ffffff; border: 1
px solid #6366f1; padding: 0.75rem; border-radius: 8px; font-weight: 700; font-size: 0.95rem; curs
```

```
or: pointer; box-shadow: 0 4px 14px rgba(99, 102, 241, 0.4);">
            ➕ Insert Record to Table
          </button>
        </form>
      </section>

      <!-- Right Panel: Dynamic DHTML Table & Stats (Inline CSS) -->
      <section style="background: #1e293b; border: 1px solid #334155; border-radius: 16px; paddin
g: 1.75rem; box-shadow: 0 10px 25px rgba(0, 0, 0, 0.3); display: flex; flex-direction: column;">

        <!-- Top Stats Summary Bar -->
        <div style="display: grid; grid-template-columns: repeat(auto-fit, minmax(130px, 1fr)); ga
p: 0.75rem; margin-bottom: 1.25rem;">
          <div style="background: #0f172a; border: 1px solid #334155; padding: 0.75rem; border-rad
ius: 8px; text-align: center;">
            <span style="font-size: 0.75rem; color: #94a3b8; display: block;">Total Records</span>
            <strong id="stat-count" style="font-size: 1.25rem; color: #38bdf8;">3</strong>
          </div>
          <div style="background: #0f172a; border: 1px solid #334155; padding: 0.75rem; border-rad
ius: 8px; text-align: center;">
            <span style="font-size: 0.75rem; color: #94a3b8; display: block;">Avg Marks</span>
            <strong id="stat-avg" style="font-size: 1.25rem; color: #34d399;">92.3%</strong>
          </div>
          <div style="background: #0f172a; border: 1px solid #334155; padding: 0.75rem; border-rad
ius: 8px; text-align: center;">
            <span style="font-size: 0.75rem; color: #94a3b8; display: block;">Highest Score</span>
            <strong id="stat-high" style="font-size: 1.25rem; color: #fbbf24;">96/100</strong>
          </div>
        </div>

        <!-- Search input for dynamic filter -->
        <div style="margin-bottom: 1rem;">
          <input
            type="text"
            id="filter-input"
            placeholder="🔍 Filter records dynamically by name or subject..."
            onkeyup="filterRecords()"
            style="width: 100%; box-sizing: border-box; background: #0f172a; border: 1px solid #47
5569; color: #ffffff; padding: 0.6rem 1rem; border-radius: 8px; font-size: 0.85rem; outline: non
e;"
          >
        </div>

        <!-- Dynamic Table -->
        <div style="overflow-x: auto; flex-grow: 1;">
          <table id="records-table" style="width: 100%; border-collapse: collapse; text-align: lef
t; font-size: 0.9rem;">
            <thead>
              <tr style="background: #1e1b4b; color: #cbd5e1; font-size: 0.8rem; text-transform: u
ppercase; border-bottom: 1px solid #334155;">
                <th style="padding: 0.75rem 1rem;">#</th>
                <th style="padding: 0.75rem 1rem;">Student Name</th>
                <th style="padding: 0.75rem 1rem;">Enrollment</th>
                <th style="padding: 0.75rem 1rem;">Subject</th>
                <th style="padding: 0.75rem 1rem;">Marks</th>
                <th style="padding: 0.75rem 1rem;">Attendance</th>
                <th style="padding: 0.75rem 1rem; text-align: center;">Actions</th>
              </tr>
            </thead>
            <tbody id="records-body">
              <!-- Default sample rows -->
              <tr style="border-bottom: 1px solid #334155;">
                <td style="padding: 0.75rem 1rem; color: #94a3b8;">1</td>
                <td style="padding: 0.75rem 1rem; font-weight: 600; color: #ffffff;">Atul Anand</t
d>
                <td style="padding: 0.75rem 1rem; color: #cbd5e1; font-family: monospace;">A230522
```

```
1001</td>
                <td style="padding: 0.75rem 1rem; color: #38bdf8;">CSIT248 (Adv. Web Tech)</td>
                <td style="padding: 0.75rem 1rem; font-weight: 700; color: #34d399;">96</td>
                <td style="padding: 0.75rem 1rem; color: #cbd5e1;">95%</td>
                <td style="padding: 0.75rem 1rem; text-align: center;">
                  <button onclick="deleteRow(this)" style="background: rgba(239, 68, 68, 0.15); co
lor: #f87171; border: 1px solid rgba(239, 68, 68, 0.3); padding: 0.3rem 0.6rem; border-radius: 4p
x; font-size: 0.75rem; cursor: pointer;">Delete</button>
                </td>
              </tr>
              <tr style="border-bottom: 1px solid #334155; background: rgba(255, 255, 255, 0.01
5);">
                <td style="padding: 0.75rem 1rem; color: #94a3b8;">2</td>
                <td style="padding: 0.75rem 1rem; font-weight: 600; color: #ffffff;">Rohan Sharma
</td>
                <td style="padding: 0.75rem 1rem; color: #cbd5e1; font-family: monospace;">A230522
1002</td>
                <td style="padding: 0.75rem 1rem; color: #38bdf8;">CSIT244 (DBMS)</td>
                <td style="padding: 0.75rem 1rem; font-weight: 700; color: #34d399;">88</td>
                <td style="padding: 0.75rem 1rem; color: #cbd5e1;">89%</td>
                <td style="padding: 0.75rem 1rem; text-align: center;">
                  <button onclick="deleteRow(this)" style="background: rgba(239, 68, 68, 0.15); co
lor: #f87171; border: 1px solid rgba(239, 68, 68, 0.3); padding: 0.3rem 0.6rem; border-radius: 4p
x; font-size: 0.75rem; cursor: pointer;">Delete</button>
                </td>
              </tr>
              <tr style="border-bottom: 1px solid #334155;">
                <td style="padding: 0.75rem 1rem; color: #94a3b8;">3</td>
                <td style="padding: 0.75rem 1rem; font-weight: 600; color: #ffffff;">Priya Verma</
td>
                <td style="padding: 0.75rem 1rem; color: #cbd5e1; font-family: monospace;">A230522
1003</td>
                <td style="padding: 0.75rem 1rem; color: #38bdf8;">CSIT240 (OS)</td>
                <td style="padding: 0.75rem 1rem; font-weight: 700; color: #34d399;">93</td>
                <td style="padding: 0.75rem 1rem; color: #cbd5e1;">91%</td>
                <td style="padding: 0.75rem 1rem; text-align: center;">
                  <button onclick="deleteRow(this)" style="background: rgba(239, 68, 68, 0.15); co
lor: #f87171; border: 1px solid rgba(239, 68, 68, 0.3); padding: 0.3rem 0.6rem; border-radius: 4p
x; font-size: 0.75rem; cursor: pointer;">Delete</button>
                </td>
              </tr>
            </tbody>
          </table>
        </div>

      </section>

    </div>

  </main>

  <!-- DHTML JavaScript Logic -->
  <script>
    let recordsData = [
      { name: 'Atul Anand', enroll: 'A2305221001', subject: 'CSIT248 (Adv. Web Tech)', marks: 96,
attendance: 95 },
      { name: 'Rohan Sharma', enroll: 'A2305221002', subject: 'CSIT244 (DBMS)', marks: 88, attenda
nce: 89 },
      { name: 'Priya Verma', enroll: 'A2305221003', subject: 'CSIT240 (OS)', marks: 93, attendanc
e: 91 }
    ];

    function renderTable() {
      const tbody = document.getElementById('records-body');
      tbody.innerHTML = '';
```

```
      let totalMarks = 0;
      let highest = 0;

      recordsData.forEach((rec, idx) => {
        totalMarks += rec.marks;
        if (rec.marks > highest) highest = rec.marks;

        const tr = document.createElement('tr');
        tr.style.borderBottom = '1px solid #334155';
        if (idx % 2 === 1) tr.style.background = 'rgba(255, 255, 255, 0.015)';

        tr.innerHTML = `
          <td style="padding: 0.75rem 1rem; color: #94a3b8;">${idx + 1}</td>
          <td style="padding: 0.75rem 1rem; font-weight: 600; color: #ffffff;">${rec.name}</td>
          <td style="padding: 0.75rem 1rem; color: #cbd5e1; font-family: monospace;">${rec.enroll}
</td>
          <td style="padding: 0.75rem 1rem; color: #38bdf8;">${rec.subject}</td>
          <td style="padding: 0.75rem 1rem; font-weight: 700; color: #34d399;">${rec.marks}</td>
          <td style="padding: 0.75rem 1rem; color: #cbd5e1;">${rec.attendance}%</td>
          <td style="padding: 0.75rem 1rem; text-align: center;">
            <button onclick="removeRecord(${idx})" style="background: rgba(239, 68, 68, 0.15); col
or: #f87171; border: 1px solid rgba(239, 68, 68, 0.3); padding: 0.3rem 0.6rem; border-radius: 4px;
font-size: 0.75rem; cursor: pointer;">Delete</button>
          </td>
        `;
        tbody.appendChild(tr);
      });

      // Update DHTML stats
      const count = recordsData.length;
      document.getElementById('stat-count').textContent = count;
      document.getElementById('stat-avg').textContent = count > 0 ? (totalMarks / count).toFixed
(1) + '%' : '0%';
      document.getElementById('stat-high').textContent = count > 0 ? `${highest}/100` : '--';
    }

    function addRecord(e) {
      e.preventDefault();
      const name = document.getElementById('rec-name').value.trim();
      const enroll = document.getElementById('rec-enroll').value.trim();
      const subject = document.getElementById('rec-subject').value;
      const marks = parseInt(document.getElementById('rec-marks').value, 10);
      const attendance = parseInt(document.getElementById('rec-attendance').value, 10);

      recordsData.push({ name, enroll, subject, marks, attendance });
      renderTable();
      document.getElementById('record-form').reset();
    }

    function removeRecord(index) {
      recordsData.splice(index, 1);
      renderTable();
    }

    function filterRecords() {
      const query = document.getElementById('filter-input').value.toLowerCase();
      const rows = document.querySelectorAll('#records-body tr');

      rows.forEach(row => {
        const text = row.textContent.toLowerCase();
        if (text.includes(query)) {
          row.style.display = '';
        } else {
          row.style.display = 'none';
        }
      });
```

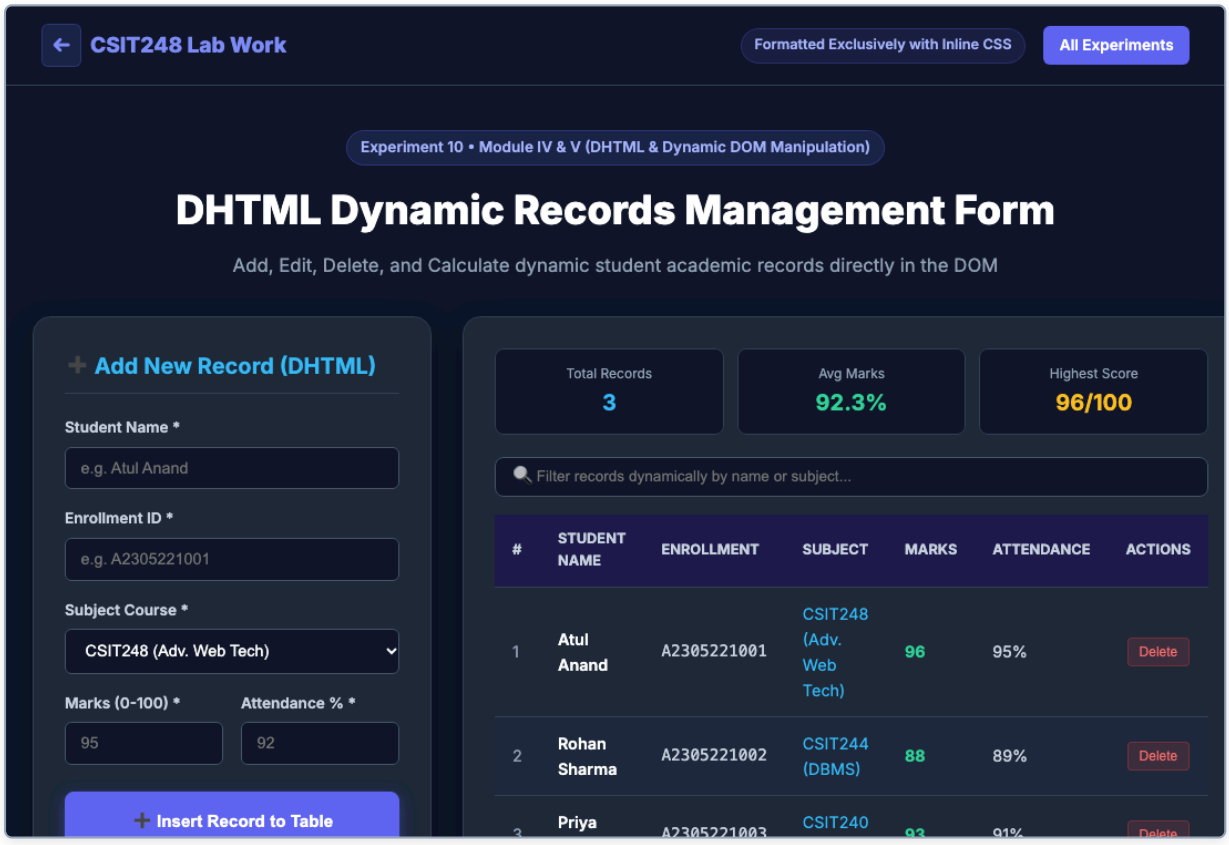
```
    }

    // Initial render
    renderTable();
  </script>

  <!-- Footer (Inline CSS) -->
  <footer style="background: #090d16; border-top: 1px solid #1e293b; padding: 2rem 1.5rem; text-al
ign: center; color: #94a3b8; font-size: 0.875rem; margin-top: 3rem;">
    <p style="margin: 0 0 0.5rem 0;">Experiment 10 &bull; CSIT248 Advanced Web Technology &bull; D
HTML Dynamic Records Form</p>
    <p style="margin: 0 0 0.75rem 0; color: #64748b; font-size: 0.8rem;">Amity University Noida &b
ull; Formatted using 100% Inline CSS</p>
    <p style="margin: 0;"><a href="../index.html" style="color: #818cf8; text-decoration: none; fo
nt-weight: 600;">&uarr; Back to Lab Work Portal</a></p>
  </footer>

</body>
</html>
```

Output :



Experiment No.: 14

Title : Computer Peripherals Shop Catalog with Live Filters.

Aim : To design an e-commerce hardware peripherals product catalog utilizing HTML tables, responsive layout, styled badges, and client-side category filtering.

Source Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Experiment 11: Create a catalog for shop which sells computer peripherals using html tables - CSIT248 Advanced Web Technology">
  <title>Experiment 11 - Computer Peripherals Catalog (Inline CSS)</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&display=swap" rel="stylesheet">
</head>
<body style="margin: 0; padding: 0; font-family: 'Inter', system-ui, -apple-system, sans-serif; background-color: #0f172a; color: #f8fafc; line-height: 1.6; min-height: 100vh;">

  <!-- Navigation (Inline CSS) -->
  <nav style="position: sticky; top: 0; z-index: 100; background: rgba(15, 23, 42, 0.92); backdrop-filter: blur(12px); border-bottom: 1px solid #334155; padding: 1rem 2rem; display: flex; justify-content: space-between; align-items: center;">
    <a href="../index.html" style="font-weight: 800; font-size: 1.2rem; color: #818cf8; text-decoration: none; display: flex; align-items: center; gap: 0.5rem;">
      <span style="background: rgba(99, 102, 241, 0.15); border: 1px solid rgba(99, 102, 241, 0.3); padding: 0.2rem 0.5rem; border-radius: 6px;">&larr;</span>
      CSIT248 Lab Work
    </a>
    <div style="display: flex; gap: 1rem; align-items: center;">
      <span style="font-size: 0.8rem; background: rgba(99, 102, 241, 0.12); color: #a5b4fc; border: 1px solid rgba(99, 102, 241, 0.25); padding: 0.3rem 0.75rem; border-radius: 9999px; font-weight: 600;">
        Formatted Exclusively with Inline CSS
      </span>
      <a href="../index.html" style="background: #6366f1; color: #ffffff; padding: 0.4rem 0.9rem; border-radius: 6px; text-decoration: none; font-size: 0.85rem; font-weight: 600;">
        All Experiments
      </a>
    </div>
  </nav>

  <main style="max-width: 1400px; margin: 0 auto; padding: 2.5rem 1.5rem;">

    <!-- Header (Inline CSS) -->
    <header style="text-align: center; margin-bottom: 2rem;">
      <h1 style="font-size: 2.25rem; font-weight: 800; color: #ffffff; margin: 0 0 0.5rem 0;">Apex Tech Computer Hardware &amp;amp; Peripherals</h1>
      <p style="color: #94a3b8; font-size: 1.05rem; margin: 0;">Official Store Catalog &bull; High-Performance Workstation &amp;amp; Gaming Equipment</p>
    </header>

    <!-- Store Statistics (Inline CSS) -->
    <section style="display: grid; grid-template-columns: repeat(auto-fit, minmax(200px, 1fr)); gap: 1rem; margin-bottom: 2rem;">
      <div style="background: #1e293b; border: 1px solid #334155; border-radius: 12px; padding: 1.25rem; text-align: center; box-shadow: 0 4px 12px rgba(0,0,0,0.2);">
```

```
        <div style="font-size: 1.75rem; font-weight: 800; color: #38bdf8;">8</div>
        <div style="font-size: 0.85rem; color: #94a3b8;">Featured Products</div>
      </div>
      <div style="background: #1e293b; border: 1px solid #334155; border-radius: 12px; padding: 1.
25rem; text-align: center; box-shadow: 0 4px 12px rgba(0,0,0,0.2);">
        <div style="font-size: 1.75rem; font-weight: 800; color: #34d399;">4</div>
        <div style="font-size: 0.85rem; color: #94a3b8;">Hardware Categories</div>
      </div>
      <div style="background: #1e293b; border: 1px solid #334155; border-radius: 12px; padding: 1.
25rem; text-align: center; box-shadow: 0 4px 12px rgba(0,0,0,0.2);">
        <div style="font-size: 1.75rem; font-weight: 800; color: #fbbf24;">98.4%</div>
        <div style="font-size: 0.85rem; color: #94a3b8;">Positive Feedback</div>
      </div>
      <div style="background: #1e293b; border: 1px solid #334155; border-radius: 12px; padding: 1.
25rem; text-align: center; box-shadow: 0 4px 12px rgba(0,0,0,0.2);">
        <div style="font-size: 1.75rem; font-weight: 800; color: #a78bfa;">2 Years</div>
        <div style="font-size: 0.85rem; color: #94a3b8;">Warranty on all Parts</div>
      </div>
    </section>

    <!-- Controls / Filters (Inline CSS) -->
    <div style="background: #1e293b; border: 1px solid #334155; border-radius: 12px; padding: 1rem
1.5rem; margin-bottom: 1.5rem; display: flex; justify-content: space-between; align-items: center;
flex-wrap: wrap; gap: 1rem;">
      <input
        type="text"
        id="search-box"
        placeholder="🔍 Search peripherals by name or specs..."
        onkeyup="filterCatalog()"
        style="background: #0f172a; border: 1px solid #475569; color: #ffffff; padding: 0.6rem 1re
m; border-radius: 8px; font-size: 0.9rem; width: 280px; outline: none;"
      >

      <div style="display: flex; gap: 0.5rem; flex-wrap: wrap;" id="filter-container">
        <button onclick="setCategory('all', this)" style="background: #6366f1; border: 1px solid #
6366f1; color: #ffffff; padding: 0.45rem 0.9rem; border-radius: 6px; font-size: 0.85rem; font-weig
ht: 600; cursor: pointer;">All Peripherals</button>
        <button onclick="setCategory('Input', this)" style="background: rgba(255, 255, 255, 0.05);
border: 1px solid #334155; color: #94a3b8; padding: 0.45rem 0.9rem; border-radius: 6px; font-size:
0.85rem; font-weight: 600; cursor: pointer;">Input Devices</button>
        <button onclick="setCategory('Display', this)" style="background: rgba(255, 255, 255, 0.0
5); border: 1px solid #334155; color: #94a3b8; padding: 0.45rem 0.9rem; border-radius: 6px; font-s
ize: 0.85rem; font-weight: 600; cursor: pointer;">Monitors</button>
        <button onclick="setCategory('Storage', this)" style="background: rgba(255, 255, 255, 0.0
5); border: 1px solid #334155; color: #94a3b8; padding: 0.45rem 0.9rem; border-radius: 6px; font-s
ize: 0.85rem; font-weight: 600; cursor: pointer;">Storage &amp;amp; GPU</button>
        <button onclick="setCategory('Audio', this)" style="background: rgba(255, 255, 255, 0.05);
border: 1px solid #334155; color: #94a3b8; padding: 0.45rem 0.9rem; border-radius: 6px; font-size:
0.85rem; font-weight: 600; cursor: pointer;">Audio &amp;amp; Video</button>
      </div>
    </div>

    <!-- Product Catalog HTML Table (Inline CSS) -->
    <div style="background: #1e293b; border: 1px solid #334155; border-radius: 14px; overflow-x: a
uto; box-shadow: 0 10px 25px rgba(0, 0, 0, 0.3);">
      <table id="product-table" style="width: 100%; border-collapse: collapse; text-align: left; m
in-width: 900px;">
        <caption style="font-size: 1.25rem; font-weight: 700; padding: 1.25rem; text-align: left;
background: #1e1b4b; color: #ffffff; border-bottom: 1px solid #334155;">
          Inventory Product Catalog (E-Commerce Tabular View)
        </caption>
        <thead>
          <tr style="background: rgba(15, 23, 42, 0.95); color: #cbd5e1; font-size: 0.85rem; text-
transform: uppercase; letter-spacing: 0.05em; border-bottom: 1px solid #334155;">
            <th style="padding: 1rem 1.25rem; width: 280px;">Item &amp;amp; Model</th>
            <th style="padding: 1rem 1.25rem; width: 140px;">Category</th>
```

```
            <th style="padding: 1rem 1.25rem;">Technical Specifications</th>
            <th style="padding: 1rem 1.25rem; width: 110px;">Rating</th>
            <th style="padding: 1rem 1.25rem; width: 140px;">Stock Status</th>
            <th style="padding: 1rem 1.25rem; width: 130px;">Unit Price</th>
            <th style="padding: 1rem 1.25rem; width: 120px; text-align: center;">Action</th>
          </tr>
        </thead>
        <tbody>

          <!-- Product 1 -->
          <tr data-category="Input" style="border-bottom: 1px solid #334155;">
            <td style="padding: 1rem 1.25rem;">
              <div style="display: flex; align-items: center; gap: 1rem;">
                <div style="width: 48px; height: 48px; background: #0f172a; border: 1px solid #334
155; border-radius: 8px; display: flex; align-items: center; justify-content: center; font-size:
1.6rem;">⌨️</div>
                <div>
                  <span style="font-weight: 700; font-size: 0.95rem; color: #ffffff; display: bloc
k;">Apex Pro TKL Mechanical Keyboard</span>
                  <span style="font-size: 0.75rem; color: #94a3b8; font-family: monospace;">SKU: K
B-APEX-87</span>
                </div>
              </div>
            </td>
            <td style="padding: 1rem 1.25rem;">
              <span style="display: inline-block; padding: 0.2rem 0.6rem; border-radius: 9999px; f
ont-size: 0.75rem; font-weight: 600; background: rgba(6, 182, 212, 0.15); color: #67e8f9; border:
1px solid rgba(6, 182, 212, 0.3);">Input Device</span>
            </td>
            <td style="padding: 1rem 1.25rem;">
              <ul style="list-style: none; margin: 0; padding: 0; font-size: 0.825rem; color: #cbd
5e1; line-height: 1.5;">
                <li>▹ OmniPoint 2.0 Adjustable HyperMagnetic Switches</li>
                <li>▹ OLED Smart Display &amp;amp; Aluminum Alloy Chassis</li>
                <li>▹ Per-Key RGB Lighting with Detachable USB-C</li>
              </ul>
            </td>
            <td style="padding: 1rem 1.25rem; color: #fbbf24; font-weight: 600; font-size: 0.9re
m;">⭐ 4.9 <small style="color: #94a3b8;">(1,240)</small></td>
            <td style="padding: 1rem 1.25rem;">
              <span style="display: inline-block; font-size: 0.8rem; font-weight: 700; padding: 0.
25rem 0.65rem; border-radius: 6px; background: rgba(16, 185, 129, 0.15); color: #34d399; border: 1
px solid rgba(16, 185, 129, 0.3);">● In Stock (42)</span>
            </td>
            <td style="padding: 1rem 1.25rem;">
              <div style="display: flex; flex-direction: column;">
                <span style="font-size: 1.15rem; font-weight: 800; color: #f8fafc;">&#8377;14,999
</span>
                <span style="font-size: 0.8rem; color: #94a3b8; text-decoration: line-through;">&#
8377;17,999</span>
                <span style="font-size: 0.75rem; color: #34d399; font-weight: 700;">16% OFF</span>
              </div>
            </td>
            <td style="padding: 1rem 1.25rem; text-align: center;">
              <button onclick="addToCart('Apex Pro TKL')" style="background: #6366f1; color: whit
e; border: none; padding: 0.5rem 0.9rem; border-radius: 6px; font-weight: 600; font-size: 0.85rem;
cursor: pointer;">Add to Cart</button>
            </td>
          </tr>

          <!-- Product 2 -->
          <tr data-category="Input" style="border-bottom: 1px solid #334155; background: rgba(255,
255,255,0.015);">
            <td style="padding: 1rem 1.25rem;">
              <div style="display: flex; align-items: center; gap: 1rem;">
                <div style="width: 48px; height: 48px; background: #0f172a; border: 1px solid #334
```

```
155; border-radius: 8px; display: flex; align-items: center; justify-content: center; font-size:
1.6rem;">🖱️</div>
                <div>
                  <span style="font-weight: 700; font-size: 0.95rem; color: #ffffff; display: bloc
k;">Superlight 2 Wireless Mouse</span>
                  <span style="font-size: 0.75rem; color: #94a3b8; font-family: monospace;">SKU: M
S-GPRO-X2</span>
                </div>
              </div>
            </td>
            <td style="padding: 1rem 1.25rem;">
              <span style="display: inline-block; padding: 0.2rem 0.6rem; border-radius: 9999px; f
ont-size: 0.75rem; font-weight: 600; background: rgba(6, 182, 212, 0.15); color: #67e8f9; border:
1px solid rgba(6, 182, 212, 0.3);">Input Device</span>
            </td>
            <td style="padding: 1rem 1.25rem;">
              <ul style="list-style: none; margin: 0; padding: 0; font-size: 0.825rem; color: #cbd
5e1; line-height: 1.5;">
                <li>▹ HERO 2 Sensor up to 32,000 DPI &amp;amp; 500+ IPS</li>
                <li>▹ Ultralight 60g Chassis with Hybrid Optical Switches</li>
                <li>▹ 95 Hours Continuous Battery Life</li>
              </ul>
            </td>
            <td style="padding: 1rem 1.25rem; color: #fbbf24; font-weight: 600; font-size: 0.9re
m;">⭐ 4.8 <small style="color: #94a3b8;">(890)</small></td>
            <td style="padding: 1rem 1.25rem;">
              <span style="display: inline-block; font-size: 0.8rem; font-weight: 700; padding: 0.
25rem 0.65rem; border-radius: 6px; background: rgba(16, 185, 129, 0.15); color: #34d399; border: 1
px solid rgba(16, 185, 129, 0.3);">● In Stock (28)</span>
            </td>
            <td style="padding: 1rem 1.25rem;">
              <div style="display: flex; flex-direction: column;">
                <span style="font-size: 1.15rem; font-weight: 800; color: #f8fafc;">&#8377;11,499
</span>
                <span style="font-size: 0.8rem; color: #94a3b8; text-decoration: line-through;">&#
8377;13,999</span>
                <span style="font-size: 0.75rem; color: #34d399; font-weight: 700;">18% OFF</span>
              </div>
            </td>
            <td style="padding: 1rem 1.25rem; text-align: center;">
              <button onclick="addToCart('Superlight 2 Mouse')" style="background: #6366f1; color:
white; border: none; padding: 0.5rem 0.9rem; border-radius: 6px; font-weight: 600; font-size: 0.85
rem; cursor: pointer;">Add to Cart</button>
            </td>
          </tr>

          <!-- Product 3 -->
          <tr data-category="Display" style="border-bottom: 1px solid #334155;">
            <td style="padding: 1rem 1.25rem;">
              <div style="display: flex; align-items: center; gap: 1rem;">
                <div style="width: 48px; height: 48px; background: #0f172a; border: 1px solid #334
155; border-radius: 8px; display: flex; align-items: center; justify-content: center; font-size:
1.6rem;">🖥️</div>
                <div>
                  <span style="font-weight: 700; font-size: 0.95rem; color: #ffffff; display: bloc
k;">Odyssey OLED G9 49" Curved Monitor</span>
                  <span style="font-size: 0.75rem; color: #94a3b8; font-family: monospace;">SKU: M
ON-ODYS-49</span>
                </div>
              </div>
            </td>
            <td style="padding: 1rem 1.25rem;">
              <span style="display: inline-block; padding: 0.2rem 0.6rem; border-radius: 9999px; f
ont-size: 0.75rem; font-weight: 600; background: rgba(59, 130, 246, 0.15); color: #93c5fd; border:
1px solid rgba(59, 130, 246, 0.3);">Display</span>
            </td>
```

```
            <td style="padding: 1rem 1.25rem;">
              <ul style="list-style: none; margin: 0; padding: 0; font-size: 0.825rem; color: #cbd
5e1; line-height: 1.5;">
                <li>▹ Dual QHD (5120 x 1440), 240Hz, 0.03ms (GtG)</li>
                <li>▹ 1800R Curvature, Quantum Dot OLED, DisplayHDR 400</li>
                <li>▹ HDMI 2.1, Micro HDMI, DisplayPort 1.4</li>
              </ul>
            </td>
            <td style="padding: 1rem 1.25rem; color: #fbbf24; font-weight: 600; font-size: 0.9re
m;">⭐ 4.9 <small style="color: #94a3b8;">(412)</small></td>
            <td style="padding: 1rem 1.25rem;">
              <span style="display: inline-block; font-size: 0.8rem; font-weight: 700; padding: 0.
25rem 0.65rem; border-radius: 6px; background: rgba(245, 158, 11, 0.15); color: #fbbf24; border: 1
px solid rgba(245, 158, 11, 0.3);">● Low Stock (3)</span>
            </td>
            <td style="padding: 1rem 1.25rem;">
              <div style="display: flex; flex-direction: column;">
                <span style="font-size: 1.15rem; font-weight: 800; color: #f8fafc;">&#8377;1,24,99
0</span>
                <span style="font-size: 0.8rem; color: #94a3b8; text-decoration: line-through;">&#
8377;1,49,900</span>
                <span style="font-size: 0.75rem; color: #34d399; font-weight: 700;">17% OFF</span>
              </div>
            </td>
            <td style="padding: 1rem 1.25rem; text-align: center;">
              <button onclick="addToCart('Odyssey OLED G9')" style="background: #6366f1; color: wh
ite; border: none; padding: 0.5rem 0.9rem; border-radius: 6px; font-weight: 600; font-size: 0.85re
m; cursor: pointer;">Add to Cart</button>
            </td>
          </tr>

          <!-- Product 4 -->
          <tr data-category="Storage" style="border-bottom: 1px solid #334155; background: rgba(25
5,255,255,0.015);">
            <td style="padding: 1rem 1.25rem;">
              <div style="display: flex; align-items: center; gap: 1rem;">
                <div style="width: 48px; height: 48px; background: #0f172a; border: 1px solid #334
155; border-radius: 8px; display: flex; align-items: center; justify-content: center; font-size:
1.6rem;">💾</div>
                <div>
                  <span style="font-weight: 700; font-size: 0.95rem; color: #ffffff; display: bloc
k;">Samsung 990 PRO 2TB NVMe SSD</span>
                  <span style="font-size: 0.75rem; color: #94a3b8; font-family: monospace;">SKU: S
SD-SAM-990</span>
                </div>
              </div>
            </td>
            <td style="padding: 1rem 1.25rem;">
              <span style="display: inline-block; padding: 0.2rem 0.6rem; border-radius: 9999px; f
ont-size: 0.75rem; font-weight: 600; background: rgba(139, 92, 246, 0.15); color: #c4b5fd; border:
1px solid rgba(139, 92, 246, 0.3);">Storage</span>
            </td>
            <td style="padding: 1rem 1.25rem;">
              <ul style="list-style: none; margin: 0; padding: 0; font-size: 0.825rem; color: #cbd
5e1; line-height: 1.5;">
                <li>▹ Sequential Read: 7,450 MB/s | Write: 6,900 MB/s</li>
                <li>▹ Thermal controller with Aluminum Heatsink</li>
                <li>▹ Optimized for DirectStorage and 4K Editing</li>
              </ul>
            </td>
            <td style="padding: 1rem 1.25rem; color: #fbbf24; font-weight: 600; font-size: 0.9re
m;">⭐ 4.9 <small style="color: #94a3b8;">(3,420)</small></td>
            <td style="padding: 1rem 1.25rem;">
              <span style="display: inline-block; font-size: 0.8rem; font-weight: 700; padding: 0.
25rem 0.65rem; border-radius: 6px; background: rgba(16, 185, 129, 0.15); color: #34d399; border: 1
px solid rgba(16, 185, 129, 0.3);">● In Stock (65)</span>
```

```
            </td>
            <td style="padding: 1rem 1.25rem;">
              <div style="display: flex; flex-direction: column;">
                <span style="font-size: 1.15rem; font-weight: 800; color: #f8fafc;">&#8377;16,999
</span>
                <span style="font-size: 0.8rem; color: #94a3b8; text-decoration: line-through;">&#
8377;21,500</span>
                <span style="font-size: 0.75rem; color: #34d399; font-weight: 700;">21% OFF</span>
              </div>
            </td>
            <td style="padding: 1rem 1.25rem; text-align: center;">
              <button onclick="addToCart('Samsung 990 PRO SSD')" style="background: #6366f1; colo
r: white; border: none; padding: 0.5rem 0.9rem; border-radius: 6px; font-weight: 600; font-size:
0.85rem; cursor: pointer;">Add to Cart</button>
            </td>
          </tr>

          <!-- Product 5 -->
          <tr data-category="Audio" style="border-bottom: 1px solid #334155;">
            <td style="padding: 1rem 1.25rem;">
              <div style="display: flex; align-items: center; gap: 1rem;">
                <div style="width: 48px; height: 48px; background: #0f172a; border: 1px solid #334
155; border-radius: 8px; display: flex; align-items: center; justify-content: center; font-size:
1.6rem;">🎧</div>
                <div>
                  <span style="font-weight: 700; font-size: 0.95rem; color: #ffffff; display: bloc
k;">Arctis Nova Pro Wireless Headset</span>
                  <span style="font-size: 0.75rem; color: #94a3b8; font-family: monospace;">SKU: A
UD-NOVA-PRO</span>
                </div>
              </div>
            </td>
            <td style="padding: 1rem 1.25rem;">
              <span style="display: inline-block; padding: 0.2rem 0.6rem; border-radius: 9999px; f
ont-size: 0.75rem; font-weight: 600; background: rgba(236, 72, 153, 0.15); color: #f9a8d4; border:
1px solid rgba(236, 72, 153, 0.3);">Audio</span>
            </td>
            <td style="padding: 1rem 1.25rem;">
              <ul style="list-style: none; margin: 0; padding: 0; font-size: 0.825rem; color: #cbd
5e1; line-height: 1.5;">
                <li>▸ Premium High Fidelity Drivers with 360° Spatial Audio</li>
                <li>▸ Active Noise Cancellation (ANC) with Transparency</li>
                <li>▸ Dual Hot-Swap Battery System</li>
              </ul>
            </td>
            <td style="padding: 1rem 1.25rem; color: #fbbf24; font-weight: 600; font-size: 0.9re
m;">⭐ 4.7 <small style="color: #94a3b8;">(760)</small></td>
            <td style="padding: 1rem 1.25rem;">
              <span style="display: inline-block; font-size: 0.8rem; font-weight: 700; padding: 0.
25rem 0.65rem; border-radius: 6px; background: rgba(16, 185, 129, 0.15); color: #34d399; border: 1
px solid rgba(16, 185, 129, 0.3);">● In Stock (19)</span>
            </td>
            <td style="padding: 1rem 1.25rem;">
              <div style="display: flex; flex-direction: column;">
                <span style="font-size: 1.15rem; font-weight: 800; color: #f8fafc;">&#8377;29,990
</span>
                <span style="font-size: 0.8rem; color: #94a3b8; text-decoration: line-through;">&#
8377;34,990</span>
                <span style="font-size: 0.75rem; color: #34d399; font-weight: 700;">14% OFF</span>
              </div>
            </td>
            <td style="padding: 1rem 1.25rem; text-align: center;">
              <button onclick="addToCart('Arctis Nova Pro')" style="background: #6366f1; color: wh
ite; border: none; padding: 0.5rem 0.9rem; border-radius: 6px; font-weight: 600; font-size: 0.85re
m; cursor: pointer;">Add to Cart</button>
            </td>
```

```
          </tr>

          <!-- Product 6 -->
          <tr data-category="Storage" style="border-bottom: 1px solid #334155; background: rgba(25
5,255,255,0.015);">
            <td style="padding: 1rem 1.25rem;">
              <div style="display: flex; align-items: center; gap: 1rem;">
                <div style="width: 48px; height: 48px; background: #0f172a; border: 1px solid #334
155; border-radius: 8px; display: flex; align-items: center; justify-content: center; font-size:
1.6rem;">⚡</div>
                <div>
                  <span style="font-weight: 700; font-size: 0.95rem; color: #ffffff; display: bloc
k;">ASUS ROG RTX 4080 Super (16GB)</span>
                  <span style="font-size: 0.75rem; color: #94a3b8; font-family: monospace;">SKU: G
PU-RTX-4080S</span>
                </div>
              </div>
            </td>
            <td style="padding: 1rem 1.25rem;">
              <span style="display: inline-block; padding: 0.2rem 0.6rem; border-radius: 9999px; f
ont-size: 0.75rem; font-weight: 600; background: rgba(139, 92, 246, 0.15); color: #c4b5fd; border:
1px solid rgba(139, 92, 246, 0.3);">GPU</span>
            </td>
            <td style="padding: 1rem 1.25rem;">
              <ul style="list-style: none; margin: 0; padding: 0; font-size: 0.825rem; color: #cbd
5e1; line-height: 1.5;">
                <li>▹ 10,240 CUDA Cores, 2550 MHz Boost Clock, DLSS 3.5</li>
                <li>▹ Vented Exoskeleton &amp;amp; Patented Vapor Chamber</li>
                <li>▹ Dual BIOS switch &amp;amp; Auto-Extreme Build</li>
              </ul>
            </td>
            <td style="padding: 1rem 1.25rem; color: #fbbf24; font-weight: 600; font-size: 0.9re
m;">⭐ 4.9 <small style="color: #94a3b8;">(310)</small></td>
            <td style="padding: 1rem 1.25rem;">
              <span style="display: inline-block; font-size: 0.8rem; font-weight: 700; padding: 0.
25rem 0.65rem; border-radius: 6px; background: rgba(99, 102, 241, 0.15); color: #a5b4fc; border: 1
px solid rgba(99, 102, 241, 0.3);">● Pre-Order (3 Days)</span>
            </td>
            <td style="padding: 1rem 1.25rem;">
              <div style="display: flex; flex-direction: column;">
                <span style="font-size: 1.15rem; font-weight: 800; color: #f8fafc;">&#8377;1,18,50
0</span>
                <span style="font-size: 0.8rem; color: #94a3b8; text-decoration: line-through;">&#
8377;1,29,000</span>
                <span style="font-size: 0.75rem; color: #34d399; font-weight: 700;">8% OFF</span>
              </div>
            </td>
            <td style="padding: 1rem 1.25rem; text-align: center;">
              <button onclick="addToCart('RTX 4080 Super')" style="background: #6366f1; color: whi
te; border: none; padding: 0.5rem 0.9rem; border-radius: 6px; font-weight: 600; font-size: 0.85re
m; cursor: pointer;">Pre-Order</button>
            </td>
          </tr>

          <!-- Product 7 -->
          <tr data-category="Audio" style="border-bottom: 1px solid #334155;">
            <td style="padding: 1rem 1.25rem;">
              <div style="display: flex; align-items: center; gap: 1rem;">
                <div style="width: 48px; height: 48px; background: #0f172a; border: 1px solid #334
155; border-radius: 8px; display: flex; align-items: center; justify-content: center; font-size:
1.6rem;">📹</div>
                <div>
                  <span style="font-weight: 700; font-size: 0.95rem; color: #ffffff; display: bloc
k;">Elgato Facecam Pro 4K60 Webcam</span>
                  <span style="font-size: 0.75rem; color: #94a3b8; font-family: monospace;">SKU: C
AM-ELG-4K60</span>
```

```
              </div>
            </div>
          </td>
          <td style="padding: 1rem 1.25rem;">
            <span style="display: inline-block; padding: 0.2rem 0.6rem; border-radius: 9999px; f
ont-size: 0.75rem; font-weight: 600; background: rgba(236, 72, 153, 0.15); color: #f9a8d4; border:
1px solid rgba(236, 72, 153, 0.3);">Video</span>
          </td>
          <td style="padding: 1rem 1.25rem;">
            <ul style="list-style: none; margin: 0; padding: 0; font-size: 0.825rem; color: #cbd
5e1; line-height: 1.5;">
              <li>▹ True 4K60fps capture with Sony STARVIS CMOS</li>
              <li>▹ f/2.0 21mm studio autofocus lens</li>
              <li>▹ USB-C uncompressed raw video stream</li>
            </ul>
          </td>
          <td style="padding: 1rem 1.25rem; color: #fbbf24; font-weight: 600; font-size: 0.9re
m;">⭐ 4.6 <small style="color: #94a3b8;">(520)</small></td>
          <td style="padding: 1rem 1.25rem;">
            <span style="display: inline-block; font-size: 0.8rem; font-weight: 700; padding: 0.
25rem 0.65rem; border-radius: 6px; background: rgba(16, 185, 129, 0.15); color: #34d399; border: 1
px solid rgba(16, 185, 129, 0.3);">● In Stock (15)</span>
          </td>
          <td style="padding: 1rem 1.25rem;">
            <div style="display: flex; flex-direction: column;">
              <span style="font-size: 1.15rem; font-weight: 800; color: #f8fafc;">&#8377;27,499
</span>
              <span style="font-size: 0.8rem; color: #94a3b8; text-decoration: line-through;">&#
8377;31,999</span>
              <span style="font-size: 0.75rem; color: #34d399; font-weight: 700;">14% OFF</span>
            </div>
          </td>
          <td style="padding: 1rem 1.25rem; text-align: center;">
            <button onclick="addToCart('Elgato Facecam Pro')" style="background: #6366f1; color:
white; border: none; padding: 0.5rem 0.9rem; border-radius: 6px; font-weight: 600; font-size: 0.85
rem; cursor: pointer;">Add to Cart</button>
          </td>
        </tr>

        <!-- Product 8 -->
        <tr data-category="Input" style="border-bottom: 1px solid #334155; background: rgba(255,
255,255,0.015);">
          <td style="padding: 1rem 1.25rem;">
            <div style="display: flex; align-items: center; gap: 1rem;">
              <div style="width: 48px; height: 48px; background: #0f172a; border: 1px solid #334
155; border-radius: 8px; display: flex; align-items: center; justify-content: center; font-size:
1.6rem;">🔌</div>
              <div>
                <span style="font-weight: 700; font-size: 0.95rem; color: #ffffff; display: bloc
k;">CalDigit TS4 Thunderbolt 4 Station</span>
                <span style="font-size: 0.75rem; color: #94a3b8; font-family: monospace;">SKU: D
OCK-CAL-TS4</span>
              </div>
            </div>
          </td>
          <td style="padding: 1rem 1.25rem;">
            <span style="display: inline-block; padding: 0.2rem 0.6rem; border-radius: 9999px; f
ont-size: 0.75rem; font-weight: 600; background: rgba(6, 182, 212, 0.15); color: #67e8f9; border:
1px solid rgba(6, 182, 212, 0.3);">Hub / Dock</span>
          </td>
          <td style="padding: 1rem 1.25rem;">
            <ul style="list-style: none; margin: 0; padding: 0; font-size: 0.825rem; color: #cbd
5e1; line-height: 1.5;">
              <li>▹ 98W Power Delivery, 2.5G Ethernet, SD 4.0</li>
              <li>▹ 3x Thunderbolt 4 ports, 5x USB-A, 3x USB-C</li>
              <li>▹ Dual 6K 60Hz or single 8K monitor support</li>
```

```
              </ul>
            </td>
            <td style="padding: 1rem 1.25rem; color: #fbbf24; font-weight: 600; font-size: 0.9re
m;">⭐ 4.8 <small style="color: #94a3b8;">(640)</small></td>
            <td style="padding: 1rem 1.25rem;">
              <span style="display: inline-block; font-size: 0.8rem; font-weight: 700; padding: 0.
25rem 0.65rem; border-radius: 6px; background: rgba(16, 185, 129, 0.15); color: #34d399; border: 1
px solid rgba(16, 185, 129, 0.3);">● In Stock (12)</span>
            </td>
            <td style="padding: 1rem 1.25rem;">
              <div style="display: flex; flex-direction: column;">
                <span style="font-size: 1.15rem; font-weight: 800; color: #f8fafc;">&#8377;36,990
</span>
                <span style="font-size: 0.8rem; color: #94a3b8; text-decoration: line-through;">&#
8377;42,000</span>
                <span style="font-size: 0.75rem; color: #34d399; font-weight: 700;">12% OFF</span>
              </div>
            </td>
            <td style="padding: 1rem 1.25rem; text-align: center;">
              <button onclick="addToCart('CalDigit TS4 Dock')" style="background: #6366f1; color:
white; border: none; padding: 0.5rem 0.9rem; border-radius: 6px; font-weight: 600; font-size: 0.85
rem; cursor: pointer;">Add to Cart</button>
            </td>
          </tr>

        </tbody>
        <tfoot>
          <tr style="background: #0f172a; font-weight: 600; color: #94a3b8; font-size: 0.85rem;">
            <td colspan="3" style="padding: 1rem 1.25rem;">
              Total Hardware SKUs Displayed: <strong style="color: #ffffff;">8 Items</strong> | Al
l items include standard GST invoice &amp;amp; OEM warranty.
            </td>
            <td colspan="4" style="padding: 1rem 1.25rem; text-align: right;">
              Free Insured Express Shipping on Orders Above &#8377;5,000 &bull; 7-Day Replacement
Policy
            </td>
          </tr>
        </tfoot>
      </table>
    </div>

  </main>

  <!-- Interactive JavaScript for Live Filter & Cart Notification -->
  <script>
    let activeCategory = 'all';

    function setCategory(cat, btn) {
      activeCategory = cat;
      const buttons = document.querySelectorAll('#filter-container button');
      buttons.forEach(b => {
        b.style.background = 'rgba(255, 255, 255, 0.05)';
        b.style.color = '#94a3b8';
        b.style.borderColor = '#334155';
      });
      btn.style.background = '#6366f1';
      btn.style.color = '#ffffff';
      btn.style.borderColor = '#6366f1';
      filterCatalog();
    }

    function filterCatalog() {
      const search = document.getElementById('search-box').value.toLowerCase();
      const rows = document.querySelectorAll('#product-table tbody tr');

      rows.forEach(row => {
```

```
        const text = row.textContent.toLowerCase();
        const category = row.getAttribute('data-category');

        const matchesCategory = (activeCategory === 'all' || category.includes(activeCategory));
        const matchesSearch = text.includes(search);

        if (matchesCategory && matchesSearch) {
          row.style.display = '';
        } else {
          row.style.display = 'none';
        }
      });
    }

    function addToCart(itemName) {
      alert(`🛒 Added to Cart:\n\n"${itemName}" has been added to your shopping cart!`);
    }
  </script>

  <!-- Footer (Inline CSS) -->
  <footer style="background: #090d16; border-top: 1px solid #1e293b; padding: 2rem 1.5rem; text-al
ign: center; color: #94a3b8; font-size: 0.875rem; margin-top: 3rem;">
    <p style="margin: 0 0 0.5rem 0;">Experiment 11 &bull; CSIT248 Advanced Web Technology &bull; C
omputer Peripherals Table Catalog</p>
    <p style="margin: 0 0 0.75rem 0; color: #64748b; font-size: 0.8rem;">Formatted using 100% Inli
ne CSS properties.</p>
    <p style="margin: 0;"><a href="../index.html" style="color: #818cf8; text-decoration: none; fo
nt-weight: 600;">&uarr; Back to Lab Work Portal</a></p>
  </footer>

</body>
</html>
```

Output :



Experiment No.: 15

Title : Home Town Heritage Showcase (Patna, Bihar) with Inline CSS.

Aim : To design an attractive web page showcasing hometown heritage, culture, landmarks, and cuisine (Patna, Bihar) utilizing 100% inline CSS properties, custom typography, and responsive layouts.

Source Code :

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description"
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link
    href="https://fonts.googleapis.com/css2?family=Cinzel:wght@600;700;800&family=Plus+Jakarta+San
s:wght@400;500;600;700&display=swap"
    rel="stylesheet">
</head>

<!-- Body styled with attractive dark gradient background and custom typography via INLINE CSS -->
<body
  style="margin: 0; padding: 0; background: linear-gradient(135deg, #090e17 0%, #151c2e 50%, #0c12
1e 100%); color: #f1f5f9; font-family: 'Plus Jakarta Sans', system-ui, -apple-system, sans-serif;
line-height: 1.7; min-height: 100vh;">

  <!-- Sticky Top Navigation Bar (Inline CSS) -->
  <nav
    style="position: sticky; top: 0; z-index: 100; background: rgba(9, 14, 23, 0.92); backdrop-fil
ter: blur(12px); border-bottom: 1px solid #1e293b; padding: 1rem 2rem; display: flex; justify-cont
ent: space-between; align-items: center;">
    <a href="../index.html"
      style="font-weight: 800; font-size: 1.15rem; color: #f59e0b; text-decoration: none; display:
flex; align-items: center; gap: 0.5rem;">
      <span
        style="background: rgba(245, 158, 11, 0.15); border: 1px solid rgba(245, 158, 11, 0.3); pa
dding: 0.2rem 0.5rem; border-radius: 6px;">&larr;</span>
      CSIT248 Lab Work
    </a>
    <div style="display: flex; gap: 1rem; align-items: center;">
      <span
        style="font-size: 0.8rem; background: rgba(245, 158, 11, 0.12); color: #fbbf24; border: 1p
x solid rgba(245, 158, 11, 0.25); padding: 0.3rem 0.75rem; border-radius: 9999px; font-weight: 60
0;">
        Formatted Exclusively with Inline CSS
      </span>
      <a href="../index.html"
        style="background: #f59e0b; color: #000; padding: 0.4rem 0.9rem; border-radius: 6px; text-
decoration: none; font-size: 0.85rem; font-weight: 700; transition: all 0.2s;">
        All Experiments
      </a>
    </div>
  </nav>

  <!-- Hero Banner Header (Inline CSS) -->
  <header
    style="padding: 4.5rem 1.5rem 3.5rem 1.5rem; text-align: center; background: radial-gradient(c
ircle at 50% 30%, rgba(245, 158, 11, 0.18) 0%, rgba(15, 23, 42, 0) 70%); border-bottom: 1px solid
```

```
#1e293b;">
    <div style="max-width: 900px; margin: 0 auto;">
      <span
        style="display: inline-block; font-size: 0.85rem; font-weight: 700; text-transform: upperc
ase; letter-spacing: 0.15em; color: #f59e0b; background: rgba(245, 158, 11, 0.1); padding: 0.35rem
1rem; border-radius: 9999px; border: 1px solid rgba(245, 158, 11, 0.25); margin-bottom: 1rem;">
        Ancient Pataliputra &bull; Capital of Bihar
      </span>
      <h1
        style="font-family: 'Cinzel', serif; font-size: 3.25rem; font-weight: 800; letter-spacing:
0.02em; color: #ffffff; margin: 0 0 1rem 0; text-shadow: 0 4px 20px rgba(245, 158, 11, 0.3);">
        PATNA (PATALIPUTRA)
      </h1>
      <p
        style="font-size: 1.2rem; color: #cbd5e1; max-width: 750px; margin: 0 auto 1.5rem auto; fo
nt-weight: 400; line-height: 1.6;">
        One of the world's oldest continuously inhabited capitals, founded in 490 BCE on the banks
of the sacred River Ganga — the historic seat of the Mauryan and Gupta Empires, home to Aryabhata,
Chanakya, and Guru Gobind Singh Ji.
      </p>

      <!-- Key Badges (Inline CSS) -->
      <div style="display: flex; justify-content: center; gap: 0.75rem; flex-wrap: wrap;">
        <span
          style="background: #1e293b; color: #94a3b8; border: 1px solid #334155; padding: 0.35rem
0.85rem; border-radius: 8px; font-size: 0.85rem; font-weight: 600;">📍
          Bihar, India</span>
        <span
          style="background: #1e293b; color: #94a3b8; border: 1px solid #334155; padding: 0.35rem
0.85rem; border-radius: 8px; font-size: 0.85rem; font-weight: 600;">⏳
          &gt;2,500 Years Heritage</span>
        <span
          style="background: #1e293b; color: #94a3b8; border: 1px solid #334155; padding: 0.35rem
0.85rem; border-radius: 8px; font-size: 0.85rem; font-weight: 600;">🌊
          River Ganga &amp; Gandak</span>
        <span
          style="background: #1e293b; color: #94a3b8; border: 1px solid #334155; padding: 0.35rem
0.85rem; border-radius: 8px; font-size: 0.85rem; font-weight: 600;">🏛️
          Seat of Ashoka the Great</span>
      </div>
    </div>
  </header>

  <main style="max-width: 1100px; margin: 0 auto; padding: 3rem 1.5rem;">

    <!-- Visual Highlight Banner / Scenic Graphic (Inline CSS) -->
    <div
      style="background: linear-gradient(135deg, #1e1b4b 0%, #312e81 40%, #0f172a 100%); border: 1
px solid #4338ca; border-radius: 16px; padding: 2.5rem; margin-bottom: 3rem; box-shadow: 0 15px 35
px rgba(0, 0, 0, 0.4); display: flex; align-items: center; justify-content: space-between; flex-wr
ap: wrap; gap: 2rem;">
      <div style="flex: 1; min-width: 280px;">
        <h2 style="font-family: 'Cinzel', serif; font-size: 1.85rem; color: #fbbf24; margin: 0 0
0.75rem 0;">
          The Iconic Golghar &amp; Ganga Riverfront
        </h2>
        <p style="color: #e0e7ff; font-size: 0.95rem; margin-bottom: 1.25rem;">
          Built in 1786, the monumental stupa-shaped granary of <strong>Golghar</strong> stands ov
erlooking the majestic Ganges, offering a 360-degree panoramic vista across the city of Patna thro
ugh its twin spiral stairways of 145 steps each.
        </p>
        <div
          style="display: inline-block; background: rgba(251, 191, 36, 0.15); border: 1px solid #f
bbf24; color: #fbbf24; padding: 0.4rem 1rem; border-radius: 6px; font-size: 0.85rem; font-weight:
700;">
          ✨ Cradle of Indian Democracy, Mathematics &amp; Statecraft
```

```
        </div>
      </div>

      <!-- Illustrated Monument SVG Graphic (Golghar & Ganga - Inline CSS) -->
      <div
        style="background: #0f172a; border: 2px solid #6366f1; border-radius: 12px; padding: 1.5re
m; display: flex; align-items: center; justify-content: center; box-shadow: 0 10px 25px rgba(99, 1
02, 241, 0.25);">
        <svg width="190" height="145" viewBox="0 0 200 150" fill="none" xmlns="http://www.w3.org/2
000/svg">
          <!-- River Ganga Waves -->
          <path d="M10 135 C50 125, 80 145, 120 135 C160 125, 180 145, 195 135" stroke="#38bdf8" s
troke-width="3" fill="none" />
          <path d="M5 142 C45 132, 75 152, 115 142 C155 132, 175 152, 195 142" stroke="#0284c7" st
roke-width="2" fill="none" />

          <!-- Base Ground -->
          <rect x="25" y="118" width="150" height="10" rx="3" fill="#334155" />

          <!-- Golghar Dome Structure -->
          <path d="M45 118 C45 60, 155 60, 155 118 Z" fill="#b45309" stroke="#f59e0b" stroke-width
="2" />
          <ellipse cx="100" cy="62" rx="14" ry="5" fill="#d97706" />
          <circle cx="100" cy="55" r="4" fill="#fbbf24" />

          <!-- Spiral Staircase Lines on Dome -->
          <path d="M48 115 Q 100 85, 152 115" stroke="#fef08a" stroke-width="1.5" fill="none" stro
ke-dasharray="3 2" />
          <path d="M55 100 Q 100 75, 145 100" stroke="#fde047" stroke-width="1.5" fill="none" stro
ke-dasharray="3 2" />

          <!-- Sun & Morning Light -->
          <circle cx="165" cy="35" r="16" fill="#f59e0b" opacity="0.85" />
          <!-- Flying Birds -->
          <path d="M30 35 Q 35 30, 40 35 Q 45 30, 50 35" stroke="#94a3b8" stroke-width="1.5" fill
="none" />
          <path d="M60 45 Q 64 40, 68 45 Q 72 40, 76 45" stroke="#94a3b8" stroke-width="1.2" fill
="none" />
        </svg>
      </div>
    </div>

    <!-- Section 1: Overview & Heritage (Inline CSS) -->
    <section
      style="background: #1e293b; border: 1px solid #334155; border-radius: 14px; padding: 2rem; m
argin-bottom: 2.5rem;">
      <h2
        style="font-family: 'Cinzel', serif; font-size: 1.6rem; color: #38bdf8; margin: 0 0 1rem
0; display: flex; align-items: center; gap: 0.5rem; border-bottom: 2px solid #334155; padding-bott
om: 0.5rem;">
        <span>🏛 </span> Historical Glory &amp;amp; Imperial Legacy
      </h2>
      <p style="color: #e2e8f0; margin-bottom: 1rem; font-size: 1rem;">
        Patna, historically known as <strong>Pataliputra</strong> and <em>Kusumpura</em>, served a
s the epicenter of power for the greatest empires of ancient India, including the <strong>Nanda Em
pire, Mauryan Empire, and Gupta Empire</strong>. Megasthenes, the Greek ambassador to the court of
Chandragupta Maurya, described Pataliputra as one of the grandest cities in the ancient civilized
world.
      </p>
      <p style="color: #cbd5e1; font-size: 0.95rem;">
        The city is the birthplace of intellectual giants: <strong>Aryabhata</strong> (who pioneer
ed the concept of zero and astronomical calculations at Taregna near Patna), <strong>Chanakya</str
ong> (author of the <em>Arthashastra</em>), and <strong>Emperor Ashoka the Great</strong>, who spr
ead the principles of Dhamma across the Asian continent.
      </p>
    </section>
```

```
    <!-- Section 2: Prominent Landmarks (Grid with Inline CSS) -->
    <section style="margin-bottom: 2.5rem;">
      <h2
        style="font-family: 'Cinzel', serif; font-size: 1.6rem; color: #fbbf24; margin: 0 0 1.5rem
0; border-bottom: 2px solid #334155; padding-bottom: 0.5rem;">
        <span>📍</span> Key Landmarks &amp;amp; Heritage Sites
      </h2>

      <div style="display: grid; grid-template-columns: repeat(auto-fit, minmax(310px, 1fr)); gap:
1.5rem;">

        <!-- Card 1: Takht Sri Patna Sahib -->
        <div
          style="background: #162032; border: 1px solid #334155; border-radius: 12px; padding: 1.5
rem; box-shadow: 0 8px 20px rgba(0, 0, 0, 0.2);">
          <div style="font-size: 2.25rem; margin-bottom: 0.5rem;">🕉️</div>
          <h3 style="color: #f8fafc; font-size: 1.2rem; margin: 0 0 0.5rem 0; font-family: 'Cinze
l', serif;">
            Takht Sri Patna Sahib
          </h3>
          <p style="color: #94a3b8; font-size: 0.9rem; margin: 0 0 1rem 0;">
            One of the Five Holy Takhts of Sikhism, marking the revered birthplace of the tenth Si
kh Guru, Guru Gobind Singh Ji, built in white marble on the bank of the Ganges.
          </p>
          <span
            style="font-size: 0.75rem; background: rgba(245, 158, 11, 0.15); color: #fcd34d; paddi
ng: 0.2rem 0.5rem; border-radius: 4px; font-weight: 700;">
            Sacred Spiritual Takht
          </span>
        </div>

        <!-- Card 2: Bihar Museum & Patna Museum -->
        <div
          style="background: #162032; border: 1px solid #334155; border-radius: 12px; padding: 1.5
rem; box-shadow: 0 8px 20px rgba(0, 0, 0, 0.2);">
          <div style="font-size: 2.25rem; margin-bottom: 0.5rem;">🏺</div>
          <h3 style="color: #f8fafc; font-size: 1.2rem; margin: 0 0 0.5rem 0; font-family: 'Cinze
l', serif;">
            Bihar Museum &amp;amp; Kumhrar
          </h3>
          <p style="color: #94a3b8; font-size: 0.9rem; margin: 0 0 1rem 0;">
            A state-of-the-art international museum preserving the world-famous Didarganj Yakshi,
Mauryan pillar relics at Kumhrar, and 3,000-year-old bronze sculptures.
          </p>
          <span
            style="font-size: 0.75rem; background: rgba(56, 189, 248, 0.15); color: #7dd3fc; paddi
ng: 0.2rem 0.5rem; border-radius: 4px; font-weight: 700;">
            World Archaeological Treasury
          </span>
        </div>

        <!-- Card 3: Mahavir Mandir & Buddha Smriti Park -->
        <div
          style="background: #162032; border: 1px solid #334155; border-radius: 12px; padding: 1.5
rem; box-shadow: 0 8px 20px rgba(0, 0, 0, 0.2);">
          <div style="font-size: 2.25rem; margin-bottom: 0.5rem;">🛕</div>
          <h3 style="color: #f8fafc; font-size: 1.2rem; margin: 0 0 0.5rem 0; font-family: 'Cinze
l', serif;">
            Mahavir Mandir &amp;amp; Buddha Smriti Park
          </h3>
          <p style="color: #94a3b8; font-size: 0.9rem; margin: 0 0 1rem 0;">
            One of the most visited Hanuman temples in North India paired with the serene 22-acre
Buddha Smriti Park housing sacred Buddhist relics inaugurated by the Dalai Lama.
          </p>
          <span
```

```
            style="font-size: 0.75rem; background: rgba(52, 211, 153, 0.15); color: #6ee7b7; paddi
ng: 0.2rem 0.5rem; border-radius: 4px; font-weight: 700;">
            Cultural &amp;amp; Peace Center
          </span>
        </div>

      </div>
    </section>

    <!-- Section 3: Culture, Craft & Authentic Cuisine (Inline CSS) -->
    <section
      style="display: grid; grid-template-columns: repeat(auto-fit, minmax(320px, 1fr)); gap: 1.5r
em; margin-bottom: 2.5rem;">

      <!-- Cuisine Box -->
      <div style="background: #1e293b; border: 1px solid #334155; border-radius: 14px; padding: 1.
75rem;">
        <h3 style="font-family: 'Cinzel', serif; color: #f472b6; font-size: 1.3rem; margin: 0 0 1r
em 0;">
          🍲 Iconic Bihari Delicacies
        </h3>
        <ul style="margin: 0; padding-left: 1.25rem; color: #cbd5e1; font-size: 0.925rem; line-hei
ght: 1.9;">
          <li><strong style="color: #fff;">Litti Chokha:</strong> Roasted whole-wheat dough balls
stuffed with spiced sattu (roasted gram), dipped in pure desi ghee, paired with baingan-aloo chokh
a.</li>
          <li><strong style="color: #fff;">Silao Khaja:</strong> Crispy, multi-layered delicacy re
cognized with a prestigious Geographical Indication (GI Tag).</li>
          <li><strong style="color: #fff;">Naivedyam &amp;amp; Anarsa:</strong> Ghee-fried rice flour
sweets coated with sesame seeds, prepared during traditional festivals.</li>
          <li><strong style="color: #fff;">Tilkut &amp;amp; Thekua:</strong> Famous traditional festiv
e sweets prepared with jaggery, whole wheat, and fennel seeds.</li>
        </ul>
      </div>

      <!-- Handicrafts Box -->
      <div style="background: #1e293b; border: 1px solid #334155; border-radius: 14px; padding: 1.
75rem;">
        <h3 style="font-family: 'Cinzel', serif; color: #a78bfa; font-size: 1.3rem; margin: 0 0 1r
em 0;">
          🎨 Art, Tikuli &amp;amp; Madhubani Heritage
        </h3>
        <p style="color: #cbd5e1; font-size: 0.925rem; margin-bottom: 1rem;">
          Patna is the cultural center for <strong>Tikuli Art</strong> (an 800-year-old indigenous
craft created on glazed hardboard with gold enamel) and the world-renowned <strong>Madhubani (Mith
ila) Paintings</strong> depicting nature and mythology.
        </p>
        <div
          style="background: rgba(167, 139, 250, 0.1); border: 1px solid rgba(167, 139, 250, 0.3);
padding: 0.75rem; border-radius: 8px; color: #ddd6fe; font-size: 0.85rem;">
          🏷️ <strong>Geographical Heritage:</strong> Renowned for Bhagalpuri Tussar Silk, Sikki G
rass Craft, and Madhubani Folk Art.
        </div>
      </div>

    </section>

    <!-- Section 4: Quick Facts Table (Inline CSS) -->
    <section
      style="background: #162032; border: 1px solid #334155; border-radius: 14px; padding: 1.75re
m; margin-bottom: 2rem;">
      <h3 style="font-family: 'Cinzel', serif; color: #38bdf8; font-size: 1.3rem; margin: 0 0 1rem
0;">
        📊 Geographic &amp;amp; Civic Profile of Patna
      </h3>
      <div style="overflow-x: auto;">
```

```
        <table style="width: 100%; border-collapse: collapse; text-align: left; font-size: 0.9re
m;">
          <tbody>
            <tr style="border-bottom: 1px solid #334155;">
              <td style="padding: 0.75rem 1rem; color: #94a3b8; font-weight: 600; width: 240px;">G
eographical Coordinates</td>
              <td style="padding: 0.75rem 1rem; color: #f8fafc;">25.5941° N, 85.1376° E</td>
            </tr>
            <tr style="border-bottom: 1px solid #334155; background: rgba(255, 255, 255, 0.02);">
              <td style="padding: 0.75rem 1rem; color: #94a3b8; font-weight: 600;">Administrative
Status</td>
              <td style="padding: 0.75rem 1rem; color: #f8fafc;">State Capital of Bihar &bull; Hea
dquarters of Patna District</td>
            </tr>
            <tr style="border-bottom: 1px solid #334155;">
              <td style="padding: 0.75rem 1rem; color: #94a3b8; font-weight: 600;">Sacred River Co
nfluences</td>
              <td style="padding: 0.75rem 1rem; color: #f8fafc;">River Ganga (South Bank), Gandak,
Sone &amp;amp; Punpun</td>
            </tr>
            <tr style="border-bottom: 1px solid #334155; background: rgba(255, 255, 255, 0.02);">
              <td style="padding: 0.75rem 1rem; color: #94a3b8; font-weight: 600;">Climate &amp; W
eather</td>
              <td style="padding: 0.75rem 1rem; color: #f8fafc;">Humid Subtropical (Pleasant Winte
rs 10°C&ndash;20°C, Warm Summers)</td>
            </tr>
            <tr>
              <td style="padding: 0.75rem 1rem; color: #94a3b8; font-weight: 600;">Major Transport
ation Hubs</td>
              <td style="padding: 0.75rem 1rem; color: #f8fafc;">Jay Prakash Narayan International
Airport (PAT) &bull; Patna Junction (PNBE) &bull; Mahatma Gandhi Setu</td>
            </tr>
          </tbody>
        </table>
      </div>
    </section>

  </main>

  <!-- Footer with Inline CSS -->
  <footer
    style="background: #090e17; border-top: 1px solid #1e293b; padding: 2rem 1.5rem; text-align: c
enter; color: #94a3b8; font-size: 0.875rem;">
    <p style="margin: 0; color: #64748b;">Formatted using 100% Inline CSS properties (background,
color, font-family, flex/grid layouts, borders).</p>
    <div style="margin-top: 1rem;">
      <a href="../index.html" style="color: #f59e0b; text-decoration: none; font-weight: 600;">&ua
rr; Return to Lab Work Navigation Portal</a>
    </div>
  </footer>

</body>

</html>
```

Output :

CSIT248 Lab Work

Formatted Exclusively with Inline CSS

All Experiments

ANCIENT PATALIPUTRA • CAPITAL OF BIHAR

PATNA (PATALIPUTRA)

One of the world's oldest continuously inhabited capitals, founded in 490 BCE on the banks of the sacred River Ganga — the historic seat of the Mauryan and Gupta Empires, home to Aryabhata, Chanakya, and Guru Gobind Singh Ji.

📍 Bihar, India

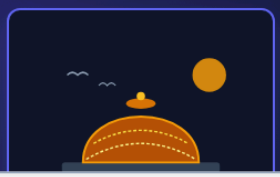
⏳ >2,500 Years Heritage

🌊 River Ganga & Gandak

🏛️ Seat of Ashoka the Great

The Iconic Golghar & Ganga Riverfront

Built in 1786, the monumental stupa-shaped granary of **Golghar** stands overlooking the majestic Ganges, offering a 360-degree panoramic vista across the city of Patna through its twin spiral stairways of 145 steps each.



Experiment No.: 16

Title : Student XML Database & Server-Side Lookup System.

Aim : To create an XML document of 10 students of SEM VI IT storing 5 subject marks, total marks, and percentage, and write a program that accepts an Enrollment No. and queries the XML file to retrieve scorecard data.

Source Code :

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description"
    content="Experiment 13: Create an XML document of 10 students of SEM VI IT and query marks, to
tal, and percentage - CSIT248 Advanced Web Technology">
  <title>Experiment 13 - Student XML Document &amp;amp; Server-Side Lookup System</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link
    href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&family=Fira+Cod
e:wght@400;500;600&display=swap"
    rel="stylesheet">
  <style>
    :root {
      --primary: #6366f1;
      --primary-hover: #4f46e5;
      --secondary: #06b6d4;
      --accent: #f43f5e;
      --success: #10b981;
      --warning: #f59e0b;
      --danger: #ef4444;
      --bg-dark: #0f172a;
      --card-bg: #1e293b;
      --card-border: #334155;
      --text-main: #f8fafc;
      --text-muted: #94a3b8;
    }

    * {
      box-sizing: border-box;
      margin: 0;
      padding: 0;
    }

    body {
      font-family: 'Inter', system-ui, -apple-system, sans-serif;
      background-color: var(--bg-dark);
      color: var(--text-main);
      line-height: 1.6;
      min-height: 100vh;
      display: flex;
      flex-direction: column;
    }

    /* Sticky Navigation */
    nav {
      position: sticky;
      top: 0;
      z-index: 100;
```

```
  background: rgba(15, 23, 42, 0.92);
  backdrop-filter: blur(12px);
  border-bottom: 1px solid var(--card-border);
  padding: 1rem 2rem;
  display: flex;
  justify-content: space-between;
  align-items: center;
}

.nav-brand {
  font-weight: 800;
  font-size: 1.15rem;
  color: #818cf8;
  text-decoration: none;
  display: flex;
  align-items: center;
  gap: 0.5rem;
}

.nav-brand span {
  background: rgba(99, 102, 241, 0.15);
  border: 1px solid rgba(99, 102, 241, 0.3);
  padding: 0.2rem 0.5rem;
  border-radius: 6px;
}

.nav-links {
  display: flex;
  align-items: center;
  gap: 1rem;
}

.nav-btn {
  background: var(--primary);
  color: #ffffff;
  padding: 0.45rem 1rem;
  border-radius: 6px;
  text-decoration: none;
  font-size: 0.85rem;
  font-weight: 600;
  transition: background 0.2s;
}

.nav-btn:hover {
  background: var(--primary-hover);
}

/* Container */
main {
  max-width: 1200px;
  width: 100%;
  margin: 0 auto;
  padding: 2.5rem 1.5rem;
  flex: 1;
}

/* Header */
.page-header {
  text-align: center;
  margin-bottom: 2rem;
}

.badge-pill {
  display: inline-block;
  background: rgba(99, 102, 241, 0.15);
  color: #a5b4fc;
```

```
  border: 1px solid rgba(99, 102, 241, 0.3);
  padding: 0.3rem 0.9rem;
  border-radius: 9999px;
  font-size: 0.825rem;
  font-weight: 600;
  margin-bottom: 0.75rem;
}

.page-title {
  font-size: 2.25rem;
  font-weight: 800;
  color: #ffffff;
  margin-bottom: 0.5rem;
}

.page-subtitle {
  color: var(--text-muted);
  font-size: 1.05rem;
  max-width: 820px;
  margin: 0 auto;
}

/* Tab Controller */
.tabs-nav {
  display: flex;
  gap: 0.5rem;
  border-bottom: 1px solid var(--card-border);
  margin-bottom: 2rem;
  overflow-x: auto;
  padding-bottom: 0.25rem;
}

.tab-btn {
  background: transparent;
  border: none;
  color: var(--text-muted);
  padding: 0.75rem 1.25rem;
  font-size: 0.95rem;
  font-weight: 600;
  cursor: pointer;
  border-radius: 8px 8px 0 0;
  display: flex;
  align-items: center;
  gap: 0.5rem;
  transition: all 0.2s;
}

.tab-btn:hover {
  color: #ffffff;
  background: rgba(255, 255, 255, 0.05);
}

.tab-btn.active {
  color: #38bdf8;
  background: var(--card-bg);
  border-bottom: 2px solid #38bdf8;
}

.tab-panel {
  display: none;
}

.tab-panel.active {
  display: block;
  animation: fadeIn 0.3s ease-in-out;
}
```

```
@keyframes fadeIn {
  from { opacity: 0; transform: translateY(6px); }
  to { opacity: 1; transform: translateY(0); }
}

/* Layout Grid */
.grid-2 {
  display: grid;
  grid-template-columns: 380px 1fr;
  gap: 1.75rem;
}

@media (max-width: 900px) {
  .grid-2 {
    grid-template-columns: 1fr;
  }
}

.card {
  background: var(--card-bg);
  border: 1px solid var(--card-border);
  border-radius: 16px;
  padding: 1.75rem;
  box-shadow: 0 10px 25px rgba(0, 0, 0, 0.3);
}

.card-title {
  font-size: 1.25rem;
  font-weight: 700;
  color: #38bdf8;
  margin-bottom: 1.25rem;
  display: flex;
  align-items: center;
  gap: 0.5rem;
  border-bottom: 1px solid var(--card-border);
  padding-bottom: 0.75rem;
}

/* Input Styles */
.form-group {
  margin-bottom: 1.25rem;
}

label {
  display: block;
  font-size: 0.85rem;
  font-weight: 600;
  color: #cbd5e1;
  margin-bottom: 0.4rem;
}

input[type="text"] {
  width: 100%;
  background: #0f172a;
  border: 1px solid #475569;
  color: #ffffff;
  padding: 0.75rem 1rem;
  border-radius: 8px;
  font-size: 0.95rem;
  outline: none;
  transition: border-color 0.2s;
}

input[type="text"]:focus {
  border-color: var(--primary);
```

```
}

.btn-submit {
  width: 100%;
  background: linear-gradient(135deg, #6366f1, #4f46e5);
  color: white;
  border: none;
  padding: 0.85rem 1.25rem;
  border-radius: 8px;
  font-size: 0.95rem;
  font-weight: 700;
  cursor: pointer;
  display: flex;
  align-items: center;
  justify-content: center;
  gap: 0.5rem;
  box-shadow: 0 4px 14px rgba(99, 102, 241, 0.3);
  transition: all 0.2s;
}

.btn-submit:hover {
  transform: translateY(-1px);
  box-shadow: 0 6px 20px rgba(99, 102, 241, 0.45);
}

/* Quick Student Selector Chips */
.quick-list {
  margin-top: 1.5rem;
  border-top: 1px solid var(--card-border);
  padding-top: 1.25rem;
}

.chips-container {
  display: flex;
  flex-wrap: wrap;
  gap: 0.5rem;
  margin-top: 0.75rem;
}

.student-chip {
  background: #0f172a;
  border: 1px solid #334155;
  color: #cbd5e1;
  padding: 0.35rem 0.65rem;
  border-radius: 6px;
  font-size: 0.8rem;
  cursor: pointer;
  display: flex;
  align-items: center;
  gap: 0.4rem;
  transition: all 0.2s;
  font-family: 'Fira Code', monospace;
}

.student-chip:hover {
  border-color: #38bdf8;
  color: #38bdf8;
  background: rgba(56, 189, 248, 0.1);
}

.student-chip.active {
  border-color: var(--primary);
  background: rgba(99, 102, 241, 0.2);
  color: #a5b4fc;
  font-weight: 600;
}
```

```
/* Scorecard Display */
.scorecard-box {
  background: #0f172a;
  border: 1px solid #334155;
  border-radius: 12px;
  padding: 1.5rem;
}

.scorecard-header {
  display: flex;
  justify-content: space-between;
  align-items: flex-start;
  margin-bottom: 1.25rem;
  flex-wrap: wrap;
  gap: 1rem;
  border-bottom: 1px solid #334155;
  padding-bottom: 1rem;
}

.student-name {
  font-size: 1.5rem;
  font-weight: 800;
  color: #ffffff;
}

.student-meta {
  font-size: 0.85rem;
  color: var(--text-muted);
  margin-top: 0.25rem;
}

.grade-badge {
  background: rgba(16, 185, 129, 0.2);
  border: 1px solid var(--success);
  color: #34d399;
  font-size: 1.15rem;
  font-weight: 800;
  padding: 0.4rem 1rem;
  border-radius: 8px;
  text-align: center;
}

/* Marks Table */
.marks-table {
  width: 100%;
  border-collapse: collapse;
  margin: 1.25rem 0;
  font-size: 0.9rem;
}

.marks-table th,
.marks-table td {
  padding: 0.75rem 0.9rem;
  text-align: left;
  border-bottom: 1px solid #1e293b;
}

.marks-table th {
  background: #1e293b;
  color: #38bdf8;
  font-weight: 600;
}

.progress-bar-bg {
  background: #1e293b;
```

```
  border-radius: 9999px;
  height: 8px;
  width: 100%;
  overflow: hidden;
  border: 1px solid #334155;
}

.progress-bar-fill {
  height: 100%;
  border-radius: 9999px;
  transition: width 0.6s cubic-bezier(0.4, 0, 0.2, 1);
}

/* Summary Stats */
.stats-grid {
  display: grid;
  grid-template-columns: repeat(3, 1fr);
  gap: 1rem;
  margin-top: 1.25rem;
}

@media (max-width: 600px) {
  .stats-grid {
    grid-template-columns: 1fr;
  }
}

.stat-card {
  background: #1e293b;
  border: 1px solid #334155;
  border-radius: 10px;
  padding: 1rem;
  text-align: center;
}

.stat-card .label {
  font-size: 0.75rem;
  color: var(--text-muted);
  text-transform: uppercase;
  letter-spacing: 0.05em;
}

.stat-card .value {
  font-size: 1.6rem;
  font-weight: 800;
  color: #ffffff;
  margin-top: 0.2rem;
}

/* Code Blocks & Pre */
.code-container {
  background: #0f172a;
  border: 1px solid var(--card-border);
  border-radius: 12px;
  overflow: hidden;
}

.code-header {
  background: #1e293b;
  padding: 0.75rem 1.25rem;
  display: flex;
  justify-content: space-between;
  align-items: center;
  border-bottom: 1px solid var(--card-border);
}
```

```
    .code-header span {
      font-family: 'Fira Code', monospace;
      font-size: 0.85rem;
      color: #38bdf8;
    }

    .btn-copy {
      background: #334155;
      color: #f8fafc;
      border: 1px solid #475569;
      padding: 0.35rem 0.75rem;
      border-radius: 6px;
      font-size: 0.8rem;
      cursor: pointer;
      transition: all 0.2s;
    }

    .btn-copy:hover {
      background: var(--primary);
      border-color: var(--primary);
    }

    pre code {
      display: block;
      padding: 1.25rem;
      font-family: 'Fira Code', monospace;
      font-size: 0.875rem;
      line-height: 1.6;
      color: #e2e8f0;
      overflow-x: auto;
      white-space: pre;
    }

    .empty-state {
      text-align: center;
      padding: 3rem 1.5rem;
      color: var(--text-muted);
    }

    .empty-state svg {
      width: 56px;
      height: 56px;
      color: #475569;
      margin-bottom: 1rem;
    }

    /* Footer */
    footer {
      background: #0b1120;
      border-top: 1px solid var(--card-border);
      padding: 1.5rem 2rem;
      text-align: center;
      font-size: 0.85rem;
      color: var(--text-muted);
      margin-top: 3rem;
    }
  </style>
</head>

<body>

  <!-- Navigation -->
  <nav>
    <a href="../index.html" class="nav-brand">
      <span>&larr;</span> CSIT248 Lab Work
    </a>
```

```
    <div class="nav-links">
      <a href="https://atulanand-web-tech-labwork.netlify.app/" target="_blank" rel="noopener nore
ferrer"
        style="font-size: 0.85rem; color: #00c7b7; background: rgba(0, 199, 183, 0.15); border: 1p
x solid rgba(0, 199, 183, 0.35); padding: 0.3rem 0.75rem; border-radius: 9999px; font-weight: 600;
text-decoration: none; display: flex; align-items: center; gap: 0.35rem;">
        🌐 Live Portfolio
      </a>
      <span style="font-size: 0.85rem; color: #a5b4fc; background: rgba(99, 102, 241, 0.15); borde
r: 1px solid rgba(99, 102, 241, 0.3); padding: 0.3rem 0.75rem; border-radius: 9999px; font-weight:
600;">
        XML &amp;amp; Server-Side Processing
      </span>
      <a href="../index.html" class="nav-btn">All Experiments</a>
    </div>
  </nav>

  <!-- Main Content -->
  <main>

    <!-- Page Header -->
    <header class="page-header">
      <span class="badge-pill">Experiment 13 &amp;bull; Module XML &amp;amp; Web Services</span>
      <h1 class="page-title">SEM VI IT Student XML &amp;amp; Server-Side Lookup System</h1>
      <p class="page-subtitle">
        Creation of a structured server XML document housing 10 student records with 5 subject mar
ks, total, and percentage, along with an interactive search system executing queries via enrollmen
t number.
      </p>
    </header>

    <!-- Tab Navigation -->
    <div class="tabs-nav">
      <button class="tab-btn active" onclick="switchTab('demo', this)">
        ⚡ Live Interactive Lookup
      </button>
      <button class="tab-btn" onclick="switchTab('xml', this)">
        📄 students.xml (Server Data)
      </button>
      <button class="tab-btn" onclick="switchTab('php', this)">
        🐘 student_lookup.php (Server Script)
      </button>
      <button class="tab-btn" onclick="switchTab('report', this)">
        📘 Academic Lab Manual &amp;amp; Theory
      </button>
    </div>

    <!-- TAB 1: Live Interactive Lookup -->
    <section id="tab-demo" class="tab-panel active">
      <div class="grid-2">

        <!-- Left Column: Search & Controls -->
        <div class="card">
          <h2 class="card-title">🔍 Search Student Record</h2>

          <form id="search-form" onsubmit="handleSearch(event)">
            <div class="form-group">
              <label for="enrollment-input">Enrollment Number</label>
              <input type="text" id="enrollment-input" placeholder="e.g. IT2023001" required value
="IT2023001">
            </div>
            <button type="submit" class="btn-submit">
              <span>Query Server XML</span> &rarr;
            </button>
          </form>
```

```
          <div class="quick-list">
            <label>Click Any of the 10 Students of SEM VI IT:</label>
            <div class="chips-container" id="chips-container">
              <!-- Dynamically populated from XML -->
            </div>
          </div>

          <div style="margin-top: 1.5rem; background: #0f172a; border: 1px solid #334155; border-r
adius: 8px; padding: 1rem; font-size: 0.8rem; color: var(--text-muted);">
            <strong style="color: #38bdf8;">Server Mechanism:</strong> The system parses <code>stu
dents.xml</code> on the server (using PHP SimpleXML / DOMDocument XPath) and returns filtered reco
rds according to the requested <code>enrollment_no</code>.
          </div>
        </div>

        <!-- Right Column: Student Result Scorecard -->
        <div class="card" id="result-card">
          <h2 class="card-title">🎓 Student Academic Scorecard</h2>

          <div id="scorecard-container">
            <!-- Populated via JavaScript after parsing XML -->
          </div>
        </div>

      </div>
    </section>

    <!-- TAB 2: students.xml Source View -->
    <section id="tab-xml" class="tab-panel">
      <div class="card">
        <div class="code-container">
          <div class="code-header">
            <span>students.xml (Saved at Server Root)</span>
            <button class="btn-copy" onclick="copyCode('code-xml', this)">📋 Copy XML</button>
          </div>
          <pre><code id="code-xml">&lt;?xml version="1.0" encoding="UTF-8"?&gt;
&lt;!-- Experiment 13: Student Academic Database XML (Sem VI IT) --&gt;
&lt;students_database semester="VI" branch="Information Technology" academic_year="2025-2026"&gt;

  &lt;student id="1"&gt;
    &lt;enrollment_no&gt;IT2023001&lt;/enrollment_no&gt;
    &lt;name&gt;Atul Anand&lt;/name&gt;
    &lt;semester&gt;VI IT&lt;/semester&gt;
    &lt;academic_batch&gt;2023-2026&lt;/academic_batch&gt;
    &lt;marks&gt;
      &lt;subject code="IT601" name="Cloud Computing" max="100"&gt;92&lt;/subject&gt;
      &lt;subject code="IT602" name="Information &amp;amp; Network Security" max="100"&gt;88&lt;/s
ubject&gt;
      &lt;subject code="IT603" name="Software Engineering &amp;amp; Agile" max="100"&gt;94&lt;/sub
ject&gt;
      &lt;subject code="IT604" name="Data Mining &amp;amp; Business Intelligence" max="100"&gt;90&
lt;/subject&gt;
      &lt;subject code="IT605" name="Mobile Computing &amp;amp; IoT" max="100"&gt;96&lt;/subject&g
t;
    &lt;/marks&gt;
    &lt;total_marks max="500"&gt;460&lt;/total_marks&gt;
    &lt;percentage&gt;92.00&lt;/percentage&gt;
    &lt;grade&gt;A+&lt;/grade&gt;
    &lt;result&gt;PASS&lt;/result&gt;
  &lt;/student&gt;

  &lt;student id="2"&gt;
    &lt;enrollment_no&gt;IT2023002&lt;/enrollment_no&gt;
    &lt;name&gt;Priya Sharma&lt;/name&gt;
    &lt;semester&gt;VI IT&lt;/semester&gt;
    &lt;academic_batch&gt;2023-2026&lt;/academic_batch&gt;
```

```
    &lt;marks&gt;
      &lt;subject code="IT601" name="Cloud Computing" max="100"&gt;85&lt;/subject&gt;
      &lt;subject code="IT602" name="Information &amp;amp; Network Security" max="100"&gt;79&lt;/s
ubject&gt;
      &lt;subject code="IT603" name="Software Engineering &amp;amp; Agile" max="100"&gt;88&lt;/sub
ject&gt;
      &lt;subject code="IT604" name="Data Mining &amp;amp; Business Intelligence" max="100"&gt;82&
lt;/subject&gt;
      &lt;subject code="IT605" name="Mobile Computing &amp;amp; IoT" max="100"&gt;91&lt;/subject&g
t;
    &lt;/marks&gt;
    &lt;total_marks max="500"&gt;425&lt;/total_marks&gt;
    &lt;percentage&gt;85.00&lt;/percentage&gt;
    &lt;grade&gt;A&lt;/grade&gt;
    &lt;result&gt;PASS&lt;/result&gt;
  &lt;/student&gt;

  &lt;!-- Records 3 to 10 follow identical XML hierarchy --&gt;
&lt;/students_database&gt;</code></pre>
        </div>
      </div>
    </section>

    <!-- TAB 3: PHP Server Script View -->
    <section id="tab-php" class="tab-panel">
      <div class="card">
        <div class="code-container">
          <div class="code-header">
            <span>student_lookup.php (PHP Server Parser)</span>
            <button class="btn-copy" onclick="copyCode('code-php', this)">📋 Copy PHP Script</but
ton>
          </div>
          <pre><code id="code-php">&lt;?php
/**
 * Experiment 13: Student Marks Retrieval from Server XML
 * Advanced Web Technology (CSIT248) - Sem VI IT
 */

$xmlFile = 'students.xml';
$enrollment = trim($_POST['enrollment_no'] ?? $_GET['enrollment_no'] ?? '');

if (empty($enrollment)) {
    die("&lt;p&gt;Please enter an Enrollment Number.&lt;/p&gt;");
}

if (!file_exists($xmlFile)) {
    die("&lt;p&gt;Error: XML Document not found on server.&lt;/p&gt;");
}

// Load and parse XML document using SimpleXML
$xml = simplexml_load_file($xmlFile);
$found = null;

// Iterate and match student enrollment number
foreach ($xml->student as $student) {
    if (strcasecmp((string)$student->enrollment_no, $enrollment) === 0) {
        $found = $student;
        break;
    }
}

if ($found) {
    echo "&lt;h2&gt;Student Name: " . htmlspecialchars($found->name) . "&lt;/h2&gt;";
    echo "&lt;p&gt;Enrollment No: " . htmlspecialchars($found->enrollment_no) . "&lt;/p&gt;";
    echo "&lt;p&gt;Semester: " . htmlspecialchars($found->semester) . "&lt;/p&gt;";
    echo "&lt;table border='1' cellpadding='8'&gt;";
```

```
    echo "&lt;tr&gt;&lt;th&gt;Code&lt;/th&gt;&lt;th&gt;Subject&lt;/th&gt;&lt;th&gt;Marks&lt;/th&g
t;&lt;/tr&gt;";

    foreach ($found->marks->subject as $subject) {
        echo "&lt;tr&gt;";
        echo "&lt;td&gt;" . htmlspecialchars($subject['code']) . "&lt;/td&gt;";
        echo "&lt;td&gt;" . htmlspecialchars($subject['name']) . "&lt;/td&gt;";
        echo "&lt;td&gt;" . htmlspecialchars($subject) . " / " . htmlspecialchars($subject['max'])
. "&lt;/td&gt;";
        echo "&lt;/tr&gt;";
    }
    echo "&lt;/table&gt;";

    echo "&lt;h3&gt;Total Marks: " . htmlspecialchars($found->total_marks) . " / 500&lt;/h3&gt;";
    echo "&lt;h3&gt;Percentage: " . htmlspecialchars($found->percentage) . "%&lt;/h3&gt;";
    echo "&lt;h3&gt;Grade: " . htmlspecialchars($found->grade) . "&lt;/h3&gt;";
} else {
    echo "&lt;p style='color:red;'&gt;No record found for Enrollment: " . htmlspecialchars($enroll
ment) . "&lt;/p&gt;";
}
?&gt;</code></pre>
        </div>
      </div>
    </section>

    <!-- TAB 4: Academic Lab Manual & Theory -->
    <section id="tab-report" class="tab-panel">
      <div class="card" style="line-height: 1.8;">
        <h2 style="color: #38bdf8; font-size: 1.5rem; margin-bottom: 1rem;">Experiment 13: Academi
c Lab Manual</h2>

        <h3 style="color: #818cf8; margin-top: 1.5rem; margin-bottom: 0.5rem;">1. Aim of the Exper
iment</h3>
        <p style="color: #cbd5e1;">
          To create a structured XML document containing records of 10 students of SEM VI IT (incl
uding Enrollment Number, Marks obtained in 5 subjects, Total Marks, and Aggregate Percentage) save
d on the web server, and develop a server-side processing script that accepts an Enrollment Number
as user input and returns individual subject marks, total marks, and percentage by querying the XM
L database.
        </p>

        <h3 style="color: #818cf8; margin-top: 1.5rem; margin-bottom: 0.5rem;">2. Theoretical Foun
dations</h3>
        <ul style="color: #cbd5e1; margin-left: 1.5rem;">
          <li><strong>XML (eXtensible Markup Language):</strong> A standard text-based markup lang
uage for encoding documents in a format that is both human-readable and machine-readable, widely u
sed for platform-independent data exchange.</li>
          <li><strong>DOM (Document Object Model) &amp;amp; SimpleXML:</strong> PHP provides native ex
tensions like <code>SimpleXML</code> and <code>DOMDocument</code> to load, parse, traverse, and qu
ery XML documents using tree-based and XPath representations.</li>
          <li><strong>XPath Expression:</strong> Syntax used to navigate and search through nodes
and attributes in an XML document (e.g., <code>//student[enrollment_no='IT2023001']</code>).</li>
        </ul>

        <h3 style="color: #818cf8; margin-top: 1.5rem; margin-bottom: 0.5rem;">3. Algorithm</h3>
        <ol style="color: #cbd5e1; margin-left: 1.5rem;">
          <li>Construct <code>students.xml</code> with root <code>&lt;students_database&gt;</code>
and 10 child <code>&lt;student&gt;</code> elements with 5 subject scores.</li>
          <li>Prompt user via an HTML form to supply their unique <code>enrollment_no</code>.</li>
          <li>On submission, the server receives the input via HTTP POST/GET.</li>
          <li>Validate the input and check whether <code>students.xml</code> exists on the server
filesystem.</li>
          <li>Load the XML tree using <code>simplexml_load_file()</code>.</li>
          <li>Traverse the student elements comparing the enrollment number using case-insensitive
string comparison.</li>
          <li>If a match is discovered, extract subject marks, total marks, percentage, and grade,
```

```
then render a formatted tabular response.</li>
          <li>If no match exists, return an informative error status.</li>
        </ol>

        <h3 style="color: #818cf8; margin-top: 1.5rem; margin-bottom: 0.5rem;">4. Five Subjects of SEM VI IT</h3>
        <table style="width: 100%; border-collapse: collapse; margin-top: 1rem; color: #cbd5e1;">
          <thead>
            <tr style="background: #0f172a; text-align: left;">
              <th style="padding: 0.6rem 1rem; border: 1px solid #334155;">Subject Code</th>
              <th style="padding: 0.6rem 1rem; border: 1px solid #334155;">Subject Name</th>
              <th style="padding: 0.6rem 1rem; border: 1px solid #334155;">Max Marks</th>
            </tr>
          </thead>
          <tbody>
            <tr><td style="padding: 0.5rem 1rem; border: 1px solid #334155;">IT601</td><td style="padding: 0.5rem 1rem; border: 1px solid #334155;">Cloud Computing</td><td style="padding: 0.5rem 1rem; border: 1px solid #334155;">100</td></tr>
            <tr><td style="padding: 0.5rem 1rem; border: 1px solid #334155;">IT602</td><td style="padding: 0.5rem 1rem; border: 1px solid #334155;">Information &amp;amp; Network Security</td><td style="padding: 0.5rem 1rem; border: 1px solid #334155;">100</td></tr>
            <tr><td style="padding: 0.5rem 1rem; border: 1px solid #334155;">IT603</td><td style="padding: 0.5rem 1rem; border: 1px solid #334155;">Software Engineering &amp;amp; Agile</td><td style="padding: 0.5rem 1rem; border: 1px solid #334155;">100</td></tr>
            <tr><td style="padding: 0.5rem 1rem; border: 1px solid #334155;">IT604</td><td style="padding: 0.5rem 1rem; border: 1px solid #334155;">Data Mining &amp;amp; Business Intelligence</td><td style="padding: 0.5rem 1rem; border: 1px solid #334155;">100</td></tr>
            <tr><td style="padding: 0.5rem 1rem; border: 1px solid #334155;">IT605</td><td style="padding: 0.5rem 1rem; border: 1px solid #334155;">Mobile Computing &amp;amp; IoT</td><td style="padding: 0.5rem 1rem; border: 1px solid #334155;">100</td></tr>
          </tbody>
        </table>
      </div>
    </section>

  </main>

  <!-- Footer -->
  <footer>
    <p style="margin-bottom: 0.4rem;">
      Designed &amp;amp; Developed by <strong style="color: #818cf8;">Atul Anand</strong> &bull; BCA (Hons.) with Research &bull; Semester III
    </p>
    <p style="color: #64748b; font-size: 0.8rem; margin-bottom: 0.35rem;">
      Amity University Noida &bull; Department of Computer Applications &bull; CSIT248: Advanced Web Technology
    </p>
    <p style="font-size: 0.8rem;">
      Live Portfolio: <a href="https://atulanand-web-tech-labwork.netlify.app/" target="_blank" rel="noopener noreferrer" style="color: #00c7b7; text-decoration: none; font-weight: 600;">atulanand-web-tech-labwork.netlify.app</a>
    </p>
  </footer>

  <!-- Script: XML Parser & Dynamic UI -->
  <script>
    // Embedded XML dataset to guarantee 100% offline & local file:// support
    const XML_SOURCE = `<?xml version="1.0" encoding="UTF-8"?>
<students_database semester="VI" branch="Information Technology" academic_year="2025-2026">
  <student id="1">
    <enrollment_no>IT2023001</enrollment_no>
    <name>Atul Anand</name>
    <semester>VI IT</semester>
    <academic_batch>2023-2026</academic_batch>
    <marks>
      <subject code="IT601" name="Cloud Computing" max="100">92</subject>
```

```
    <subject code="IT602" name="Information &amp;amp; Network Security" max="100">88</subject>
    <subject code="IT603" name="Software Engineering &amp;amp; Agile" max="100">94</subject>
    <subject code="IT604" name="Data Mining &amp;amp; Business Intelligence" max="100">90</subject>
    <subject code="IT605" name="Mobile Computing &amp;amp; IoT" max="100">96</subject>
  </marks>
  <total_marks max="500">460</total_marks>
  <percentage>92.00</percentage>
  <grade>A+</grade>
  <result>PASS</result>
</student>
<student id="2">
  <enrollment_no>IT2023002</enrollment_no>
  <name>Priya Sharma</name>
  <semester>VI IT</semester>
  <academic_batch>2023-2026</academic_batch>
  <marks>
    <subject code="IT601" name="Cloud Computing" max="100">85</subject>
    <subject code="IT602" name="Information &amp;amp; Network Security" max="100">79</subject>
    <subject code="IT603" name="Software Engineering &amp;amp; Agile" max="100">88</subject>
    <subject code="IT604" name="Data Mining &amp;amp; Business Intelligence" max="100">82</subject>
    <subject code="IT605" name="Mobile Computing &amp;amp; IoT" max="100">91</subject>
  </marks>
  <total_marks max="500">425</total_marks>
  <percentage>85.00</percentage>
  <grade>A</grade>
  <result>PASS</result>
</student>
<student id="3">
  <enrollment_no>IT2023003</enrollment_no>
  <name>Rohan Verma</name>
  <semester>VI IT</semester>
  <academic_batch>2023-2026</academic_batch>
  <marks>
    <subject code="IT601" name="Cloud Computing" max="100">78</subject>
    <subject code="IT602" name="Information &amp;amp; Network Security" max="100">84</subject>
    <subject code="IT603" name="Software Engineering &amp;amp; Agile" max="100">76</subject>
    <subject code="IT604" name="Data Mining &amp;amp; Business Intelligence" max="100">80</subject>
    <subject code="IT605" name="Mobile Computing &amp;amp; IoT" max="100">82</subject>
  </marks>
  <total_marks max="500">400</total_marks>
  <percentage>80.00</percentage>
  <grade>A</grade>
  <result>PASS</result>
</student>
<student id="4">
  <enrollment_no>IT2023004</enrollment_no>
  <name>Ananya Sengupta</name>
  <semester>VI IT</semester>
  <academic_batch>2023-2026</academic_batch>
  <marks>
    <subject code="IT601" name="Cloud Computing" max="100">95</subject>
    <subject code="IT602" name="Information &amp;amp; Network Security" max="100">91</subject>
    <subject code="IT603" name="Software Engineering &amp;amp; Agile" max="100">96</subject>
    <subject code="IT604" name="Data Mining &amp;amp; Business Intelligence" max="100">94</subject>
    <subject code="IT605" name="Mobile Computing &amp;amp; IoT" max="100">98</subject>
  </marks>
  <total_marks max="500">474</total_marks>
  <percentage>94.80</percentage>
  <grade>A+</grade>
  <result>PASS</result>
</student>
<student id="5">
  <enrollment_no>IT2023005</enrollment_no>
  <name>Vikramaditya Rao</name>
  <semester>VI IT</semester>
  <academic_batch>2023-2026</academic_batch>
```

```
  <marks>
    <subject code="IT601" name="Cloud Computing" max="100">70</subject>
    <subject code="IT602" name="Information &amp; Network Security" max="100">68</subject>
    <subject code="IT603" name="Software Engineering &amp; Agile" max="100">74</subject>
    <subject code="IT604" name="Data Mining &amp; Business Intelligence" max="100">72</subject>
    <subject code="IT605" name="Mobile Computing &amp; IoT" max="100">76</subject>
  </marks>
  <total_marks max="500">360</total_marks>
  <percentage>72.00</percentage>
  <grade>B+</grade>
  <result>PASS</result>
</student>
<student id="6">
  <enrollment_no>IT2023006</enrollment_no>
  <name>Sneha Kulkarni</name>
  <semester>VI IT</semester>
  <academic_batch>2023-2026</academic_batch>
  <marks>
    <subject code="IT601" name="Cloud Computing" max="100">89</subject>
    <subject code="IT602" name="Information &amp; Network Security" max="100">86</subject>
    <subject code="IT603" name="Software Engineering &amp; Agile" max="100">91</subject>
    <subject code="IT604" name="Data Mining &amp; Business Intelligence" max="100">88</subject>
    <subject code="IT605" name="Mobile Computing &amp; IoT" max="100">93</subject>
  </marks>
  <total_marks max="500">447</total_marks>
  <percentage>89.40</percentage>
  <grade>A</grade>
  <result>PASS</result>
</student>
<student id="7">
  <enrollment_no>IT2023007</enrollment_no>
  <name>Arjun Nair</name>
  <semester>VI IT</semester>
  <academic_batch>2023-2026</academic_batch>
  <marks>
    <subject code="IT601" name="Cloud Computing" max="100">62</subject>
    <subject code="IT602" name="Information &amp; Network Security" max="100">65</subject>
    <subject code="IT603" name="Software Engineering &amp; Agile" max="100">68</subject>
    <subject code="IT604" name="Data Mining &amp; Business Intelligence" max="100">64</subject>
    <subject code="IT605" name="Mobile Computing &amp; IoT" max="100">71</subject>
  </marks>
  <total_marks max="500">330</total_marks>
  <percentage>66.00</percentage>
  <grade>B</grade>
  <result>PASS</result>
</student>
<student id="8">
  <enrollment_no>IT2023008</enrollment_no>
  <name>Meera Iyer</name>
  <semester>VI IT</semester>
  <academic_batch>2023-2026</academic_batch>
  <marks>
    <subject code="IT601" name="Cloud Computing" max="100">91</subject>
    <subject code="IT602" name="Information &amp; Network Security" max="100">89</subject>
    <subject code="IT603" name="Software Engineering &amp; Agile" max="100">87</subject>
    <subject code="IT604" name="Data Mining &amp; Business Intelligence" max="100">93</subject>
    <subject code="IT605" name="Mobile Computing &amp; IoT" max="100">95</subject>
  </marks>
  <total_marks max="500">455</total_marks>
  <percentage>91.00</percentage>
  <grade>A+</grade>
  <result>PASS</result>
</student>
<student id="9">
  <enrollment_no>IT2023009</enrollment_no>
  <name>Kabir Das</name>
```

```
    <semester>VI IT</semester>
    <academic_batch>2023-2026</academic_batch>
    <marks>
      <subject code="IT601" name="Cloud Computing" max="100">75</subject>
      <subject code="IT602" name="Information &amp;amp; Network Security" max="100">72</subject>
      <subject code="IT603" name="Software Engineering &amp;amp; Agile" max="100">78</subject>
      <subject code="IT604" name="Data Mining &amp;amp; Business Intelligence" max="100">70</subject>
      <subject code="IT605" name="Mobile Computing &amp;amp; IoT" max="100">80</subject>
    </marks>
    <total_marks max="500">375</total_marks>
    <percentage>75.00</percentage>
    <grade>B+</grade>
    <result>PASS</result>
  </student>
  <student id="10">
    <enrollment_no>IT2023010</enrollment_no>
    <name>Tanvi Singhania</name>
    <semester>VI IT</semester>
    <academic_batch>2023-2026</academic_batch>
    <marks>
      <subject code="IT601" name="Cloud Computing" max="100">88</subject>
      <subject code="IT602" name="Information &amp;amp; Network Security" max="100">85</subject>
      <subject code="IT603" name="Software Engineering &amp;amp; Agile" max="100">90</subject>
      <subject code="IT604" name="Data Mining &amp;amp; Business Intelligence" max="100">87</subject>
      <subject code="IT605" name="Mobile Computing &amp;amp; IoT" max="100">92</subject>
    </marks>
    <total_marks max="500">442</total_marks>
    <percentage>88.40</percentage>
    <grade>A</grade>
    <result>PASS</result>
  </student>
</students_database>`;

    let xmlDoc = null;
    let studentsArray = [];

    // Parse XML Document
    function initXML() {
      const parser = new DOMParser();
      xmlDoc = parser.parseFromString(XML_SOURCE, "application/xml");

      const studentNodes = xmlDoc.getElementsByTagName("student");
      studentsArray = [];

      const chipsContainer = document.getElementById("chips-container");
      chipsContainer.innerHTML = "";

      for (let i = 0; i < studentNodes.length; i++) {
        const node = studentNodes[i];
        const enroll = node.getElementsByTagName("enrollment_no")[0].textContent;
        const name = node.getElementsByTagName("name")[0].textContent;
        const total = node.getElementsByTagName("total_marks")[0].textContent;
        const pct = node.getElementsByTagName("percentage")[0].textContent;
        const grade = node.getElementsByTagName("grade")[0].textContent;
        const result = node.getElementsByTagName("result")[0].textContent;
        const semester = node.getElementsByTagName("semester")[0].textContent;
        const batch = node.getElementsByTagName("academic_batch")[0].textContent;

        const subjectNodes = node.getElementsByTagName("subject");
        const subjects = [];
        for (let j = 0; j < subjectNodes.length; j++) {
          subjects.push({
            code: subjectNodes[j].getAttribute("code"),
            name: subjectNodes[j].getAttribute("name"),
            max: parseInt(subjectNodes[j].getAttribute("max")),
            score: parseInt(subjectNodes[j].textContent)
```

```
      });
    }

    const studentObj = {
      enroll,
      name,
      semester,
      batch,
      total,
      pct,
      grade,
      result,
      subjects
    };

    studentsArray.push(studentObj);

    // Create Quick Chip
    const chip = document.createElement("button");
    chip.type = "button";
    chip.className = "student-chip" + (i === 0 ? " active" : "");
    chip.innerHTML = `<span>👤</span> ${enroll}`;
    chip.title = `${name} (${enroll})`;
    chip.onclick = () => selectStudent(enroll);
    chipsContainer.appendChild(chip);
  }

  // Render First Student by default
  renderScorecard(studentsArray[0]);
}

function selectStudent(enroll) {
  document.getElementById("enrollment-input").value = enroll;

  // Update active chip
  const chips = document.querySelectorAll(".student-chip");
  chips.forEach(c => {
    if (c.textContent.includes(enroll)) {
      c.classList.add("active");
    } else {
      c.classList.remove("active");
    }
  });

  const found = studentsArray.find(s => s.enroll.toLowerCase() === enroll.toLowerCase());
  if (found) {
    renderScorecard(found);
  }
}

function handleSearch(e) {
  e.preventDefault();
  const inputVal = document.getElementById("enrollment-input").value.trim();
  const found = studentsArray.find(s => s.enroll.toLowerCase() === inputVal.toLowerCase());

  const chips = document.querySelectorAll(".student-chip");
  chips.forEach(c => {
    if (c.textContent.includes(inputVal)) {
      c.classList.add("active");
    } else {
      c.classList.remove("active");
    }
  });

  if (found) {
    renderScorecard(found);
```

```
      } else {
        renderNotFound(inputVal);
      }
    }

    function renderScorecard(s) {
      const container = document.getElementById("scorecard-container");

      let rowsHtml = s.subjects.map(sub => {
        const pct = (sub.score / sub.max) * 100;
        const barColor = sub.score >= 85 ? '#10b981' : (sub.score >= 70 ? '#38bdf8' : '#f59e0b');
        return `
          <tr>
            <td style="font-family: 'Fira Code', monospace; color: #a5b4fc;">${sub.code}</td>
            <td><strong style="color: #f1f5f9;">${sub.name}</strong></td>
            <td style="color: #94a3b8;">${sub.max}</td>
            <td style="font-weight: 700; color: #38bdf8;">${sub.score}</td>
            <td style="width: 28%;">
              <div style="display: flex; align-items: center; gap: 0.5rem;">
                <div class="progress-bar-bg">
                  <div class="progress-bar-fill" style="width: ${pct}%; background: ${barColor};">
</div>
                </div>
                <span style="font-size: 0.75rem; color: #cbd5e1; font-family: 'Fira Code', monospa
ce;">${pct}%</span>
              </div>
            </td>
          </tr>
        `;
      }).join('');

      container.innerHTML = `
        <div class="scorecard-box">
          <div class="scorecard-header">
            <div>
              <div class="student-name">${s.name}</div>
              <div class="student-meta">
                Enrollment No: <strong style="color: #38bdf8; font-family: 'Fira Code', monospac
e;">${s.enroll}</strong> &bull;
                Semester: <strong>${s.semester}</strong> &bull;
                Batch: <strong>${s.batch}</strong>
              </div>
            </div>
            <div class="grade-badge">
              Grade ${s.grade} &bull; ${s.result}
            </div>
          </div>

          <table class="marks-table">
            <thead>
              <tr>
                <th>Code</th>
                <th>Subject Name (5 Core Subjects)</th>
                <th>Max</th>
                <th>Marks</th>
                <th>Performance</th>
              </tr>
            </thead>
            <tbody>
              ${rowsHtml}
            </tbody>
          </table>

          <div class="stats-grid">
            <div class="stat-card">
              <div class="label">Total Marks Obtained</div>
```

```
              <div class="value" style="color: #818cf8;">${s.total} <span style="font-size: 0.9re
m; color: #94a3b8;">/ 500</span></div>
            </div>
            <div class="stat-card">
              <div class="label">Aggregate Percentage</div>
              <div class="value" style="color: #10b981;">${parseFloat(s.pct).toFixed(2)}%</div>
            </div>
            <div class="stat-card">
              <div class="label">Final Evaluation</div>
              <div class="value" style="color: #38bdf8;">${s.result} (${s.grade})</div>
            </div>
          </div>
        </div>
      `;
    }

    function renderNotFound(enroll) {
      const container = document.getElementById("scorecard-container");
      container.innerHTML = `
        <div class="scorecard-box" style="border-color: #ef4444; background: rgba(239, 68, 68, 0.0
8); text-align: center; padding: 2.5rem 1.5rem;">
          <div style="font-size: 2.5rem; margin-bottom: 0.5rem;">⚠️</div>
          <h3 style="color: #f87171; font-size: 1.25rem; margin-bottom: 0.5rem;">Student Record No
t Found</h3>
          <p style="color: #cbd5e1; font-size: 0.95rem; margin-bottom: 1.25rem;">
            No student matching Enrollment Number "<strong style="color: #ffffff;">${enroll}</stro
ng>" exists in the SEM VI IT XML database.
          </p>
          <button type="button" class="btn-copy" onclick="selectStudent('IT2023001')">Load Default
Student (IT2023001)</button>
        </div>
      `;
    }

    // Tab Switching
    function switchTab(tabId, btn) {
      document.querySelectorAll('.tab-panel').forEach(p => p.classList.remove('active'));
      document.querySelectorAll('.tab-btn').forEach(b => b.classList.remove('active'));
      document.getElementById('tab-' + tabId).classList.add('active');
      btn.classList.add('active');
    }

    // Clipboard Copy
    function copyCode(elementId, btn) {
      const text = document.getElementById(elementId).innerText;
      navigator.clipboard.writeText(text).then(() => {
        const orig = btn.innerText;
        btn.innerText = "Copied! ✅";
        btn.style.background = "#10b981";
        setTimeout(() => {
          btn.innerText = orig;
          btn.style.background = "";
        }, 2000);
      });
    }

    // Initialize on load
    window.addEventListener("DOMContentLoaded", initXML);
  </script>

</body>

</html>
```

Supplementary Source File: students.xml

```xml
<?xml version="1.0" encoding="UTF-8"?>
<!--
  Experiment 13: Student Academic Database XML
  Course: Advanced Web Technology (CSIT248) - Sem VI IT
  Amity University Noida
  Contains records of 10 students with 5 subject marks, total marks, percentage, and grade.
-->
<students_database semester="VI" branch="Information Technology" academic_year="2025-2026">

  <!-- Student 1 -->
  <student id="1">
    <enrollment_no>IT2023001</enrollment_no>
    <name>Atul Anand</name>
    <semester>VI IT</semester>
    <academic_batch>2023-2026</academic_batch>
    <marks>
      <subject code="IT601" name="Cloud Computing" max="100">92</subject>
      <subject code="IT602" name="Information &amp; Network Security" max="100">88</subject>
      <subject code="IT603" name="Software Engineering &amp; Agile" max="100">94</subject>
      <subject code="IT604" name="Data Mining &amp; Business Intelligence" max="100">90</subject>
      <subject code="IT605" name="Mobile Computing &amp; IoT" max="100">96</subject>
    </marks>
    <total_marks max="500">460</total_marks>
    <percentage>92.00</percentage>
    <grade>A+</grade>
    <result>PASS</result>
  </student>

  <!-- Student 2 -->
  <student id="2">
    <enrollment_no>IT2023002</enrollment_no>
    <name>Priya Sharma</name>
    <semester>VI IT</semester>
    <academic_batch>2023-2026</academic_batch>
    <marks>
      <subject code="IT601" name="Cloud Computing" max="100">85</subject>
      <subject code="IT602" name="Information &amp; Network Security" max="100">79</subject>
      <subject code="IT603" name="Software Engineering &amp; Agile" max="100">88</subject>
      <subject code="IT604" name="Data Mining &amp; Business Intelligence" max="100">82</subject>
      <subject code="IT605" name="Mobile Computing &amp; IoT" max="100">91</subject>
    </marks>
    <total_marks max="500">425</total_marks>
    <percentage>85.00</percentage>
    <grade>A</grade>
    <result>PASS</result>
  </student>

  <!-- Student 3 -->
  <student id="3">
    <enrollment_no>IT2023003</enrollment_no>
    <name>Rohan Verma</name>
    <semester>VI IT</semester>
    <academic_batch>2023-2026</academic_batch>
    <marks>
      <subject code="IT601" name="Cloud Computing" max="100">78</subject>
      <subject code="IT602" name="Information &amp; Network Security" max="100">84</subject>
      <subject code="IT603" name="Software Engineering &amp; Agile" max="100">76</subject>
      <subject code="IT604" name="Data Mining &amp; Business Intelligence" max="100">80</subject>
      <subject code="IT605" name="Mobile Computing &amp; IoT" max="100">82</subject>
    </marks>
    <total_marks max="500">400</total_marks>
    <percentage>80.00</percentage>
    <grade>A</grade>
    <result>PASS</result>
  </student>
```

```
<!-- Student 4 -->
<student id="4">
  <enrollment_no>IT2023004</enrollment_no>
  <name>Ananya Sengupta</name>
  <semester>VI IT</semester>
  <academic_batch>2023-2026</academic_batch>
  <marks>
    <subject code="IT601" name="Cloud Computing" max="100">95</subject>
    <subject code="IT602" name="Information &amp;amp; Network Security" max="100">91</subject>
    <subject code="IT603" name="Software Engineering &amp;amp; Agile" max="100">96</subject>
    <subject code="IT604" name="Data Mining &amp;amp; Business Intelligence" max="100">94</subject>
    <subject code="IT605" name="Mobile Computing &amp;amp; IoT" max="100">98</subject>
  </marks>
  <total_marks max="500">474</total_marks>
  <percentage>94.80</percentage>
  <grade>A+</grade>
  <result>PASS</result>
</student>

<!-- Student 5 -->
<student id="5">
  <enrollment_no>IT2023005</enrollment_no>
  <name>Vikramaditya Rao</name>
  <semester>VI IT</semester>
  <academic_batch>2023-2026</academic_batch>
  <marks>
    <subject code="IT601" name="Cloud Computing" max="100">70</subject>
    <subject code="IT602" name="Information &amp;amp; Network Security" max="100">68</subject>
    <subject code="IT603" name="Software Engineering &amp;amp; Agile" max="100">74</subject>
    <subject code="IT604" name="Data Mining &amp;amp; Business Intelligence" max="100">72</subject>
    <subject code="IT605" name="Mobile Computing &amp;amp; IoT" max="100">76</subject>
  </marks>
  <total_marks max="500">360</total_marks>
  <percentage>72.00</percentage>
  <grade>B+</grade>
  <result>PASS</result>
</student>

<!-- Student 6 -->
<student id="6">
  <enrollment_no>IT2023006</enrollment_no>
  <name>Sneha Kulkarni</name>
  <semester>VI IT</semester>
  <academic_batch>2023-2026</academic_batch>
  <marks>
    <subject code="IT601" name="Cloud Computing" max="100">89</subject>
    <subject code="IT602" name="Information &amp;amp; Network Security" max="100">86</subject>
    <subject code="IT603" name="Software Engineering &amp;amp; Agile" max="100">91</subject>
    <subject code="IT604" name="Data Mining &amp;amp; Business Intelligence" max="100">88</subject>
    <subject code="IT605" name="Mobile Computing &amp;amp; IoT" max="100">93</subject>
  </marks>
  <total_marks max="500">447</total_marks>
  <percentage>89.40</percentage>
  <grade>A</grade>
  <result>PASS</result>
</student>

<!-- Student 7 -->
<student id="7">
  <enrollment_no>IT2023007</enrollment_no>
  <name>Arjun Nair</name>
  <semester>VI IT</semester>
  <academic_batch>2023-2026</academic_batch>
  <marks>
    <subject code="IT601" name="Cloud Computing" max="100">62</subject>
    <subject code="IT602" name="Information &amp;amp; Network Security" max="100">65</subject>
```

```
    <subject code="IT603" name="Software Engineering &amp; Agile" max="100">68</subject>
    <subject code="IT604" name="Data Mining &amp; Business Intelligence" max="100">64</subject>
    <subject code="IT605" name="Mobile Computing &amp; IoT" max="100">71</subject>
  </marks>
  <total_marks max="500">330</total_marks>
  <percentage>66.00</percentage>
  <grade>B</grade>
  <result>PASS</result>
</student>

<!-- Student 8 -->
<student id="8">
  <enrollment_no>IT2023008</enrollment_no>
  <name>Meera Iyer</name>
  <semester>VI IT</semester>
  <academic_batch>2023-2026</academic_batch>
  <marks>
    <subject code="IT601" name="Cloud Computing" max="100">91</subject>
    <subject code="IT602" name="Information &amp; Network Security" max="100">89</subject>
    <subject code="IT603" name="Software Engineering &amp; Agile" max="100">87</subject>
    <subject code="IT604" name="Data Mining &amp; Business Intelligence" max="100">93</subject>
    <subject code="IT605" name="Mobile Computing &amp; IoT" max="100">95</subject>
  </marks>
  <total_marks max="500">455</total_marks>
  <percentage>91.00</percentage>
  <grade>A+</grade>
  <result>PASS</result>
</student>

<!-- Student 9 -->
<student id="9">
  <enrollment_no>IT2023009</enrollment_no>
  <name>Kabir Das</name>
  <semester>VI IT</semester>
  <academic_batch>2023-2026</academic_batch>
  <marks>
    <subject code="IT601" name="Cloud Computing" max="100">75</subject>
    <subject code="IT602" name="Information &amp; Network Security" max="100">72</subject>
    <subject code="IT603" name="Software Engineering &amp; Agile" max="100">78</subject>
    <subject code="IT604" name="Data Mining &amp; Business Intelligence" max="100">70</subject>
    <subject code="IT605" name="Mobile Computing &amp; IoT" max="100">80</subject>
  </marks>
  <total_marks max="500">375</total_marks>
  <percentage>75.00</percentage>
  <grade>B+</grade>
  <result>PASS</result>
</student>

<!-- Student 10 -->
<student id="10">
  <enrollment_no>IT2023010</enrollment_no>
  <name>Tanvi Singhania</name>
  <semester>VI IT</semester>
  <academic_batch>2023-2026</academic_batch>
  <marks>
    <subject code="IT601" name="Cloud Computing" max="100">88</subject>
    <subject code="IT602" name="Information &amp; Network Security" max="100">85</subject>
    <subject code="IT603" name="Software Engineering &amp; Agile" max="100">90</subject>
    <subject code="IT604" name="Data Mining &amp; Business Intelligence" max="100">87</subject>
    <subject code="IT605" name="Mobile Computing &amp; IoT" max="100">92</subject>
  </marks>
  <total_marks max="500">442</total_marks>
  <percentage>88.40</percentage>
  <grade>A</grade>
  <result>PASS</result>
</student>
```

```
</students_database>
```

Supplementary Source File: student_lookup.php

```php
<?php
/**
 * Experiment 13: Student Information Lookup from Server-side XML Document
 * Course: Advanced Web Technology (CSIT248) - Sem VI IT
 * Amity University Noida
 *
 * Aim:
 *   Create an XML document of 10 students of SEM VI IT containing Enrollment No.,
 *   marks obtained in 5 subjects, total marks, and percentage saved at the server.
 *   Write a program that accepts student Enrollment No. as an input and returns
 *   marks, total, and percentage by querying the server XML document.
 */

// Define path to the XML document saved at the server
$xmlFilePath = __DIR__ . '/students.xml';

$searchedEnrollment = '';
$studentData = null;
$errorMessage = '';
$hasSearched = false;

// Process lookup upon Form Submission (POST or GET)
if ($_SERVER['REQUEST_METHOD'] === 'POST' || isset($_GET['enrollment_no'])) {
    $searchedEnrollment = trim($_POST['enrollment_no'] ?? $_GET['enrollment_no'] ?? '');
    $hasSearched = true;

    if (empty($searchedEnrollment)) {
        $errorMessage = "Please enter a valid Enrollment Number (e.g., IT2023001).";
    } else {
        if (!file_exists($xmlFilePath)) {
            $errorMessage = "Server Error: XML database file ('students.xml') not found on serve
r.";
        } else {
            // Load and parse XML document using PHP SimpleXML
            $xml = simplexml_load_file($xmlFilePath);

            if ($xml === false) {
                $errorMessage = "Failed to parse the server XML document.";
            } else {
                // Search for matching student record by enrollment number (case-insensitive)
                foreach ($xml->student as $student) {
                    if (strcasecmp((string)$student->enrollment_no, $searchedEnrollment) === 0) {
                        $studentData = [
                            'enrollment_no' => (string)$student->enrollment_no,
                            'name'          => (string)$student->name,
                            'semester'      => (string)$student->semester,
                            'academic_batch'=> (string)$student->academic_batch,
                            'total_marks'   => (int)$student->total_marks,
                            'percentage'    => (float)$student->percentage,
                            'grade'         => (string)$student->grade,
                            'result'        => (string)$student->result,
                            'subjects'      => []
                        ];

                        foreach ($student->marks->subject as $sub) {
                            $studentData['subjects'][] = [
                                'code'  => (string)$sub['code'],
                                'name'  => (string)$sub['name'],
                                'max'   => (int)$sub['max'],
                                'score' => (int)$sub
                            ];
                        }
```

```
                        break;
                    }
                }

                if (!$studentData) {
                    $errorMessage = "No record found for Enrollment Number: '" . htmlspecialchars
($searchedEnrollment) . "'.";
                }
            }
        }
    }
}
?>
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Experiment 13 - Student XML Lookup System (CSIT248)</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&family=Fira+
Code:wght@400;600&display=swap" rel="stylesheet">
  <style>
    :root {
      --primary: #6366f1;
      --primary-hover: #4f46e5;
      --secondary: #06b6d4;
      --success: #10b981;
      --warning: #f59e0b;
      --danger: #ef4444;
      --bg-dark: #0f172a;
      --card-bg: #1e293b;
      --border-color: #334155;
      --text-main: #f8fafc;
      --text-muted: #94a3b8;
    }
    * { box-sizing: border-box; margin: 0; padding: 0; }
    body {
      font-family: 'Inter', system-ui, -apple-system, sans-serif;
      background-color: var(--bg-dark);
      color: var(--text-main);
      line-height: 1.6;
      min-height: 100vh;
      padding-bottom: 3rem;
    }
    nav {
      background: rgba(15, 23, 42, 0.95);
      border-bottom: 1px solid var(--border-color);
      padding: 1rem 2rem;
      display: flex;
      justify-content: space-between;
      align-items: center;
      position: sticky;
      top: 0;
      z-index: 50;
      backdrop-filter: blur(10px);
    }
    .nav-brand {
      color: #818cf8;
      font-weight: 800;
      text-decoration: none;
      font-size: 1.15rem;
    }
    .container {
      max-width: 950px;
```

```
  margin: 2rem auto;
  padding: 0 1.5rem;
}
.badge {
  display: inline-block;
  background: rgba(99, 102, 241, 0.15);
  color: #a5b4fc;
  border: 1px solid rgba(99, 102, 241, 0.3);
  padding: 0.25rem 0.75rem;
  border-radius: 9999px;
  font-size: 0.8rem;
  font-weight: 600;
  margin-bottom: 0.5rem;
}
h1 {
  font-size: 2rem;
  font-weight: 800;
  margin-bottom: 0.5rem;
}
p.subtitle {
  color: var(--text-muted);
  margin-bottom: 2rem;
  font-size: 0.95rem;
}
.card {
  background: var(--card-bg);
  border: 1px solid var(--border-color);
  border-radius: 12px;
  padding: 2rem;
  margin-bottom: 2rem;
  box-shadow: 0 10px 25px rgba(0,0,0,0.3);
}
.form-group {
  display: flex;
  gap: 1rem;
  flex-wrap: wrap;
}
input[type="text"] {
  flex: 1;
  min-width: 260px;
  background: #0f172a;
  border: 1px solid var(--border-color);
  color: #fff;
  padding: 0.85rem 1rem;
  border-radius: 8px;
  font-size: 1rem;
  outline: none;
  transition: border-color 0.2s;
}
input[type="text"]:focus {
  border-color: var(--primary);
}
button {
  background: var(--primary);
  color: white;
  border: none;
  padding: 0.85rem 1.75rem;
  border-radius: 8px;
  font-size: 1rem;
  font-weight: 600;
  cursor: pointer;
  transition: background 0.2s;
}
button:hover {
  background: var(--primary-hover);
}
```

```
.sample-chips {
  margin-top: 1rem;
  display: flex;
  align-items: center;
  gap: 0.5rem;
  flex-wrap: wrap;
  font-size: 0.85rem;
  color: var(--text-muted);
}
.chip {
  background: #0f172a;
  border: 1px solid var(--border-color);
  color: #38bdf8;
  padding: 0.25rem 0.6rem;
  border-radius: 6px;
  cursor: pointer;
  text-decoration: none;
  font-family: 'Fira Code', monospace;
  font-size: 0.8rem;
}
.chip:hover {
  background: #334155;
}
.alert-error {
  background: rgba(239, 68, 68, 0.15);
  border: 1px solid var(--danger);
  color: #fca5a5;
  padding: 1rem 1.25rem;
  border-radius: 8px;
  margin-top: 1.5rem;
}
.scorecard {
  margin-top: 2rem;
  border-top: 1px solid var(--border-color);
  padding-top: 2rem;
}
.scorecard-header {
  display: flex;
  justify-content: space-between;
  align-items: flex-start;
  margin-bottom: 1.5rem;
  flex-wrap: wrap;
  gap: 1rem;
}
.badge-grade {
  background: rgba(16, 185, 129, 0.2);
  border: 1px solid var(--success);
  color: #34d399;
  font-size: 1.25rem;
  font-weight: 800;
  padding: 0.5rem 1.25rem;
  border-radius: 8px;
}
table {
  width: 100%;
  border-collapse: collapse;
  margin: 1.5rem 0;
  font-size: 0.95rem;
}
th, td {
  padding: 0.85rem 1rem;
  text-align: left;
  border-bottom: 1px solid var(--border-color);
}
th {
  background: #0f172a;
```

```
      color: #38bdf8;
      font-weight: 600;
    }
    tr:hover td {
      background: rgba(255, 255, 255, 0.02);
    }
    .summary-grid {
      display: grid;
      grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
      gap: 1rem;
      margin-top: 1.5rem;
    }
    .summary-box {
      background: #0f172a;
      border: 1px solid var(--border-color);
      border-radius: 8px;
      padding: 1rem 1.25rem;
    }
    .summary-box .label {
      font-size: 0.8rem;
      color: var(--text-muted);
      text-transform: uppercase;
      letter-spacing: 0.05em;
    }
    .summary-box .val {
      font-size: 1.6rem;
      font-weight: 800;
      color: #ffffff;
      margin-top: 0.25rem;
    }
  </style>
</head>
<body>

  <nav>
    <a href="index.html" class="nav-brand">&larr; CSIT248 Lab Experiments</a>
    <div style="display: flex; align-items: center; gap: 0.75rem;">
      <a href="https://atulanand-web-tech-labwork.netlify.app/" target="_blank" rel="noopener nore
ferrer"
        style="font-size: 0.85rem; color: #00c7b7; text-decoration: none; background: rgba(0, 199,
183, 0.15); border: 1px solid rgba(0, 199, 183, 0.35); padding: 0.3rem 0.75rem; border-radius: 999
9px; font-weight: 600; display: inline-flex; align-items: center; gap: 0.35rem;">
        🌐 Live Portfolio
      </a>
      <span style="font-size: 0.85rem; color: var(--text-muted);">Experiment 13: XML Server Proces
sing</span>
    </div>
  </nav>

  <div class="container">
    <div style="text-align: center; margin-bottom: 2rem;">
      <span class="badge">Experiment 13 &bull; Module XML &amp; Server-Side Processing</span>
      <h1>Student Information Lookup (XML Document)</h1>
      <p class="subtitle">Search and retrieve Semester VI IT student academic records from server-
side XML</p>
    </div>

    <div class="card">
      <form method="POST" action="student_lookup.php">
        <div class="form-group">
          <input type="text" name="enrollment_no" placeholder="Enter Enrollment No. (e.g., IT20230
01)" value="<?php echo htmlspecialchars($searchedEnrollment); ?>" required autofocus>
          <button type="submit">🔍 Fetch Marks</button>
        </div>
      </form>
```

```
      <div class="sample-chips">
        <span>Quick Samples:</span>
        <a class="chip" href="student_lookup.php?enrollment_no=IT2023001">IT2023001</a>
        <a class="chip" href="student_lookup.php?enrollment_no=IT2023002">IT2023002</a>
        <a class="chip" href="student_lookup.php?enrollment_no=IT2023004">IT2023004</a>
        <a class="chip" href="student_lookup.php?enrollment_no=IT2023008">IT2023008</a>
        <a class="chip" href="student_lookup.php?enrollment_no=IT2023010">IT2023010</a>
      </div>

      <?php if (!empty($errorMessage)): ?>
        <div class="alert-error">
          ⚠️ <?php echo htmlspecialchars($errorMessage); ?>
        </div>
      <?php endif; ?>

      <?php if ($studentData): ?>
        <div class="scorecard">
          <div class="scorecard-header">
            <div>
              <h2 style="color: #ffffff; font-size: 1.5rem;"><?php echo htmlspecialchars($studentD
ata['name']); ?></h2>
              <p style="color: var(--text-muted); font-size: 0.9rem;">
                Enrollment No: <strong style="color: #38bdf8; font-family: 'Fira Code', monospac
e;"><?php echo htmlspecialchars($studentData['enrollment_no']); ?></strong> &bull;
                Semester: <strong><?php echo htmlspecialchars($studentData['semester']); ?></stron
g> &bull;
                Batch: <strong><?php echo htmlspecialchars($studentData['academic_batch']); ?></st
rong>
              </p>
            </div>
            <div class="badge-grade">
              Grade <?php echo htmlspecialchars($studentData['grade']); ?> (<?php echo htmlspecial
chars($studentData['result']); ?>)
            </div>
          </div>

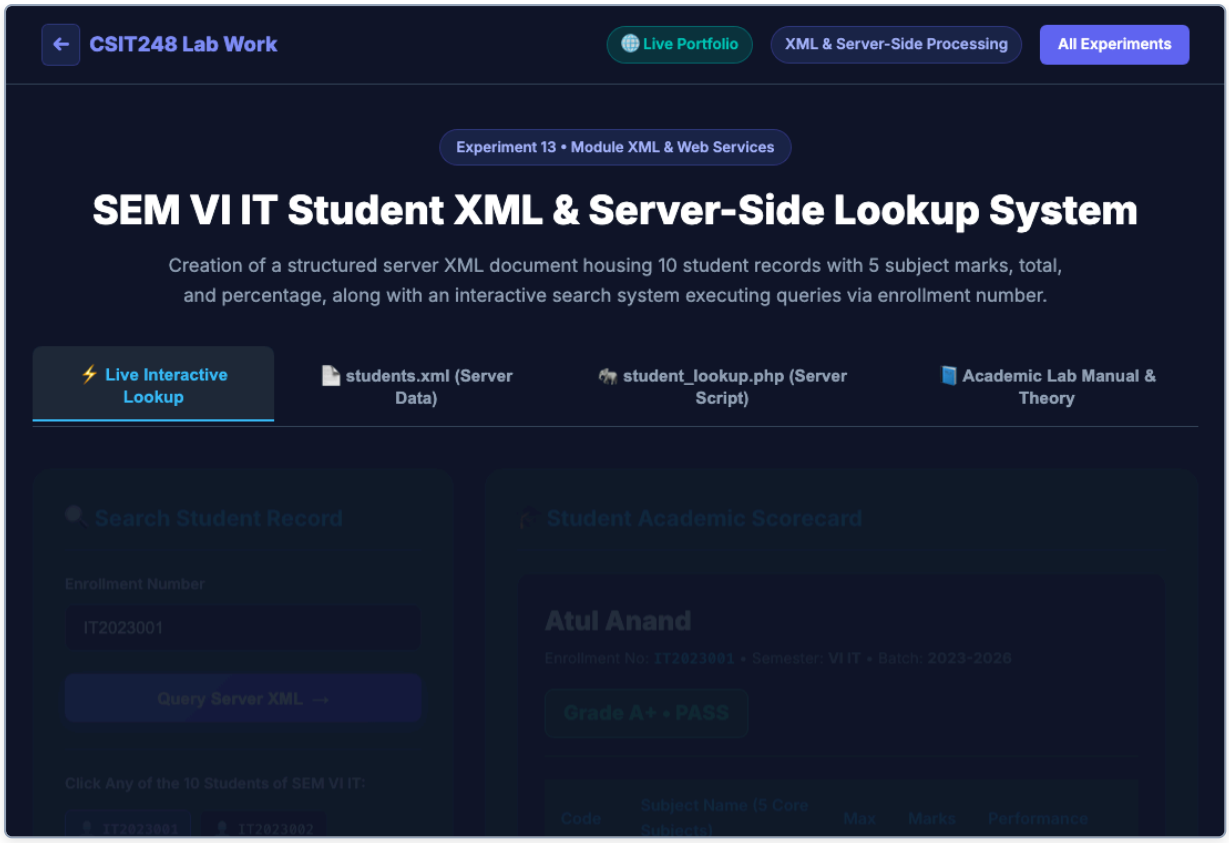
          <table>
            <thead>
              <tr>
                <th>Code</th>
                <th>Subject Name</th>
                <th>Max Marks</th>
                <th>Marks Obtained</th>
                <th>Performance</th>
              </tr>
            </thead>
            <tbody>
              <?php foreach ($studentData['subjects'] as $sub): ?>
                <tr>
                  <td style="font-family: 'Fira Code', monospace; color: #a5b4fc;"><?php echo html
specialchars($sub['code']); ?></td>
                  <td><strong><?php echo htmlspecialchars($sub['name']); ?></strong></td>
                  <td><?php echo $sub['max']; ?></td>
                  <td style="font-weight: 700; color: #38bdf8;"><?php echo $sub['score']; ?></td>
                  <td style="width: 25%;">
                    <div style="background: #0f172a; border-radius: 9999px; height: 10px; width: 1
00%; overflow: hidden; border: 1px solid #334155;">
                      <div style="background: <?php echo $sub['score'] >= 80 ? '#10b981' : ($sub
['score'] >= 60 ? '#f59e0b' : '#ef4444'); ?>; height: 100%; width: <?php echo ($sub['score'] / $su
b['max']) * 100; ?>%;"></div>
                    </div>
                  </td>
                </tr>
              <?php endforeach; ?>
            </tbody>
          </table>
```

```
          <div class="summary-grid">
            <div class="summary-box">
              <div class="label">Total Marks (Out of 500)</div>
              <div class="val" style="color: #818cf8;"><?php echo $studentData['total_marks']; ?>
<span style="font-size: 0.9rem; color: var(--text-muted);">/ 500</span></div>
            </div>
            <div class="summary-box">
              <div class="label">Aggregate Percentage</div>
              <div class="val" style="color: #10b981;"><?php echo number_format($studentData['perc
entage'], 2); ?>%</div>
            </div>
            <div class="summary-box">
              <div class="label">Academic Result</div>
              <div class="val" style="color: #38bdf8;"><?php echo htmlspecialchars($studentData['r
esult']); ?></div>
            </div>
          </div>
        </div>
      <?php endif; ?>
    </div>
  </div>

  <footer style="text-align: center; margin-top: 3rem; padding: 1.5rem; border-top: 1px solid var
(--border-color); color: var(--text-muted); font-size: 0.85rem;">
    <p style="margin-bottom: 0.35rem;">Designed &amp;amp; Developed by <strong style="color: #818cf
8;">Atul Anand</strong> &bull; Amity University Noida &bull; CSIT248</p>
    <p style="margin: 0;">
      Live Portfolio: <a href="https://atulanand-web-tech-labwork.netlify.app/" target="_blank" re
l="noopener noreferrer" style="color: #00c7b7; text-decoration: none; font-weight: 600;">atulanand
-web-tech-labwork.netlify.app</a>
    </p>
  </footer>

</body>
</html>
```

Output :


CSIT248 Lab Work
Live Portfolio
XML & Server-Side Processing
All Experiments
Experiment 13 • Module XML & Web Services
SEM VI IT Student XML & Server-Side Lookup System
Creation of a structured server XML document housing 10 student records with 5 subject marks, total, and percentage, along with an interactive search system executing queries via enrollment number.
⚡ Live Interactive Lookup
students.xml (Server Data)
student_lookup.php (Server Script)
Academic Lab Manual & Theory
Search Student Record
Enrollment Number
IT2023001
Query Server XML →
Click Any of the 10 Students of SEM VI IT:
Student Academic Scorecard
Atul Anand
Enrollment No: IT2023001 • Semester: VI IT • Batch: 2023-2026
Grade A+ • PASS
Code
Subject Name (5 Core Subjects)
Max
Marks
Performance

Experiment No.: 17

Title : Interactive Login Portal with Real-Time Validation.

Aim : To design an institutional student/faculty login portal implementing real-time client-side credential verification, show/hide password toggle, and entropy strength meter.

Source Code :

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description"
    content="Experiment 14: Design a web page for Login form and perform validation using JavaScri
pt - CSIT248 Advanced Web Technology">
  <title>Experiment 14 - Login Form with JavaScript Validation</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&display=swa
p" rel="stylesheet">
  <style>
    :root {
      --primary: #6366f1;
      --primary-hover: #4f46e5;
      --secondary: #06b6d4;
      --accent: #f43f5e;
      --success: #10b981;
      --warning: #f59e0b;
      --bg-dark: #0f172a;
      --card-bg: #1e293b;
      --card-border: #334155;
      --text-main: #f8fafc;
      --text-muted: #94a3b8;
    }

    * {
      box-sizing: border-box;
      margin: 0;
      padding: 0;
    }

    body {
      font-family: 'Inter', system-ui, -apple-system, sans-serif;
      background-color: var(--bg-dark);
      color: var(--text-main);
      line-height: 1.6;
      min-height: 100vh;
      display: flex;
      flex-direction: column;
    }

    /* Navigation Bar */
    nav {
      position: sticky;
      top: 0;
      z-index: 100;
      background: rgba(15, 23, 42, 0.92);
      backdrop-filter: blur(12px);
      border-bottom: 1px solid #334155;
      padding: 1rem 2rem;
```

```
  display: flex;
  justify-content: space-between;
  align-items: center;
}

.nav-brand {
  font-weight: 800;
  font-size: 1.15rem;
  color: #818cf8;
  text-decoration: none;
  display: flex;
  align-items: center;
  gap: 0.5rem;
}

.nav-brand span {
  background: rgba(99, 102, 241, 0.15);
  border: 1px solid rgba(99, 102, 241, 0.3);
  padding: 0.2rem 0.5rem;
  border-radius: 6px;
}

.nav-links {
  display: flex;
  gap: 0.75rem;
  align-items: center;
}

.badge-pill {
  font-size: 0.8rem;
  background: rgba(99, 102, 241, 0.12);
  color: #a5b4fc;
  border: 1px solid rgba(99, 102, 241, 0.25);
  padding: 0.3rem 0.75rem;
  border-radius: 9999px;
  font-weight: 600;
}

.btn-nav {
  background: #6366f1;
  color: #ffffff;
  padding: 0.4rem 0.9rem;
  border-radius: 6px;
  text-decoration: none;
  font-size: 0.85rem;
  font-weight: 600;
  transition: all 0.2s ease;
}

.btn-nav:hover {
  background: #4f46e5;
  transform: translateY(-1px);
}

/* Main Container */
main {
  flex: 1;
  max-width: 900px;
  width: 100%;
  margin: 0 auto;
  padding: 2.5rem 1.25rem;
}

.exp-header {
  text-align: center;
  margin-bottom: 2rem;
```

```
}

.exp-pill {
  display: inline-block;
  background: rgba(99, 102, 241, 0.15);
  color: #a5b4fc;
  border: 1px solid rgba(99, 102, 241, 0.3);
  padding: 0.25rem 0.85rem;
  border-radius: 9999px;
  font-size: 0.85rem;
  font-weight: 600;
  margin-bottom: 0.75rem;
}

.exp-header h1 {
  font-size: 2.25rem;
  font-weight: 800;
  color: #ffffff;
  margin-bottom: 0.5rem;
  letter-spacing: -0.02em;
}

.exp-header p {
  color: #94a3b8;
  font-size: 1.05rem;
}

/* Auth Wrapper & Card */
.auth-card {
  background: #1e293b;
  border: 1px solid #334155;
  border-radius: 20px;
  padding: 2.25rem;
  box-shadow: 0 20px 40px rgba(0, 0, 0, 0.4), 0 0 30px rgba(99, 102, 241, 0.1);
  position: relative;
  overflow: hidden;
}

.auth-card::before {
  content: '';
  position: absolute;
  top: 0;
  left: 0;
  right: 0;
  height: 4px;
  background: linear-gradient(90deg, #6366f1, #06b6d4, #ec4899);
}

/* Portal Tabs */
.portal-tabs {
  display: flex;
  background: #0f172a;
  padding: 0.35rem;
  border-radius: 12px;
  border: 1px solid #334155;
  margin-bottom: 1.75rem;
  gap: 0.35rem;
}

.portal-tab {
  flex: 1;
  padding: 0.65rem 1rem;
  border: none;
  background: transparent;
  color: #94a3b8;
  font-weight: 600;
```

```
  font-size: 0.9rem;
  border-radius: 8px;
  cursor: pointer;
  display: flex;
  align-items: center;
  justify-content: center;
  gap: 0.5rem;
  transition: all 0.2s ease;
}

.portal-tab.active {
  background: #6366f1;
  color: #ffffff;
  box-shadow: 0 4px 12px rgba(99, 102, 241, 0.3);
}

/* Quick Demo Helper */
.demo-bar {
  background: rgba(99, 102, 241, 0.1);
  border: 1px dashed rgba(99, 102, 241, 0.35);
  border-radius: 10px;
  padding: 0.75rem 1rem;
  margin-bottom: 1.5rem;
  display: flex;
  justify-content: space-between;
  align-items: center;
  flex-wrap: wrap;
  gap: 0.5rem;
}

.demo-bar span {
  font-size: 0.85rem;
  color: #c7d2fe;
}

.btn-autofill {
  background: rgba(99, 102, 241, 0.25);
  border: 1px solid #6366f1;
  color: #e0e7ff;
  padding: 0.3rem 0.75rem;
  font-size: 0.8rem;
  font-weight: 600;
  border-radius: 6px;
  cursor: pointer;
  transition: all 0.2s ease;
}

.btn-autofill:hover {
  background: #6366f1;
  color: #ffffff;
}

/* Form Fields */
.form-group {
  margin-bottom: 1.35rem;
}

.form-label {
  display: flex;
  justify-content: space-between;
  align-items: center;
  font-size: 0.875rem;
  font-weight: 600;
  color: #cbd5e1;
  margin-bottom: 0.4rem;
}
```

```
    .form-label .req {
      color: #f43f5e;
    }

    .input-wrap {
      position: relative;
      display: flex;
      align-items: center;
    }

    .input-icon {
      position: absolute;
      left: 1rem;
      color: #64748b;
      pointer-events: none;
      display: flex;
      align-items: center;
    }

    .form-control {
      width: 100%;
      background: #0f172a;
      border: 1.5px solid #334155;
      color: #f8fafc;
      padding: 0.8rem 1rem 0.8rem 2.75rem;
      font-size: 0.95rem;
      border-radius: 10px;
      outline: none;
      font-family: inherit;
      transition: all 0.2s ease;
    }

    .form-control:focus {
      border-color: #6366f1;
      box-shadow: 0 0 0 3px rgba(99, 102, 241, 0.2);
    }

    .form-control.is-valid {
      border-color: var(--success);
      background-image: radial-gradient(circle at right 12px center, rgba(16, 185, 129, 0.15) 0%,
transparent 70%);
    }

    .form-control.is-invalid {
      border-color: var(--accent);
      background-image: radial-gradient(circle at right 12px center, rgba(244, 63, 94, 0.15) 0%, t
ransparent 70%);
    }

    .toggle-pass-btn {
      position: absolute;
      right: 0.85rem;
      background: none;
      border: none;
      color: #94a3b8;
      cursor: pointer;
      padding: 0.25rem;
      display: flex;
      align-items: center;
    }

    .toggle-pass-btn:hover {
      color: #f8fafc;
    }
```

```
.feedback-msg {
  font-size: 0.8rem;
  margin-top: 0.35rem;
  display: flex;
  align-items: center;
  gap: 0.35rem;
}

.feedback-msg.error {
  color: #fb7185;
}

.feedback-msg.success {
  color: #34d399;
}

/* Password Strength Meter */
.strength-wrapper {
  margin-top: 0.5rem;
  background: #0f172a;
  border: 1px solid #334155;
  padding: 0.65rem 0.85rem;
  border-radius: 8px;
}

.strength-bar-container {
  height: 6px;
  background: #334155;
  border-radius: 9999px;
  overflow: hidden;
  margin-bottom: 0.5rem;
}

.strength-bar {
  height: 100%;
  width: 0%;
  transition: all 0.3s ease;
}

.strength-criteria {
  display: grid;
  grid-template-columns: repeat(2, 1fr);
  gap: 0.35rem 0.75rem;
  font-size: 0.75rem;
  color: #94a3b8;
}

.criteria-item {
  display: flex;
  align-items: center;
  gap: 0.3rem;
}

.criteria-item.met {
  color: #34d399;
}

/* Captcha Box */
.captcha-container {
  background: #0f172a;
  border: 1px solid #334155;
  border-radius: 10px;
  padding: 0.85rem 1rem;
  display: flex;
  align-items: center;
  gap: 1rem;
```

```
  margin-bottom: 0.5rem;
}

.captcha-badge {
  background: linear-gradient(135deg, #312e81, #1e1b4b);
  border: 1px dashed #6366f1;
  padding: 0.5rem 1rem;
  border-radius: 8px;
  font-family: 'Courier New', Courier, monospace;
  font-size: 1.15rem;
  font-weight: 800;
  letter-spacing: 0.2em;
  color: #a5b4fc;
  user-select: none;
}

.btn-refresh-captcha {
  background: #334155;
  border: none;
  color: #cbd5e1;
  padding: 0.45rem 0.75rem;
  border-radius: 6px;
  cursor: pointer;
  font-size: 0.8rem;
  display: flex;
  align-items: center;
  gap: 0.35rem;
  transition: all 0.2s ease;
}

.btn-refresh-captcha:hover {
  background: #475569;
  color: #ffffff;
}

/* Options Row */
.options-row {
  display: flex;
  justify-content: space-between;
  align-items: center;
  font-size: 0.85rem;
  margin-bottom: 1.5rem;
}

.checkbox-label {
  display: flex;
  align-items: center;
  gap: 0.5rem;
  color: #cbd5e1;
  cursor: pointer;
}

.forgot-link {
  color: #818cf8;
  text-decoration: none;
  cursor: pointer;
}

.forgot-link:hover {
  text-decoration: underline;
}

/* Submit Button */
.btn-submit {
  width: 100%;
  background: linear-gradient(135deg, #6366f1 0%, #4f46e5 100%);
```

```
  color: #ffffff;
  border: none;
  padding: 0.9rem;
  font-size: 1rem;
  font-weight: 700;
  border-radius: 10px;
  cursor: pointer;
  display: flex;
  align-items: center;
  justify-content: center;
  gap: 0.5rem;
  box-shadow: 0 4px 15px rgba(99, 102, 241, 0.35);
  transition: all 0.25s ease;
}

.btn-submit:hover {
  background: linear-gradient(135deg, #4f46e5 0%, #4338ca 100%);
  transform: translateY(-2px);
  box-shadow: 0 6px 20px rgba(99, 102, 241, 0.45);
}

.btn-submit:active {
  transform: translateY(0);
}

/* Dashboard Card (Shown upon successful login) */
#dashboard-view {
  display: none;
  background: #1e293b;
  border: 1px solid #10b981;
  border-radius: 20px;
  padding: 2.25rem;
  box-shadow: 0 20px 40px rgba(0, 0, 0, 0.4), 0 0 30px rgba(16, 185, 129, 0.15);
  animation: fadeIn 0.4s ease-out;
}

@keyframes fadeIn {
  from {
    opacity: 0;
    transform: translateY(10px);
  }
  to {
    opacity: 1;
    transform: translateY(0);
  }
}

.dash-header {
  display: flex;
  align-items: center;
  gap: 1.25rem;
  padding-bottom: 1.5rem;
  border-bottom: 1px solid #334155;
  margin-bottom: 1.5rem;
}

.dash-avatar {
  width: 68px;
  height: 68px;
  border-radius: 50%;
  background: linear-gradient(135deg, #10b981, #06b6d4);
  display: flex;
  align-items: center;
  justify-content: center;
  font-size: 1.8rem;
  font-weight: 800;
```

```
  color: white;
}

.dash-stats-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(180px, 1fr));
  gap: 1rem;
  margin-bottom: 1.5rem;
}

.dash-stat-card {
  background: #0f172a;
  border: 1px solid #334155;
  padding: 1rem;
  border-radius: 12px;
}

.dash-stat-card h4 {
  font-size: 0.8rem;
  color: #94a3b8;
  text-transform: uppercase;
  margin-bottom: 0.25rem;
}

.dash-stat-card p {
  font-size: 1.15rem;
  font-weight: 700;
  color: #f8fafc;
}

/* Explanation Section */
.info-section {
  background: #1e293b;
  border: 1px solid #334155;
  border-radius: 16px;
  padding: 1.75rem;
  margin-top: 2rem;
}

.info-section h3 {
  font-size: 1.25rem;
  color: #ffffff;
  margin-bottom: 0.85rem;
  display: flex;
  align-items: center;
  gap: 0.5rem;
}

.tech-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(240px, 1fr));
  gap: 1rem;
  margin-top: 1rem;
}

.tech-card {
  background: #0f172a;
  border: 1px solid #334155;
  padding: 1rem;
  border-radius: 10px;
}

.tech-card h4 {
  font-size: 0.9rem;
  color: #818cf8;
  margin-bottom: 0.3rem;
```

```
    }

    .tech-card p {
      font-size: 0.825rem;
      color: #94a3b8;
    }

    /* Toast Notification */
    #toast {
      position: fixed;
      bottom: 2rem;
      right: 2rem;
      background: #1e293b;
      color: #fff;
      padding: 0.85rem 1.35rem;
      border-radius: 10px;
      border: 1px solid #334155;
      box-shadow: 0 10px 25px rgba(0, 0, 0, 0.4);
      display: flex;
      align-items: center;
      gap: 0.65rem;
      font-size: 0.9rem;
      font-weight: 600;
      z-index: 9999;
      transform: translateY(100px);
      opacity: 0;
      transition: all 0.3s cubic-bezier(0.4, 0, 0.2, 1);
    }

    #toast.show {
      transform: translateY(0);
      opacity: 1;
    }

    #toast.success {
      border-color: #10b981;
      background: #064e3b;
    }

    #toast.error {
      border-color: #ef4444;
      background: #7f1d1d;
    }

    footer {
      background: #090d16;
      border-top: 1px solid #334155;
      padding: 1.5rem;
      text-align: center;
      color: #94a3b8;
      font-size: 0.875rem;
      margin-top: auto;
    }
  </style>
</head>

<body>

  <!-- Navigation -->
  <nav>
    <a href="../index.html" class="nav-brand">
      <span>&larr;</span> CSIT248 Lab Work
    </a>
    <div class="nav-links">
      <span class="badge-pill">JavaScript Client Validation</span>
      <a href="../index.html" class="btn-nav">All Experiments</a>
```

```
    </div>
  </nav>

  <main>

    <!-- Experiment Header -->
    <header class="exp-header">
      <div class="exp-pill">Experiment 14 &bull; Module IV (JavaScript Form Validation)</div>
      <h1>Interactive Portal Login &amp;amp; Validation</h1>
      <p>Robust real-time validation, password strength analysis, and security verification using
JavaScript</p>
    </header>

    <!-- Login Card -->
    <section class="auth-card" id="login-view">

      <!-- Portal Tabs -->
      <div class="portal-tabs">
        <button type="button" class="portal-tab active" onclick="switchPortal('student', this)">
          🎓 Student Portal
        </button>
        <button type="button" class="portal-tab" onclick="switchPortal('faculty', this)">
          🏛️ Faculty / Admin
        </button>
      </div>

      <!-- Quick Demo Credentials Helper -->
      <div class="demo-bar">
        <span>💡 <strong>Quick Test Credentials:</strong> <code>student@amity.edu</code> / <code>
Pass@2024!</code></span>
        <button type="button" class="btn-autofill" onclick="autoFillDemo()">⚡ Autofill Demo Creden
tials</button>
      </div>

      <!-- Main Login Form -->
      <form id="loginForm" onsubmit="handleLoginSubmit(event)" novalidate>

        <!-- Username / Email Field -->
        <div class="form-group">
          <label for="userIdentifier" class="form-label">
            <span id="identifierLabel">Student Email or Enrollment ID</span>
            <span class="req">*</span>
          </label>
          <div class="input-wrap">
            <span class="input-icon">
              <svg width="18" height="18" fill="none" stroke="currentColor" stroke-width="2" viewB
ox="0 0 24 24">
                <path d="M20 21v-2a4 4 0 0 0-4-4H8a4 4 0 0 0-4 4v2"></path>
                <circle cx="12" cy="7" r="4"></circle>
              </svg>
            </span>
            <input type="text" id="userIdentifier" class="form-control"
              placeholder="e.g. atul.anand@amity.edu or AU2024101" autocomplete="username"
              oninput="validateIdentifier()" onblur="validateIdentifier()">
          </div>
          <div id="identifierFeedback" class="feedback-msg"></div>
        </div>

        <!-- Password Field -->
        <div class="form-group">
          <label for="password" class="form-label">
            <span>Account Password</span>
            <span class="req">*</span>
          </label>
          <div class="input-wrap">
            <span class="input-icon">
```

```
              <svg width="18" height="18" fill="none" stroke="currentColor" stroke-width="2" viewB
ox="0 0 24 24">
                <rect x="3" y="11" width="18" height="11" rx="2" ry="2"></rect>
                <path d="M7 11V7a5 5 0 0 1 10 0v4"></path>
              </svg>
            </span>
            <input type="password" id="password" class="form-control" placeholder="Enter secure pa
ssword"
              autocomplete="current-password" oninput="validatePassword()" onblur="validatePasswor
d()">
            <button type="button" class="toggle-pass-btn" onclick="togglePasswordVisibility()" tit
le="Toggle visibility">
              <svg id="eyeIcon" width="18" height="18" fill="none" stroke="currentColor" stroke-wi
dth="2" viewBox="0 0 24 24">
                <path d="M1 12s4-8 11-8 11 8 11 8-4 8-11 8-11-8-11-8z"></path>
                <circle cx="12" cy="12" r="3"></circle>
              </svg>
            </button>
          </div>

          <!-- Password Strength Meter -->
          <div class="strength-wrapper" id="strengthBox">
            <div style="display: flex; justify-content: space-between; font-size: 0.75rem; margin-
bottom: 0.35rem;">
              <span style="color: #94a3b8;">Password Strength:</span>
              <strong id="strengthLabel" style="color: #f43f5e;">Too Short</strong>
            </div>
            <div class="strength-bar-container">
              <div id="strengthBar" class="strength-bar"></div>
            </div>
            <div class="strength-criteria">
              <span id="critLength" class="criteria-item">⚪ Min 8 chars</span>
              <span id="critUpper" class="criteria-item">⚪ Uppercase (A-Z)</span>
              <span id="critLower" class="criteria-item">⚪ Lowercase (a-z)</span>
              <span id="critSpecial" class="criteria-item">⚪ Number &amp;amp; Symbol</span>
            </div>
          </div>
          <div id="passwordFeedback" class="feedback-msg"></div>
        </div>

        <!-- Interactive Captcha Verification -->
        <div class="form-group">
          <label class="form-label">
            <span>Security Verification (Anti-Bot)</span>
            <span class="req">*</span>
          </label>
          <div class="captcha-container">
            <div id="captchaText" class="captcha-badge">9 + 5 = ?</div>
            <button type="button" class="btn-refresh-captcha" onclick="generateCaptcha()">
              🔄 Refresh Code
            </button>
          </div>
          <div class="input-wrap">
            <input type="text" id="captchaInput" class="form-control" style="padding-left: 1rem;"
              placeholder="Enter result above" oninput="validateCaptcha()">
          </div>
          <div id="captchaFeedback" class="feedback-msg"></div>
        </div>

        <!-- Options: Remember me & Forgot Password -->
        <div class="options-row">
          <label class="checkbox-label">
            <input type="checkbox" id="rememberMe" style="accent-color: #6366f1; width: 16px; heig
ht: 16px;">
            <span>Remember session on this device</span>
          </label>
```

```
          <a href="javascript:void(0)" class="forgot-link" onclick="handleForgotPassword()">Forgot
Password?</a>
        </div>

        <!-- Submit Button -->
        <button type="submit" class="btn-submit" id="submitBtn">
          <span>Authenticate &amp; Sign In &rarr;</span>
        </button>

      </form>
    </section>

    <!-- Successful Authenticated Dashboard View (Simulated) -->
    <section id="dashboard-view">
      <div class="dash-header">
        <div class="dash-avatar" id="dashAvatar">A</div>
        <div>
          <span style="background: rgba(16, 185, 129, 0.15); color: #34d399; border: 1px solid rgb
a(16, 185, 129, 0.3); padding: 0.2rem 0.65rem; border-radius: 9999px; font-size: 0.75rem; font-wei
ght: 700;">
            SESSION ACTIVE &bull; AUTHENTICATED
          </span>
          <h2 style="font-size: 1.6rem; font-weight: 800; color: #ffffff; margin: 0.35rem 0 0.2re
m;" id="dashWelcome">
            Welcome back, Atul Anand!
          </h2>
          <p style="color: #94a3b8; font-size: 0.9rem;" id="dashSubtitle">
            BCA (Hons.) with Research &bull; Semester III &bull; CSIT248
          </p>
        </div>
      </div>

      <div class="dash-stats-grid">
        <div class="dash-stat-card">
          <h4>Enrolled Course</h4>
          <p>CSIT248 Adv Web Tech</p>
        </div>
        <div class="dash-stat-card">
          <h4>Lab Progress</h4>
          <p style="color: #34d399;">100% Completed</p>
        </div>
        <div class="dash-stat-card">
          <h4>Authentication</h4>
          <p style="color: #38bdf8;">Client Validation Valid</p>
        </div>
        <div class="dash-stat-card">
          <h4>Security Token</h4>
          <p style="font-family: monospace; font-size: 0.9rem;">#AU-JWT-SEC-9238</p>
        </div>
      </div>

      <div style="background: #0f172a; border: 1px solid #334155; border-radius: 12px; padding: 1.
25rem; margin-bottom: 1.5rem;">
        <h4 style="color: #cbd5e1; margin-bottom: 0.5rem; font-size: 0.95rem;">Authentication Veri
fication Summary:</h4>
        <ul style="color: #94a3b8; font-size: 0.85rem; padding-left: 1.25rem; line-height: 1.8;">
          <li>Identifier verified using strict Regular Expression matching.</li>
          <li>Password entropy calculated: Score &ge; 4 (Length, Casing, Numerical, Special Charac
ter).</li>
          <li>Mathematical Anti-Bot challenge solved correctly.</li>
          <li>Session token initialized in client memory without server round-trip latency.</li>
        </ul>
      </div>

      <button type="button" class="btn-submit" onclick="logoutSession()" style="background: #33415
5; border: 1px solid #475569;">
```

```
        Sign Out &amp;amp; Return to Login Form
      </button>
    </section>

    <!-- Academic & Technical Documentation -->
    <section class="info-section">
      <h3>
        <svg width="20" height="20" fill="none" stroke="#818cf8" stroke-width="2" viewBox="0 0 24
24">
          <circle cx="12" cy="12" r="10"></circle>
          <line x1="12" y1="16" x2="12" y2="12"></line>
          <line x1="12" y1="8" x2="12.01" y2="8"></line>
        </svg>
        Experiment 14 &bull; Technical Lab Report &amp;amp; Validation Architecture
      </h3>
      <p style="color: #cbd5e1; font-size: 0.9rem;">
        This experiment illustrates comprehensive client-side form validation using native JavaScr
ipt DOM manipulation. Real-time validation improves user experience by alerting users to syntax er
rors prior to costly network transactions.
      </p>

      <div class="tech-grid">
        <div class="tech-card">
          <h4>1. RegEx Matching</h4>
          <p>Email format is verified with <code>/^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}
$/</code> and Student ID with <code>/^AU[0-9]{6,8}$/i</code>.</p>
        </div>
        <div class="tech-card">
          <h4>2. Password Entropy Meter</h4>
          <p>Evaluates string length, uppercase, lowercase, numbers, and special symbols dynamical
ly on the <code>oninput</code> event.</p>
        </div>
        <div class="tech-card">
          <h4>3. Dynamic Captcha</h4>
          <p>Randomly generates arithmetic operations to prevent automated brute-force attacks dir
ectly within browser memory.</p>
        </div>
        <div class="tech-card">
          <h4>4. Event Prevention</h4>
          <p>Uses <code>event.preventDefault()</code> to intercept form submission and execute mul
ti-factor integrity checks.</p>
        </div>
      </div>
    </section>

  </main>

  <!-- Toast Notification element -->
  <div id="toast"></div>

  <!-- Footer -->
  <footer>
    <p style="margin: 0 0 0.5rem 0; font-weight: 600; color: #cbd5e1;">
      Designed &amp;amp; Developed by <strong style="color: #818cf8;">Atul Anand</strong> &bull; BCA
(Hons.) with Research (Semester III)
    </p>
    <p style="margin: 0; color: #64748b; font-size: 0.8rem;">
      Amity University Noida &bull; CSIT248: Advanced Web Technology &bull; Experiment 14 Complete
    </p>
  </footer>

  <!-- JavaScript Validation Logic -->
  <script>
    // State variables
    let currentRole = 'student';
    let captchaExpectedResult = null;
```

```
    // Toast utility
    function showToast(message, type = 'info') {
      const toast = document.getElementById('toast');
      toast.textContent = message;
      toast.className = `show ${type}`;
      setTimeout(() => {
        toast.className = '';
      }, 3500);
    }

    // Switch between Student and Faculty tabs
    function switchPortal(role, button) {
      currentRole = role;
      document.querySelectorAll('.portal-tab').forEach(b => b.classList.remove('active'));
      button.classList.add('active');

      const label = document.getElementById('identifierLabel');
      const input = document.getElementById('userIdentifier');
      if (role === 'student') {
        label.textContent = 'Student Email or Enrollment ID';
        input.placeholder = 'e.g. atul.anand@amity.edu or AU2024101';
      } else {
        label.textContent = 'Faculty Official Email or Staff ID';
        input.placeholder = 'e.g. professor.sharma@amity.edu or FAC9041';
      }
      validateIdentifier();
    }

    // Generate dynamic arithmetic captcha
    function generateCaptcha() {
      const num1 = Math.floor(Math.random() * 12) + 3;
      const num2 = Math.floor(Math.random() * 10) + 2;
      captchaExpectedResult = num1 + num2;
      document.getElementById('captchaText').textContent = `${num1} + ${num2} = ?`;
      const captchaInput = document.getElementById('captchaInput');
      captchaInput.value = '';
      captchaInput.classList.remove('is-valid', 'is-invalid');
      document.getElementById('captchaFeedback').textContent = '';
    }

    // Toggle Password Visibility
    function togglePasswordVisibility() {
      const passInput = document.getElementById('password');
      const eyeIcon = document.getElementById('eyeIcon');
      if (passInput.type === 'password') {
        passInput.type = 'text';
        eyeIcon.innerHTML = `
          <path d="M17.94 17.94A10.07 10.07 0 0 1 12 20c-7 0-11-8-11-8a18.45 18.45 0 0 1 5.06-5.94
M9.9 4.24A9.12 9.12 0 0 1 12 4c7 0 11 8 11 8a18.5 18.5 0 0 1-2.16 3.19m-6.72-1.07a3 3 0 1 1-4.24-
4.24"></path>
          <line x1="1" y1="1" x2="23" y2="23"></line>
        `;
      } else {
        passInput.type = 'password';
        eyeIcon.innerHTML = `
          <path d="M1 12s4-8 11-8 11 8 11 8-4 8-11 8-11-8-11-8z"></path>
          <circle cx="12" cy="12" r="3"></circle>
        `;
      }
    }

    // 1. Validate Username / Email
    function validateIdentifier() {
      const input = document.getElementById('userIdentifier');
      const feedback = document.getElementById('identifierFeedback');
```

```
      const val = input.value.trim();

      if (!val) {
        input.classList.remove('is-valid');
        input.classList.add('is-invalid');
        feedback.className = 'feedback-msg error';
        feedback.textContent = '❌ Please enter your email or institutional ID.';
        return false;
      }

      const emailRegex = /^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$/;
      const studentIdRegex = /^AU[0-9]{6,8}$/i;
      const facultyIdRegex = /^FAC[0-9]{4,6}$/i;

      const isValidEmail = emailRegex.test(val);
      const isValidId = currentRole === 'student' ? studentIdRegex.test(val) : facultyIdRegex.test
(val);

      if (isValidEmail || isValidId) {
        input.classList.remove('is-invalid');
        input.classList.add('is-valid');
        feedback.className = 'feedback-msg success';
        feedback.textContent = `✅ Valid ${isValidEmail ? 'Email format' : 'Institutional ID'}.`;
        return true;
      } else {
        input.classList.remove('is-valid');
        input.classList.add('is-invalid');
        feedback.className = 'feedback-msg error';
        feedback.textContent = currentRole === 'student'
          ? '❌ Enter a valid email or student ID starting with AU (e.g. AU2024101).'
          : '❌ Enter a valid email or faculty ID starting with FAC (e.g. FAC9041).';
        return false;
      }
    }

    // 2. Validate Password and Strength Meter
    function validatePassword() {
      const input = document.getElementById('password');
      const feedback = document.getElementById('passwordFeedback');
      const strengthBar = document.getElementById('strengthBar');
      const strengthLabel = document.getElementById('strengthLabel');
      const val = input.value;

      // Criteria elements
      const cLength = document.getElementById('critLength');
      const cUpper = document.getElementById('critUpper');
      const cLower = document.getElementById('critLower');
      const cSpecial = document.getElementById('critSpecial');

      const hasLength = val.length >= 8;
      const hasUpper = /[A-Z]/.test(val);
      const hasLower = /[a-z]/.test(val);
      const hasNumber = /[0-9]/.test(val);
      const hasSpecial = /[@$!%*?&#^()_+\-=[\]{};':"\\|,.<>/?]/.test(val);

      // Update criteria indicators
      cLength.innerHTML = hasLength ? '✅ Min 8 chars' : '⚪ Min 8 chars';
      cLength.className = 'criteria-item' + (hasLength ? ' met' : '');

      cUpper.innerHTML = hasUpper ? '✅ Uppercase (A-Z)' : '⚪ Uppercase (A-Z)';
      cUpper.className = 'criteria-item' + (hasUpper ? ' met' : '');

      cLower.innerHTML = hasLower ? '✅ Lowercase (a-z)' : '⚪ Lowercase (a-z)';
      cLower.className = 'criteria-item' + (hasLower ? ' met' : '');

      cSpecial.innerHTML = (hasNumber && hasSpecial) ? '✅ Number &amp;amp; Symbol' : '⚪ Number &am
```

```
p; Symbol';
      cSpecial.className = 'criteria-item' + (hasNumber && hasSpecial ? ' met' : '');

      // Score 0 to 4
      let score = 0;
      if (hasLength) score++;
      if (hasUpper) score++;
      if (hasLower) score++;
      if (hasNumber && hasSpecial) score++;

      // Bar coloring
      if (!val) {
        strengthBar.style.width = '0%';
        strengthLabel.textContent = 'Too Short';
        strengthLabel.style.color = '#f43f5e';
      } else if (score <= 1) {
        strengthBar.style.width = '25%';
        strengthBar.style.background = '#ef4444';
        strengthLabel.textContent = 'Weak';
        strengthLabel.style.color = '#ef4444';
      } else if (score === 2) {
        strengthBar.style.width = '50%';
        strengthBar.style.background = '#f59e0b';
        strengthLabel.textContent = 'Moderate';
        strengthLabel.style.color = '#f59e0b';
      } else if (score === 3) {
        strengthBar.style.width = '75%';
        strengthBar.style.background = '#3b82f6';
        strengthLabel.textContent = 'Good';
        strengthLabel.style.color = '#3b82f6';
      } else {
        strengthBar.style.width = '100%';
        strengthBar.style.background = '#10b981';
        strengthLabel.textContent = 'Strong';
        strengthLabel.style.color = '#10b981';
      }

      if (hasLength && score >= 3) {
        input.classList.remove('is-invalid');
        input.classList.add('is-valid');
        feedback.className = 'feedback-msg success';
        feedback.textContent = '✅ Password satisfies security policy.';
        return true;
      } else {
        input.classList.remove('is-valid');
        input.classList.add('is-invalid');
        feedback.className = 'feedback-msg error';
        feedback.textContent = '❌ Password must be at least 8 characters with letters, numbers,
and symbols.';
        return false;
      }
    }

    // 3. Validate Captcha
    function validateCaptcha() {
      const input = document.getElementById('captchaInput');
      const feedback = document.getElementById('captchaFeedback');
      const val = parseInt(input.value.trim(), 10);

      if (val === captchaExpectedResult) {
        input.classList.remove('is-invalid');
        input.classList.add('is-valid');
        feedback.className = 'feedback-msg success';
        feedback.textContent = '✅ Security verification passed.';
        return true;
      } else {
```

```
        input.classList.remove('is-valid');
        input.classList.add('is-invalid');
        feedback.className = 'feedback-msg error';
        feedback.textContent = '❌ Verification answer is incorrect.';
        return false;
      }
    }

    // Auto-fill demo credentials
    function autoFillDemo() {
      document.getElementById('userIdentifier').value = 'student@amity.edu';
      document.getElementById('password').value = 'Pass@2024!';
      document.getElementById('captchaInput').value = captchaExpectedResult;

      validateIdentifier();
      validatePassword();
      validateCaptcha();
      showToast('Demo credentials autofilled!', 'success');
    }

    // Handle Forgot Password
    function handleForgotPassword() {
      const email = prompt('Enter your registered Institutional Email to receive password reset li
nk:');
      if (email) {
        const emailRegex = /^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$/;
        if (emailRegex.test(email.trim())) {
          alert(`Success! Password reset instructions sent to ${email.trim()}`);
        } else {
          alert('Invalid email format provided. Please check and try again.');
        }
      }
    }

    // Form Submission Handler
    function handleLoginSubmit(event) {
      event.preventDefault();

      const isIdValid = validateIdentifier();
      const isPassValid = validatePassword();
      const isCaptchaValid = validateCaptcha();

      if (!isIdValid || !isPassValid || !isCaptchaValid) {
        showToast('⚠️ Please correct the highlighted errors before submitting.', 'error');
        return;
      }

      // Simulate loading state
      const submitBtn = document.getElementById('submitBtn');
      const originalText = submitBtn.innerHTML;
      submitBtn.innerHTML = `
        <svg class="spin" width="18" height="18" viewBox="0 0 24 24" fill="none" stroke="currentCo
lor" stroke-width="2" style="animation: spin 0.8s linear infinite;">
          <circle cx="12" cy="12" r="10" stroke-opacity="0.25"></circle>
          <path d="M12 2a10 10 0 0 1 10 10"></path>
        </svg>
        <span>Authenticating with Amity Auth Server...</span>
      `;
      submitBtn.disabled = true;

      setTimeout(() => {
        submitBtn.innerHTML = originalText;
        submitBtn.disabled = false;

        // Transition to authenticated dashboard view
        const identifierVal = document.getElementById('userIdentifier').value;
```

```
        const displayName = identifierVal.split('@')[0].replace('.', ' ').toUpperCase();

        document.getElementById('login-view').style.display = 'none';
        const dashView = document.getElementById('dashboard-view');
        dashView.style.display = 'block';

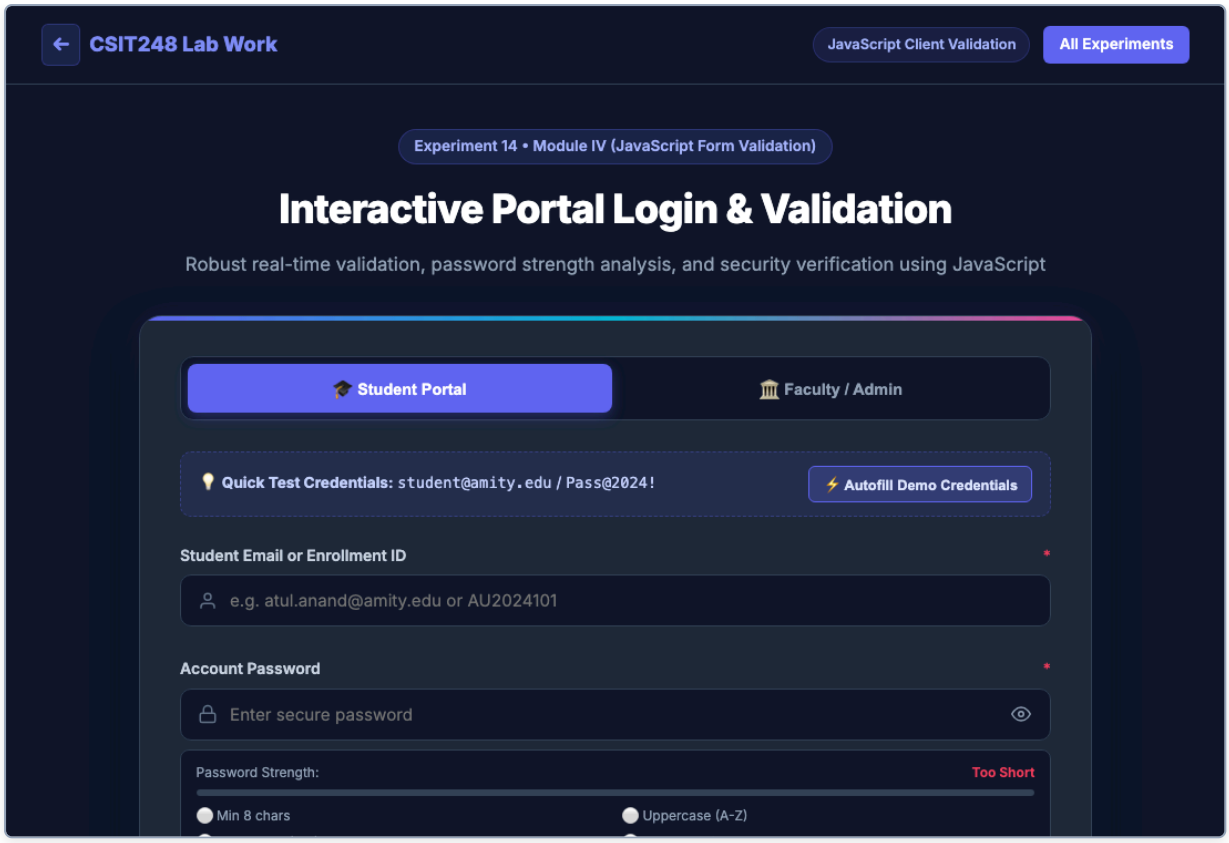
        document.getElementById('dashWelcome').textContent = `Welcome back, ${displayName}!`;
        document.getElementById('dashAvatar').textContent = displayName.charAt(0);
        showToast('🎉 Authentication Successful! Session token generated.', 'success');
      }, 1000);
    }

    // Logout session
    function logoutSession() {
      document.getElementById('dashboard-view').style.display = 'none';
      document.getElementById('login-view').style.display = 'block';
      document.getElementById('loginForm').reset();
      document.getElementById('userIdentifier').classList.remove('is-valid', 'is-invalid');
      document.getElementById('password').classList.remove('is-valid', 'is-invalid');
      document.getElementById('captchaInput').classList.remove('is-valid', 'is-invalid');
      document.getElementById('identifierFeedback').textContent = '';
      document.getElementById('passwordFeedback').textContent = '';
      document.getElementById('captchaFeedback').textContent = '';
      generateCaptcha();
      showToast('You have been logged out securely.', 'info');
    }

    // Initialize captcha on page load
    window.addEventListener('DOMContentLoaded', () => {
      generateCaptcha();
    });
  </script>
</body>

</html>
```

Output :


CSIT248 Lab Work
JavaScript Client Validation
All Experiments
Experiment 14 • Module IV (JavaScript Form Validation)
Interactive Portal Login & Validation
Robust real-time validation, password strength analysis, and security verification using JavaScript
Student Portal
Faculty / Admin
Quick Test Credentials: student@amity.edu / Pass@2024!
Autofill Demo Credentials
Student Email or Enrollment ID
e.g. atul.anand@amity.edu or AU2024101
Account Password
Enter secure password
Password Strength:
Too Short
Min 8 chars
Uppercase (A-Z)

Experiment No.: 18

Title : Web Technology Quiz Engine and Dynamic Score Calculation.

Aim : To build an interactive web technology quiz system featuring multiple choice questions, timer countdown, dynamic answer evaluation, and immediate percentage scorecard display.

Source Code :

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description"
    content="Experiment 15: Design a web page for Quiz and calculate the result using JavaScript - CSIT248 Advanced Web Technology">
  <title>Experiment 15 - Web Technology Quiz &amp;amp; Result Calculation</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&display=swap" rel="stylesheet">
  <style>
    :root {
      --primary: #6366f1;
      --primary-hover: #4f46e5;
      --secondary: #06b6d4;
      --accent: #f43f5e;
      --success: #10b981;
      --warning: #f59e0b;
      --bg-dark: #0f172a;
      --card-bg: #1e293b;
      --card-border: #334155;
      --text-main: #f8fafc;
      --text-muted: #94a3b8;
    }

    * {
      box-sizing: border-box;
      margin: 0;
      padding: 0;
    }

    body {
      font-family: 'Inter', system-ui, -apple-system, sans-serif;
      background-color: var(--bg-dark);
      color: var(--text-main);
      line-height: 1.6;
      min-height: 100vh;
      display: flex;
      flex-direction: column;
    }

    /* Top Sticky Navigation */
    nav {
      position: sticky;
      top: 0;
      z-index: 100;
      background: rgba(15, 23, 42, 0.92);
      backdrop-filter: blur(12px);
      border-bottom: 1px solid #334155;
```

```
  padding: 1rem 2rem;
  display: flex;
  justify-content: space-between;
  align-items: center;
}

.nav-brand {
  font-weight: 800;
  font-size: 1.15rem;
  color: #818cf8;
  text-decoration: none;
  display: flex;
  align-items: center;
  gap: 0.5rem;
}

.nav-brand span {
  background: rgba(99, 102, 241, 0.15);
  border: 1px solid rgba(99, 102, 241, 0.3);
  padding: 0.2rem 0.5rem;
  border-radius: 6px;
}

.nav-links {
  display: flex;
  gap: 0.75rem;
  align-items: center;
}

.badge-pill {
  font-size: 0.8rem;
  background: rgba(99, 102, 241, 0.12);
  color: #a5b4fc;
  border: 1px solid rgba(99, 102, 241, 0.25);
  padding: 0.3rem 0.75rem;
  border-radius: 9999px;
  font-weight: 600;
}

.btn-nav {
  background: #6366f1;
  color: #ffffff;
  padding: 0.4rem 0.9rem;
  border-radius: 6px;
  text-decoration: none;
  font-size: 0.85rem;
  font-weight: 600;
  transition: all 0.2s ease;
}

.btn-nav:hover {
  background: #4f46e5;
}

/* Main Container */
main {
  flex: 1;
  max-width: 1000px;
  width: 100%;
  margin: 0 auto;
  padding: 2.5rem 1.25rem;
}

.exp-header {
  text-align: center;
  margin-bottom: 2rem;
```

```
}

.exp-pill {
  display: inline-block;
  background: rgba(99, 102, 241, 0.15);
  color: #a5b4fc;
  border: 1px solid rgba(99, 102, 241, 0.3);
  padding: 0.25rem 0.85rem;
  border-radius: 9999px;
  font-size: 0.85rem;
  font-weight: 600;
  margin-bottom: 0.75rem;
}

.exp-header h1 {
  font-size: 2.25rem;
  font-weight: 800;
  color: #ffffff;
  margin-bottom: 0.5rem;
  letter-spacing: -0.02em;
}

.exp-header p {
  color: #94a3b8;
  font-size: 1.05rem;
}

/* Quiz Stats Bar */
.quiz-stats-bar {
  background: #1e293b;
  border: 1px solid #334155;
  border-radius: 14px;
  padding: 1rem 1.5rem;
  display: flex;
  justify-content: space-between;
  align-items: center;
  flex-wrap: wrap;
  gap: 1rem;
  margin-bottom: 1.5rem;
}

.stat-pill {
  display: flex;
  align-items: center;
  gap: 0.5rem;
  font-size: 0.9rem;
  font-weight: 600;
}

.timer-display {
  background: rgba(244, 63, 94, 0.15);
  border: 1px solid rgba(244, 63, 94, 0.3);
  color: #fb7185;
  padding: 0.35rem 0.85rem;
  border-radius: 8px;
  font-family: 'Courier New', monospace;
  font-size: 1.1rem;
  font-weight: 700;
}

.timer-display.pulse {
  animation: pulseAnim 1s infinite alternate;
}

@keyframes pulseAnim {
  from { transform: scale(1); }
```

```
  to { transform: scale(1.05); border-color: #f43f5e; color: #f43f5e; }
}

/* Progress bar */
.progress-track {
  width: 100%;
  height: 8px;
  background: #0f172a;
  border-radius: 9999px;
  overflow: hidden;
  border: 1px solid #334155;
  margin-bottom: 1.5rem;
}

.progress-fill {
  height: 100%;
  background: linear-gradient(90deg, #6366f1, #06b6d4);
  width: 10%;
  transition: width 0.3s ease;
}

/* Quiz Layout: Question Card + Sidebar */
.quiz-layout {
  display: grid;
  grid-template-columns: 1fr 280px;
  gap: 1.5rem;
}

@media (max-width: 850px) {
  .quiz-layout {
    grid-template-columns: 1fr;
  }
}

/* Question Card */
.question-card {
  background: #1e293b;
  border: 1px solid #334155;
  border-radius: 16px;
  padding: 2rem;
  box-shadow: 0 10px 30px rgba(0, 0, 0, 0.3);
  display: flex;
  flex-direction: column;
  justify-content: space-between;
  min-height: 440px;
}

.q-meta {
  display: flex;
  justify-content: space-between;
  align-items: center;
  margin-bottom: 1rem;
}

.q-badge {
  font-size: 0.75rem;
  font-weight: 700;
  text-transform: uppercase;
  letter-spacing: 0.05em;
  background: rgba(99, 102, 241, 0.15);
  color: #818cf8;
  border: 1px solid rgba(99, 102, 241, 0.3);
  padding: 0.25rem 0.65rem;
  border-radius: 6px;
}
```

```
.q-points {
  font-size: 0.8rem;
  color: #34d399;
  font-weight: 600;
}

.q-title {
  font-size: 1.25rem;
  font-weight: 700;
  color: #ffffff;
  margin-bottom: 1.25rem;
  line-height: 1.45;
}

.code-snippet {
  background: #0f172a;
  border: 1px solid #334155;
  border-radius: 8px;
  padding: 1rem;
  font-family: 'Courier New', Courier, monospace;
  font-size: 0.9rem;
  color: #38bdf8;
  margin-bottom: 1.25rem;
  white-space: pre-wrap;
}

/* Options List */
.options-list {
  display: flex;
  flex-direction: column;
  gap: 0.75rem;
  margin-bottom: 1.5rem;
}

.option-item {
  background: #0f172a;
  border: 1.5px solid #334155;
  border-radius: 10px;
  padding: 0.85rem 1.15rem;
  display: flex;
  align-items: center;
  gap: 0.85rem;
  cursor: pointer;
  transition: all 0.2s ease;
  user-select: none;
}

.option-item:hover {
  border-color: #6366f1;
  background: rgba(99, 102, 241, 0.08);
  transform: translateX(4px);
}

.option-item.selected {
  border-color: #6366f1;
  background: rgba(99, 102, 241, 0.2);
}

.option-item input[type="radio"],
.option-item input[type="checkbox"] {
  accent-color: #6366f1;
  width: 18px;
  height: 18px;
  cursor: pointer;
}
```

```
.option-text {
  font-size: 0.95rem;
  color: #e2e8f0;
  flex: 1;
}

/* Action Controls */
.controls-row {
  display: flex;
  justify-content: space-between;
  align-items: center;
  gap: 0.75rem;
  flex-wrap: wrap;
  padding-top: 1.25rem;
  border-top: 1px solid #334155;
}

.btn-quiz {
  padding: 0.65rem 1.25rem;
  font-size: 0.9rem;
  font-weight: 600;
  border-radius: 8px;
  border: none;
  cursor: pointer;
  display: inline-flex;
  align-items: center;
  gap: 0.4rem;
  transition: all 0.2s ease;
}

.btn-secondary {
  background: #334155;
  color: #f8fafc;
}

.btn-secondary:hover:not(:disabled) {
  background: #475569;
}

.btn-secondary:disabled {
  opacity: 0.4;
  cursor: not-allowed;
}

.btn-review {
  background: rgba(245, 158, 11, 0.15);
  border: 1px solid rgba(245, 158, 11, 0.35);
  color: #fcd34d;
}

.btn-review:hover {
  background: rgba(245, 158, 11, 0.25);
}

.btn-primary {
  background: #6366f1;
  color: #ffffff;
}

.btn-primary:hover {
  background: #4f46e5;
  transform: translateY(-1px);
}

.btn-submit-quiz {
  background: #10b981;
```

```
  color: #ffffff;
}

.btn-submit-quiz:hover {
  background: #059669;
  transform: translateY(-1px);
}

/* Sidebar Palette */
.sidebar-card {
  background: #1e293b;
  border: 1px solid #334155;
  border-radius: 16px;
  padding: 1.5rem;
  display: flex;
  flex-direction: column;
  gap: 1.25rem;
}

.sidebar-title {
  font-size: 1rem;
  font-weight: 700;
  color: #ffffff;
  padding-bottom: 0.5rem;
  border-bottom: 1px solid #334155;
}

.palette-grid {
  display: grid;
  grid-template-columns: repeat(4, 1fr);
  gap: 0.6rem;
}

.palette-btn {
  width: 100%;
  aspect-ratio: 1;
  border-radius: 8px;
  border: 1.5px solid #334155;
  background: #0f172a;
  color: #cbd5e1;
  font-weight: 700;
  font-size: 0.85rem;
  cursor: pointer;
  display: flex;
  align-items: center;
  justify-content: center;
  transition: all 0.2s ease;
}

.palette-btn:hover {
  border-color: #6366f1;
  transform: scale(1.05);
}

.palette-btn.current {
  border-color: #6366f1;
  outline: 2px solid #6366f1;
  outline-offset: 2px;
}

.palette-btn.answered {
  background: #10b981;
  color: #ffffff;
  border-color: #10b981;
}
```

```
.palette-btn.marked {
  background: #f59e0b;
  color: #ffffff;
  border-color: #f59e0b;
}

/* Legend */
.palette-legend {
  display: flex;
  flex-direction: column;
  gap: 0.4rem;
  font-size: 0.75rem;
  color: #94a3b8;
}

.legend-item {
  display: flex;
  align-items: center;
  gap: 0.5rem;
}

.legend-dot {
  width: 10px;
  height: 10px;
  border-radius: 3px;
}

/* Result View Card */
#result-view {
  display: none;
  background: #1e293b;
  border: 1px solid #334155;
  border-radius: 20px;
  padding: 2.5rem;
  box-shadow: 0 20px 40px rgba(0, 0, 0, 0.4);
  animation: fadeIn 0.4s ease-out;
}

@keyframes fadeIn {
  from { opacity: 0; transform: translateY(12px); }
  to { opacity: 1; transform: translateY(0); }
}

.result-score-hero {
  text-align: center;
  padding-bottom: 2rem;
  border-bottom: 1px solid #334155;
  margin-bottom: 2rem;
}

.grade-badge {
  display: inline-block;
  padding: 0.4rem 1.25rem;
  border-radius: 9999px;
  font-size: 0.95rem;
  font-weight: 800;
  margin-bottom: 1rem;
}

.grade-badge.excellent {
  background: rgba(16, 185, 129, 0.2);
  color: #34d399;
  border: 1px solid #10b981;
}

.grade-badge.good {
```

```
  background: rgba(59, 130, 246, 0.2);
  color: #60a5fa;
  border: 1px solid #3b82f6;
}

.grade-badge.fair {
  background: rgba(245, 158, 11, 0.2);
  color: #fbbf24;
  border: 1px solid #f59e0b;
}

.grade-badge.poor {
  background: rgba(244, 63, 94, 0.2);
  color: #fb7185;
  border: 1px solid #f43f5e;
}

.score-number {
  font-size: 4rem;
  font-weight: 800;
  color: #ffffff;
  line-height: 1;
  margin-bottom: 0.5rem;
}

.score-number span {
  font-size: 2rem;
  color: #94a3b8;
}

.score-breakdown-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(180px, 1fr));
  gap: 1rem;
  margin-bottom: 2rem;
}

.score-card {
  background: #0f172a;
  border: 1px solid #334155;
  border-radius: 12px;
  padding: 1.25rem;
  text-align: center;
}

.score-card h5 {
  font-size: 0.8rem;
  color: #94a3b8;
  text-transform: uppercase;
  margin-bottom: 0.4rem;
}

.score-card p {
  font-size: 1.5rem;
  font-weight: 800;
  color: #ffffff;
}

/* Answers Review Section */
.review-accordion-title {
  font-size: 1.25rem;
  font-weight: 700;
  color: #ffffff;
  margin-bottom: 1rem;
  display: flex;
  align-items: center;
```

```
  gap: 0.5rem;
}

.review-item {
  background: #0f172a;
  border: 1px solid #334155;
  border-radius: 12px;
  padding: 1.25rem;
  margin-bottom: 1rem;
}

.review-item.correct {
  border-left: 5px solid #10b981;
}

.review-item.incorrect {
  border-left: 5px solid #f43f5e;
}

.review-item.unanswered {
  border-left: 5px solid #f59e0b;
}

.review-header {
  display: flex;
  justify-content: space-between;
  margin-bottom: 0.5rem;
  font-size: 0.85rem;
}

.explanation-box {
  background: rgba(99, 102, 241, 0.1);
  border: 1px dashed rgba(99, 102, 241, 0.3);
  padding: 0.75rem;
  border-radius: 8px;
  margin-top: 0.75rem;
  font-size: 0.85rem;
  color: #cbd5e1;
}

/* Technical Info Section */
.info-section {
  background: #1e293b;
  border: 1px solid #334155;
  border-radius: 16px;
  padding: 1.75rem;
  margin-top: 2rem;
}

.info-section h3 {
  font-size: 1.25rem;
  color: #ffffff;
  margin-bottom: 0.85rem;
  display: flex;
  align-items: center;
  gap: 0.5rem;
}

.tech-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(220px, 1fr));
  gap: 1rem;
  margin-top: 1rem;
}

.tech-card {
```

```
      background: #0f172a;
      border: 1px solid #334155;
      padding: 1rem;
      border-radius: 10px;
    }

    .tech-card h4 {
      font-size: 0.9rem;
      color: #818cf8;
      margin-bottom: 0.3rem;
    }

    .tech-card p {
      font-size: 0.825rem;
      color: #94a3b8;
    }

    footer {
      background: #090d16;
      border-top: 1px solid #334155;
      padding: 1.5rem;
      text-align: center;
      color: #94a3b8;
      font-size: 0.875rem;
      margin-top: auto;
    }
  </style>
</head>

<body>

  <!-- Navigation -->
  <nav>
    <a href="../index.html" class="nav-brand">
      <span>&larr;</span> CSIT248 Lab Work
    </a>
    <div class="nav-links">
      <span class="badge-pill">JavaScript Quiz Engine</span>
      <a href="../index.html" class="btn-nav">All Experiments</a>
    </div>
  </nav>

  <main>

    <!-- Header -->
    <header class="exp-header">
      <div class="exp-pill">Experiment 15 &bull; Module IV (JavaScript Quiz &amp;amp; Score Calculatio
n)</div>
      <h1>CSIT248 Web Technology Assessment Quiz</h1>
      <p>Interactive testing interface with dynamic DOM rendering, live countdown timer, and autom
ated result calculation</p>
    </header>

    <!-- Active Quiz Container -->
    <div id="quiz-active-view">

      <!-- Quiz Stats Bar -->
      <div class="quiz-stats-bar">
        <div class="stat-pill">
          <span>📚 Subject:</span>
          <strong style="color: #818cf8;">Advanced Web Technology</strong>
        </div>
        <div class="stat-pill">
          <span>⏱️ Time Remaining:</span>
          <div id="timerDisplay" class="timer-display">05:00</div>
        </div>
```

```
        <div class="stat-pill">
          <span>🎯 Question:</span>
          <strong id="qTracker" style="color: #34d399;">1 of 8</strong>
        </div>
      </div>

      <!-- Progress Track -->
      <div class="progress-track">
        <div id="progressFill" class="progress-fill"></div>
      </div>

      <!-- Main Layout -->
      <div class="quiz-layout">

        <!-- Question View Box -->
        <section class="question-card">
          <div>
            <div class="q-meta">
              <span id="qBadge" class="q-badge">HTML5 Semantics</span>
              <span id="qPoints" class="q-points">+4 Marks / -1 Negative</span>
            </div>

            <h2 id="qTitle" class="q-title">Loading Question...</h2>

            <div id="codeSnippetBox" class="code-snippet" style="display: none;"></div>

            <!-- Options -->
            <div id="optionsContainer" class="options-list"></div>
          </div>

          <!-- Controls -->
          <div class="controls-row">
            <button type="button" id="prevBtn" class="btn-quiz btn-secondary" onclick="prevQuestio
n()" disabled>
              &larr; Previous
            </button>

            <div style="display: flex; gap: 0.5rem;">
              <button type="button" class="btn-quiz btn-review" onclick="toggleMarkReview()">
                🚩 Mark for Review
              </button>
              <button type="button" class="btn-quiz btn-secondary" onclick="clearOption()">
                ✖ Clear
              </button>
            </div>

            <button type="button" id="nextBtn" class="btn-quiz btn-primary" onclick="nextQuestion
()">
              Next Question &rarr;
            </button>
          </div>
        </section>

        <!-- Sidebar Palette -->
        <aside class="sidebar-card">
          <div class="sidebar-title">Question Palette</div>

          <div id="paletteGrid" class="palette-grid">
            <!-- Rendered by JS -->
          </div>

          <div class="palette-legend">
            <div class="legend-item">
              <div class="legend-dot" style="background: #10b981;"></div>
              <span>Answered</span>
            </div>
```

```
            <div class="legend-item">
              <div class="legend-dot" style="background: #f59e0b;"></div>
              <span>Marked for Review</span>
            </div>
            <div class="legend-item">
              <div class="legend-dot" style="background: #0f172a; border: 1px solid #334155;"></di
v>
              <span>Unanswered / Not Visited</span>
            </div>
          </div>

          <button type="button" class="btn-quiz btn-submit-quiz" onclick="confirmSubmitQuiz()" sty
le="width: 100%; justify-content: center; margin-top: 0.5rem;">
             Submit Quiz Now
          </button>
        </aside>

      </div>

    </div>

    <!-- Final Result View (Hidden until submit) -->
    <div id="result-view">
      <div class="result-score-hero">
        <div id="gradeBadge" class="grade-badge excellent">GRADE A+ &bull; DISTINCTION</div>
        <div class="score-number" id="finalScoreDisplay">32 <span>/ 32</span></div>
        <p style="color: #cbd5e1; font-size: 1.1rem; max-width: 500px; margin: 0 auto;" id="result
Feedback">
          Outstanding performance! You displayed exceptional mastery across HTML5, CSS3, and JavaS
cript fundamentals.
        </p>
      </div>

      <div class="score-breakdown-grid">
        <div class="score-card">
          <h5>Total Questions</h5>
          <p id="statTotal">8</p>
        </div>
        <div class="score-card">
          <h5>Correct Answers</h5>
          <p id="statCorrect" style="color: #10b981;">8</p>
        </div>
        <div class="score-card">
          <h5>Incorrect Answers</h5>
          <p id="statWrong" style="color: #f43f5e;">0</p>
        </div>
        <div class="score-card">
          <h5>Accuracy Rate</h5>
          <p id="statAccuracy" style="color: #38bdf8;">100%</p>
        </div>
        <div class="score-card">
          <h5>Time Taken</h5>
          <p id="statTime" style="color: #a78bfa;">01:42</p>
        </div>
      </div>

      <div style="display: flex; gap: 1rem; justify-content: center; margin-bottom: 2rem;">
        <button type="button" class="btn-quiz btn-primary" onclick="restartQuiz()">
           Retake Quiz
        </button>
        <button type="button" class="btn-quiz btn-secondary" onclick="window.print()">
           Print Score Report
        </button>
      </div>

      <!-- Detailed Question-by-Question Review -->
```

```
      <div>
        <h3 class="review-accordion-title">
          Detailed Solutions &amp;amp; Pedagogical Review
        </h3>
        <div id="detailedReviewContainer">
          <!-- Populated by JavaScript -->
        </div>
      </div>
    </div>

    <!-- Academic Documentation -->
    <section class="info-section">
      <h3>
        <svg width="20" height="20" fill="none" stroke="#818cf8" stroke-width="2" viewBox="0 0 24
24">
          <circle cx="12" cy="12" r="10"></circle>
          <line x1="12" y1="16" x2="12" y2="12"></line>
          <line x1="12" y1="8" x2="12.01" y2="8"></line>
        </svg>
        Experiment 15 &bull; Technical Architecture &amp;amp; Calculation Algorithm
      </h3>
      <p style="color: #cbd5e1; font-size: 0.9rem;">
        This experiment highlights dynamic client-side state manipulation using JavaScript object
s, event listeners, array methods, and real-time score calculation algorithms without requiring ex
ternal server calls.
      </p>

      <div class="tech-grid">
        <div class="tech-card">
          <h4>1. State Data Structure</h4>
          <p>Questions are structured in an immutable array of objects storing stems, code snippet
s, multiple choices, correct indices, and explanation strings.</p>
        </div>
        <div class="tech-card">
          <h4>2. Score Algorithm</h4>
          <p>Calculates score using <code>Total = (Correct &times; 4) - (Incorrect &times; 1)</cod
e>. Computes accuracy percentage: <code>(Correct / Attempted) &times; 100</code>.</p>
        </div>
        <div class="tech-card">
          <h4>3. Real-Time Timer</h4>
          <p>Utilizes <code>setInterval()</code> to decrement a countdown timestamp and automatica
lly triggers submission when <code>remainingSeconds === 0</code>.</p>
        </div>
        <div class="tech-card">
          <h4>4. Dynamic DOM Reconstruction</h4>
          <p>Re-renders option nodes dynamically on each navigation event, preserving state select
ions across the user response matrix.</p>
        </div>
      </div>
    </section>

  </main>

  <!-- Footer -->
  <footer>
    <p style="margin: 0 0 0.5rem 0; font-weight: 600; color: #cbd5e1;">
      Designed &amp;amp; Developed by <strong style="color: #818cf8;">Atul Anand</strong> &bull; BCA
(Hons.) with Research (Semester III)
    </p>
    <p style="margin: 0; color: #64748b; font-size: 0.8rem;">
      Amity University Noida &bull; CSIT248: Advanced Web Technology &bull; Experiment 15 Complete
    </p>
  </footer>

  <!-- JavaScript Quiz Engine -->
  <script>
```

```
    // Quiz Question Bank
    const quizData = [
      {
        id: 1,
        topic: "HTML5 Semantics",
        question: "Which of the following HTML5 elements is specifically designed to contain tange
ntial content such as sidebars, glossaries, or related links?",
        code: null,
        options: [
          "<section>",
          "<aside>",
          "<article>",
          "<nav>"
        ],
        correct: 1,
        explanation: "The <aside> element in HTML5 represents a portion of a document whose conten
t is only indirectly related to the main content, typically rendered as a sidebar."
      },
      {
        id: 2,
        topic: "JavaScript Types & Coercion",
        question: "What will be the exact evaluated output of the following JavaScript expressio
n?",
        code: "console.log(1 + '2' + 3 - 1);",
        options: [
          "122",
          "5",
          "123",
          "NaN"
        ],
        correct: 0,
        explanation: "First, '1' + '2' undergoes string concatenation to become '12'. Then '12' +
3 gives '123'. Finally, '123' - 1 triggers numeric subtraction coercion, giving the number 122."
      },
      {
        id: 3,
        topic: "CSS Box Model",
        question: "When applying `box-sizing: border-box;` to an element with width 300px, padding
20px, and border 2px, what is the total rendered width?",
        code: ".box {\n  box-sizing: border-box;\n  width: 300px;\n  padding: 20px;\n  border: 2px
solid #fff;\n}",
        options: [
          "344px",
          "300px",
          "322px",
          "256px"
        ],
        correct: 1,
        explanation: "With `box-sizing: border-box`, padding and border are absorbed inside the sp
ecified width, ensuring the element's total outer footprint remains exactly 300px."
      },
      {
        id: 4,
        topic: "JavaScript Scope & Closures",
        question: "Which keyword declares a block-scoped variable that cannot be re-declared withi
n the same scope but can be mutated if it is an object/array?",
        code: null,
        options: [
          "var",
          "let",
          "const",
          "function"
        ],
        correct: 2,
        explanation: "`const` enforces block scope and constant reference binding. While the varia
ble identifier cannot be reassigned, mutable composite data (such as properties of an object or el
```

```
ements of an array) can still be altered."
      },
      {
        id: 5,
        topic: "HTML Forms & History",
        question: "Which JavaScript History API method allows changing the browser URL without tri
ggering a full page reload or network request?",
        code: null,
        options: [
          "history.back()",
          "history.forward()",
          "history.pushState()",
          "location.reload()"
        ],
        correct: 2,
        explanation: "`history.pushState(state, title, url)` adds a new entry to the browser sessi
on history stack without initiating a network round-trip, essential for Single Page Applications
(SPAs)."
      },
      {
        id: 6,
        topic: "DOM Event Delegation",
        question: "What property of the Event object indicates the precise DOM node that originall
y dispatched the event?",
        code: null,
        options: [
          "event.currentTarget",
          "event.target",
          "event.origin",
          "event.sender"
        ],
        correct: 1,
        explanation: "`event.target` references the specific innermost DOM element that originated
the user action, whereas `event.currentTarget` points to the element whose listener is currently h
andling the event."
      },
      {
        id: 7,
        topic: "CSS Pseudo-Classes",
        question: "Which CSS pseudo-class matches an element if ANY of its descendant children cur
rently have active user focus?",
        code: null,
        options: [
          ":focus",
          ":focus-visible",
          ":focus-within",
          ":has(:active)"
        ],
        correct: 2,
        explanation: "`:focus-within` allows styling an ancestor container (such as an entire `<fo
rm>` or `<div class='input-wrap'>`) when any child element inside it gains focus."
      },
      {
        id: 8,
        topic: "JavaScript Equality",
        question: "What is the result of evaluating `[] == ![]` in JavaScript?",
        code: "console.log([] == ![]);",
        options: [
          "false",
          "true",
          "TypeError",
          "undefined"
        ],
        correct: 1,
        explanation: "In JavaScript, `![]` evaluates to `false` because an empty array is truthy.
Then `[] == false` converts both to primitives (`'' == 0`), which converts to `0 == 0`, evaluating
```

```
strictly to `true`!"
      }
    ];

    // State Variables
    let currentIdx = 0;
    const userAnswers = new Array(quizData.length).fill(null);
    const markedReview = new Array(quizData.length).fill(false);
    let timeRemaining = 300; // 5 minutes in seconds
    let timerInterval = null;
    let quizStartTime = Date.now();

    // DOM Elements
    const qTracker = document.getElementById('qTracker');
    const qBadge = document.getElementById('qBadge');
    const qTitle = document.getElementById('qTitle');
    const codeSnippetBox = document.getElementById('codeSnippetBox');
    const optionsContainer = document.getElementById('optionsContainer');
    const prevBtn = document.getElementById('prevBtn');
    const nextBtn = document.getElementById('nextBtn');
    const progressFill = document.getElementById('progressFill');
    const paletteGrid = document.getElementById('paletteGrid');
    const timerDisplay = document.getElementById('timerDisplay');

    // Initialize Quiz
    function initQuiz() {
      renderPalette();
      loadQuestion(0);
      startTimer();
    }

    // Render Question Palette buttons
    function renderPalette() {
      paletteGrid.innerHTML = '';
      quizData.forEach((q, idx) => {
        const btn = document.createElement('button');
        btn.type = 'button';
        btn.className = 'palette-btn';
        btn.textContent = idx + 1;
        btn.onclick = () => loadQuestion(idx);

        if (idx === currentIdx) {
          btn.classList.add('current');
        }
        if (userAnswers[idx] !== null) {
          btn.classList.add('answered');
        }
        if (markedReview[idx]) {
          btn.classList.add('marked');
        }

        paletteGrid.appendChild(btn);
      });
    }

    // Load Specific Question
    function loadQuestion(idx) {
      currentIdx = idx;
      const q = quizData[idx];

      qTracker.textContent = `${idx + 1} of ${quizData.length}`;
      qBadge.textContent = q.topic;
      qTitle.textContent = `${idx + 1}. ${q.question}`;

      // Code snippet
      if (q.code) {
```

```
    codeSnippetBox.style.display = 'block';
    codeSnippetBox.textContent = q.code;
  } else {
    codeSnippetBox.style.display = 'none';
  }

  // Render Options
  optionsContainer.innerHTML = '';
  q.options.forEach((optText, optIdx) => {
    const optDiv = document.createElement('div');
    optDiv.className = 'option-item';
    if (userAnswers[idx] === optIdx) {
      optDiv.classList.add('selected');
    }

    const radio = document.createElement('input');
    radio.type = 'radio';
    radio.name = `q_${idx}`;
    radio.id = `opt_${idx}_${optIdx}`;
    radio.checked = userAnswers[idx] === optIdx;

    const label = document.createElement('label');
    label.className = 'option-text';
    label.htmlFor = `opt_${idx}_${optIdx}`;
    label.textContent = optText;

    optDiv.onclick = (e) => {
      selectOption(optIdx);
    };

    optDiv.appendChild(radio);
    optDiv.appendChild(label);
    optionsContainer.appendChild(optDiv);
  });

  // Update Navigation Buttons
  prevBtn.disabled = idx === 0;
  if (idx === quizData.length - 1) {
    nextBtn.textContent = 'Review & Submit 🏁';
    nextBtn.className = 'btn-quiz btn-submit-quiz';
  } else {
    nextBtn.textContent = 'Next Question →';
    nextBtn.className = 'btn-quiz btn-primary';
  }

  // Update Progress Bar
  const answeredCount = userAnswers.filter(a => a !== null).length;
  const progressPercent = ((answeredCount / quizData.length) * 100);
  progressFill.style.width = `${Math.max(progressPercent, 12)}%`;

  renderPalette();
}

// Select Option
function selectOption(optionIndex) {
  userAnswers[currentIdx] = optionIndex;
  loadQuestion(currentIdx);
}

// Clear Option
function clearOption() {
  userAnswers[currentIdx] = null;
  loadQuestion(currentIdx);
}

// Toggle Marked for Review
```

```
    function toggleMarkReview() {
      markedReview[currentIdx] = !markedReview[currentIdx];
      renderPalette();
    }

    // Navigation Handlers
    function prevQuestion() {
      if (currentIdx > 0) {
        loadQuestion(currentIdx - 1);
      }
    }

    function nextQuestion() {
      if (currentIdx < quizData.length - 1) {
        loadQuestion(currentIdx + 1);
      } else {
        confirmSubmitQuiz();
      }
    }

    // Timer logic
    function startTimer() {
      timerInterval = setInterval(() => {
        timeRemaining--;
        if (timeRemaining <= 0) {
          clearInterval(timerInterval);
          alert('⏰ Time has expired! Your quiz will now be submitted automatically.');
          calculateAndDisplayResult();
          return;
        }

        const mins = Math.floor(timeRemaining / 60);
        const secs = timeRemaining % 60;
        timerDisplay.textContent = `${String(mins).padStart(2, '0')}:${String(secs).padStart(2,
'0')}`;

        if (timeRemaining <= 60) {
          timerDisplay.classList.add('pulse');
        }
      }, 1000);
    }

    // Confirm Submission
    function confirmSubmitQuiz() {
      const answered = userAnswers.filter(a => a !== null).length;
      const unattempted = quizData.length - answered;

      let msg = `You have answered ${answered} of ${quizData.length} questions.`;
      if (unattempted > 0) {
        msg += `\n⚠️ You have ${unattempted} unattempted question(s). Are you sure you wish to su
bmit now?`;
      } else {
        msg += '\nReady to calculate your score?';
      }

      if (confirm(msg)) {
        calculateAndDisplayResult();
      }
    }

    // Calculate Result and Display Comprehensive Report
    function calculateAndDisplayResult() {
      clearInterval(timerInterval);

      let correctCount = 0;
      let incorrectCount = 0;
```

```
      let unattemptedCount = 0;

      quizData.forEach((q, idx) => {
        if (userAnswers[idx] === null) {
          unattemptedCount++;
        } else if (userAnswers[idx] === q.correct) {
          correctCount++;
        } else {
          incorrectCount++;
        }
      });

      // Marking Scheme: +4 per correct, -1 per incorrect
      const rawScore = (correctCount * 4) - (incorrectCount * 1);
      const totalPossible = quizData.length * 4;
      const accuracy = answeredCount = (correctCount + incorrectCount) > 0
        ? Math.round((correctCount / (correctCount + incorrectCount)) * 100)
        : 0;

      const timeElapsedSeconds = Math.floor((Date.now() - quizStartTime) / 1000);
      const elapsedMins = Math.floor(timeElapsedSeconds / 60);
      const elapsedSecs = timeElapsedSeconds % 60;
      const timeFormatted = `${String(elapsedMins).padStart(2, '0')}:${String(elapsedSecs).padStar
t(2, '0')}`;

      // Grade classification
      const percentage = Math.max(0, Math.round((rawScore / totalPossible) * 100));
      const gradeBadge = document.getElementById('gradeBadge');
      const resultFeedback = document.getElementById('resultFeedback');

      if (percentage >= 85) {
        gradeBadge.className = 'grade-badge excellent';
        gradeBadge.textContent = 'GRADE A+ • DISTINCTION (TOP TIER)';
        resultFeedback.textContent = 'Brilliant work! You demonstrated comprehensive mastery acros
s modern web development, DOM semantics, and JavaScript internals.';
      } else if (percentage >= 70) {
        gradeBadge.className = 'grade-badge good';
        gradeBadge.textContent = 'GRADE A • EXCELLENT PROFICIENCY';
        resultFeedback.textContent = 'Great achievement! You have a solid grasp of core web concep
ts with minor room for brush-ups in tricky JavaScript edge cases.';
      } else if (percentage >= 50) {
        gradeBadge.className = 'grade-badge fair';
        gradeBadge.textContent = 'GRADE B • SATISFACTORY';
        resultFeedback.textContent = 'Fair performance. Review the detailed solutions below to rei
nforce weak topics in CSS Box Model and Closure scopes.';
      } else {
        gradeBadge.className = 'grade-badge poor';
        gradeBadge.textContent = 'GRADE C • NEEDS IMPROVEMENT';
        resultFeedback.textContent = 'Keep practicing! Review the detailed question explanations b
elow and retake the quiz to improve your score.';
      }

      // Populate Stats
      document.getElementById('finalScoreDisplay').innerHTML = `${Math.max(0, rawScore)} <span>/
${totalPossible}</span>`;
      document.getElementById('statTotal').textContent = quizData.length;
      document.getElementById('statCorrect').textContent = correctCount;
      document.getElementById('statWrong').textContent = incorrectCount;
      document.getElementById('statAccuracy').textContent = `${accuracy}%`;
      document.getElementById('statTime').textContent = timeFormatted;

      // Populate Detailed Question Review
      const reviewContainer = document.getElementById('detailedReviewContainer');
      reviewContainer.innerHTML = '';

      quizData.forEach((q, idx) => {
```

```
        const itemDiv = document.createElement('div');
        const userChoice = userAnswers[idx];
        const isCorrect = userChoice === q.correct;
        const isUnanswered = userChoice === null;

        itemDiv.className = `review-item ${isCorrect ? 'correct' : (isUnanswered ? 'unanswered' :
'incorrect')}`;

        const statusTag = isCorrect
          ? '<span style="color: #34d399; font-weight: 700;">✅ Correct (+4)</span>'
          : (isUnanswered
              ? '<span style="color: #fbbf24; font-weight: 700;">⚠️ Unanswered (0)</span>'
              : '<span style="color: #fb7185; font-weight: 700;">❌ Incorrect (-1)</span>');

        itemDiv.innerHTML = `
          <div class="review-header">
            <span style="color: #818cf8; font-weight: 700;">Question ${idx + 1} &bull; ${q.topic}
</span>
            ${statusTag}
          </div>
          <div style="font-weight: 600; color: #ffffff; margin-bottom: 0.5rem;">${q.question}</div
>
          ${q.code ? `<pre class="code-snippet">${q.code}</pre>` : ''}
          <div style="font-size: 0.85rem; color: #cbd5e1; margin-top: 0.5rem; line-height: 1.6;">
            <div><strong>Your Answer:</strong> ${userChoice !== null ? q.options[userChoice] : '<e
m style="color: #94a3b8;">None Selected</em>'}</div>
            <div style="color: #34d399; margin-top: 0.2rem;"><strong>Correct Answer:</strong> ${q.
options[q.correct]}</div>
          </div>
          <div class="explanation-box">
            💡 <strong>Explanation:</strong> ${q.explanation}
          </div>
        `;
        reviewContainer.appendChild(itemDiv);
      });

      // Switch View
      document.getElementById('quiz-active-view').style.display = 'none';
      document.getElementById('result-view').style.display = 'block';
      window.scrollTo({ top: 0, behavior: 'smooth' });
    }

    // Restart Quiz
    function restartQuiz() {
      for (let i = 0; i < userAnswers.length; i++) {
        userAnswers[i] = null;
        markedReview[i] = false;
      }
      currentIdx = 0;
      timeRemaining = 300;
      quizStartTime = Date.now();

      document.getElementById('result-view').style.display = 'none';
      document.getElementById('quiz-active-view').style.display = 'block';
      initQuiz();
    }

    // Start on DOM ready
    window.addEventListener('DOMContentLoaded', () => {
      initQuiz();
    });
  </script>
</body>

</html>
```

Output :


CSIT248 Lab Work
JavaScript Quiz Engine
All Experiments
Experiment 15 • Module IV (JavaScript Quiz & Score Calculation)
CSIT248 Web Technology Assessment Quiz
Interactive testing interface with dynamic DOM rendering, live countdown timer, and automated result calculation
Subject: Advanced Web Technology
Time Remaining: 05:00
Question: 1 of 8
HTML5 SEMANTICS
+4 Marks / -1 Negative
1. Which of the following HTML5 elements is specifically designed to contain tangential content such as sidebars, glossaries, or related links?
<section>
<aside>
Question Palette
1
2
3
4
5
6
7
8
Answered
Marked for Review
Unanswered / Not Visited
Submit Quiz Now

Experiment No.: 19

Title : Proportional HTML Frames Layout Using Percentage Dimensions.

Aim : To create a multi-window web layout using HTML <frameset> and <frame> tags partitioned into proportional percentage rows and columns with hyperlinked target navigation.

Source Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description" content="Experiment 16: Design a web page to create frames using percen
tage - CSIT248 Advanced Web Technology">
  <title>Experiment 16 - HTML Frames Using Percentage (CSIT248)</title>
</head>

<!-- ========================================================================
     Master Frameset Layout Using Percentage Dimensions:
     - Outer Rows: 20% (Header), 70% (Middle Content Area), 10% (Footer)
     - Inner Columns (inside middle row): 25% (Left Menu), 75% (Right Display Area)
     ======================================================================== -->
<frameset rows="20%, 70%, 10%" border="3" bordercolor="#334155" framespacing="3">

  <!-- 1. Top Header Frame (20% Height) -->
  <frame src="frame_header.html" name="top_frame" noresize="noresize" scrolling="no" title="Top He
ader Frame (20%)">

  <!-- 2. Middle Nested Frameset: Split into 25% Navigation and 75% Content -->
  <frameset cols="25%, 75%" border="3" bordercolor="#334155" framespacing="3">
    <!-- Left Navigation Frame (25% Width) -->
    <frame src="frame_menu.html" name="menu_frame" scrolling="auto" title="Left Navigation Menu Fr
ame (25%)">

    <!-- Right Main Content Frame (75% Width, Target of menu links) -->
    <frame src="frame_content.html" name="content_frame" scrolling="auto" title="Right Main Conten
t Frame (75%)">
  </frameset>

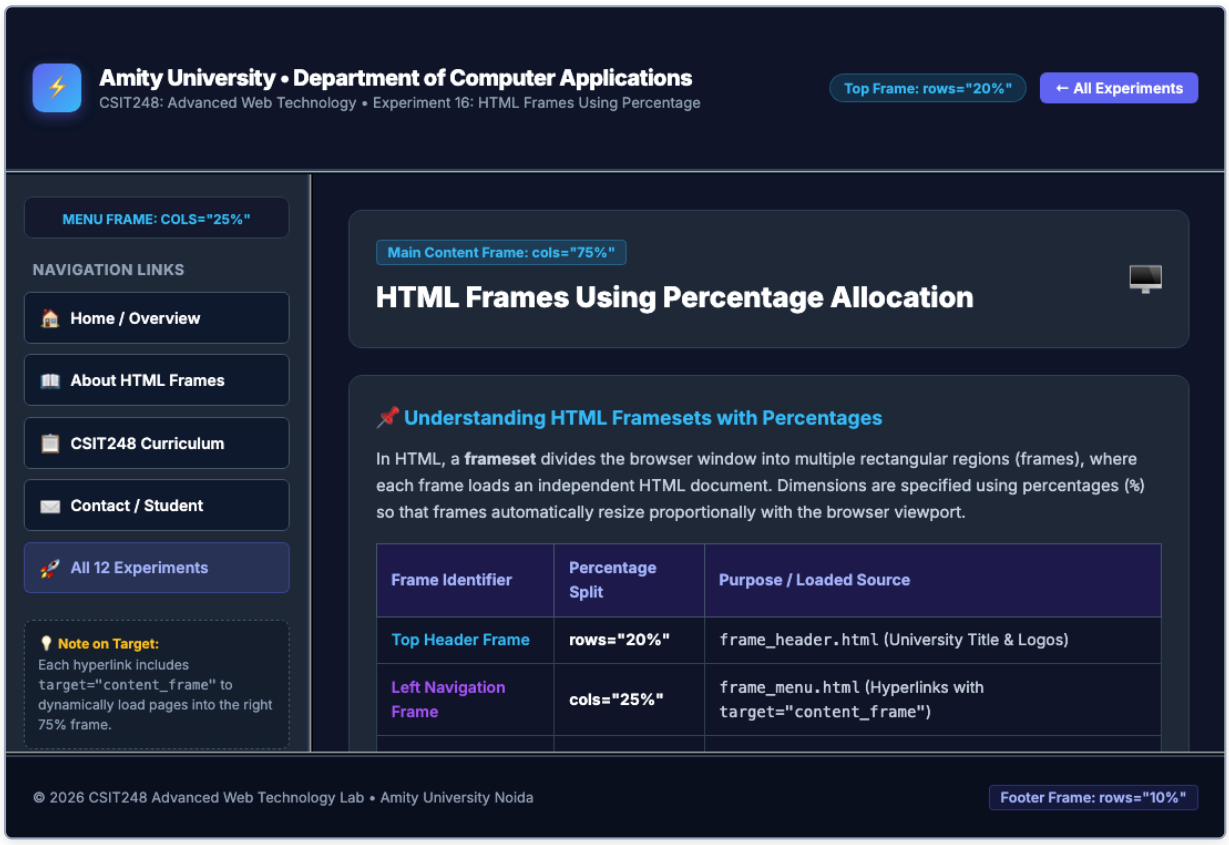
  <!-- 3. Bottom Footer Frame (10% Height) -->
  <frame src="frame_footer.html" name="footer_frame" noresize="noresize" scrolling="no" title="Bot
tom Footer Frame (10%)">

  <!-- Fallback for non-supporting browsers -->
  <noframes>
    <body style="margin: 0; padding: 2rem; font-family: 'Inter', system-ui, sans-serif; background
-color: #0f172a; color: #f8fafc; text-align: center;">
      <div style="max-width: 600px; margin: 3rem auto; background: #1e293b; border: 1px solid #334
155; border-radius: 12px; padding: 2rem;">
        <h2 style="color: #38bdf8; margin: 0 0 1rem 0;">HTML Frameset Notice</h2>
        <p style="color: #cbd5e1; font-size: 0.95rem; margin-bottom: 1.5rem;">
          Your current browser does not support traditional HTML <code>&lt;frameset&gt;</code> tag
s. You can access individual frame components below:
        </p>
        <div style="display: flex; flex-direction: column; gap: 0.75rem;">
          <a href="frame_header.html" style="color: #818cf8; text-decoration: none; font-weight: 6
00;">1. Top Header Frame (20%)</a>
          <a href="frame_menu.html" style="color: #818cf8; text-decoration: none; font-weight: 60
0;">2. Navigation Menu Frame (25%)</a>
          <a href="frame_content.html" style="color: #818cf8; text-decoration: none; font-weight:
```

```
600;">3. Main Content Frame (75%)</a>
          <a href="frame_footer.html" style="color: #818cf8; text-decoration: none; font-weight: 6
00;">4. Footer Frame (10%)</a>
        </div>
      </div>
    </body>
  </noframes>

</frameset>
</html>
```

Output :



Experiment No.: 20

Title : Responsive Menu Bar, DIV Partitions and IFRAME Sandboxing.

Aim : To construct a modern responsive web portal featuring a top horizontal menu bar, multi-column layout partitions, mobile hamburger toggle, and sandboxed <iframe> content loaders.

Source Code :

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description"
    content="Experiment 17: Design a web page with menu bar, different sections using DIV and IFRA
ME, and responsive layout - CSIT248 Advanced Web Technology">
  <title>Experiment 17 - Responsive Web Page with Menu Bar, DIV &amp;amp; IFRAME</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&display=swa
p" rel="stylesheet">
  <style>
    :root {
      --primary: #6366f1;
      --primary-hover: #4f46e5;
      --secondary: #06b6d4;
      --accent: #f43f5e;
      --success: #10b981;
      --bg-dark: #0f172a;
      --card-bg: #1e293b;
      --card-border: #334155;
      --text-main: #f8fafc;
      --text-muted: #94a3b8;
    }

    * {
      box-sizing: border-box;
      margin: 0;
      padding: 0;
    }

    html {
      scroll-behavior: smooth;
    }

    body {
      font-family: 'Inter', system-ui, -apple-system, sans-serif;
      background-color: var(--bg-dark);
      color: var(--text-main);
      line-height: 1.6;
      min-height: 100vh;
      display: flex;
      flex-direction: column;
    }

    /* Requirement a: MENU BAR (Responsive Navigation) */
    .site-nav {
      position: sticky;
      top: 0;
      z-index: 1000;
```

```
  background: rgba(15, 23, 42, 0.95);
  backdrop-filter: blur(14px);
  border-bottom: 1px solid #334155;
  padding: 0.85rem 2rem;
}

.nav-container {
  max-width: 1200px;
  margin: 0 auto;
  display: flex;
  justify-content: space-between;
  align-items: center;
}

.nav-brand {
  font-weight: 800;
  font-size: 1.15rem;
  color: #818cf8;
  text-decoration: none;
  display: flex;
  align-items: center;
  gap: 0.5rem;
}

.nav-brand span {
  background: rgba(99, 102, 241, 0.15);
  border: 1px solid rgba(99, 102, 241, 0.3);
  padding: 0.2rem 0.5rem;
  border-radius: 6px;
}

/* Desktop Navigation Links */
.menu-bar {
  display: flex;
  align-items: center;
  gap: 0.5rem;
  list-style: none;
}

.menu-item {
  position: relative;
}

.menu-link {
  color: #cbd5e1;
  text-decoration: none;
  font-size: 0.875rem;
  font-weight: 600;
  padding: 0.5rem 0.85rem;
  border-radius: 8px;
  transition: all 0.2s ease;
  display: flex;
  align-items: center;
  gap: 0.35rem;
}

.menu-link:hover,
.menu-link.active {
  color: #ffffff;
  background: rgba(99, 102, 241, 0.15);
}

/* Dropdown inside Menu Bar */
.dropdown-menu {
  display: none;
  position: absolute;
```

```
  top: 100%;
  left: 0;
  background: #1e293b;
  border: 1px solid #334155;
  border-radius: 10px;
  padding: 0.5rem;
  min-width: 190px;
  box-shadow: 0 10px 25px rgba(0, 0, 0, 0.4);
  z-index: 100;
}

.menu-item:hover .dropdown-menu {
  display: block;
}

.dropdown-item {
  display: block;
  color: #cbd5e1;
  text-decoration: none;
  font-size: 0.825rem;
  padding: 0.5rem 0.75rem;
  border-radius: 6px;
  font-weight: 500;
}

.dropdown-item:hover {
  background: rgba(99, 102, 241, 0.2);
  color: #ffffff;
}

/* Mobile Hamburger Button */
.mobile-toggle {
  display: none;
  background: transparent;
  border: 1px solid #334155;
  color: #ffffff;
  padding: 0.5rem;
  border-radius: 8px;
  cursor: pointer;
}

/* Mobile Nav Drawer */
.mobile-drawer {
  display: none;
  background: #0f172a;
  border-bottom: 1px solid #334155;
  padding: 1rem 1.5rem;
}

.mobile-drawer.open {
  display: flex;
  flex-direction: column;
  gap: 0.75rem;
}

.mobile-drawer a {
  color: #cbd5e1;
  text-decoration: none;
  font-size: 0.95rem;
  font-weight: 600;
  padding: 0.5rem 0;
  border-bottom: 1px solid #1e293b;
}

.mobile-drawer a:hover {
  color: #818cf8;
```

```
}

/* Viewport Preview Control Bar */
.viewport-bar {
  background: #1e1b4b;
  border-bottom: 1px solid #4338ca;
  padding: 0.5rem 1rem;
  display: flex;
  justify-content: center;
  align-items: center;
  gap: 1rem;
  flex-wrap: wrap;
  font-size: 0.8rem;
}

.viewport-btn {
  background: #312e81;
  border: 1px solid #6366f1;
  color: #e0e7ff;
  padding: 0.25rem 0.65rem;
  border-radius: 6px;
  font-size: 0.75rem;
  font-weight: 600;
  cursor: pointer;
  transition: all 0.2s;
}

.viewport-btn:hover,
.viewport-btn.active {
  background: #6366f1;
  color: #ffffff;
}

/* Main Container (Responsive Width) */
#page-wrapper {
  max-width: 1200px;
  width: 100%;
  margin: 0 auto;
  padding: 2rem 1.25rem;
  transition: max-width 0.3s ease;
  flex: 1;
}

/* Requirement b: DIFFERENT SECTIONS USING DIV */

/* Section 1: Hero Banner DIV */
.hero-section {
  background: linear-gradient(135deg, rgba(30, 41, 59, 0.8) 0%, rgba(15, 23, 42, 0.95) 100%);
  border: 1px solid #334155;
  border-radius: 20px;
  padding: 3rem 2rem;
  margin-bottom: 2.5rem;
  position: relative;
  overflow: hidden;
  box-shadow: 0 10px 30px rgba(0, 0, 0, 0.3);
}

.hero-badge {
  display: inline-block;
  background: rgba(99, 102, 241, 0.15);
  color: #a5b4fc;
  border: 1px solid rgba(99, 102, 241, 0.3);
  padding: 0.25rem 0.85rem;
  border-radius: 9999px;
  font-size: 0.85rem;
  font-weight: 600;
```

```
  margin-bottom: 1rem;
}

.hero-title {
  font-size: 2.5rem;
  font-weight: 800;
  color: #ffffff;
  line-height: 1.2;
  margin-bottom: 0.85rem;
  letter-spacing: -0.02em;
}

.hero-desc {
  color: #94a3b8;
  font-size: 1.1rem;
  max-width: 750px;
  margin-bottom: 1.75rem;
}

.hero-cta-group {
  display: flex;
  gap: 1rem;
  flex-wrap: wrap;
}

.btn-hero-primary {
  background: #6366f1;
  color: #ffffff;
  text-decoration: none;
  padding: 0.75rem 1.5rem;
  border-radius: 10px;
  font-weight: 700;
  font-size: 0.95rem;
  box-shadow: 0 4px 14px rgba(99, 102, 241, 0.4);
  transition: all 0.2s ease;
}

.btn-hero-primary:hover {
  background: #4f46e5;
  transform: translateY(-2px);
}

.btn-hero-secondary {
  background: #1e293b;
  border: 1px solid #475569;
  color: #f8fafc;
  text-decoration: none;
  padding: 0.75rem 1.5rem;
  border-radius: 10px;
  font-weight: 600;
  font-size: 0.95rem;
  transition: all 0.2s ease;
}

.btn-hero-secondary:hover {
  background: #334155;
}

/* Section 2: Course Modules Grid DIV */
.section-header {
  display: flex;
  justify-content: space-between;
  align-items: flex-end;
  margin-bottom: 1.5rem;
  flex-wrap: wrap;
  gap: 0.75rem;
```

```
}

.section-heading {
  font-size: 1.75rem;
  font-weight: 800;
  color: #ffffff;
  display: flex;
  align-items: center;
  gap: 0.5rem;
}

.section-heading span {
  color: var(--secondary);
}

.modules-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(320px, 1fr));
  gap: 1.5rem;
  margin-bottom: 3rem;
}

.module-card {
  background: #1e293b;
  border: 1px solid #334155;
  border-radius: 16px;
  padding: 1.75rem;
  transition: all 0.25s ease;
  display: flex;
  flex-direction: column;
  justify-content: space-between;
}

.module-card:hover {
  transform: translateY(-4px);
  border-color: #6366f1;
  box-shadow: 0 12px 25px rgba(99, 102, 241, 0.15);
}

.module-icon {
  width: 48px;
  height: 48px;
  border-radius: 12px;
  background: rgba(99, 102, 241, 0.15);
  border: 1px solid rgba(99, 102, 241, 0.3);
  display: flex;
  align-items: center;
  justify-content: center;
  font-size: 1.5rem;
  margin-bottom: 1rem;
}

.module-title {
  font-size: 1.2rem;
  font-weight: 700;
  color: #ffffff;
  margin-bottom: 0.5rem;
}

.module-desc {
  color: #94a3b8;
  font-size: 0.9rem;
  margin-bottom: 1rem;
}

.module-tags {
```

```
  display: flex;
  gap: 0.4rem;
  flex-wrap: wrap;
}

.tag-pill {
  font-size: 0.75rem;
  background: #0f172a;
  border: 1px solid #334155;
  color: #cbd5e1;
  padding: 0.15rem 0.55rem;
  border-radius: 4px;
}

/* Requirement b: DEDICATED IFRAME SECTION */
.iframe-showcase-section {
  background: #1e293b;
  border: 1px solid #334155;
  border-radius: 20px;
  padding: 2rem;
  margin-bottom: 3rem;
  box-shadow: 0 15px 35px rgba(0, 0, 0, 0.35);
}

.iframe-controls {
  display: flex;
  justify-content: space-between;
  align-items: center;
  flex-wrap: wrap;
  gap: 1rem;
  background: #0f172a;
  padding: 0.85rem 1.25rem;
  border-radius: 12px;
  border: 1px solid #334155;
  margin-bottom: 1.25rem;
}

.frame-tab-group {
  display: flex;
  gap: 0.5rem;
  flex-wrap: wrap;
}

.frame-tab-btn {
  background: #1e293b;
  border: 1px solid #334155;
  color: #cbd5e1;
  padding: 0.45rem 0.9rem;
  border-radius: 6px;
  font-size: 0.825rem;
  font-weight: 600;
  cursor: pointer;
  transition: all 0.2s;
}

.frame-tab-btn:hover {
  border-color: #6366f1;
  color: #ffffff;
}

.frame-tab-btn.active {
  background: #6366f1;
  color: #ffffff;
  border-color: #6366f1;
}
```

```
.frame-action-btns {
  display: flex;
  gap: 0.5rem;
}

.btn-frame-action {
  background: #334155;
  border: none;
  color: #cbd5e1;
  padding: 0.45rem 0.75rem;
  border-radius: 6px;
  font-size: 0.8rem;
  font-weight: 600;
  cursor: pointer;
  display: flex;
  align-items: center;
  gap: 0.3rem;
}

.btn-frame-action:hover {
  background: #475569;
  color: #ffffff;
}

/* The IFRAME element */
.iframe-container {
  width: 100%;
  height: 380px;
  border-radius: 12px;
  overflow: hidden;
  border: 2px solid #334155;
  background: #0f172a;
  box-shadow: inset 0 2px 10px rgba(0, 0, 0, 0.5);
}

.iframe-container iframe {
  width: 100%;
  height: 100%;
  border: none;
  display: block;
}

/* Section 4: Responsive Contact / Lab Queries DIV */
.contact-grid {
  display: grid;
  grid-template-columns: 1fr 1fr;
  gap: 1.5rem;
  margin-bottom: 2.5rem;
}

.contact-card {
  background: #1e293b;
  border: 1px solid #334155;
  border-radius: 16px;
  padding: 1.75rem;
}

.contact-card h3 {
  font-size: 1.25rem;
  color: #ffffff;
  margin-bottom: 1rem;
}

.contact-info-list {
  display: flex;
  flex-direction: column;
```

```
  gap: 0.85rem;
  color: #cbd5e1;
  font-size: 0.9rem;
}

.contact-item {
  display: flex;
  align-items: center;
  gap: 0.75rem;
}

.contact-form-group {
  margin-bottom: 1rem;
}

.contact-form-group label {
  display: block;
  font-size: 0.85rem;
  color: #cbd5e1;
  font-weight: 600;
  margin-bottom: 0.35rem;
}

.contact-input {
  width: 100%;
  background: #0f172a;
  border: 1px solid #334155;
  color: #ffffff;
  padding: 0.7rem 0.9rem;
  border-radius: 8px;
  font-family: inherit;
  font-size: 0.9rem;
}

.contact-input:focus {
  outline: none;
  border-color: #6366f1;
}

/* Requirement c: RESPONSIVE MEDIA QUERIES */
@media (max-width: 900px) {
  .menu-bar {
    display: none;
  }

  .mobile-toggle {
    display: block;
  }

  .contact-grid {
    grid-template-columns: 1fr;
  }

  .hero-title {
    font-size: 2rem;
  }
}

@media (max-width: 600px) {
  .hero-section {
    padding: 2rem 1.25rem;
  }

  .hero-title {
    font-size: 1.75rem;
  }
```

```
  .modules-grid {
    grid-template-columns: 1fr;
  }

  .iframe-container {
    height: 320px;
  }

  .site-nav {
    padding: 0.75rem 1rem;
  }
}

/* Technical Report Section */
.info-section {
  background: #1e293b;
  border: 1px solid #334155;
  border-radius: 16px;
  padding: 1.75rem;
  margin-top: 1rem;
}

.info-section h3 {
  font-size: 1.25rem;
  color: #ffffff;
  margin-bottom: 0.85rem;
  display: flex;
  align-items: center;
  gap: 0.5rem;
}

.tech-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(220px, 1fr));
  gap: 1rem;
  margin-top: 1rem;
}

.tech-card {
  background: #0f172a;
  border: 1px solid #334155;
  padding: 1rem;
  border-radius: 10px;
}

.tech-card h4 {
  font-size: 0.9rem;
  color: #818cf8;
  margin-bottom: 0.3rem;
}

.tech-card p {
  font-size: 0.825rem;
  color: #94a3b8;
}

footer {
  background: #090d16;
  border-top: 1px solid #334155;
  padding: 1.5rem;
  text-align: center;
  color: #94a3b8;
  font-size: 0.875rem;
  margin-top: auto;
}
```

```
  </style>
</head>

<body>

  <!-- Viewport Preview Simulator Bar -->
  <div class="viewport-bar">
    <span>📱 <strong>Test Responsiveness:</strong></span>
    <button type="button" class="viewport-btn active" onclick="setSimulatedWidth('100%', this)">
🖥️ Full Desktop (100%)</button>
    <button type="button" class="viewport-btn" onclick="setSimulatedWidth('768px', this)">📟 Tabl
et (768px)</button>
    <button type="button" class="viewport-btn" onclick="setSimulatedWidth('420px', this)">📱 Mobi
le (420px)</button>
    <span style="color: #a5b4fc; font-size: 0.75rem;">(Demonstrates requirement c: Fluid responsiv
eness)</span>
  </div>

  <!-- Requirement a: RESPONSIVE MENU BAR -->
  <header class="site-nav">
    <div class="nav-container">
      <a href="../index.html" class="nav-brand">
        <span>&larr;</span> CSIT248 Lab Work
      </a>

      <!-- Desktop Menu Bar -->
      <nav>
        <ul class="menu-bar">
          <li class="menu-item"><a href="#hero" class="menu-link active">Home</a></li>
          <li class="menu-item"><a href="#modules" class="menu-link">Course Modules</a></li>
          <li class="menu-item"><a href="#iframe-section" class="menu-link">IFRAME Workspace</a></
li>
          <li class="menu-item">
            <a href="javascript:void(0)" class="menu-link">
              Quick Nav ▾
            </a>
            <div class="dropdown-menu">
              <a href="#hero" class="dropdown-item">Top Banner</a>
              <a href="#modules" class="dropdown-item">Core Curriculum</a>
              <a href="#iframe-section" class="dropdown-item">Nested IFRAME</a>
              <a href="#contact" class="dropdown-item">Campus Office</a>
            </div>
          </li>
          <li class="menu-item"><a href="#contact" class="menu-link">Contact</a></li>
          <li class="menu-item">
            <a href="../index.html" class="menu-link" style="background: #6366f1; color: white;">A
ll Experiments</a>
          </li>
        </ul>
      </nav>

      <!-- Mobile Hamburger Toggle -->
      <button type="button" class="mobile-toggle" onclick="toggleMobileMenu()" aria-label="Toggle
navigation menu">
        <svg width="24" height="24" fill="none" stroke="currentColor" stroke-width="2" viewBox="0
0 24 24">
          <line x1="3" y1="12" x2="21" y2="12"></line>
          <line x1="3" y1="6" x2="21" y2="6"></line>
          <line x1="3" y1="18" x2="21" y2="18"></line>
        </svg>
      </button>
    </div>
  </header>

  <!-- Mobile Drawer Menu -->
  <div id="mobileDrawer" class="mobile-drawer">
```

```
    <a href="#hero" onclick="toggleMobileMenu()">🏠 Home</a>
    <a href="#modules" onclick="toggleMobileMenu()">📚 Course Modules (DIV Section)</a>
    <a href="#iframe-section" onclick="toggleMobileMenu()">🖼️ IFRAME Workspace Section</a>
    <a href="#contact" onclick="toggleMobileMenu()">📍 Contact &amp; Department Info</a>
    <a href="../index.html" style="color: #818cf8;">&larr; Return to All Experiments</a>
  </div>

  <!-- Page Wrapper with dynamic responsiveness testing -->
  <div id="page-wrapper">

    <!-- Requirement b: SECTION 1 (DIV: Hero Section) -->
    <div id="hero" class="hero-section">
      <span class="hero-badge">Experiment 17 &bull; Module V (Responsive Layout &amp; Frame Elements)</span>
      <h1 class="hero-title">Multi-Section Responsive Web Architecture</h1>
      <p class="hero-desc">
        Implementing a persistent responsive menu bar (with mobile collapse), distinct content divisions partitioned with <code>&lt;div&gt;</code>, and encapsulated sub-windows embedded via <code>&lt;iframe&gt;</code>.
      </p>
      <div class="hero-cta-group">
        <a href="#iframe-section" class="btn-hero-primary">Explore IFRAME Workspace &darr;</a>
        <a href="#modules" class="btn-hero-secondary">View DIV Content Sections</a>
      </div>
    </div>

    <!-- Requirement b: SECTION 2 (DIV: Structured Course Modules Grid) -->
    <div id="modules">
      <div class="section-header">
        <div>
          <h2 class="section-heading"><span>#</span> Section Divided with DIVs: Curriculum Grid</h2>
          <p style="color: #94a3b8; font-size: 0.95rem;">Structured divisions adapting dynamically to viewport bounds using CSS Grid</p>
        </div>
        <span style="font-size: 0.8rem; background: rgba(6, 182, 212, 0.15); color: #22d3ee; padding: 0.25rem 0.75rem; border-radius: 9999px; border: 1px solid rgba(6, 182, 212, 0.3);">
          &lt;div class="modules-grid"&gt;
        </span>
      </div>

      <div class="modules-grid">
        <div class="module-card">
          <div>
            <div class="module-icon">📄</div>
            <h3 class="module-title">1. Form Controls &amp; Validation</h3>
            <p class="module-desc">Client-level form data validation utilizing JavaScript RegEx and DOM manipulation to ensure front-line data integrity.</p>
          </div>
          <div class="module-tags">
            <span class="tag-pill">&lt;form&gt;</span>
            <span class="tag-pill">JavaScript RegEx</span>
            <span class="tag-pill">Exp 14</span>
          </div>
        </div>

        <div class="module-card">
          <div>
            <div class="module-icon">🎯</div>
            <h3 class="module-title">2. Dynamic Quiz Engines</h3>
            <p class="module-desc">Interactive assessment interfaces calculating scores, grade boundaries, and timing metrics entirely within client memory.</p>
          </div>
          <div class="module-tags">
            <span class="tag-pill">DOM Timers</span>
            <span class="tag-pill">Score Logic</span>
```

```
            <span class="tag-pill">Exp 15</span>
          </div>
        </div>

        <div class="module-card">
          <div>
            <div class="module-icon">🎨</div>
            <h3 class="module-title">3. CSS Pseudo-Classes</h3>
            <p class="module-desc">Styling elements based on user interaction, state, and structur
al positions without additional JavaScript overhead.</p>
          </div>
          <div class="module-tags">
            <span class="tag-pill">:hover</span>
            <span class="tag-pill">:focus-within</span>
            <span class="tag-pill">Exp 18</span>
          </div>
        </div>
      </div>
    </div>

    <!-- Requirement b: SECTION 3 (IFRAME: Dedicated Embedded Showcase Section) -->
    <div id="iframe-section" class="iframe-showcase-section">
      <div class="section-header">
        <div>
          <h2 class="section-heading"><span>#</span> Section Rendered with IFRAME: Embedded Sub-Wi
ndow</h2>
          <p style="color: #94a3b8; font-size: 0.95rem;">Encapsulated HTML document context embedd
ed inside the parent page using <code>&lt;iframe&gt;</code></p>
        </div>
        <span style="font-size: 0.8rem; background: rgba(16, 185, 129, 0.15); color: #34d399; padd
ing: 0.25rem 0.75rem; border-radius: 9999px; border: 1px solid rgba(16, 185, 129, 0.3);">
          &lt;iframe src="..."&gt;
        </span>
      </div>

      <!-- Iframe Controls Bar -->
      <div class="iframe-controls">
        <div class="frame-tab-group">
          <button type="button" class="frame-tab-btn active" onclick="switchFrame('frame_syllabus.
html', this)">
            📖 Syllabus Matrix Frame
          </button>
          <button type="button" class="frame-tab-btn" onclick="switchFrame('frame_sandbox.html', t
his)">
            ⚡ Interactive Sandbox Frame
          </button>
          <button type="button" class="frame-tab-btn" onclick="switchFrame('frame_notice.html', th
is)">
            📌 Department Circulars Frame
          </button>
        </div>

        <div class="frame-action-btns">
          <button type="button" class="btn-frame-action" onclick="reloadCurrentFrame()">
            🔄 Reload Frame
          </button>
          <button type="button" class="btn-frame-action" onclick="openFrameNewTab()">
            ↗️ Popout
          </button>
        </div>
      </div>

      <!-- Live IFRAME Window -->
      <div class="iframe-container">
        <iframe id="mainIframe" src="frame_syllabus.html" title="CSIT248 Embedded Content Frame">
          <p>Your browser does not support IFRAMEs. Please visit <a href="frame_syllabus.html">fra
```

```
me content</a> directly.</p>
        </iframe>
      </div>
      <p style="color: #64748b; font-size: 0.8rem; margin-top: 0.75rem; text-align: center;">
        Currently displaying: <code id="activeFrameLabel" style="color: #a5b4fc;">frame_syllabus.h
tml</code> &bull; Sandboxed &amp;amp; fully styled
      </p>
    </div>

    <!-- Requirement b: SECTION 4 (DIV: Responsive Contact & Department Section) -->
    <div id="contact" class="contact-grid">
      <div class="contact-card">
        <h3>📍 Department &amp;amp; Office Hours</h3>
        <p style="color: #94a3b8; font-size: 0.9rem; margin-bottom: 1.25rem;">
          Department of Computer Applications, Amity University Noida. Academic guidance and pract
ical laboratory sessions.
        </p>
        <div class="contact-info-list">
          <div class="contact-item">
            <span>🏛️</span>
            <span><strong>Location:</strong> E-3 Block, Amity University Campus, Sector 125, Noida
</span>
          </div>
          <div class="contact-item">
            <span>👨‍🎓</span>
            <span><strong>Student:</strong> Atul Anand (BCA Hons. with Research &bull; Sem III)</s
pan>
          </div>
          <div class="contact-item">
            <span>📧</span>
            <span><strong>Institutional Email:</strong> atulll0923@gmail.com</span>
          </div>
          <div class="contact-item">
            <span>⏰</span>
            <span><strong>Lab Timings:</strong> Monday to Friday, 09:00 AM - 05:00 PM</span>
          </div>
        </div>
      </div>

      <div class="contact-card">
        <h3>💬 Send Lab Query</h3>
        <form onsubmit="handleContactSubmit(event)">
          <div class="contact-form-group">
            <label for="cName">Your Name</label>
            <input type="text" id="cName" class="contact-input" placeholder="e.g. Rahul Sharma" re
quired>
          </div>
          <div class="contact-form-group">
            <label for="cEmail">Email Address</label>
            <input type="email" id="cEmail" class="contact-input" placeholder="e.g. rahul@amity.ed
u" required>
          </div>
          <div class="contact-form-group">
            <label for="cMsg">Question / Feedback</label>
            <textarea id="cMsg" class="contact-input" rows="3" placeholder="Enter inquiry regardin
g experiment 17..." required></textarea>
          </div>
          <button type="submit" class="btn-hero-primary" style="width: 100%; border: none; cursor:
pointer;">
            Dispatch Query &rarr;
          </button>
        </form>
      </div>
    </div>

    <!-- Academic Documentation -->
```

```
    <section class="info-section">
      <h3>
        <svg width="20" height="20" fill="none" stroke="#818cf8" stroke-width="2" viewBox="0 0 24
24">
          <circle cx="12" cy="12" r="10"></circle>
          <line x1="12" y1="16" x2="12" y2="12"></line>
          <line x1="12" y1="8" x2="12.01" y2="8"></line>
        </svg>
        Experiment 17 &bull; Technical Lab Report (Menu Bar, DIVs &amp; IFRAMEs)
      </h3>
      <p style="color: #cbd5e1; font-size: 0.9rem;">
        This experiment satisfies all 3 sub-requirements: a persistent responsive menu bar, struct
ured content partitions using <code>&lt;div&gt;</code>, embedded sandboxed sub-documents using <co
de>&lt;iframe&gt;</code>, and fluid responsive adaptability across viewports.
      </p>

      <div class="tech-grid">
        <div class="tech-card">
          <h4>a. Responsive Menu Bar</h4>
          <p>Constructed with semantic <code>&lt;nav&gt;</code>, <code>&lt;ul&gt;</code>, and <cod
e>&lt;li&gt;</code> with CSS dropdowns, collapsing into a slide drawer via media queries and JS to
ggle.</p>
        </div>
        <div class="tech-card">
          <h4>b. DIV vs IFRAME Partitioning</h4>
          <p>DIVs provide semantic, DOM-accessible page divisions within the host document. IFRAME
s isolate separate HTML browsing contexts with independent CSS and JS execution.</p>
        </div>
        <div class="tech-card">
          <h4>c. Responsive Architecture</h4>
          <p>Utilizes CSS Flexbox, modern CSS Grid with <code>minmax(320px, 1fr)</code>, and media
query breakpoints (900px and 600px) ensuring fluid mobile and desktop layouts.</p>
        </div>
      </div>
    </section>

  </div>

  <!-- Footer -->
  <footer>
    <p style="margin: 0 0 0.5rem 0; font-weight: 600; color: #cbd5e1;">
      Designed &amp; Developed by <strong style="color: #818cf8;">Atul Anand</strong> &bull; BCA
(Hons.) with Research (Semester III)
    </p>
    <p style="margin: 0; color: #64748b; font-size: 0.8rem;">
      Amity University Noida &bull; CSIT248: Advanced Web Technology &bull; Experiment 17 Complete
    </p>
  </footer>

  <!-- JavaScript for Interactive Controls -->
  <script>
    // 1. Mobile Menu Toggle
    function toggleMobileMenu() {
      const drawer = document.getElementById('mobileDrawer');
      drawer.classList.toggle('open');
    }

    // 2. Viewport Simulator
    function setSimulatedWidth(width, btn) {
      document.querySelectorAll('.viewport-btn').forEach(b => b.classList.remove('active'));
      btn.classList.add('active');
      const wrapper = document.getElementById('page-wrapper');
      wrapper.style.maxWidth = width;
    }

    // 3. IFRAME Tab Switching
```

```
    function switchFrame(url, btn) {
      document.querySelectorAll('.frame-tab-btn').forEach(b => b.classList.remove('active'));
      btn.classList.add('active');
      const iframe = document.getElementById('mainIframe');
      iframe.src = url;
      document.getElementById('activeFrameLabel').textContent = url;
    }

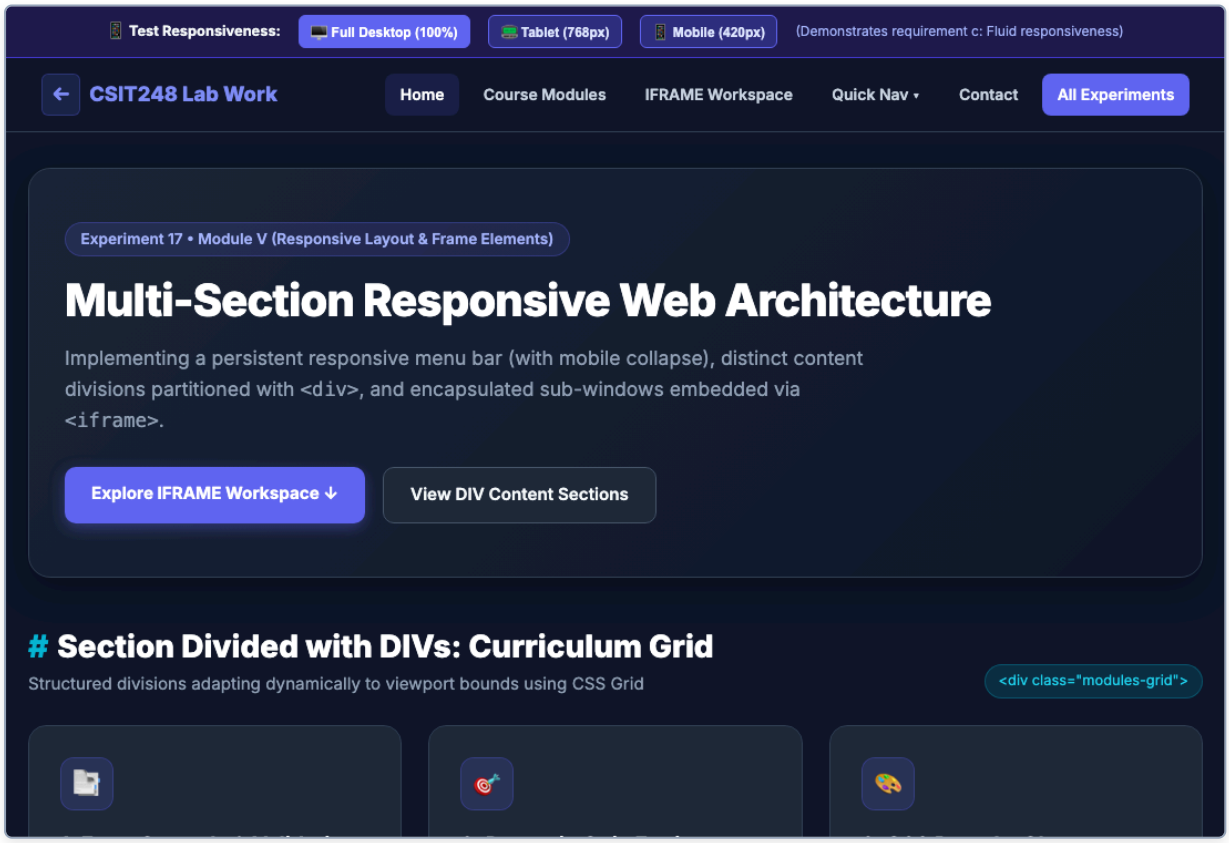
    // 4. Reload Current Frame
    function reloadCurrentFrame() {
      const iframe = document.getElementById('mainIframe');
      iframe.src = iframe.src;
    }

    // 5. Open Frame in New Window
    function openFrameNewTab() {
      const iframe = document.getElementById('mainIframe');
      window.open(iframe.src, '_blank');
    }

    // 6. Contact Form Submission
    function handleContactSubmit(e) {
      e.preventDefault();
      const name = document.getElementById('cName').value;
      alert(`Thank you, ${name}! Your query regarding Experiment 17 has been submitted successfull
y.`);
      e.target.reset();
    }
  </script>
</body>

</html>
```

Output :



Experiment No.: 21

Title : CSS Pseudo-Classes Interactive Studio & Sandbox.

Aim : To demonstrate interactive user-action and structural CSS pseudo-classes (:hover, :active, :focus, :first-child, :nth-child, :disabled) through a real-time interactive styling sandbox.

Source Code :

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description"
    content="Experiment 18: Design a webpage using CSS Pseudo Classes - CSIT248 Advanced Web Techn
ology">
  <title>Experiment 18 - CSS Pseudo-Classes Interactive Studio</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link
    href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&family=Fira+Cod
e:wght@400;600&display=swap"
    rel="stylesheet">
  <style>
    :root {
      --primary: #6366f1;
      --primary-hover: #4f46e5;
      --secondary: #06b6d4;
      --accent: #f43f5e;
      --success: #10b981;
      --warning: #f59e0b;
      --bg-dark: #0f172a;
      --card-bg: #1e293b;
      --card-border: #334155;
      --text-main: #f8fafc;
      --text-muted: #94a3b8;
    }

    * {
      box-sizing: border-box;
      margin: 0;
      padding: 0;
    }

    body {
      font-family: 'Inter', system-ui, -apple-system, sans-serif;
      background-color: var(--bg-dark);
      color: var(--text-main);
      line-height: 1.6;
      min-height: 100vh;
      display: flex;
      flex-direction: column;
    }

    /* Top Sticky Navigation */
    nav {
      position: sticky;
      top: 0;
      z-index: 100;
      background: rgba(15, 23, 42, 0.92);
```

```
  backdrop-filter: blur(12px);
  border-bottom: 1px solid #334155;
  padding: 1rem 2rem;
  display: flex;
  justify-content: space-between;
  align-items: center;
}

.nav-brand {
  font-weight: 800;
  font-size: 1.15rem;
  color: #818cf8;
  text-decoration: none;
  display: flex;
  align-items: center;
  gap: 0.5rem;
}

.nav-brand span {
  background: rgba(99, 102, 241, 0.15);
  border: 1px solid rgba(99, 102, 241, 0.3);
  padding: 0.2rem 0.5rem;
  border-radius: 6px;
}

.nav-links {
  display: flex;
  gap: 0.75rem;
  align-items: center;
}

.badge-pill {
  font-size: 0.8rem;
  background: rgba(99, 102, 241, 0.12);
  color: #a5b4fc;
  border: 1px solid rgba(99, 102, 241, 0.25);
  padding: 0.3rem 0.75rem;
  border-radius: 9999px;
  font-weight: 600;
}

.btn-nav {
  background: #6366f1;
  color: #ffffff;
  padding: 0.4rem 0.9rem;
  border-radius: 6px;
  text-decoration: none;
  font-size: 0.85rem;
  font-weight: 600;
  transition: all 0.2s ease;
}

.btn-nav:hover {
  background: #4f46e5;
}

/* Main Container */
main {
  flex: 1;
  max-width: 1100px;
  width: 100%;
  margin: 0 auto;
  padding: 2.5rem 1.25rem;
}

.exp-header {
```

```
  text-align: center;
  margin-bottom: 2.5rem;
}

.exp-pill {
  display: inline-block;
  background: rgba(99, 102, 241, 0.15);
  color: #a5b4fc;
  border: 1px solid rgba(99, 102, 241, 0.3);
  padding: 0.25rem 0.85rem;
  border-radius: 9999px;
  font-size: 0.85rem;
  font-weight: 600;
  margin-bottom: 0.75rem;
}

.exp-header h1 {
  font-size: 2.25rem;
  font-weight: 800;
  color: #ffffff;
  margin-bottom: 0.5rem;
  letter-spacing: -0.02em;
}

.exp-header p {
  color: #94a3b8;
  font-size: 1.05rem;
  max-width: 750px;
  margin: 0 auto;
}

/* Section Headings */
.group-title {
  font-size: 1.4rem;
  font-weight: 800;
  color: #ffffff;
  margin: 2.5rem 0 1rem;
  display: flex;
  align-items: center;
  gap: 0.65rem;
  padding-bottom: 0.5rem;
  border-bottom: 1px solid #334155;
}

.group-title span {
  background: rgba(99, 102, 241, 0.2);
  color: #818cf8;
  padding: 0.2rem 0.6rem;
  border-radius: 6px;
  font-size: 0.85rem;
}

/* Grid of Demo Cards */
.demo-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(320px, 1fr));
  gap: 1.5rem;
  margin-bottom: 2rem;
}

.demo-card {
  background: #1e293b;
  border: 1px solid #334155;
  border-radius: 16px;
  padding: 1.5rem;
  display: flex;
```

```
  flex-direction: column;
  justify-content: space-between;
  gap: 1rem;
  box-shadow: 0 4px 20px rgba(0, 0, 0, 0.25);
}

.demo-card-header {
  display: flex;
  justify-content: space-between;
  align-items: center;
}

.code-badge {
  font-family: 'Fira Code', monospace;
  font-size: 0.8rem;
  font-weight: 600;
  color: #38bdf8;
  background: #0f172a;
  border: 1px solid #334155;
  padding: 0.2rem 0.6rem;
  border-radius: 6px;
}

.demo-desc {
  font-size: 0.875rem;
  color: #94a3b8;
  margin-top: 0.25rem;
}

.demo-stage {
  background: #0f172a;
  border: 1px dashed #334155;
  border-radius: 12px;
  padding: 1.25rem;
  display: flex;
  flex-direction: column;
  align-items: center;
  justify-content: center;
  min-height: 120px;
  gap: 0.75rem;
}

/* ------------------------------------------------- */
/* 1. USER ACTION PSEUDO-CLASSES */
/* ------------------------------------------------- */

/* :hover demonstration */
.btn-hover-demo {
  background: #6366f1;
  color: #ffffff;
  border: none;
  padding: 0.75rem 1.5rem;
  border-radius: 8px;
  font-weight: 700;
  font-size: 0.95rem;
  cursor: pointer;
  transition: all 0.3s cubic-bezier(0.4, 0, 0.2, 1);
}

.btn-hover-demo:hover {
  background: #ec4899;
  transform: translateY(-4px) scale(1.05);
  box-shadow: 0 10px 25px rgba(236, 72, 153, 0.45);
}

/* :active demonstration */
```

```
.btn-active-demo {
  background: #10b981;
  color: #ffffff;
  border: none;
  padding: 0.75rem 1.5rem;
  border-radius: 8px;
  font-weight: 700;
  font-size: 0.95rem;
  cursor: pointer;
  box-shadow: 0 6px 0 #047857;
  transition: all 0.1s ease;
}

.btn-active-demo:active {
  transform: translateY(6px);
  box-shadow: 0 0 0 #047857;
  background: #059669;
}

/* :focus & :focus-visible demonstration */
.input-focus-demo {
  width: 100%;
  background: #1e293b;
  border: 2px solid #334155;
  color: #f8fafc;
  padding: 0.75rem 1rem;
  border-radius: 8px;
  outline: none;
  font-size: 0.9rem;
  font-family: inherit;
  transition: all 0.25s ease;
}

.input-focus-demo:focus {
  border-color: #38bdf8;
  background: #0f172a;
  box-shadow: 0 0 0 4px rgba(56, 189, 248, 0.25);
}

/* :focus-within demonstration */
.container-focus-within {
  width: 100%;
  background: #1e293b;
  border: 2px solid #334155;
  border-radius: 10px;
  padding: 1rem;
  transition: all 0.3s ease;
}

.container-focus-within:focus-within {
  border-color: #f59e0b;
  background: rgba(245, 158, 11, 0.08);
  box-shadow: 0 0 20px rgba(245, 158, 11, 0.2);
}

/* ------------------------------------------------- */
/* 2. FORM STATE PSEUDO-CLASSES */
/* ------------------------------------------------- */

/* :checked demonstration (pure CSS toggle switch) */
.toggle-switch {
  position: relative;
  display: inline-block;
  width: 60px;
  height: 32px;
}
```

```
.toggle-switch input {
  opacity: 0;
  width: 0;
  height: 0;
}

.toggle-slider {
  position: absolute;
  cursor: pointer;
  top: 0;
  left: 0;
  right: 0;
  bottom: 0;
  background-color: #334155;
  transition: 0.3s;
  border-radius: 34px;
}

.toggle-slider:before {
  position: absolute;
  content: "";
  height: 24px;
  width: 24px;
  left: 4px;
  bottom: 4px;
  background-color: white;
  transition: 0.3s;
  border-radius: 50%;
}

.toggle-switch input:checked+.toggle-slider {
  background-color: #6366f1;
}

.toggle-switch input:checked+.toggle-slider:before {
  transform: translateX(28px);
}

/* :disabled & :enabled demonstration */
.btn-disabled-demo {
  padding: 0.6rem 1.2rem;
  border-radius: 6px;
  font-weight: 600;
  font-size: 0.85rem;
  border: none;
}

.btn-disabled-demo:enabled {
  background: #3b82f6;
  color: white;
  cursor: pointer;
}

.btn-disabled-demo:disabled {
  background: #334155;
  color: #64748b;
  cursor: not-allowed;
  opacity: 0.6;
}

/* :valid & :invalid demonstration */
.form-val-demo {
  width: 100%;
  background: #1e293b;
  border: 2px solid #334155;
```

```
  color: #f8fafc;
  padding: 0.7rem 1rem;
  border-radius: 8px;
  outline: none;
  font-size: 0.9rem;
  transition: border-color 0.2s;
}

.form-val-demo:valid {
  border-color: #10b981;
  background: rgba(16, 185, 129, 0.08);
}

.form-val-demo:invalid:not(:placeholder-shown) {
  border-color: #f43f5e;
  background: rgba(244, 63, 94, 0.08);
}

/* :placeholder-shown demonstration */
.floating-box {
  position: relative;
  width: 100%;
}

.floating-input {
  width: 100%;
  background: #1e293b;
  border: 1.5px solid #334155;
  color: #fff;
  padding: 0.9rem 1rem 0.5rem;
  border-radius: 8px;
  outline: none;
  font-size: 0.95rem;
}

.floating-label {
  position: absolute;
  left: 1rem;
  top: 0.75rem;
  color: #94a3b8;
  font-size: 0.85rem;
  pointer-events: none;
  transition: all 0.2s ease;
}

.floating-input:not(:placeholder-shown)+.floating-label,
.floating-input:focus+.floating-label {
  top: 0.25rem;
  font-size: 0.65rem;
  color: #818cf8;
  font-weight: 700;
}

/* ------------------------------------------------- */
/* 3. STRUCTURAL PSEUDO-CLASSES */
/* ------------------------------------------------- */

/* Zebra stripe table with :nth-child(even) & :nth-child(odd) */
.zebra-table {
  width: 100%;
  border-collapse: collapse;
  font-size: 0.825rem;
  border-radius: 8px;
  overflow: hidden;
  border: 1px solid #334155;
}
```

```
.zebra-table th {
  background: #1e1b4b;
  color: #cbd5e1;
  padding: 0.6rem 0.85rem;
  text-align: left;
}

.zebra-table td {
  padding: 0.55rem 0.85rem;
  color: #e2e8f0;
}

.zebra-table tbody tr:nth-child(odd) {
  background-color: #0f172a;
}

.zebra-table tbody tr:nth-child(even) {
  background-color: #1e293b;
}

.zebra-table tbody tr:hover {
  background-color: rgba(99, 102, 241, 0.2);
}

/* :first-child & :last-child terminal list */
.terminal-list {
  list-style: none;
  width: 100%;
  display: flex;
  flex-direction: column;
  gap: 0.4rem;
}

.terminal-list li {
  padding: 0.4rem 0.75rem;
  background: #1e293b;
  border-radius: 6px;
  font-size: 0.8rem;
  font-family: 'Fira Code', monospace;
  color: #cbd5e1;
  display: flex;
  justify-content: space-between;
}

.terminal-list li:first-child {
  border-left: 4px solid #10b981;
  background: rgba(16, 185, 129, 0.1);
  color: #34d399;
  font-weight: 700;
}

.terminal-list li:last-child {
  border-left: 4px solid #f43f5e;
  background: rgba(244, 63, 94, 0.1);
  color: #fb7185;
  font-weight: 700;
}

/* :not() selector demonstration */
.not-list {
  display: flex;
  gap: 0.5rem;
  width: 100%;
  justify-content: center;
}
```

```css
.not-item {
  padding: 0.6rem 1rem;
  border-radius: 8px;
  font-size: 0.85rem;
  font-weight: 600;
}

.not-item:not(.special) {
  background: #0f172a;
  border: 1px dashed #64748b;
  color: #94a3b8;
}

.not-item.special {
  background: linear-gradient(135deg, #6366f1, #06b6d4);
  color: #ffffff;
  box-shadow: 0 4px 12px rgba(99, 102, 241, 0.35);
}

/* ------------------------------------------------- */
/* 4. LOCATION & :TARGET PSEUDO-CLASS (Pure CSS Modal) */
/* ------------------------------------------------- */
.btn-target-trigger {
  display: inline-block;
  background: #06b6d4;
  color: #0f172a;
  text-decoration: none;
  padding: 0.65rem 1.25rem;
  border-radius: 8px;
  font-weight: 700;
  font-size: 0.9rem;
  transition: all 0.2s;
}

.btn-target-trigger:hover {
  background: #22d3ee;
  transform: translateY(-2px);
}

.css-modal {
  display: none;
  position: fixed;
  top: 0;
  left: 0;
  right: 0;
  bottom: 0;
  background: rgba(15, 23, 42, 0.85);
  backdrop-filter: blur(8px);
  z-index: 9999;
  align-items: center;
  justify-content: center;
  padding: 1.5rem;
}

/* The core :target magic! Displays popup when URL hash matches #target-modal */
.css-modal:target {
  display: flex;
}

.css-modal-content {
  background: #1e293b;
  border: 1px solid #6366f1;
  border-radius: 16px;
  padding: 2rem;
  max-width: 500px;
```

```
  width: 100%;
  box-shadow: 0 25px 50px -12px rgba(0, 0, 0, 0.5);
  position: relative;
}

.css-modal-close {
  position: absolute;
  top: 1rem;
  right: 1.25rem;
  color: #94a3b8;
  text-decoration: none;
  font-size: 1.5rem;
  font-weight: 700;
}

.css-modal-close:hover {
  color: #ffffff;
}

/* Interactive Live Sandbox */
.sandbox-container {
  background: #1e293b;
  border: 1px solid #6366f1;
  border-radius: 16px;
  padding: 2rem;
  margin: 2.5rem 0;
  box-shadow: 0 10px 30px rgba(99, 102, 241, 0.15);
}

.sandbox-preview-box {
  background: #0f172a;
  border: 2px dashed #475569;
  border-radius: 12px;
  padding: 2rem;
  display: flex;
  justify-content: center;
  align-items: center;
  margin: 1.25rem 0;
}

.test-box {
  width: 220px;
  padding: 1.25rem;
  background: #334155;
  color: #ffffff;
  border-radius: 10px;
  text-align: center;
  font-weight: 700;
  font-size: 0.95rem;
  cursor: pointer;
  transition: all 0.3s ease;
  user-select: none;
}

/* Technical Info Section */
.info-section {
  background: #1e293b;
  border: 1px solid #334155;
  border-radius: 16px;
  padding: 1.75rem;
  margin-top: 2rem;
}

.info-section h3 {
  font-size: 1.25rem;
  color: #ffffff;
```

```
      margin-bottom: 0.85rem;
      display: flex;
      align-items: center;
      gap: 0.5rem;
    }

    .tech-grid {
      display: grid;
      grid-template-columns: repeat(auto-fit, minmax(220px, 1fr));
      gap: 1rem;
      margin-top: 1rem;
    }

    .tech-card {
      background: #0f172a;
      border: 1px solid #334155;
      padding: 1rem;
      border-radius: 10px;
    }

    .tech-card h4 {
      font-size: 0.9rem;
      color: #818cf8;
      margin-bottom: 0.3rem;
    }

    .tech-card p {
      font-size: 0.825rem;
      color: #94a3b8;
    }

    footer {
      background: #090d16;
      border-top: 1px solid #334155;
      padding: 1.5rem;
      text-align: center;
      color: #94a3b8;
      font-size: 0.875rem;
      margin-top: auto;
    }
  </style>
</head>

<body>

  <!-- Navigation -->
  <nav>
    <a href="../index.html" class="nav-brand">
      <span>&larr;</span> CSIT248 Lab Work
    </a>
    <div class="nav-links">
      <span class="badge-pill">CSS3 Selectors &amp; States</span>
      <a href="../index.html" class="btn-nav">All Experiments</a>
    </div>
  </nav>

  <main>

    <!-- Header -->
    <header class="exp-header">
      <div class="exp-pill">Experiment 18 &bull; Module III (CSS Pseudo-Classes &amp; State Select
ors)</div>
      <h1>CSS Pseudo-Classes Interactive Studio</h1>
      <p>Exploring dynamic UI state transformations, user actions, structural queries, and pure CS
S state machines</p>
    </header>
```

```
    <!-- 1. USER ACTION PSEUDO-CLASSES -->
    <h2 class="group-title">
      <span>Category 1</span> User Action &amp;amp; Interaction Pseudo-Classes
    </h2>
    <div class="demo-grid">

      <!-- :hover -->
      <div class="demo-card">
        <div>
          <div class="demo-card-header">
            <h3 style="font-size: 1.1rem; color: #ffffff;">1. Pointer Hover</h3>
            <span class="code-badge">:hover</span>
          </div>
          <p class="demo-desc">Applies styling when the user hovers over an element with a pointin
g device.</p>
        </div>
        <div class="demo-stage">
          <button type="button" class="btn-hover-demo">Hover Over Me ✨</button>
        </div>
        <div style="font-size: 0.75rem; color: #64748b; font-family: monospace;">button:hover { tr
ansform:
          translateY(-4px); }</div>
      </div>

      <!-- :active -->
      <div class="demo-card">
        <div>
          <div class="demo-card-header">
            <h3 style="font-size: 1.1rem; color: #ffffff;">2. Element Activation</h3>
            <span class="code-badge">:active</span>
          </div>
          <p class="demo-desc">Matches while the mouse button is actively held down (press down).
</p>
        </div>
        <div class="demo-stage">
          <button type="button" class="btn-active-demo">Press &amp;amp; Hold Me 🔘</button>
        </div>
        <div style="font-size: 0.75rem; color: #64748b; font-family: monospace;">button:active { t
ransform:
          translateY(6px); }</div>
      </div>

      <!-- :focus -->
      <div class="demo-card">
        <div>
          <div class="demo-card-header">
            <h3 style="font-size: 1.1rem; color: #ffffff;">3. Input Focus</h3>
            <span class="code-badge">:focus</span>
          </div>
          <p class="demo-desc">Triggers when an interactive field accepts keyboard or mouse focus.
</p>
        </div>
        <div class="demo-stage">
          <input type="text" class="input-focus-demo" placeholder="Click or Tab here to focus...">
        </div>
        <div style="font-size: 0.75rem; color: #64748b; font-family: monospace;">input:focus { bor
der-color: #38bdf8; }
        </div>
      </div>

      <!-- :focus-within -->
      <div class="demo-card">
        <div>
          <div class="demo-card-header">
            <h3 style="font-size: 1.1rem; color: #ffffff;">4. Ancestor Focus</h3>
```

```
            <span class="code-badge">:focus-within</span>
          </div>
          <p class="demo-desc">Highlights the parent wrapper if ANY internal descendant has focus.
</p>
        </div>
        <div class="demo-stage">
          <div class="container-focus-within">
            <div style="font-size: 0.75rem; color: #94a3b8; margin-bottom: 0.35rem;">Parent Contai
ner</div>
            <input type="text" class="input-focus-demo" placeholder="Focus child input..."
              style="padding: 0.4rem 0.75rem;">
          </div>
        </div>
        <div style="font-size: 0.75rem; color: #64748b; font-family: monospace;">.parent:focus-wit
hin { border-color:
          #f59e0b; }</div>
      </div>

    </div>

    <!-- 2. FORM STATE PSEUDO-CLASSES -->
    <h2 class="group-title">
      <span>Category 2</span> Form State &amp;amp; Validation Pseudo-Classes
    </h2>
    <div class="demo-grid">

      <!-- :checked -->
      <div class="demo-card">
        <div>
          <div class="demo-card-header">
            <h3 style="font-size: 1.1rem; color: #ffffff;">5. Selection State</h3>
            <span class="code-badge">:checked</span>
          </div>
          <p class="demo-desc">Pure CSS toggle switch created without JavaScript using checkbox <c
ode>:checked</code>.
          </p>
        </div>
        <div class="demo-stage">
          <label class="toggle-switch">
            <input type="checkbox">
            <span class="toggle-slider"></span>
          </label>
          <span style="font-size: 0.8rem; color: #cbd5e1;">Click toggle to toggle state</span>
        </div>
        <div style="font-size: 0.75rem; color: #64748b; font-family: monospace;">input:checked + .
slider { background:
          #6366f1; }</div>
      </div>

      <!-- :disabled & :enabled -->
      <div class="demo-card">
        <div>
          <div class="demo-card-header">
            <h3 style="font-size: 1.1rem; color: #ffffff;">6. Availability States</h3>
            <span class="code-badge">:disabled / :enabled</span>
          </div>
          <p class="demo-desc">Styles inputs based on user interactability and form enabling.</p>
        </div>
        <div class="demo-stage" style="flex-direction: row;">
          <button type="button" class="btn-disabled-demo">Enabled</button>
          <button type="button" class="btn-disabled-demo" disabled>Disabled</button>
        </div>
        <div style="font-size: 0.75rem; color: #64748b; font-family: monospace;">button:disabled {
cursor: not-allowed;
          }</div>
      </div>
```

```
      <!-- :valid & :invalid -->
      <div class="demo-card">
        <div>
          <div class="demo-card-header">
            <h3 style="font-size: 1.1rem; color: #ffffff;">7. Format Validation</h3>
            <span class="code-badge">:valid / :invalid</span>
          </div>
          <p class="demo-desc">Validates email syntax in real-time purely through CSS selectors.</
p>
        </div>
        <div class="demo-stage">
          <input type="email" class="form-val-demo" placeholder="Type email: test@domain.com" requ
ired>
        </div>
        <div style="font-size: 0.75rem; color: #64748b; font-family: monospace;">input:valid { bor
der-color: #10b981; }
        </div>
      </div>

      <!-- :placeholder-shown -->
      <div class="demo-card">
        <div>
          <div class="demo-card-header">
            <h3 style="font-size: 1.1rem; color: #ffffff;">8. Floating Label</h3>
            <span class="code-badge">:placeholder-shown</span>
          </div>
          <p class="demo-desc">CSS-only floating label detecting whether a placeholder is visible.
</p>
        </div>
        <div class="demo-stage">
          <div class="floating-box">
            <input type="text" id="floatingInput" class="floating-input" placeholder=" ">
            <label for="floatingInput" class="floating-label">Enter Student Roll Number</label>
          </div>
        </div>
        <div style="font-size: 0.75rem; color: #64748b; font-family: monospace;">input:not(:placeh
older-shown) + label {
          top: 2px; }</div>
      </div>


    </div>

    <!-- 3. STRUCTURAL PSEUDO-CLASSES -->
    <h2 class="group-title">
      <span>Category 3</span> Tree-Structural &amp;amp; Filter Pseudo-Classes
    </h2>
    <div class="demo-grid">

      <!-- :nth-child(even) & (odd) -->
      <div class="demo-card">
        <div>
          <div class="demo-card-header">
            <h3 style="font-size: 1.1rem; color: #ffffff;">9. Zebra Striping</h3>
            <span class="code-badge">:nth-child(even/odd)</span>
          </div>
          <p class="demo-desc">Alternates table background shading dynamically without hardcoded c
lasses.</p>
        </div>
        <div class="demo-stage" style="padding: 0.75rem;">
          <table class="zebra-table">
            <thead>
              <tr>
                <th>Course</th>
                <th>Semester</th>
```

```
                <th>Marks</th>
              </tr>
            </thead>
            <tbody>
              <tr>
                <td>HTML5 Forms</td>
                <td>Sem III</td>
                <td>98%</td>
              </tr>
              <tr>
                <td>CSS Selectors</td>
                <td>Sem III</td>
                <td>95%</td>
              </tr>
              <tr>
                <td>JavaScript DOM</td>
                <td>Sem III</td>
                <td>94%</td>
              </tr>
              <tr>
                <td>DHTML Records</td>
                <td>Sem III</td>
                <td>96%</td>
              </tr>
            </tbody>
          </table>
        </div>
        <div style="font-size: 0.75rem; color: #64748b; font-family: monospace;">tr:nth-child(eve
n) { background:
          #1e293b; }</div>
      </div>

      <!-- :first-child & :last-child -->
      <div class="demo-card">
        <div>
          <div class="demo-card-header">
            <h3 style="font-size: 1.1rem; color: #ffffff;">10. Boundary Selectors</h3>
            <span class="code-badge">:first-child / :last-child</span>
          </div>
          <p class="demo-desc">Customizes boundary items (first item = green, last item = red).</p
>
        </div>
        <div class="demo-stage" style="padding: 0.75rem;">
          <ul class="terminal-list">
            <li><span>Init System (1)</span> <span>[FIRST-CHILD]</span></li>
            <li><span>Load Modules (2)</span> <span>[INTERMEDIATE]</span></li>
            <li><span>Validate State (3)</span> <span>[INTERMEDIATE]</span></li>
            <li><span>Shutdown Routine (4)</span> <span>[LAST-CHILD]</span></li>
          </ul>
        </div>
        <div style="font-size: 0.75rem; color: #64748b; font-family: monospace;">li:first-child {
border-left-color:
          #10b981; }</div>
      </div>

      <!-- :not() -->
      <div class="demo-card">
        <div>
          <div class="demo-card-header">
            <h3 style="font-size: 1.1rem; color: #ffffff;">11. Negation Selector</h3>
            <span class="code-badge">:not(.special)</span>
          </div>
          <p class="demo-desc">Selects every sibling element that does NOT match the specified cla
ss.</p>
        </div>
        <div class="demo-stage">
```

```
          <div class="not-list">
            <div class="not-item">Item 1</div>
            <div class="not-item special">Special ★</div>
            <div class="not-item">Item 3</div>
          </div>
        </div>
        <div style="font-size: 0.75rem; color: #64748b; font-family: monospace;">.item:not(.specia
l) { border: 1px
          dashed; }</div>
      </div>

      <!-- :target -->
      <div class="demo-card">
        <div>
          <div class="demo-card-header">
            <h3 style="font-size: 1.1rem; color: #ffffff;">12. Hash Anchor Target</h3>
            <span class="code-badge">:target</span>
          </div>
          <p class="demo-desc">Pure CSS modal overlay activated when URL fragment matches element
ID.</p>
        </div>
        <div class="demo-stage">
          <a href="#target-modal" class="btn-target-trigger">🚀 Open Pure CSS Modal</a>
        </div>
        <div style="font-size: 0.75rem; color: #64748b; font-family: monospace;">.modal:target { d
isplay: flex; }</div>
      </div>

    </div>

    <!-- PURE CSS MODAL POWERED BY :target (No JS Required) -->
    <div id="target-modal" class="css-modal">
      <div class="css-modal-content">
        <a href="#" class="css-modal-close" title="Close Modal">&times;</a>
        <span
          style="background: rgba(6, 182, 212, 0.15); color: #22d3ee; padding: 0.2rem 0.65rem; bor
der-radius: 9999px; font-size: 0.75rem; font-weight: 700;">
          STATE: :TARGET ACTIVATED
        </span>
        <h3 style="font-size: 1.4rem; color: #ffffff; margin: 0.5rem 0;">Pure CSS Modal Activated!
</h3>
        <p style="color: #cbd5e1; font-size: 0.9rem; margin-bottom: 1rem;">
          This dialog appears entirely without executing a single line of JavaScript. When you cli
cked the button, the
          browser URL updated to <code>#target-modal</code>, matching the CSS selector
          <code>#target-modal:target</code>.
        </p>
        <div
          style="background: #0f172a; padding: 0.75rem; border-radius: 8px; border: 1px solid #334
155; font-family: monospace; font-size: 0.8rem; color: #818cf8; margin-bottom: 1.25rem;">
          #target-modal:target { display: flex; }
        </div>
        <a href="#" class="btn-nav" style="display: inline-block; padding: 0.6rem 1.25rem;">
          Close Modal &larr;
        </a>
      </div>
    </div>

    <!-- INTERACTIVE CSS SANDBOX -->
    <section class="sandbox-container">
      <div style="text-align: center; margin-bottom: 1.5rem;">
        <span
          style="background: rgba(99, 102, 241, 0.2); color: #a5b4fc; padding: 0.25rem 0.75rem; bo
rder-radius: 9999px; font-size: 0.8rem; font-weight: 700;">
          LABORATORY WORKBENCH
        </span>
```

```
        <h2 style="font-size: 1.75rem; color: #ffffff; margin: 0.5rem 0 0.25rem;">Live Pseudo-Clas
s Sandbox</h2>
        <p style="color: #94a3b8; font-size: 0.95rem;">Test how different pseudo-classes transform
this target element
          in real time</p>
      </div>

      <div style="display: flex; gap: 0.5rem; justify-content: center; flex-wrap: wrap;">
        <button type="button" class="btn-nav" onclick="applySandboxClass('hover-mode')">Simulate :
hover Glow</button>
        <button type="button" class="btn-nav" onclick="applySandboxClass('active-mode')"
          style="background: #10b981;">Simulate :active Press</button>
        <button type="button" class="btn-nav" onclick="applySandboxClass('focus-mode')"
          style="background: #06b6d4;">Simulate :focus Ring</button>
        <button type="button" class="btn-nav" onclick="resetSandbox()" style="background: #47556
9;">Reset
          Element</button>
      </div>

      <div class="sandbox-preview-box">
        <div id="sandboxBox" class="test-box">
          Interactive Node
        </div>
      </div>

      <p id="sandboxStatus" style="text-align: center; color: #cbd5e1; font-size: 0.85rem; font-fa
mily: monospace;">
        Status: Standard default state (No pseudo-class active)
      </p>
    </section>

    <!-- Technical & Academic Documentation -->
    <section class="info-section">
      <h3>
        <svg width="20" height="20" fill="none" stroke="#818cf8" stroke-width="2" viewBox="0 0 24
24">
          <circle cx="12" cy="12" r="10"></circle>
          <line x1="12" y1="16" x2="12" y2="12"></line>
          <line x1="12" y1="8" x2="12.01" y2="8"></line>
        </svg>
        Experiment 18 &bull; Technical Lab Report (CSS Pseudo-Classes Architecture)
      </h3>
      <p style="color: #cbd5e1; font-size: 0.9rem;">
        CSS Pseudo-classes allow developers to dynamically style elements based on their runtime s
tate, user
        interactions, or structural position in the DOM hierarchy without injecting redundant inli
ne classes or DOM
        mutation scripts.
      </p>

      <div class="tech-grid">
        <div class="tech-card">
          <h4>1. Dynamic State Selectors</h4>
          <p>Pseudo-classes like <code>:hover</code>, <code>:active</code>, and <code>:focus</code
> listen directly to
            browser input events, enabling fluid reactive UI styling with zero JavaScript overhea
d.</p>
        </div>
        <div class="tech-card">
          <h4>2. Tree Structural Selectors</h4>
          <p>Patterns like <code>:nth-child(even)</code>, <code>:first-child</code>, and <code>:no
t()</code> compute
            document tree positions mathematically, perfect for tables and alternating lists.</p>
        </div>
        <div class="tech-card">
          <h4>3. Pure CSS State Machines</h4>
```

```
          <p>By leveraging <code>:checked</code> and <code>:target</code>, developers can implemen
t accessible toggles,
            modals, and tabs purely through declarative CSS rules.</p>
        </div>
        <div class="tech-card">
          <h4>4. Specificity &amp;amp; Performance</h4>
          <p>Pseudo-classes contribute 0,1,0 to CSS specificity (equivalent to a class selector),
and are heavily
            accelerated by modern GPU styling pipelines.</p>
        </div>
      </div>
    </section>

  </main>

  <!-- Footer -->
  <footer>
    <p style="margin: 0 0 0.5rem 0; font-weight: 600; color: #cbd5e1;">
      Designed &amp;amp; Developed by <strong style="color: #818cf8;">Atul Anand</strong> &bull; BCA
(Hons.) with Research
      (Semester III)
    </p>
    <p style="margin: 0; color: #64748b; font-size: 0.8rem;">
      Amity University Noida &bull; CSIT248: Advanced Web Technology &bull; Experiment 18 Complete
    </p>
  </footer>

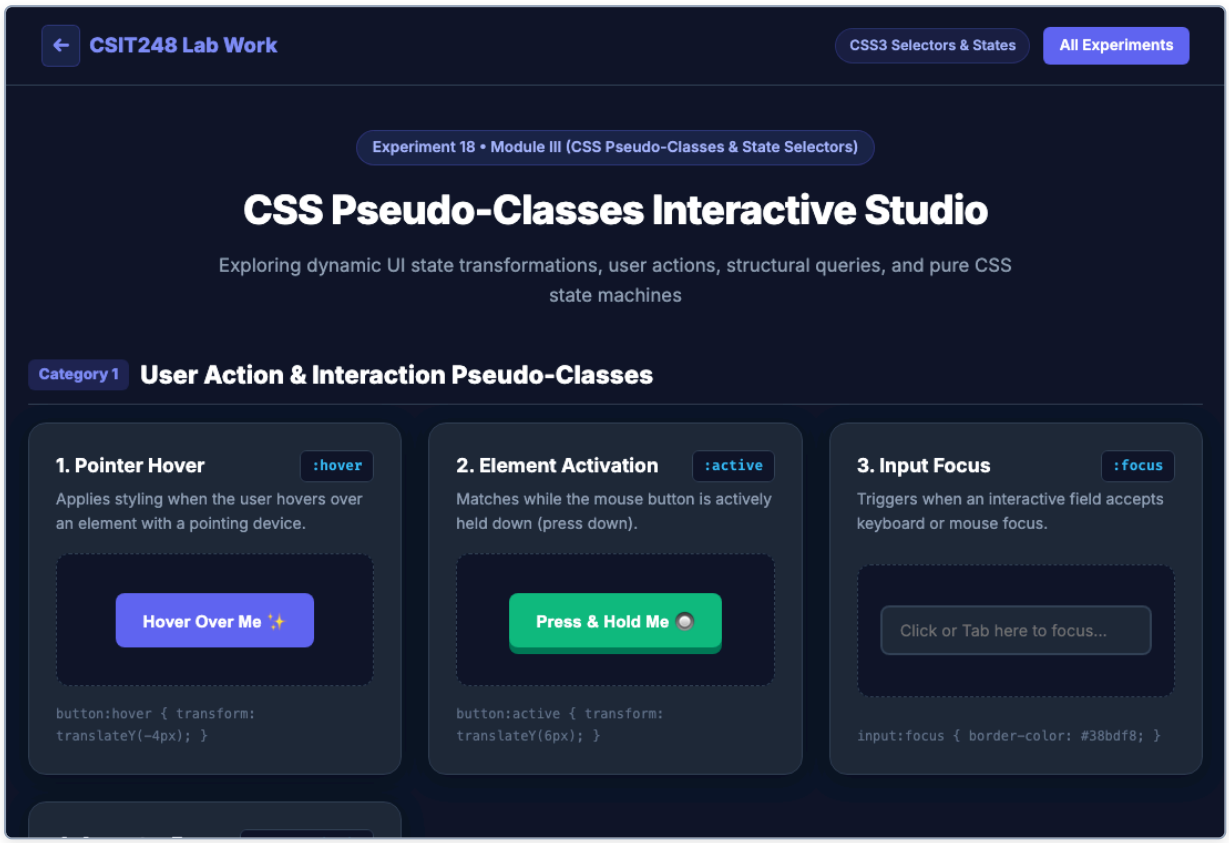
  <script>
    function applySandboxClass(mode) {
      const box = document.getElementById('sandboxBox');
      const status = document.getElementById('sandboxStatus');

      if (mode === 'hover-mode') {
        box.style.transform = 'translateY(-8px) scale(1.06)';
        box.style.background = 'linear-gradient(135deg, #6366f1, #ec4899)';
        box.style.boxShadow = '0 15px 30px rgba(236, 72, 153, 0.45)';
        status.textContent = 'Active Rule: element:hover { transform: translateY(-8px); backgroun
d: #ec4899; }';
      } else if (mode === 'active-mode') {
        box.style.transform = 'translateY(6px) scale(0.95)';
        box.style.background = '#059669';
        box.style.boxShadow = '0 2px 5px rgba(0,0,0,0.5)';
        status.textContent = 'Active Rule: element:active { transform: translateY(6px); backgroun
d: #059669; }';
      } else if (mode === 'focus-mode') {
        box.style.transform = 'none';
        box.style.background = '#0f172a';
        box.style.border = '2px solid #38bdf8';
        box.style.boxShadow = '0 0 0 5px rgba(56, 189, 248, 0.35)';
        status.textContent = 'Active Rule: element:focus { outline: none; box-shadow: 0 0 0 5px #3
8bdf8; }';
      }
    }

    function resetSandbox() {
      const box = document.getElementById('sandboxBox');
      box.style.transform = '';
      box.style.background = '';
      box.style.border = '';
      box.style.boxShadow = '';
      document.getElementById('sandboxStatus').textContent = 'Status: Standard default state (No p
seudo-class active)';
    }
  </script>
</body>
```

```
</html>
```

Output :


CSIT248 Lab Work
CSS3 Selectors & States
All Experiments
Experiment 18 • Module III (CSS Pseudo-Classes & State Selectors)
CSS Pseudo-Classes Interactive Studio
Exploring dynamic UI state transformations, user actions, structural queries, and pure CSS state machines
Category 1
User Action & Interaction Pseudo-Classes
1. Pointer Hover
:hover
Applies styling when the user hovers over an element with a pointing device.
Hover Over Me ✨
button:hover { transform: translateY(-4px); }
2. Element Activation
:active
Matches while the mouse button is actively held down (press down).
Press & Hold Me
button:active { transform: translateY(6px); }
3. Input Focus
:focus
Triggers when an interactive field accepts keyboard or mouse focus.
Click or Tab here to focus...
input:focus { border-color: #38bdf8; }

Experiment No.: 22

Title : Add New Record in CUSTOMER Database using PHP and MySQL.

Aim : To write and execute a secure PHP script to insert new customer records into a MySQL database utilizing prepared statements, input sanitization, and SQL injection defense.

Source Code :

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description"
    content="Experiment 19: Write a PHP Script to add new record in CUSTOMER database - CSIT248 Ad
vanced Web Technology">
  <title>Experiment 19 - Add New Record in CUSTOMER Database (PHP &amp;amp; MySQL)</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link
    href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&family=Fira+Cod
e:wght@400;500;600&display=swap"
    rel="stylesheet">
  <style>
    :root {
      --primary: #6366f1;
      --primary-hover: #4f46e5;
      --secondary: #06b6d4;
      --accent: #f43f5e;
      --success: #10b981;
      --warning: #f59e0b;
      --danger: #ef4444;
      --bg-dark: #0f172a;
      --card-bg: #1e293b;
      --card-border: #334155;
      --text-main: #f8fafc;
      --text-muted: #94a3b8;
    }

    * {
      box-sizing: border-box;
      margin: 0;
      padding: 0;
    }

    body {
      font-family: 'Inter', system-ui, -apple-system, sans-serif;
      background-color: var(--bg-dark);
      color: var(--text-main);
      line-height: 1.6;
      min-height: 100vh;
      display: flex;
      flex-direction: column;
    }

    /* Sticky Navigation */
    nav {
      position: sticky;
      top: 0;
      z-index: 100;
```

```
  background: rgba(15, 23, 42, 0.92);
  backdrop-filter: blur(12px);
  border-bottom: 1px solid var(--card-border);
  padding: 1rem 2rem;
  display: flex;
  justify-content: space-between;
  align-items: center;
}

.nav-brand {
  font-weight: 800;
  font-size: 1.15rem;
  color: #818cf8;
  text-decoration: none;
  display: flex;
  align-items: center;
  gap: 0.5rem;
}

.nav-brand span {
  background: rgba(99, 102, 241, 0.15);
  border: 1px solid rgba(99, 102, 241, 0.3);
  padding: 0.2rem 0.5rem;
  border-radius: 6px;
}

.nav-links {
  display: flex;
  align-items: center;
  gap: 1rem;
}

.nav-btn {
  background: var(--primary);
  color: #ffffff;
  padding: 0.45rem 1rem;
  border-radius: 6px;
  text-decoration: none;
  font-size: 0.85rem;
  font-weight: 600;
  transition: background 0.2s;
}

.nav-btn:hover {
  background: var(--primary-hover);
}

/* Main Container */
main {
  max-width: 1200px;
  width: 100%;
  margin: 0 auto;
  padding: 2.5rem 1.5rem;
  flex: 1;
}

/* Page Header */
.page-header {
  text-align: center;
  margin-bottom: 2rem;
}

.badge-pill {
  display: inline-block;
  background: rgba(99, 102, 241, 0.15);
  color: #a5b4fc;
```

```
  border: 1px solid rgba(99, 102, 241, 0.3);
  padding: 0.3rem 0.9rem;
  border-radius: 9999px;
  font-size: 0.825rem;
  font-weight: 600;
  margin-bottom: 0.75rem;
}

.page-title {
  font-size: 2.25rem;
  font-weight: 800;
  color: #ffffff;
  margin-bottom: 0.5rem;
}

.page-subtitle {
  color: var(--text-muted);
  font-size: 1.05rem;
  max-width: 820px;
  margin: 0 auto;
}

/* Tab Controller */
.tabs-nav {
  display: flex;
  gap: 0.5rem;
  border-bottom: 1px solid var(--card-border);
  margin-bottom: 2rem;
  overflow-x: auto;
  padding-bottom: 0.25rem;
}

.tab-btn {
  background: transparent;
  border: none;
  color: var(--text-muted);
  padding: 0.75rem 1.25rem;
  font-size: 0.95rem;
  font-weight: 600;
  cursor: pointer;
  border-radius: 8px 8px 0 0;
  display: flex;
  align-items: center;
  gap: 0.5rem;
  transition: all 0.2s;
}

.tab-btn:hover {
  color: #ffffff;
  background: rgba(255, 255, 255, 0.05);
}

.tab-btn.active {
  color: #38bdf8;
  background: var(--card-bg);
  border-bottom: 2px solid #38bdf8;
}

.tab-panel {
  display: none;
}

.tab-panel.active {
  display: block;
  animation: fadeIn 0.3s ease-in-out;
}
```

```
@keyframes fadeIn {
  from { opacity: 0; transform: translateY(6px); }
  to { opacity: 1; transform: translateY(0); }
}

/* Layout */
.grid-2 {
  display: grid;
  grid-template-columns: 400px 1fr;
  gap: 1.75rem;
}

@media (max-width: 950px) {
  .grid-2 {
    grid-template-columns: 1fr;
  }
}

.card {
  background: var(--card-bg);
  border: 1px solid var(--card-border);
  border-radius: 16px;
  padding: 1.75rem;
  box-shadow: 0 10px 25px rgba(0, 0, 0, 0.3);
}

.card-title {
  font-size: 1.25rem;
  font-weight: 700;
  color: #38bdf8;
  margin-bottom: 1.25rem;
  display: flex;
  align-items: center;
  gap: 0.5rem;
  border-bottom: 1px solid var(--card-border);
  padding-bottom: 0.75rem;
}

/* Form Fields */
.form-group {
  margin-bottom: 1rem;
}

label {
  display: block;
  font-size: 0.85rem;
  font-weight: 600;
  color: #cbd5e1;
  margin-bottom: 0.35rem;
}

input[type="text"],
input[type="email"],
textarea {
  width: 100%;
  background: #0f172a;
  border: 1px solid #475569;
  color: #ffffff;
  padding: 0.65rem 0.85rem;
  border-radius: 6px;
  font-size: 0.9rem;
  outline: none;
  transition: border-color 0.2s;
}
```

```
input:focus,
textarea:focus {
  border-color: var(--primary);
}

.btn-primary {
  width: 100%;
  background: linear-gradient(135deg, #6366f1, #4f46e5);
  color: white;
  border: none;
  padding: 0.85rem 1.25rem;
  border-radius: 8px;
  font-size: 0.95rem;
  font-weight: 700;
  cursor: pointer;
  display: flex;
  align-items: center;
  justify-content: center;
  gap: 0.5rem;
  box-shadow: 0 4px 14px rgba(99, 102, 241, 0.3);
  transition: all 0.2s;
}

.btn-primary:hover {
  transform: translateY(-1px);
  box-shadow: 0 6px 20px rgba(99, 102, 241, 0.45);
}

.alert-box {
  padding: 0.85rem 1rem;
  border-radius: 8px;
  font-size: 0.9rem;
  margin-bottom: 1.25rem;
  display: none;
}

.alert-box.success {
  display: block;
  background: rgba(16, 185, 129, 0.15);
  border: 1px solid var(--success);
  color: #6ee7b7;
}

.alert-box.error {
  display: block;
  background: rgba(239, 68, 68, 0.15);
  border: 1px solid var(--danger);
  color: #fca5a5;
}

/* Table Styles */
.table-container {
  overflow-x: auto;
  margin-top: 1rem;
}

table {
  width: 100%;
  border-collapse: collapse;
  font-size: 0.875rem;
}

th,
td {
  padding: 0.75rem 0.9rem;
  text-align: left;
```

```
  border-bottom: 1px solid #334155;
}

th {
  background: #0f172a;
  color: #38bdf8;
  font-weight: 600;
}

tr:hover td {
  background: rgba(255, 255, 255, 0.02);
}

/* SQL Live Query Box */
.sql-preview-box {
  margin-top: 1.25rem;
  background: #0f172a;
  border: 1px solid #334155;
  border-radius: 8px;
  padding: 1rem;
  font-family: 'Fira Code', monospace;
  font-size: 0.8rem;
  color: #e2e8f0;
}

.sql-preview-box .query-label {
  color: #a5b4fc;
  font-weight: 600;
  margin-bottom: 0.35rem;
  display: block;
}

/* Code Viewers */
.code-container {
  background: #0f172a;
  border: 1px solid var(--card-border);
  border-radius: 12px;
  overflow: hidden;
}

.code-header {
  background: #1e293b;
  padding: 0.75rem 1.25rem;
  display: flex;
  justify-content: space-between;
  align-items: center;
  border-bottom: 1px solid var(--card-border);
}

.code-header span {
  font-family: 'Fira Code', monospace;
  font-size: 0.85rem;
  color: #38bdf8;
}

.btn-copy {
  background: #334155;
  color: #f8fafc;
  border: 1px solid #475569;
  padding: 0.35rem 0.75rem;
  border-radius: 6px;
  font-size: 0.8rem;
  cursor: pointer;
  transition: all 0.2s;
}
```

```
    .btn-copy:hover {
      background: var(--primary);
      border-color: var(--primary);
    }

    pre code {
      display: block;
      padding: 1.25rem;
      font-family: 'Fira Code', monospace;
      font-size: 0.875rem;
      line-height: 1.6;
      color: #e2e8f0;
      overflow-x: auto;
      white-space: pre;
    }

    footer {
      background: #0b1120;
      border-top: 1px solid var(--card-border);
      padding: 1.5rem 2rem;
      text-align: center;
      font-size: 0.85rem;
      color: var(--text-muted);
      margin-top: 3rem;
    }
  </style>
</head>

<body>

  <!-- Navigation -->
  <nav>
    <a href="../index.html" class="nav-brand">
      <span>&larr;</span> CSIT248 Lab Work
    </a>
    <div class="nav-links">
      <span style="font-size: 0.85rem; color: #a5b4fc; background: rgba(99, 102, 241, 0.15); borde
r: 1px solid rgba(99, 102, 241, 0.3); padding: 0.3rem 0.75rem; border-radius: 9999px; font-weight:
600;">
        MySQL &amp;amp; PHP Connectivity
      </span>
      <a href="../index.html" class="nav-btn">All Experiments</a>
    </div>
  </nav>

  <!-- Main Container -->
  <main>

    <!-- Header -->
    <header class="page-header">
      <span class="badge-pill">Experiment 19 &bull; Module PHP &amp;amp; MySQL Connectivity</span>
      <h1 class="page-title">Add New Record in CUSTOMER Database</h1>
      <p class="page-subtitle">
        Design and implementation of a PHP script utilizing PDO/MySQL prepared statements to inser
t new customer records into a relational database with strict validation and error handling.
      </p>
    </header>

    <!-- Tabs Nav -->
    <div class="tabs-nav">
      <button class="tab-btn active" onclick="switchTab('demo', this)">
        ⚡ Live Interactive Studio
      </button>
      <button class="tab-btn" onclick="switchTab('php', this)">
        🐘 add_customer.php (Server Script)
      </button>
```

```
      <button class="tab-btn" onclick="switchTab('sql', this)">
        🗄️ customer_schema.sql (Database DDL)
      </button>
      <button class="tab-btn" onclick="switchTab('theory', this)">
        📘 Theory &amp;amp; Lab Manual
      </button>
    </div>

    <!-- TAB 1: Live Interactive Studio -->
    <section id="tab-demo" class="tab-panel active">
      <div class="grid-2">

        <!-- Form Column -->
        <div class="card">
          <h2 class="card-title">➕ Customer Registration Form</h2>

          <div id="status-alert" class="alert-box"></div>

          <form id="customer-form" onsubmit="handleCustomerInsert(event)">
            <div class="form-group">
              <label for="c-name">Full Name *</label>
              <input type="text" id="c-name" required placeholder="e.g. Atul Anand">
            </div>

            <div class="form-group">
              <label for="c-email">Email Address *</label>
              <input type="email" id="c-email" required placeholder="e.g. atul@example.com">
            </div>

            <div class="form-group">
              <label for="c-phone">Phone Number *</label>
              <input type="text" id="c-phone" required placeholder="e.g. +91 98765 43210">
            </div>

            <div class="form-group">
              <label for="c-city">City *</label>
              <input type="text" id="c-city" required placeholder="e.g. Noida / New Delhi">
            </div>

            <div class="form-group">
              <label for="c-address">Street Address *</label>
              <textarea id="c-address" rows="2" required placeholder="e.g. Sector 125, Amity Campu
s"></textarea>
            </div>

            <button type="submit" class="btn-primary">
              💾 Execute SQL INSERT Statement
            </button>
          </form>

          <div class="sql-preview-box">
            <span class="query-label">Generated Prepared SQL:</span>
            <code>INSERT INTO customers (customer_name, email, phone, address, city)<br>VALUES (?,
?, ?, ?, ?);</code>
          </div>
        </div>

        <!-- Table Column -->
        <div class="card">
          <div style="display: flex; justify-content: space-between; align-items: center; margin-b
ottom: 1.25rem; border-bottom: 1px solid var(--card-border); padding-bottom: 0.75rem;">
            <h2 class="card-title" style="margin-bottom: 0; border-bottom: none; padding-bottom:
0;">
              📋 Relational Database Records (Table: customers)
            </h2>
            <button type="button" class="btn-copy" onclick="resetDefaultCustomers()">🔄 Reset</bu
```

```
tton>
          </div>

          <div class="table-container">
            <table>
              <thead>
                <tr>
                  <th>ID</th>
                  <th>Customer Name</th>
                  <th>Email</th>
                  <th>Phone</th>
                  <th>City</th>
                </tr>
              </thead>
              <tbody id="customers-table-body">
                <!-- Dynamically rendered -->
              </tbody>
            </table>
          </div>

          <div style="margin-top: 1.5rem; display: flex; justify-content: space-between; align-ite
ms: center; font-size: 0.85rem; color: var(--text-muted);">
            <span>Database: <strong style="color: #38bdf8;">customer_db</strong></span>
            <span>Total Records: <strong style="color: #10b981;" id="record-count">4</strong></spa
n>
          </div>
        </div>

      </div>
    </section>

    <!-- TAB 2: add_customer.php Source -->
    <section id="tab-php" class="tab-panel">
      <div class="card">
        <div class="code-container">
          <div class="code-header">
            <span>add_customer.php (PHP Prepared Statement &amp;amp; PDO)</span>
            <button class="btn-copy" onclick="copyCode('code-php', this)">📋 Copy Script</button>
          </div>
          <pre><code id="code-php">&lt;?php
/**
 * Experiment 19: PHP Script to Add New Record in CUSTOMER Database
 * Course: Advanced Web Technology (CSIT248) - Amity University Noida
 */

$host = 'localhost';
$user = 'root';
$pass = ''; // Default XAMPP password
$db   = 'customer_db';

try {
    // 1. Establish PDO Connection
    $pdo = new PDO("mysql:host=$host;dbname=$db;charset=utf8mb4", $user, $pass, [
        PDO::ATTR_ERRMODE            => PDO::ERRMODE_EXCEPTION,
        PDO::ATTR_DEFAULT_FETCH_MODE => PDO::FETCH_ASSOC
    ]);

    // 2. Handle Form POST Submission
    if ($_SERVER['REQUEST_METHOD'] === 'POST') {
        $name    = trim($_POST['customer_name'] ?? '');
        $email   = trim($_POST['email'] ?? '');
        $phone   = trim($_POST['phone'] ?? '');
        $address = trim($_POST['address'] ?? '');
        $city    = trim($_POST['city'] ?? '');

        // Validation
```

```
        if (empty($name) || empty($email) || empty($phone) || empty($address) || empty($city)) {
            die("Error: All fields are required.");
        }

        if (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
            die("Error: Invalid email format.");
        }

        // 3. Prepare and Execute INSERT query
        $sql = "INSERT INTO customers (customer_name, email, phone, address, city)
                VALUES (:name, :email, :phone, :address, :city)";
        $stmt = $pdo->prepare($sql);
        $stmt->execute([
            ':name'    => $name,
            ':email'   => $email,
            ':phone'   => $phone,
            ':address' => $address,
            ':city'    => $city
        ]);

        $insertedId = $pdo->lastInsertId();
        echo "Customer successfully created with ID: #" . $insertedId;
    }
} catch (PDOException $e) {
    if ($e->getCode() == 23000) {
        echo "Error: A customer with this email already exists.";
    } else {
        echo "Database Connection Error: " . $e->getMessage();
    }
}
?&gt;</code></pre>
        </div>
      </div>
    </section>

    <!-- TAB 3: customer_schema.sql Source -->
    <section id="tab-sql" class="tab-panel">
      <div class="card">
        <div class="code-container">
          <div class="code-header">
            <span>customer_schema.sql (MySQL Table DDL &amp;amp; Seed Data)</span>
            <button class="btn-copy" onclick="copyCode('code-sql', this)">📋 Copy SQL</button>
          </div>
          <pre><code id="code-sql">-- Experiment 19: Database Schema for CUSTOMER Database
CREATE DATABASE IF NOT EXISTS customer_db;
USE customer_db;

-- Create CUSTOMERS Table
CREATE TABLE IF NOT EXISTS customers (
    customer_id INT AUTO_INCREMENT PRIMARY KEY,
    customer_name VARCHAR(100) NOT NULL,
    email VARCHAR(120) NOT NULL UNIQUE,
    phone VARCHAR(20) NOT NULL,
    address TEXT NOT NULL,
    city VARCHAR(60) NOT NULL,
    registered_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- Seed Sample Records
INSERT INTO customers (customer_name, email, phone, address, city) VALUES
('Atul Anand', 'atul.anand@example.com', '+91 98765 43210', 'Sector 125, Express Highway', 'Noid
a'),
('Priya Sharma', 'priya.sharma@example.com', '+91 98123 45678', 'Connaught Place, Central Delhi',
'New Delhi'),
('Rohan Verma', 'rohan.verma@example.com', '+91 97234 56789', 'Boring Road, Near Canal', 'Patna'),
('Ananya Sengupta', 'ananya.s@example.com', '+91 96345 67890', 'Park Street, South Block', 'Kolkat
```

```
a');</code></pre>
        </div>
      </div>
    </section>

    <!-- TAB 4: Theory & Lab Manual -->
    <section id="tab-theory" class="tab-panel">
      <div class="card" style="line-height: 1.8;">
        <h2 style="color: #38bdf8; font-size: 1.5rem; margin-bottom: 1rem;">Experiment 19: Academi
c Lab Manual</h2>

        <h3 style="color: #818cf8; margin-top: 1.5rem; margin-bottom: 0.5rem;">1. Aim</h3>
        <p style="color: #cbd5e1;">
          To write a robust PHP script that connects to MySQL, accepts customer details via an int
eractive HTML form, sanitizes and validates input data, and adds a new record to the <code>custome
rs</code> table using prepared statements to prevent SQL Injection attacks.
        </p>

        <h3 style="color: #818cf8; margin-top: 1.5rem; margin-bottom: 0.5rem;">2. Core Concepts</h
3>
        <ul style="color: #cbd5e1; margin-left: 1.5rem;">
          <li><strong>PDO (PHP Data Objects):</strong> A lightweight, consistent interface for acc
essing databases in PHP. Offers prepared statements, transactions, and robust error modes.</li>
          <li><strong>SQL Injection Prevention:</strong> Direct string interpolation allows attack
ers to manipulate queries. Prepared statements separate the SQL template from data parameters, gua
ranteeing immunity from SQL injection.</li>
          <li><strong>Unique Constraint Handling:</strong> The <code>email</code> column has a <co
de>UNIQUE</code> index; attempting to insert duplicate emails triggers error code <code>23000</cod
e>, which is caught and gracefully handled.</li>
        </ul>

        <h3 style="color: #818cf8; margin-top: 1.5rem; margin-bottom: 0.5rem;">3. Algorithm</h3>
        <ol style="color: #cbd5e1; margin-left: 1.5rem;">
          <li>Create MySQL database <code>customer_db</code> and table <code>customers</code>.</li
>
          <li>Render an HTML form with input fields for Name, Email, Phone, City, and Address.</li
>
          <li>Capture user form submission via HTTP POST.</li>
          <li>Validate that required fields are not empty and verify email syntax via <code>filter
_var()</code>.</li>
          <li>Establish a PDO connection with exception error mode enabled.</li>
          <li>Prepare the SQL <code>INSERT INTO customers ...</code> query with bound placeholder
s.</li>
          <li>Execute the query passing sanitized input values.</li>
          <li>Fetch the auto-incremented primary key using <code>lastInsertId()</code> and inform
the user.</li>
          <li>Query all records using <code>SELECT * FROM customers</code> and display in an HTML
table.</li>
        </ol>
      </div>
    </section>

  </main>

  <!-- Footer -->
  <footer>
    <p style="margin-bottom: 0.4rem;">
      Designed &amp;amp; Developed by <strong style="color: #818cf8;">Atul Anand</strong> &bull; BCA
(Hons.) with Research &bull; Semester III
    </p>
    <p style="color: #64748b; font-size: 0.8rem;">
      Amity University Noida &bull; Department of Computer Applications &bull; CSIT248: Advanced W
eb Technology
    </p>
  </footer>
```

```
  <!-- Script for Live Simulation -->
  <script>
    const INITIAL_CUSTOMERS = [
      { id: 101, name: "Atul Anand", email: "atul.anand@example.com", phone: "+91 98765 43210", ci
ty: "Noida", address: "Sector 125, Amity Campus" },
      { id: 102, name: "Priya Sharma", email: "priya.sharma@example.com", phone: "+91 98123 4567
8", city: "New Delhi", address: "Connaught Place, Central Block" },
      { id: 103, name: "Rohan Verma", email: "rohan.verma@example.com", phone: "+91 97234 56789",
city: "Patna", address: "Boring Road, Near Canal" },
      { id: 104, name: "Ananya Sengupta", email: "ananya.s@example.com", phone: "+91 96345 67890",
city: "Kolkata", address: "Park Street, South Block" }
    ];

    let customers = JSON.parse(JSON.stringify(INITIAL_CUSTOMERS));
    let nextId = 105;

    function renderCustomersTable() {
      const tbody = document.getElementById("customers-table-body");
      tbody.innerHTML = "";

      customers.forEach(c => {
        const tr = document.createElement("tr");
        tr.innerHTML = `
          <td style="font-family: 'Fira Code', monospace; color: #a5b4fc;">#${c.id}</td>
          <td><strong style="color: #ffffff;">${escapeHtml(c.name)}</strong></td>
          <td style="color: #94a3b8;">${escapeHtml(c.email)}</td>
          <td style="font-family: 'Fira Code', monospace; color: #cbd5e1;">${escapeHtml(c.phone)}
</td>
          <td><span style="background: rgba(56, 189, 248, 0.12); color: #38bdf8; padding: 0.2rem
0.5rem; border-radius: 4px; font-weight: 500;">${escapeHtml(c.city)}</span></td>
        `;
        tbody.appendChild(tr);
      });

      document.getElementById("record-count").textContent = customers.length;
    }

    function handleCustomerInsert(e) {
      e.preventDefault();
      const alertBox = document.getElementById("status-alert");
      alertBox.className = "alert-box";

      const name = document.getElementById("c-name").value.trim();
      const email = document.getElementById("c-email").value.trim();
      const phone = document.getElementById("c-phone").value.trim();
      const city = document.getElementById("c-city").value.trim();
      const address = document.getElementById("c-address").value.trim();

      // Check duplicate email
      const exists = customers.some(c => c.email.toLowerCase() === email.toLowerCase());
      if (exists) {
        alertBox.className = "alert-box error";
        alertBox.innerHTML = `⚠ <strong>Integrity Constraint Violation:</strong> A customer with
email '<em>${escapeHtml(email)}</em>' already exists.`;
        return;
      }

      // Add to array
      const newCustomer = {
        id: nextId++,
        name,
        email,
        phone,
        city,
        address
      };
```

```
      customers.unshift(newCustomer);
      renderCustomersTable();

      alertBox.className = "alert-box success";
      alertBox.innerHTML = `✅ <strong>Success:</strong> Record added! Assigned Customer ID <code
>#${newCustomer.id}</code> in database.`;

      // Reset form fields
      document.getElementById("customer-form").reset();

      setTimeout(() => {
        alertBox.style.display = "none";
      }, 5000);
    }

    function resetDefaultCustomers() {
      customers = JSON.parse(JSON.stringify(INITIAL_CUSTOMERS));
      nextId = 105;
      renderCustomersTable();
      const alertBox = document.getElementById("status-alert");
      alertBox.className = "alert-box success";
      alertBox.innerHTML = "🔄 Customer database reset to default state.";
      setTimeout(() => { alertBox.style.display = "none"; }, 3000);
    }

    function escapeHtml(str) {
      return str.replace(/&/g, "&amp;").replace(/</g, "&lt;").replace(/>/g, "&gt;").replace(/"/g,
"&quot;");
    }

    function switchTab(tabId, btn) {
      document.querySelectorAll('.tab-panel').forEach(p => p.classList.remove('active'));
      document.querySelectorAll('.tab-btn').forEach(b => b.classList.remove('active'));
      document.getElementById('tab-' + tabId).classList.add('active');
      btn.classList.add('active');
    }

    function copyCode(elementId, btn) {
      const text = document.getElementById(elementId).innerText;
      navigator.clipboard.writeText(text).then(() => {
        const orig = btn.innerText;
        btn.innerText = "Copied! ✅";
        btn.style.background = "#10b981";
        setTimeout(() => {
          btn.innerText = orig;
          btn.style.background = "";
        }, 2000);
      });
    }

    window.addEventListener("DOMContentLoaded", renderCustomersTable);
  </script>

</body>

</html>
```

Supplementary Source File: customer_schema.sql

```
-- ============================================================================
-- Experiment 19: Customer Database Schema
-- Course: Advanced Web Technology (CSIT248) - Sem III / VI
-- Amity University Noida
-- ============================================================================

-- 1. Create Database
```

```
CREATE DATABASE IF NOT EXISTS customer_db;
USE customer_db;

-- 2. Create CUSTOMER Table
CREATE TABLE IF NOT EXISTS customers (
    customer_id INT AUTO_INCREMENT PRIMARY KEY,
    customer_name VARCHAR(100) NOT NULL,
    email VARCHAR(120) NOT NULL UNIQUE,
    phone VARCHAR(20) NOT NULL,
    address TEXT NOT NULL,
    city VARCHAR(60) NOT NULL,
    registered_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- 3. Insert Initial Sample Records
INSERT INTO customers (customer_name, email, phone, address, city) VALUES
('Atul Anand', 'atul.anand@example.com', '+91 98765 43210', 'Sector 125, Express Highway', 'Noid
a'),
('Priya Sharma', 'priya.sharma@example.com', '+91 98123 45678', 'Connaught Place, Central Delhi',
'New Delhi'),
('Rohan Verma', 'rohan.verma@example.com', '+91 97234 56789', 'Boring Road, Near Canal', 'Patna'),
('Ananya Sengupta', 'ananya.s@example.com', '+91 96345 67890', 'Park Street, South Block', 'Kolkat
a');
```

Supplementary Source File: add_customer.php

```
<?php
/**
 * Experiment 19: PHP Script to Add New Record in CUSTOMER Database
 * Course: Advanced Web Technology (CSIT248)
 * Amity University Noida
 *
 * Objective:
 *   Establish a secure MySQL database connection using PDO/Prepared Statements,
 *   collect customer details via an HTML form, validate inputs, execute an INSERT query,
 *   handle exceptions (such as duplicate entries), and display existing records.
 */

// Database Configuration
$db_host = 'localhost';
$db_user = 'root';
$db_pass = ''; // Default XAMPP/WAMP password is empty
$db_name = 'customer_db';

$success_msg = '';
$error_msg = '';
$customers_list = [];

try {
    // 1. Establish PDO Connection
    $pdo = new PDO("mysql:host=$db_host;dbname=$db_name;charset=utf8mb4", $db_user, $db_pass, [
        PDO::ATTR_ERRMODE => PDO::ERRMODE_EXCEPTION,
        PDO::ATTR_DEFAULT_FETCH_MODE => PDO::FETCH_ASSOC
    ]);

    // 2. Handle Form Submission for Adding Customer
    if ($_SERVER['REQUEST_METHOD'] === 'POST' && isset($_POST['action']) && $_POST['action'] ===
'add_customer') {
        $name    = trim($_POST['customer_name'] ?? '');
        $email   = trim($_POST['email'] ?? '');
        $phone   = trim($_POST['phone'] ?? '');
        $address = trim($_POST['address'] ?? '');
        $city    = trim($_POST['city'] ?? '');

        // Server-side validation
        if (empty($name) || empty($email) || empty($phone) || empty($address) || empty($city)) {
            $error_msg = "All fields marked with an asterisk (*) are mandatory.";
```

```
        } elseif (!filter_var($email, FILTER_VALIDATE_EMAIL)) {
            $error_msg = "Invalid email address format.";
        } else {
            // Prepared statement to prevent SQL Injection
            $sql = "INSERT INTO customers (customer_name, email, phone, address, city) VALUES (:na
me, :email, :phone, :address, :city)";
            $stmt = $pdo->prepare($sql);

            $stmt->execute([
                ':name'    => htmlspecialchars($name),
                ':email'   => htmlspecialchars($email),
                ':phone'   => htmlspecialchars($phone),
                ':address' => htmlspecialchars($address),
                ':city'    => htmlspecialchars($city)
            ]);

            $new_id = $pdo->lastInsertId();
            $success_msg = "Customer record successfully created! Assigned Customer ID: #$new_id";
        }
    }

    // 3. Fetch all customer records to display in the table
    $stmt_all = $pdo->query("SELECT * FROM customers ORDER BY customer_id DESC");
    $customers_list = $stmt_all->fetchAll();

} catch (PDOException $e) {
    if ($e->getCode() == 23000) {
        $error_msg = "A customer with email '$email' already exists in the database.";
    } else {
        $error_msg = "Database Error: " . $e->getMessage();
    }
}
?>
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Experiment 19 - Add Customer Record (PHP &amp;amp; MySQL)</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&family=Fira+
Code:wght@400;600&display=swap" rel="stylesheet">
  <style>
    :root {
      --primary: #6366f1;
      --primary-hover: #4f46e5;
      --success: #10b981;
      --danger: #ef4444;
      --bg: #0f172a;
      --card-bg: #1e293b;
      --border: #334155;
      --text: #f8fafc;
      --muted: #94a3b8;
    }
    * { box-sizing: border-box; margin: 0; padding: 0; }
    body {
      font-family: 'Inter', system-ui, sans-serif;
      background: var(--bg);
      color: var(--text);
      line-height: 1.6;
      padding-bottom: 3rem;
    }
    nav {
      background: rgba(15, 23, 42, 0.95);
      border-bottom: 1px solid var(--border);
```

```
      padding: 1rem 2rem;
      display: flex;
      justify-content: space-between;
      align-items: center;
      position: sticky;
      top: 0;
      z-index: 50;
    }
    .nav-brand { color: #818cf8; text-decoration: none; font-weight: 800; font-size: 1.15rem; }
    .container { max-width: 1100px; margin: 2rem auto; padding: 0 1.5rem; }
    .header { text-align: center; margin-bottom: 2rem; }
    .badge {
      display: inline-block;
      background: rgba(99, 102, 241, 0.15);
      color: #a5b4fc;
      border: 1px solid rgba(99, 102, 241, 0.3);
      padding: 0.25rem 0.75rem;
      border-radius: 9999px;
      font-size: 0.8rem;
      font-weight: 600;
      margin-bottom: 0.5rem;
    }
    h1 { font-size: 2rem; font-weight: 800; margin-bottom: 0.5rem; }
    .grid { display: grid; grid-template-columns: 380px 1fr; gap: 2rem; }
    @media (max-width: 860px) { .grid { grid-template-columns: 1fr; } }
    .card {
      background: var(--card-bg);
      border: 1px solid var(--border);
      border-radius: 12px;
      padding: 1.75rem;
      box-shadow: 0 10px 25px rgba(0,0,0,0.3);
    }
    .card h2 {
      font-size: 1.25rem;
      color: #38bdf8;
      margin-bottom: 1.25rem;
      border-bottom: 1px solid var(--border);
      padding-bottom: 0.5rem;
    }
    .form-group { margin-bottom: 1rem; }
    label { display: block; font-size: 0.85rem; font-weight: 600; color: #cbd5e1; margin-bottom:
0.35rem; }
    input[type="text"], input[type="email"], textarea {
      width: 100%;
      background: #0f172a;
      border: 1px solid var(--border);
      color: #fff;
      padding: 0.65rem 0.85rem;
      border-radius: 6px;
      font-size: 0.9rem;
      outline: none;
    }
    input:focus, textarea:focus { border-color: var(--primary); }
    button {
      width: 100%;
      background: var(--primary);
      color: white;
      border: none;
      padding: 0.85rem;
      border-radius: 6px;
      font-size: 0.95rem;
      font-weight: 700;
      cursor: pointer;
      transition: background 0.2s;
    }
    button:hover { background: var(--primary-hover); }
```

```
    .alert-success { background: rgba(16, 185, 129, 0.15); border: 1px solid var(--success); colo
r: #6ee7b7; padding: 0.85rem 1rem; border-radius: 6px; margin-bottom: 1.25rem; font-size: 0.9rem;
}
    .alert-error { background: rgba(239, 68, 68, 0.15); border: 1px solid var(--danger); color: #f
ca5a5; padding: 0.85rem 1rem; border-radius: 6px; margin-bottom: 1.25rem; font-size: 0.9rem; }
    table { width: 100%; border-collapse: collapse; font-size: 0.85rem; }
    th, td { padding: 0.75rem 0.85rem; text-align: left; border-bottom: 1px solid var(--border); }
    th { background: #0f172a; color: #38bdf8; font-weight: 600; }
  </style>
</head>
<body>

  <nav>
    <a href="index.html" class="nav-brand">&larr; CSIT248 Lab Experiments</a>
    <span style="font-size: 0.85rem; color: var(--muted);">Experiment 19: Database Operations</spa
n>
  </nav>

  <div class="container">
    <div class="header">
      <span class="badge">Experiment 19 &bull; Module PHP &amp; MySQL Connectivity</span>
      <h1>Customer Database Record Management</h1>
      <p style="color: var(--muted); font-size: 0.95rem;">Insert new customer entries and view rea
l-time database records</p>
    </div>

    <?php if ($success_msg): ?>
      <div class="alert-success">✅ <?php echo htmlspecialchars($success_msg); ?></div>
    <?php endif; ?>

    <?php if ($error_msg): ?>
      <div class="alert-error">⚠️ <?php echo htmlspecialchars($error_msg); ?></div>
    <?php endif; ?>

    <div class="grid">
      <!-- Form Panel -->
      <div class="card">
        <h2>➕ Add New Customer</h2>
        <form method="POST" action="add_customer.php">
          <input type="hidden" name="action" value="add_customer">

          <div class="form-group">
            <label for="customer_name">Customer Full Name *</label>
            <input type="text" id="customer_name" name="customer_name" required placeholder="e.g.
Atul Anand">
          </div>

          <div class="form-group">
            <label for="email">Email Address *</label>
            <input type="email" id="email" name="email" required placeholder="e.g. atul@example.co
m">
          </div>

          <div class="form-group">
            <label for="phone">Phone Number *</label>
            <input type="text" id="phone" name="phone" required placeholder="e.g. +91 98765 4321
0">
          </div>

          <div class="form-group">
            <label for="city">City *</label>
            <input type="text" id="city" name="city" required placeholder="e.g. Noida / New Delh
i">
          </div>

          <div class="form-group">
```

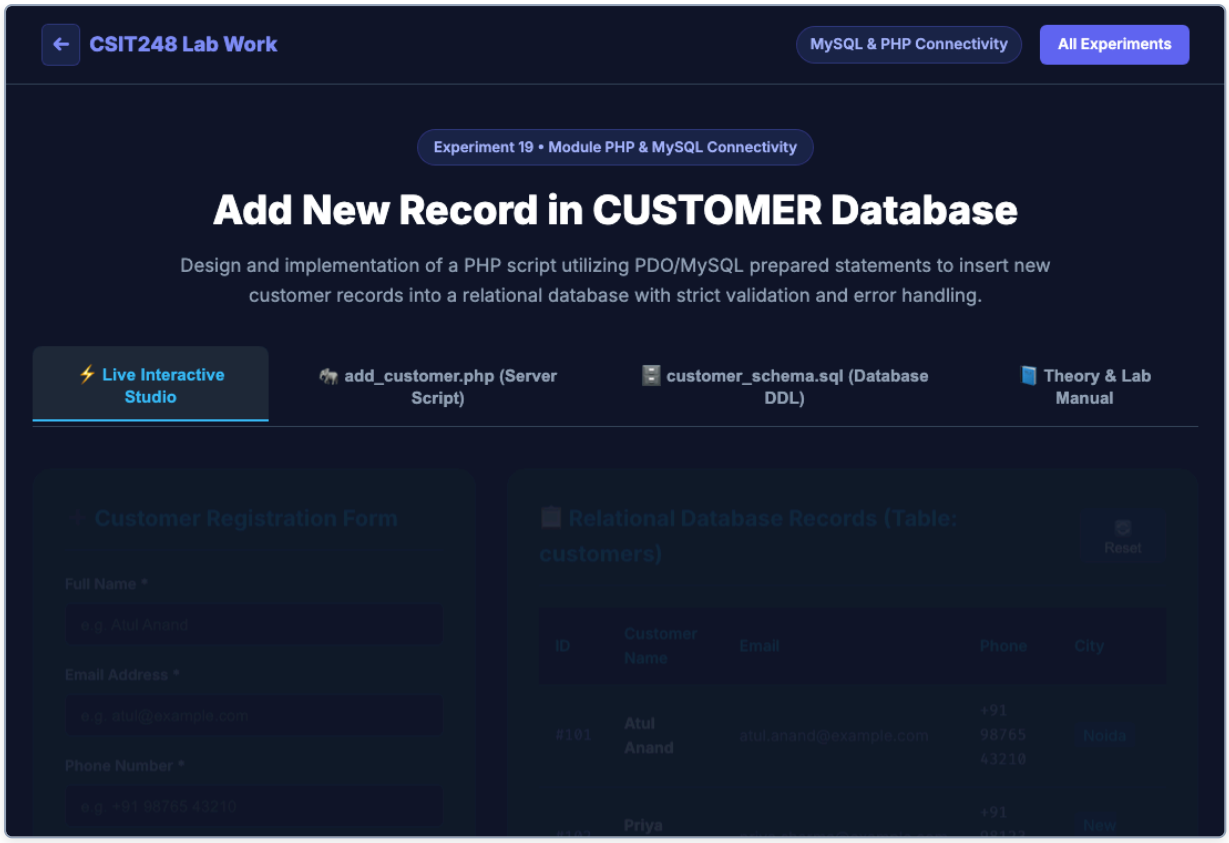
```
            <label for="address">Street Address *</label>
            <textarea id="address" name="address" rows="3" required placeholder="e.g. Sector 125,
Amity Campus"></textarea>
          </div>

          <button type="submit">Insert Record into Database</button>
        </form>
      </div>

      <!-- Customer Records List -->
      <div class="card" style="overflow-x: auto;">
        <h2>  Customer Database Records (Table: customers)</h2>
        <?php if (!empty($customers_list)): ?>
          <table>
            <thead>
              <tr>
                <th>ID</th>
                <th>Name</th>
                <th>Email</th>
                <th>Phone</th>
                <th>City</th>
              </tr>
            </thead>
            <tbody>
              <?php foreach ($customers_list as $cust): ?>
                <tr>
                  <td style="font-family: 'Fira Code', monospace; color: #a5b4fc;">#<?php echo $cu
st['customer_id']; ?></td>
                  <td><strong><?php echo htmlspecialchars($cust['customer_name']); ?></strong></td
>
                  <td><?php echo htmlspecialchars($cust['email']); ?></td>
                  <td style="font-family: 'Fira Code', monospace;"><?php echo htmlspecialchars($cu
st['phone']); ?></td>
                  <td><span style="background: rgba(56, 189, 248, 0.1); color: #38bdf8; padding:
0.2rem 0.5rem; border-radius: 4px;"><?php echo htmlspecialchars($cust['city']); ?></span></td>
                </tr>
              <?php endforeach; ?>
            </tbody>
          </table>
        <?php else: ?>
          <p style="color: var(--muted); text-align: center; padding: 2rem;">No customer records f
ound or database connection pending.</p>
        <?php endif; ?>
      </div>
    </div>
  </div>

</body>
</html>
```

Output :


CSIT248 Lab Work
MySQL & PHP Connectivity
All Experiments
Experiment 19 • Module PHP & MySQL Connectivity
Add New Record in CUSTOMER Database
Design and implementation of a PHP script utilizing PDO/MySQL prepared statements to insert new customer records into a relational database with strict validation and error handling.
⚡ Live Interactive Studio
add_customer.php (Server Script)
customer_schema.sql (Database DDL)
Theory & Lab Manual
Customer Registration Form
Full Name *
e.g. Atul Anand
Email Address *
e.g. atul@example.com
Phone Number *
e.g. +91 98765 43210
Relational Database Records (Table: customers)
Reset
ID
Customer Name
Email
Phone
City
#101
Atul Anand
atul.anand@example.com
+91 98765 43210
Noida
Priya
New

Experiment No.: 23

Title : Implementation of include( ) and require( ) Mechanics in PHP.

Aim : To design a modular PHP architecture demonstrating the functionality, differences, and error handling behaviors (E_WARNING vs E_COMPILE_ERROR) of include(), require(), include_once(), and require_once().

Source Code :

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description"
    content="Experiment 20: Write a PHP Script to implement function include() and require() - CSI T248 Advanced Web Technology">
  <title>Experiment 20 - PHP include() and require() Implementation</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link
    href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&family=Fira+Code:wght@400;500;600&display=swap"
    rel="stylesheet">
  <style>
    :root {
      --primary: #6366f1;
      --primary-hover: #4f46e5;
      --secondary: #06b6d4;
      --accent: #f43f5e;
      --success: #10b981;
      --warning: #f59e0b;
      --danger: #ef4444;
      --bg-dark: #0f172a;
      --card-bg: #1e293b;
      --card-border: #334155;
      --text-main: #f8fafc;
      --text-muted: #94a3b8;
    }

    * {
      box-sizing: border-box;
      margin: 0;
      padding: 0;
    }

    body {
      font-family: 'Inter', system-ui, -apple-system, sans-serif;
      background-color: var(--bg-dark);
      color: var(--text-main);
      line-height: 1.6;
      min-height: 100vh;
      display: flex;
      flex-direction: column;
    }

    nav {
      position: sticky;
      top: 0;
      z-index: 100;
      background: rgba(15, 23, 42, 0.92);
```

```
  backdrop-filter: blur(12px);
  border-bottom: 1px solid var(--card-border);
  padding: 1rem 2rem;
  display: flex;
  justify-content: space-between;
  align-items: center;
}

.nav-brand {
  font-weight: 800;
  font-size: 1.15rem;
  color: #818cf8;
  text-decoration: none;
  display: flex;
  align-items: center;
  gap: 0.5rem;
}

.nav-brand span {
  background: rgba(99, 102, 241, 0.15);
  border: 1px solid rgba(99, 102, 241, 0.3);
  padding: 0.2rem 0.5rem;
  border-radius: 6px;
}

.nav-links {
  display: flex;
  align-items: center;
  gap: 1rem;
}

.nav-btn {
  background: var(--primary);
  color: #ffffff;
  padding: 0.45rem 1rem;
  border-radius: 6px;
  text-decoration: none;
  font-size: 0.85rem;
  font-weight: 600;
}

main {
  max-width: 1200px;
  width: 100%;
  margin: 0 auto;
  padding: 2.5rem 1.5rem;
  flex: 1;
}

.page-header {
  text-align: center;
  margin-bottom: 2rem;
}

.badge-pill {
  display: inline-block;
  background: rgba(99, 102, 241, 0.15);
  color: #a5b4fc;
  border: 1px solid rgba(99, 102, 241, 0.3);
  padding: 0.3rem 0.9rem;
  border-radius: 9999px;
  font-size: 0.825rem;
  font-weight: 600;
  margin-bottom: 0.75rem;
}
```

```
.page-title {
  font-size: 2.25rem;
  font-weight: 800;
  color: #ffffff;
  margin-bottom: 0.5rem;
}

.page-subtitle {
  color: var(--text-muted);
  font-size: 1.05rem;
  max-width: 820px;
  margin: 0 auto;
}

.tabs-nav {
  display: flex;
  gap: 0.5rem;
  border-bottom: 1px solid var(--card-border);
  margin-bottom: 2rem;
  overflow-x: auto;
  padding-bottom: 0.25rem;
}

.tab-btn {
  background: transparent;
  border: none;
  color: var(--text-muted);
  padding: 0.75rem 1.25rem;
  font-size: 0.95rem;
  font-weight: 600;
  cursor: pointer;
  border-radius: 8px 8px 0 0;
  display: flex;
  align-items: center;
  gap: 0.5rem;
  transition: all 0.2s;
}

.tab-btn:hover {
  color: #ffffff;
  background: rgba(255, 255, 255, 0.05);
}

.tab-btn.active {
  color: #38bdf8;
  background: var(--card-bg);
  border-bottom: 2px solid #38bdf8;
}

.tab-panel {
  display: none;
}

.tab-panel.active {
  display: block;
  animation: fadeIn 0.3s ease-in-out;
}

@keyframes fadeIn {
  from { opacity: 0; transform: translateY(6px); }
  to { opacity: 1; transform: translateY(0); }
}

.card {
  background: var(--card-bg);
  border: 1px solid var(--card-border);
```

```
  border-radius: 16px;
  padding: 1.75rem;
  margin-bottom: 1.5rem;
  box-shadow: 0 10px 25px rgba(0, 0, 0, 0.3);
}

.card-title {
  font-size: 1.25rem;
  font-weight: 700;
  color: #38bdf8;
  margin-bottom: 1.25rem;
  display: flex;
  align-items: center;
  gap: 0.5rem;
  border-bottom: 1px solid var(--card-border);
  padding-bottom: 0.75rem;
}

/* Scenario Buttons */
.scenario-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(240px, 1fr));
  gap: 1rem;
  margin-bottom: 1.5rem;
}

.scenario-btn {
  background: #0f172a;
  border: 1px solid var(--card-border);
  color: #f8fafc;
  padding: 1rem 1.25rem;
  border-radius: 10px;
  cursor: pointer;
  text-align: left;
  transition: all 0.2s;
}

.scenario-btn:hover {
  border-color: var(--primary);
  transform: translateY(-2px);
}

.scenario-btn.active {
  border-color: #38bdf8;
  background: rgba(56, 189, 248, 0.08);
  box-shadow: 0 0 15px rgba(56, 189, 248, 0.2);
}

.scenario-btn .title {
  font-weight: 700;
  font-size: 0.95rem;
  display: flex;
  align-items: center;
  gap: 0.5rem;
}

.scenario-btn .desc {
  font-size: 0.8rem;
  color: var(--text-muted);
  margin-top: 0.35rem;
}

/* Simulation Output Console */
.terminal-console {
  background: #090d16;
  border: 1px solid #1e293b;
```

```
  border-radius: 12px;
  overflow: hidden;
  font-family: 'Fira Code', monospace;
  margin-top: 1rem;
}

.terminal-header {
  background: #111827;
  padding: 0.6rem 1rem;
  display: flex;
  align-items: center;
  gap: 0.5rem;
  border-bottom: 1px solid #1f2937;
}

.terminal-dot {
  width: 11px;
  height: 11px;
  border-radius: 50%;
}

.terminal-body {
  padding: 1.25rem;
  font-size: 0.875rem;
  line-height: 1.7;
}

/* Comparison Table */
table {
  width: 100%;
  border-collapse: collapse;
  margin-top: 1rem;
  font-size: 0.9rem;
}

th, td {
  padding: 0.85rem 1rem;
  text-align: left;
  border-bottom: 1px solid var(--card-border);
}

th {
  background: #0f172a;
  color: #38bdf8;
  font-weight: 600;
}

/* Code Viewers */
.code-container {
  background: #0f172a;
  border: 1px solid var(--card-border);
  border-radius: 12px;
  overflow: hidden;
  margin-bottom: 1.5rem;
}

.code-header {
  background: #1e293b;
  padding: 0.75rem 1.25rem;
  display: flex;
  justify-content: space-between;
  align-items: center;
  border-bottom: 1px solid var(--card-border);
}

.code-header span {
```

```
      font-family: 'Fira Code', monospace;
      font-size: 0.85rem;
      color: #38bdf8;
    }

    .btn-copy {
      background: #334155;
      color: #f8fafc;
      border: 1px solid #475569;
      padding: 0.35rem 0.75rem;
      border-radius: 6px;
      font-size: 0.8rem;
      cursor: pointer;
    }

    pre code {
      display: block;
      padding: 1.25rem;
      font-family: 'Fira Code', monospace;
      font-size: 0.875rem;
      line-height: 1.6;
      color: #e2e8f0;
      overflow-x: auto;
      white-space: pre;
    }

    footer {
      background: #0b1120;
      border-top: 1px solid var(--card-border);
      padding: 1.5rem 2rem;
      text-align: center;
      font-size: 0.85rem;
      color: var(--text-muted);
      margin-top: 3rem;
    }
  </style>
</head>

<body>

  <!-- Navigation -->
  <nav>
    <a href="../index.html" class="nav-brand">
      <span>&larr;</span> CSIT248 Lab Work
    </a>
    <div class="nav-links">
      <span style="font-size: 0.85rem; color: #a5b4fc; background: rgba(99, 102, 241, 0.15); borde
r: 1px solid rgba(99, 102, 241, 0.3); padding: 0.3rem 0.75rem; border-radius: 9999px; font-weight:
600;">
        Modular PHP Architecture
      </span>
      <a href="../index.html" class="nav-btn">All Experiments</a>
    </div>
  </nav>

  <!-- Main Container -->
  <main>

    <header class="page-header">
      <span class="badge-pill">Experiment 20 &bull; Module PHP File Inclusions</span>
      <h1 class="page-title">Implementation of include( ) and require( ) in PHP</h1>
      <p class="page-subtitle">
        Explore modular web development, file loading mechanics, error handling paradigms (Warning
vs Fatal Error), and single-execution safeguards with include_once() and require_once().
      </p>
    </header>
```

```
    <!-- Tabs Nav -->
    <div class="tabs-nav">
      <button class="tab-btn active" onclick="switchTab('simulator', this)">
        ⚡ Live Lifecycle &amp;amp; Error Simulator
      </button>
      <button class="tab-btn" onclick="switchTab('matrix', this)">
        📊 include vs require Comparison
      </button>
      <button class="tab-btn" onclick="switchTab('scripts', this)">
        🐘 Complete PHP Codebase (4 Files)
      </button>
      <button class="tab-btn" onclick="switchTab('manual', this)">
        📘 Academic Lab Manual &amp;amp; Viva
      </button>
    </div>

    <!-- TAB 1: Live Lifecycle & Error Simulator -->
    <section id="tab-simulator" class="tab-panel active">
      <div class="card">
        <h2 class="card-title">🧪 Interactive Execution Simulator: Select Test Scenario</h2>
        <p style="color: var(--text-muted); font-size: 0.95rem; margin-bottom: 1.25rem;">
          Test how the PHP runtime responds when evaluating valid files versus missing files under
<code>include()</code> and <code>require()</code>.
        </p>

        <div class="scenario-grid">
          <button type="button" class="scenario-btn active" onclick="runScenario('normal', this)">
            <div class="title" style="color: #34d399;">✅ Scenario A: Success Run</div>
            <div class="desc">All files exist: config.php + header.php + footer.php</div>
          </button>

          <button type="button" class="scenario-btn" onclick="runScenario('missing_include', thi
s)">
            <div class="title" style="color: #f59e0b;">⚠️ Scenario B: Missing include()</div>
            <div class="desc">Generates E_WARNING, but script execution continues</div>
          </button>

          <button type="button" class="scenario-btn" onclick="runScenario('missing_require', thi
s)">
            <div class="title" style="color: #ef4444;">🛑 Scenario C: Missing require()</div>
            <div class="desc">Generates E_COMPILE_ERROR; halts immediately</div>
          </button>

          <button type="button" class="scenario-btn" onclick="runScenario('once_guard', this)">
            <div class="title" style="color: #818cf8;">🛡️ Scenario D: require_once() Guard</div>
            <div class="desc">Prevents duplicate function/constant declaration collision</div>
          </button>
        </div>

        <div class="terminal-console">
          <div class="terminal-header">
            <span class="terminal-dot" style="background: #ef4444;"></span>
            <span class="terminal-dot" style="background: #f59e0b;"></span>
            <span class="terminal-dot" style="background: #10b981;"></span>
            <span style="font-size: 0.75rem; color: #94a3b8; margin-left: 0.5rem;" id="terminal-ti
tle">PHP 8.2 Execution Simulator &bull; STDOUT / STDERR</span>
          </div>
          <div class="terminal-body" id="terminal-content">
            <!-- Dynamically populated -->
          </div>
        </div>
      </div>
    </section>

    <!-- TAB 2: Comparison Matrix -->
```

```
    <section id="tab-matrix" class="tab-panel">
      <div class="card">
        <h2 class="card-title">📊 Architectural Comparison Matrix</h2>
        <div style="overflow-x: auto;">
          <table>
            <thead>
              <tr>
                <th>Feature / Parameter</th>
                <th>include( )</th>
                <th>require( )</th>
                <th>include_once( )</th>
                <th>require_once( )</th>
              </tr>
            </thead>
            <tbody>
              <tr>
                <td><strong>Error on Missing File</strong></td>
                <td><span style="color: #f59e0b; font-weight: 600;">E_WARNING</span></td>
                <td><span style="color: #ef4444; font-weight: 700;">E_COMPILE_ERROR (Fatal)</span>
</td>
                <td><span style="color: #f59e0b; font-weight: 600;">E_WARNING</span></td>
                <td><span style="color: #ef4444; font-weight: 700;">E_COMPILE_ERROR (Fatal)</span>
</td>
              </tr>
              <tr>
                <td><strong>Execution Flow</strong></td>
                <td>Script <em>continues</em> execution</td>
                <td>Script <em>terminates immediately</em></td>
                <td>Script <em>continues</em> execution</td>
                <td>Script <em>terminates immediately</em></td>
              </tr>
              <tr>
                <td><strong>Duplicate Inclusion Handling</strong></td>
                <td>Re-includes file every time called</td>
                <td>Re-includes file every time called</td>
                <td>Checks cache; skips if already included</td>
                <td>Checks cache; skips if already included</td>
              </tr>
              <tr>
                <td><strong>Recommended Use Cases</strong></td>
                <td>Page headers, footers, ad banners, non-essential widgets</td>
                <td>Database connections, auth guard, SSL certificates, critical APIs</td>
                <td>Helper functions, utility libraries where re-declaration may occur</td>
                <td>Global site constants (config.php), database drivers, class definitions</td>
              </tr>
            </tbody>
          </table>
        </div>
      </div>
    </section>

    <!-- TAB 3: Complete PHP Codebase -->
    <section id="tab-scripts" class="tab-panel">
      <!-- File 1: include_require_demo.php -->
      <div class="code-container">
        <div class="code-header">
          <span>1. include_require_demo.php (Main Driver Script)</span>
          <button class="btn-copy" onclick="copyCode('c-demo', this)">📋 Copy</button>
        </div>
        <pre><code id="c-demo">&lt;?php
// Step 1: Load critical config via require_once (Halt if missing)
require_once __DIR__ . '/config.php';

// Step 2: Include reusable view header via include (Render header)
include __DIR__ . '/header.php';
?&gt;
```

```
&lt;main&gt;
  &lt;h1&gt;Welcome to &lt;?php echo APP_NAME; ?&gt;&lt;/h1&gt;
  &lt;p&gt;Institution: &lt;?php echo INSTITUTION; ?&gt;&lt;/p&gt;

  &lt;?php
  // Demonstration of missing include vs require:
  // include 'optional_widget.php'; // Warning only, page keeps working
  // require 'critical_security.php'; // Fatal error, page stops here
  ?&gt;
&lt;/main&gt;

&lt;?php
// Step 3: Include reusable footer via include
include __DIR__ . '/footer.php';
?&gt;</code></pre>
      </div>

      <!-- File 2: config.php -->
      <div class="code-container">
        <div class="code-header">
          <span>2. config.php (Application Configuration Constants)</span>
          <button class="btn-copy" onclick="copyCode('c-config', this)">Copy</button>
        </div>
        <pre><code id="c-config">&lt;?php
// config.php
define('APP_NAME', 'EduPortal Portal System');
define('APP_VERSION', '3.4.1');
define('INSTITUTION', 'Amity University Noida');
define('DEPARTMENT', 'Department of Computer Applications');

function get_formatted_title($page) {
    return $page . ' | ' . APP_NAME;
}
?&gt;</code></pre>
      </div>

      <!-- File 3: header.php -->
      <div class="code-container">
        <div class="code-header">
          <span>3. header.php (Modular Layout Header)</span>
          <button class="btn-copy" onclick="copyCode('c-header', this)">Copy</button>
        </div>
        <pre><code id="c-header">&lt;!-- header.php --&gt;
&lt;header style="background: #1e293b; padding: 1rem 2rem; border-bottom: 1px solid #334155;"&gt;
  &lt;h2 style="margin: 0; color: #fff;"&gt;&lt;?php echo APP_NAME; ?&gt;&lt;/h2&gt;
  &lt;nav&gt;
    &lt;a href="#home" style="color: #38bdf8;"&gt;Home&lt;/a&gt; |
    &lt;a href="#about" style="color: #cbd5e1;"&gt;About&lt;/a&gt; |
    &lt;a href="#contact" style="color: #cbd5e1;"&gt;Contact&lt;/a&gt;
  &lt;/nav&gt;
&lt;/header&gt;</code></pre>
      </div>

      <!-- File 4: footer.php -->
      <div class="code-container">
        <div class="code-header">
          <span>4. footer.php (Modular Layout Footer)</span>
          <button class="btn-copy" onclick="copyCode('c-footer', this)">Copy</button>
        </div>
        <pre><code id="c-footer">&lt;!-- footer.php --&gt;
&lt;footer style="background: #0b1120; padding: 1.5rem; text-align: center; border-top: 1px solid
#334155;"&gt;
  &lt;p style="color: #94a3b8;"&gt;&amp;copy; &lt;?php echo date('Y'); ?&gt; &lt;?php echo APP_NAM
E; ?&gt;. All Rights Reserved.&lt;/p&gt;
  &lt;p style="color: #64748b; font-size: 0.8rem;"&gt;&lt;?php echo INSTITUTION; ?&gt;&lt;/p&gt;
```

```
&lt;/footer&gt;</code></pre>
      </div>
    </section>

    <!-- TAB 4: Academic Lab Manual & Viva -->
    <section id="tab-manual" class="tab-panel">
      <div class="card" style="line-height: 1.8;">
        <h2 style="color: #38bdf8; font-size: 1.5rem; margin-bottom: 1rem;">Experiment 20: Academi
c Lab Manual</h2>

        <h3 style="color: #818cf8; margin-top: 1.5rem; margin-bottom: 0.5rem;">1. Aim</h3>
        <p style="color: #cbd5e1;">
          To design and execute a modular PHP application demonstrating the implementation and com
parative functional behavior of <code>include()</code> and <code>require()</code>, with specific f
ocus on file dependency handling, error generation levels, and execution flow.
        </p>

        <h3 style="color: #818cf8; margin-top: 1.5rem; margin-bottom: 0.5rem;">2. Viva-Voce Questi
ons &amp; Answers</h3>
        <div style="background: #0f172a; border: 1px solid #334155; border-radius: 8px; padding:
1.25rem; margin-top: 1rem;">
          <p style="color: #38bdf8; font-weight: 700;">Q1: What is the main difference between inc
lude() and require()?</p>
          <p style="color: #cbd5e1; font-size: 0.9rem; margin-bottom: 1rem;">
            <strong>Ans:</strong> When a file cannot be found, <code>include()</code> produces a w
arning (<code>E_WARNING</code>) and allows script execution to continue, whereas <code>require()</
code> throws a fatal error (<code>E_COMPILE_ERROR</code>) and immediately halts the script.
          </p>

          <p style="color: #38bdf8; font-weight: 700;">Q2: When should require_once() be preferred
over require()?</p>
          <p style="color: #cbd5e1; font-size: 0.9rem; margin-bottom: 1rem;">
            <strong>Ans:</strong> When including files that declare functions, classes, or databas
e connections, to avoid fatal errors caused by duplicate declarations if the file is accidentally
referenced multiple times.
          </p>

          <p style="color: #38bdf8; font-weight: 700;">Q3: Does include() return a value?</p>
          <p style="color: #cbd5e1; font-size: 0.9rem;">
            <strong>Ans:</strong> Yes, <code>include()</code> returns <code>FALSE</code> on failur
e and raises a warning. If the included file uses a <code>return</code> statement, the returned va
lue is provided to the caller.
          </p>
        </div>
      </div>
    </section>

  </main>

  <!-- Footer -->
  <footer>
    <p style="margin-bottom: 0.4rem;">
      Designed &amp; Developed by <strong style="color: #818cf8;">Atul Anand</strong> &bull; BCA
(Hons.) with Research &bull; Semester III
    </p>
    <p style="color: #64748b; font-size: 0.8rem;">
      Amity University Noida &bull; Department of Computer Applications &bull; CSIT248: Advanced W
eb Technology
    </p>
  </footer>

  <!-- Script for Interactive Simulation -->
  <script>
    const SCENARIOS = {
      normal: `
<span style="color: #10b981; font-weight: bold;">[PHP BOOTSTRAP]</span> Initializing engine &bull;
```

```
Parsing include_require_demo.php
<span style="color: #38bdf8;">[REQUIRE_ONCE]</span> Resolving file: /var/www/html/experiment-20/co
nfig.php ...
<span style="color: #10b981;">&gt;&gt; File Found &amp; Evaluated Successfully (0.8ms)</span>
<span style="color: #a5b4fc;">    ├── Constant APP_NAME => "EduPortal Portal System"</span>
<span style="color: #a5b4fc;">    ├── Constant INSTITUTION => "Amity University Noida"</span>
<span style="color: #a5b4fc;">    └── Registered helper function: get_formatted_title()</span>

<span style="color: #38bdf8;">[INCLUDE]</span> Resolving file: /var/www/html/experiment-20/header.
php ...
<span style="color: #10b981;">&gt;&gt; File Found &amp; HTML Header Rendered to Output Buffer (0.4
ms)</span>

<span style="color: #38bdf8;">[EXECUTION]</span> Executing body content &bull; Evaluating student
dashboard logic ...
<span style="color: #a5b4fc;">    └── System Status: ONLINE | Academic Year: 2025-2026</span>

<span style="color: #38bdf8;">[INCLUDE]</span> Resolving file: /var/www/html/experiment-20/footer.
php ...
<span style="color: #10b981;">&gt;&gt; File Found &amp; Footer Rendered (0.3ms)</span>
<span style="color: #10b981; font-weight: bold;">[COMPLETE]</span> HTTP 200 OK &bull; Script execu
ted with 0 errors, 0 warnings.
`,

      missing_include: `
<span style="color: #10b981; font-weight: bold;">[PHP BOOTSTRAP]</span> Parsing script ...
<span style="color: #10b981;">&gt;&gt; require_once 'config.php' loaded successfully.</span>
<span style="color: #10b981;">&gt;&gt; include 'header.php' loaded successfully.</span>

<span style="color: #f59e0b; font-weight: bold;">[INCLUDE CALL]</span> include 'optional_promo_ban
ner.php';
<span style="color: #f59e0b; font-weight: bold;">⚠️ PHP Warning (E_WARNING):</span>
   include(optional_promo_banner.php): Failed to open stream: No such file or directory
   in /var/www/html/experiment-20/include_require_demo.php on line 24

<span style="color: #f59e0b; font-weight: bold;">⚠️ PHP Warning (E_WARNING):</span>
   include(): Failed opening 'optional_promo_banner.php' for inclusion
   in /var/www/html/experiment-20/include_require_demo.php on line 24

<span style="color: #38bdf8; font-weight: bold;">[RESILIENT FLOW]</span>
Execution does NOT halt! PHP logs the non-fatal warning and immediately proceeds to line 25:
<span style="color: #10b981;">&gt;&gt; Main content body renders successfully.</span>
<span style="color: #10b981;">&gt;&gt; include 'footer.php' executed and rendered successfully.</s
pan>
<span style="color: #38bdf8;">[SUMMARY]</span> Page delivered intact to user with minor warning lo
gged.
`,

      missing_require: `
<span style="color: #10b981; font-weight: bold;">[PHP BOOTSTRAP]</span> Parsing script ...
<span style="color: #ef4444; font-weight: bold;">[REQUIRE CALL]</span> require 'auth_security_engi
ne.php';
<span style="color: #f87171;">[PROBING]</span> File path 'auth_security_engine.php' does not exist
in include_path.

<span style="color: #ef4444; font-weight: bold;">⚠️ PHP Warning (E_WARNING):</span>
   require(auth_security_engine.php): Failed to open stream: No such file or directory
   in /var/www/html/experiment-20/include_require_demo.php on line 12

<span style="color: #ef4444; font-size: 1rem; font-weight: bold;">🛑 PHP Fatal Error (E_COMPILE_E
RROR):</span>
   Fatal error: require(): Failed opening required 'auth_security_engine.php' (include_path='.:/ph
p/includes')
   in /var/www/html/experiment-20/include_require_demo.php on line 12

<span style="color: #ef4444; font-weight: bold;">[EXECUTION TERMINATED]</span>
```

```
The PHP engine immediately stops!
❌ Lines 13 to 45 are NOT executed.
❌ HTML body is NOT sent.
❌ Footer is NOT rendered.
<span style="color: #ef4444;">[STATUS]</span> Process exited with Fatal Error (Code 255).
`,

      once_guard: `
<span style="color: #10b981; font-weight: bold;">[TESTING]</span> Testing multiple inclusions of l
ibrary file declaring function <code>calculate_tax()</code>:

<span style="color: #ef4444;">❌ Case 1: Using include('tax_lib.php') twice:</span>
   1st include('tax_lib.php') => Function calculate_tax() defined.
   2nd include('tax_lib.php') =>
   <span style="color: #f87171;">Fatal error: Cannot redeclare calculate_tax() (previously declare
d in tax_lib.php:5)</span>
   Execution Crashes!

<span style="color: #10b981;">✅ Case 2: Using require_once('tax_lib.php') twice:</span>
   1st require_once('tax_lib.php') => File loaded, function calculate_tax() registered in symbol t
able.
   2nd require_once('tax_lib.php') => PHP detects file is already loaded &bull; SKIPS execution!
   <span style="color: #34d399;">Result: Zero collisions, clean memory, flawless execution.</span>
`
    };

    function runScenario(type, btn) {
      document.querySelectorAll('.scenario-btn').forEach(b => b.classList.remove('active'));
      btn.classList.add('active');
      document.getElementById('terminal-content').innerHTML = SCENARIOS[type];
    }

    function switchTab(tabId, btn) {
      document.querySelectorAll('.tab-panel').forEach(p => p.classList.remove('active'));
      document.querySelectorAll('.tab-btn').forEach(b => b.classList.remove('active'));
      document.getElementById('tab-' + tabId).classList.add('active');
      btn.classList.add('active');
    }

    function copyCode(elementId, btn) {
      const text = document.getElementById(elementId).innerText;
      navigator.clipboard.writeText(text).then(() => {
        const orig = btn.innerText;
        btn.innerText = "Copied! ✅";
        btn.style.background = "#10b981";
        setTimeout(() => {
          btn.innerText = orig;
          btn.style.background = "";
        }, 2000);
      });
    }

    window.addEventListener("DOMContentLoaded", () => {
      document.getElementById('terminal-content').innerHTML = SCENARIOS.normal;
    });
  </script>

</body>

</html>
```

Supplementary Source File: include_require_demo.php

```
<?php
/**
 * Experiment 20: Comprehensive Implementation of include() and require()
 * Course: Advanced Web Technology (CSIT248) - Amity University Noida
```

```
 *
 * Key Difference:
 * 1. require() / require_once():
 *    - Used for mission-critical files (DB configuration, authorization).
 *    - If the file is not found, PHP throws a FATAL ERROR (E_COMPILE_ERROR)
 *      and completely STOPS the execution of the script.
 *
 * 2. include() / include_once():
 *    - Used for presentation or non-critical components (header, footer, banners).
 *    - If the file is not found, PHP raises a WARNING (E_WARNING)
 *      and CONTINUES executing the remaining script.
 */

// Step 1: Load Critical Configuration via require_once
// If config.php is missing, application must not boot!
require_once __DIR__ . '/config.php';

$testMode = $_GET['test'] ?? 'normal';
?>
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title><?php echo get_formatted_title('Include vs Require Demonstration'); ?></title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&family=Fira+
Code:wght@400;600&display=swap" rel="stylesheet">
  <style>
    body {
      font-family: 'Inter', system-ui, sans-serif;
      background-color: #0f172a;
      color: #f8fafc;
      margin: 0;
      padding: 0;
      min-height: 100vh;
      display: flex;
      flex-direction: column;
    }
    main {
      max-width: 1000px;
      margin: 2rem auto;
      padding: 0 1.5rem;
      flex: 1;
      width: 100%;
    }
    .badge {
      display: inline-block;
      background: rgba(99, 102, 241, 0.15);
      color: #a5b4fc;
      border: 1px solid rgba(99, 102, 241, 0.3);
      padding: 0.25rem 0.75rem;
      border-radius: 9999px;
      font-size: 0.8rem;
      font-weight: 600;
      margin-bottom: 0.5rem;
    }
    .card {
      background: #1e293b;
      border: 1px solid #334155;
      border-radius: 12px;
      padding: 1.75rem;
      margin-bottom: 1.5rem;
    }
    .grid-2 {
```

```
      display: grid;
      grid-template-columns: 1fr 1fr;
      gap: 1.5rem;
      margin: 1.5rem 0;
    }
    @media (max-width: 768px) { .grid-2 { grid-template-columns: 1fr; } }
    .box-include {
      border-left: 4px solid #f59e0b;
      background: #0f172a;
      padding: 1.25rem;
      border-radius: 0 8px 8px 0;
    }
    .box-require {
      border-left: 4px solid #ef4444;
      background: #0f172a;
      padding: 1.25rem;
      border-radius: 0 8px 8px 0;
    }
    .btn {
      display: inline-block;
      padding: 0.6rem 1.1rem;
      border-radius: 6px;
      font-size: 0.85rem;
      font-weight: 600;
      text-decoration: none;
      margin-right: 0.5rem;
      margin-top: 0.5rem;
      cursor: pointer;
    }
    .btn-neutral { background: #334155; color: #fff; }
    .btn-warn { background: #d97706; color: #fff; }
    .btn-danger { background: #dc2626; color: #fff; }
    .output-console {
      background: #000;
      border: 1px solid #334155;
      border-radius: 8px;
      padding: 1rem;
      font-family: 'Fira Code', monospace;
      font-size: 0.85rem;
      color: #38bdf8;
      margin-top: 1rem;
    }
  </style>
</head>
<body>

  <!-- Step 2: Include UI Header via include() -->
  <?php include __DIR__ . '/header.php'; ?>

  <main>
    <div style="text-align: center; margin-bottom: 2rem;">
      <span class="badge">Experiment 20 &bull; Modular PHP Architecture</span>
      <h1 style="font-size: 2rem; font-weight: 800; margin: 0 0 0.5rem 0;">Demonstration of includ
e( ) and require( )</h1>
      <p style="color: #94a3b8; font-size: 0.95rem;">Understand the execution lifecycle, warning v
ersus fatal error handling, and component reusability</p>
    </div>

    <!-- Interactive Mode Switcher -->
    <div class="card">
      <h3 style="color: #38bdf8; margin: 0 0 0.75rem 0;">🧪 Interactive Test Bench: Choose Execut
ion Scenario</h3>
      <p style="color: #cbd5e1; font-size: 0.9rem; margin-bottom: 1rem;">
        Click a button below to trigger normal execution or test missing file behaviors:
      </p>
      <div>
```

```
        <a href="include_require_demo.php?test=normal" class="btn btn-neutral">Scenario 1: Standar
d Execution (All Files Exist)</a>
        <a href="include_require_demo.php?test=missing_include" class="btn btn-warn">Scenario 2: M
issing include() File (Throws Warning &amp;amp; Continues)</a>
        <a href="include_require_demo.php?test=missing_require" class="btn btn-danger">Scenario 3:
Missing require() File (Fatal Error &amp;amp; Halts)</a>
      </div>
    </div>

    <!-- Active Scenario Result Box -->
    <div class="card">
      <h3 style="color: #f8fafc; margin: 0 0 0.5rem 0;">Execution Console Output:</h3>

      <?php
      if ($testMode === 'normal') {
          echo "<div class='output-console'>";
          echo "<span style='color: #10b981;'>[OK]</span> require_once 'config.php' executed succe
ssfully.<br>";
          echo "<span style='color: #10b981;'>[OK]</span> include 'header.php' executed successful
ly.<br>";
          echo "<span style='color: #38bdf8;'>[INFO]</span> Application Name: " . APP_NAME . "<br
>";
          echo "<span style='color: #38bdf8;'>[INFO]</span> Institution: " . INSTITUTION . "<br>";
          $sys = get_system_status();
          echo "<span style='color: #38bdf8;'>[INFO]</span> Server Timestamp: " . $sys['server_tim
e'] . " | PHP Version: " . $sys['php_version'] . "<br>";
          echo "<span style='color: #10b981;'>[OK]</span> All files loaded without interruption.";
          echo "</div>";
      } elseif ($testMode === 'missing_include') {
          echo "<div class='output-console'>";
          echo "<span style='color: #f59e0b;'>[TEST TRIGGERED]</span> Calling: <code>include 'non_
existent_optional_widget.php';</code><br><br>";

          // Execute include on missing file
          include 'non_existent_optional_widget.php';

          echo "<br><span style='color: #10b981;'>[KEY TAKEAWAY]</span> <strong>Script execution C
ONTINUES!</strong> Even though the file was missing and an E_WARNING was generated, the page body
and footer continue to render successfully.";
          echo "</div>";
      } elseif ($testMode === 'missing_require') {
          echo "<div class='output-console'>";
          echo "<span style='color: #ef4444;'>[TEST TRIGGERED]</span> Calling: <code>require 'non_
existent_critical_auth.php';</code><br>";
          echo "<span style='color: #f87171;'>[NOTICE]</span> In standard PHP, the line below gene
rates a <strong>Fatal error: E_COMPILE_ERROR</strong> and the remaining script (including this pag
e and the footer) will NOT execute!<br><br>";

          // Require missing critical file
          require 'non_existent_critical_auth.php';

          // The line below is intentionally unreachable due to fatal error
          echo "<span style='color: red;'>This line will never print because require halts executi
on!</span>";
          echo "</div>";
      }
      ?>
    </div>

    <!-- Comparative Theory Grid -->
    <div class="grid-2">
      <div class="box-include">
        <h4 style="color: #f59e0b; margin: 0 0 0.5rem 0;">1. include( ) &amp;amp; include_once( )</h4>
        <ul style="color: #cbd5e1; font-size: 0.85rem; padding-left: 1.25rem; line-height: 1.7;">
          <li><strong>Severity:</strong> <code>E_WARNING</code> if file missing.</li>
          <li><strong>Behavior:</strong> Does <em>not</em> stop script execution.</li>
```

```
          <li><strong>Best For:</strong> Reusable view components, page footers, banners, optional sidebar panels.</li>
          <li><strong>Once variant:</strong> <code>include_once()</code> ensures the file is included exactly once per request.</li>
        </ul>
      </div>

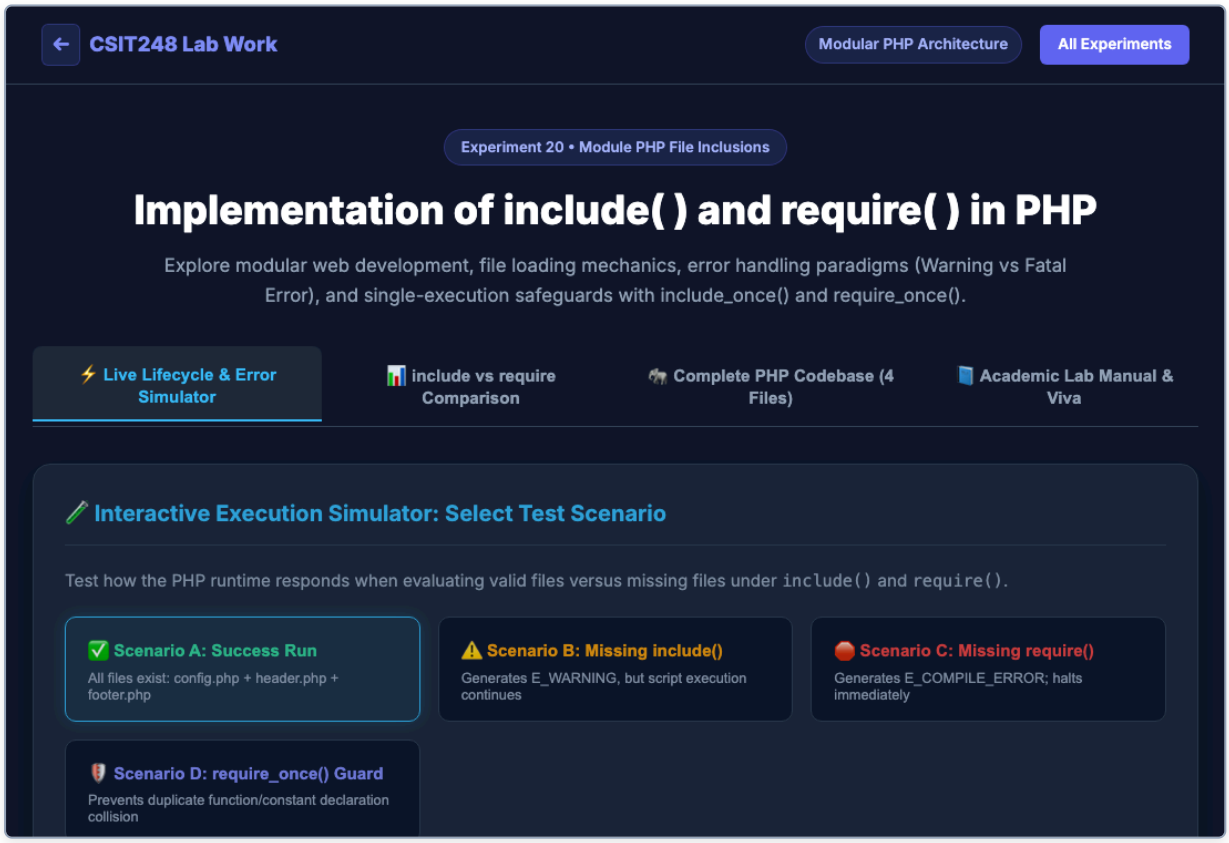
      <div class="box-require">
        <h4 style="color: #ef4444; margin: 0 0 0.5rem 0;">2. require( ) &amp;amp; require_once( )</h4>
        <ul style="color: #cbd5e1; font-size: 0.85rem; padding-left: 1.25rem; line-height: 1.7;">
          <li><strong>Severity:</strong> <code>E_COMPILE_ERROR</code> (Fatal Error).</li>
          <li><strong>Behavior:</strong> <em>Immediately halts</em> script execution.</li>
          <li><strong>Best For:</strong> Mission-critical files like database connection, authentication, encryption keys, core libraries.</li>
          <li><strong>Once variant:</strong> <code>require_once()</code> prevents function and class re-declaration collisions.</li>
        </ul>
      </div>
    </div>

  </main>

  <!-- Step 3: Include UI Footer via include() -->
  <?php include __DIR__ . '/footer.php'; ?>

</body>
</html>
```

Output :



Experiment No.: 24

Title : Demonstration of Array, String and Numeric Functions in PHP.

Aim : To write a PHP script to demonstrate and execute native array functions (array_push, sort), string functions (strlen, strpos, str_replace), and numeric functions (round, sqrt, rand).

Source Code :

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description"
    content="Experiment 21: Write PHP Script to demonstrate array, string and numeric function - C
SIT248 Advanced Web Technology">
  <title>Experiment 21 - PHP Array, String &amp;amp; Numeric Functions Sandbox</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link
    href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&family=Fira+Cod
e:wght@400;500;600&display=swap"
    rel="stylesheet">
  <style>
    :root {
      --primary: #6366f1;
      --primary-hover: #4f46e5;
      --secondary: #06b6d4;
      --accent: #f43f5e;
      --success: #10b981;
      --warning: #f59e0b;
      --danger: #ef4444;
      --bg-dark: #0f172a;
      --card-bg: #1e293b;
      --card-border: #334155;
      --text-main: #f8fafc;
      --text-muted: #94a3b8;
    }

    * {
      box-sizing: border-box;
      margin: 0;
      padding: 0;
    }

    body {
      font-family: 'Inter', system-ui, -apple-system, sans-serif;
      background-color: var(--bg-dark);
      color: var(--text-main);
      line-height: 1.6;
      min-height: 100vh;
      display: flex;
      flex-direction: column;
    }

    /* Sticky Navigation */
    nav {
      position: sticky;
      top: 0;
      z-index: 100;
```

```
  background: rgba(15, 23, 42, 0.92);
  backdrop-filter: blur(12px);
  border-bottom: 1px solid var(--card-border);
  padding: 1rem 2rem;
  display: flex;
  justify-content: space-between;
  align-items: center;
}

.nav-brand {
  font-weight: 800;
  font-size: 1.15rem;
  color: #818cf8;
  text-decoration: none;
  display: flex;
  align-items: center;
  gap: 0.5rem;
}

.nav-brand span {
  background: rgba(99, 102, 241, 0.15);
  border: 1px solid rgba(99, 102, 241, 0.3);
  padding: 0.2rem 0.5rem;
  border-radius: 6px;
}

.nav-links {
  display: flex;
  align-items: center;
  gap: 1rem;
}

.nav-btn {
  background: var(--primary);
  color: #ffffff;
  padding: 0.45rem 1rem;
  border-radius: 6px;
  text-decoration: none;
  font-size: 0.85rem;
  font-weight: 600;
  transition: background 0.2s;
}

.nav-btn:hover {
  background: var(--primary-hover);
}

/* Main Container */
main {
  max-width: 1200px;
  width: 100%;
  margin: 0 auto;
  padding: 2.5rem 1.5rem;
  flex: 1;
}

/* Header */
.page-header {
  text-align: center;
  margin-bottom: 2rem;
}

.badge-pill {
  display: inline-block;
  background: rgba(99, 102, 241, 0.15);
  color: #a5b4fc;
```

```
  border: 1px solid rgba(99, 102, 241, 0.3);
  padding: 0.3rem 0.9rem;
  border-radius: 9999px;
  font-size: 0.825rem;
  font-weight: 600;
  margin-bottom: 0.75rem;
}

.page-title {
  font-size: 2.25rem;
  font-weight: 800;
  color: #ffffff;
  margin-bottom: 0.5rem;
}

.page-subtitle {
  color: var(--text-muted);
  font-size: 1.05rem;
  max-width: 820px;
  margin: 0 auto;
}

/* Tabs Nav */
.tabs-nav {
  display: flex;
  gap: 0.5rem;
  border-bottom: 1px solid var(--card-border);
  margin-bottom: 2rem;
  overflow-x: auto;
  padding-bottom: 0.25rem;
}

.tab-btn {
  background: transparent;
  border: none;
  color: var(--text-muted);
  padding: 0.75rem 1.25rem;
  font-size: 0.95rem;
  font-weight: 600;
  cursor: pointer;
  border-radius: 8px 8px 0 0;
  display: flex;
  align-items: center;
  gap: 0.5rem;
  transition: all 0.2s;
}

.tab-btn:hover {
  color: #ffffff;
  background: rgba(255, 255, 255, 0.05);
}

.tab-btn.active {
  color: #38bdf8;
  background: var(--card-bg);
  border-bottom: 2px solid #38bdf8;
}

.tab-panel {
  display: none;
}

.tab-panel.active {
  display: block;
  animation: fadeIn 0.3s ease-in-out;
}
```

```
@keyframes fadeIn {
  from { opacity: 0; transform: translateY(6px); }
  to { opacity: 1; transform: translateY(0); }
}

.card {
  background: var(--card-bg);
  border: 1px solid var(--card-border);
  border-radius: 16px;
  padding: 1.75rem;
  margin-bottom: 1.5rem;
  box-shadow: 0 10px 25px rgba(0, 0, 0, 0.3);
}

.card-title {
  font-size: 1.25rem;
  font-weight: 700;
  color: #38bdf8;
  margin-bottom: 1.25rem;
  display: flex;
  align-items: center;
  gap: 0.5rem;
  border-bottom: 1px solid var(--card-border);
  padding-bottom: 0.75rem;
}

/* Sub-tabs for Sandbox (Array, String, Numeric) */
.sandbox-tabs {
  display: flex;
  gap: 0.5rem;
  margin-bottom: 1.5rem;
}

.sub-tab-btn {
  background: #0f172a;
  border: 1px solid var(--card-border);
  color: #cbd5e1;
  padding: 0.6rem 1.25rem;
  border-radius: 8px;
  font-size: 0.9rem;
  font-weight: 600;
  cursor: pointer;
  transition: all 0.2s;
}

.sub-tab-btn.active {
  background: var(--primary);
  color: #ffffff;
  border-color: var(--primary);
  box-shadow: 0 4px 12px rgba(99, 102, 241, 0.3);
}

/* Sandbox Form & Controls */
.form-group {
  margin-bottom: 1.25rem;
}

label {
  display: block;
  font-size: 0.85rem;
  font-weight: 600;
  color: #cbd5e1;
  margin-bottom: 0.4rem;
}
```

```
input[type="text"],
input[type="number"],
select {
  width: 100%;
  background: #0f172a;
  border: 1px solid #475569;
  color: #ffffff;
  padding: 0.75rem 1rem;
  border-radius: 8px;
  font-size: 0.95rem;
  outline: none;
  transition: border-color 0.2s;
}

input:focus,
select:focus {
  border-color: var(--primary);
}

.output-grid {
  display: grid;
  grid-template-columns: repeat(auto-fill, minmax(260px, 1fr));
  gap: 1rem;
  margin-top: 1.5rem;
}

.output-card {
  background: #0f172a;
  border: 1px solid var(--card-border);
  border-radius: 10px;
  padding: 1.15rem;
  transition: transform 0.2s;
}

.output-card:hover {
  border-color: #38bdf8;
  transform: translateY(-2px);
}

.output-card .func-name {
  font-family: 'Fira Code', monospace;
  font-size: 0.85rem;
  color: #818cf8;
  font-weight: 700;
  margin-bottom: 0.25rem;
}

.output-card .func-desc {
  font-size: 0.75rem;
  color: var(--text-muted);
  margin-bottom: 0.65rem;
}

.output-card .func-result {
  font-family: 'Fira Code', monospace;
  font-size: 1rem;
  color: #34d399;
  font-weight: 700;
  background: #1e293b;
  padding: 0.5rem 0.75rem;
  border-radius: 6px;
  word-break: break-all;
}

/* Code Viewers */
.code-container {
```

```
      background: #0f172a;
      border: 1px solid var(--card-border);
      border-radius: 12px;
      overflow: hidden;
    }

    .code-header {
      background: #1e293b;
      padding: 0.75rem 1.25rem;
      display: flex;
      justify-content: space-between;
      align-items: center;
      border-bottom: 1px solid var(--card-border);
    }

    .code-header span {
      font-family: 'Fira Code', monospace;
      font-size: 0.85rem;
      color: #38bdf8;
    }

    .btn-copy {
      background: #334155;
      color: #f8fafc;
      border: 1px solid #475569;
      padding: 0.35rem 0.75rem;
      border-radius: 6px;
      font-size: 0.8rem;
      cursor: pointer;
      transition: all 0.2s;
    }

    .btn-copy:hover {
      background: var(--primary);
      border-color: var(--primary);
    }

    pre code {
      display: block;
      padding: 1.25rem;
      font-family: 'Fira Code', monospace;
      font-size: 0.875rem;
      line-height: 1.6;
      color: #e2e8f0;
      overflow-x: auto;
      white-space: pre;
    }

    /* Footer */
    footer {
      background: #0b1120;
      border-top: 1px solid var(--card-border);
      padding: 1.5rem 2rem;
      text-align: center;
      font-size: 0.85rem;
      color: var(--text-muted);
      margin-top: 3rem;
    }
  </style>
</head>

<body>

  <!-- Navigation -->
  <nav>
    <a href="../index.html" class="nav-brand">
```

```
      <span>&larr;</span> CSIT248 Lab Work
    </a>
    <div class="nav-links">
      <span style="font-size: 0.85rem; color: #a5b4fc; background: rgba(99, 102, 241, 0.15); borde
r: 1px solid rgba(99, 102, 241, 0.3); padding: 0.3rem 0.75rem; border-radius: 9999px; font-weight:
600;">
        PHP Standard Library
      </span>
      <a href="../index.html" class="nav-btn">All Experiments</a>
    </div>
  </nav>

  <!-- Main Container -->
  <main>

    <header class="page-header">
      <span class="badge-pill">Experiment 21 &bull; Module PHP Core Functions</span>
      <h1 class="page-title">Demonstration of Array, String and Numeric Functions</h1>
      <p class="page-subtitle">
        Interactive testing studio executing and evaluating PHP's built-in array manipulation algo
rithms, string formatting and pattern operations, and mathematical precision functions.
      </p>
    </header>

    <!-- Tabs Nav -->
    <div class="tabs-nav">
      <button class="tab-btn active" onclick="switchTab('sandbox', this)">
        ⚡ Live Interactive Sandbox
      </button>
      <button class="tab-btn" onclick="switchTab('code', this)">
        🐘 functions_demo.php (Server Script)
      </button>
      <button class="tab-btn" onclick="switchTab('manual', this)">
        📘 Academic Lab Manual &amp;amp; Theory
      </button>
    </div>

    <!-- TAB 1: Live Interactive Sandbox -->
    <section id="tab-sandbox" class="tab-panel active">
      <div class="card">

        <!-- Category Toggle -->
        <div class="sandbox-tabs">
          <button type="button" class="sub-tab-btn active" onclick="switchCategory('array', thi
s)">
            📚 1. Array Functions
          </button>
          <button type="button" class="sub-tab-btn" onclick="switchCategory('string', this)">
            🔤 2. String Functions
          </button>
          <button type="button" class="sub-tab-btn" onclick="switchCategory('numeric', this)">
            🔢 3. Numeric &amp;amp; Math Functions
          </button>
        </div>

        <!-- 1. Array Sandbox -->
        <div id="sandbox-array" class="category-panel">
          <div class="form-group">
            <label for="array-input">Comma-Separated Array Elements:</label>
            <input type="text" id="array-input" value="Apple, Banana, Cherry, Mango, Banana, Orang
e" oninput="computeArrayFunctions()">
          </div>
          <div class="output-grid" id="array-results">
            <!-- Dynamically populated -->
          </div>
        </div>
```

```
        <!-- 2. String Sandbox -->
        <div id="sandbox-string" class="category-panel" style="display: none;">
          <div class="form-group">
            <label for="string-input">Input String for Analysis:</label>
            <input type="text" id="string-input" value="Advanced Web Technology in PHP" oninput="c
omputeStringFunctions()">
          </div>
          <div class="output-grid" id="string-results">
            <!-- Dynamically populated -->
          </div>
        </div>

        <!-- 3. Numeric Sandbox -->
        <div id="sandbox-numeric" class="category-panel" style="display: none;">
          <div style="display: grid; grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
gap: 1rem;">
            <div class="form-group">
              <label for="num-1">Number X:</label>
              <input type="number" id="num-1" value="-49.65" step="any" oninput="computeNumericFun
ctions()">
            </div>
            <div class="form-group">
              <label for="num-2">Number Y:</label>
              <input type="number" id="num-2" value="8" step="any" oninput="computeNumericFunction
s()">
            </div>
            <div class="form-group">
              <label for="num-large">Financial Value Z:</label>
              <input type="number" id="num-large" value="1250500.85" step="any" oninput="computeNu
mericFunctions()">
            </div>
          </div>
          <div class="output-grid" id="numeric-results">
            <!-- Dynamically populated -->
          </div>
        </div>

      </div>
    </section>

    <!-- TAB 2: PHP Script View -->
    <section id="tab-code" class="tab-panel">
      <div class="card">
        <div class="code-container">
          <div class="code-header">
            <span>functions_demo.php (Complete Implementation)</span>
            <button class="btn-copy" onclick="copyCode('code-functions', this)">📋 Copy Script</b
utton>
          </div>
          <pre><code id="code-functions">&lt;?php
/**
 * Experiment 21: Demonstration of Array, String, and Numeric Functions in PHP
 * Course: Advanced Web Technology (CSIT248) - Amity University Noida
 */

// 1. Array Functions
$fruits = ["Apple", "Banana", "Cherry", "Mango", "Banana"];
$numbers = [45, 12, 89, 34, 67, 23];

echo "Count: " . count($fruits) . "\n";
echo "Unique: " . implode(", ", array_unique($fruits)) . "\n";
echo "Search: " . (in_array("Cherry", $fruits) ? "Found" : "Not Found") . "\n";
echo "Reversed: " . implode(", ", array_reverse($numbers)) . "\n";

// Sorting
```

```
sort($numbers);
echo "Sorted Asc: " . implode(", ", $numbers) . "\n";
rsort($numbers);
echo "Sorted Desc: " . implode(", ", $numbers) . "\n";

// 2. String Functions
$str = "Advanced Web Technology in PHP";
echo "Length: " . strlen($str) . "\n";
echo "Word Count: " . str_word_count($str) . "\n";
echo "Uppercase: " . strtoupper($str) . "\n";
echo "Lowercase: " . strtolower($str) . "\n";
echo "Position of PHP: " . strpos($str, "PHP") . "\n";
echo "Replaced: " . str_replace("PHP", "Modern Web", $str) . "\n";
echo "Substring (0, 8): " . substr($str, 0, 8) . "\n";
echo "Reversed: " . strrev("Amity") . "\n";

// 3. Numeric / Math Functions
$val = -42.85;
echo "Absolute: " . abs($val) . "\n";
echo "Round (1 dec): " . round($val, 1) . "\n";
echo "Ceil: " . ceil($val) . "\n";
echo "Floor: " . floor($val) . "\n";
echo "Square Root: " . sqrt(144) . "\n";
echo "Power 2^8: " . pow(2, 8) . "\n";
echo "Min: " . min(15, 82, 4, 99) . " | Max: " . max(15, 82, 4, 99) . "\n";
echo "Random (100-999): " . rand(100, 999) . "\n";
echo "Formatted: " . number_format(1250000.75, 2) . "\n";
?&gt;</code></pre>
        </div>
      </div>
    </section>

    <!-- TAB 3: Academic Lab Manual & Theory -->
    <section id="tab-manual" class="tab-panel">
      <div class="card" style="line-height: 1.8;">
        <h2 style="color: #38bdf8; font-size: 1.5rem; margin-bottom: 1rem;">Experiment 21: Academi
c Lab Manual</h2>

        <h3 style="color: #818cf8; margin-top: 1.5rem; margin-bottom: 0.5rem;">1. Aim</h3>
        <p style="color: #cbd5e1;">
          To design and execute a comprehensive PHP script demonstrating the syntax, operational s
emantics, and practical usage of key built-in functions spanning indexed/associative arrays, strin
g manipulation, and mathematical/numeric calculations.
        </p>

        <h3 style="color: #818cf8; margin-top: 1.5rem; margin-bottom: 0.5rem;">2. Core Function Ta
xonomy</h3>
        <div style="display: grid; grid-template-columns: repeat(auto-fit, minmax(280px, 1fr)); ga
p: 1rem; margin-top: 1rem;">
          <div style="background: #0f172a; padding: 1.25rem; border: 1px solid #334155; border-rad
ius: 8px;">
            <h4 style="color: #38bdf8; margin-bottom: 0.5rem;">Array Functions</h4>
            <ul style="color: #cbd5e1; font-size: 0.85rem; padding-left: 1.25rem; line-height: 1.
7;">
              <li><code>count()</code>: Counts array items</li>
              <li><code>sort()</code> &amp;amp; <code>rsort()</code>: Asc/Desc sorting</li>
              <li><code>in_array()</code>: Value existence search</li>
              <li><code>array_unique()</code>: Deduplication</li>
              <li><code>array_reverse()</code>: Order inversion</li>
              <li><code>array_merge()</code>: Combining arrays</li>
            </ul>
          </div>

          <div style="background: #0f172a; padding: 1.25rem; border: 1px solid #334155; border-rad
ius: 8px;">
            <h4 style="color: #a5b4fc; margin-bottom: 0.5rem;">String Functions</h4>
```

```
            <ul style="color: #cbd5e1; font-size: 0.85rem; padding-left: 1.25rem; line-height: 1.
7;">
              <li><code>strlen()</code>: Character length</li>
              <li><code>str_word_count()</code>: Total words</li>
              <li><code>strtoupper()</code> &amp;amp; <code>strtolower()</code></li>
              <li><code>strpos()</code>: Substring position finder</li>
              <li><code>str_replace()</code>: Search and replace</li>
              <li><code>substr()</code>: Extraction of substring</li>
            </ul>
          </div>

          <div style="background: #0f172a; padding: 1.25rem; border: 1px solid #334155; border-rad
ius: 8px;">
            <h4 style="color: #34d399; margin-bottom: 0.5rem;">Numeric / Math</h4>
            <ul style="color: #cbd5e1; font-size: 0.85rem; padding-left: 1.25rem; line-height: 1.
7;">
              <li><code>abs()</code>: Absolute non-negative value</li>
              <li><code>round()</code>, <code>ceil()</code>, <code>floor()</code></li>
              <li><code>sqrt()</code> &amp;amp; <code>pow()</code>: Exponent/root</li>
              <li><code>min()</code> &amp;amp; <code>max()</code>: Bounds detection</li>
              <li><code>rand()</code>: Pseudorandom integer</li>
              <li><code>number_format()</code>: Currency/comma notation</li>
            </ul>
          </div>
        </div>
      </div>
    </section>

  </main>

  <!-- Footer -->
  <footer>
    <p style="margin-bottom: 0.4rem;">
      Designed &amp;amp; Developed by <strong style="color: #818cf8;">Atul Anand</strong> &bull; BCA
(Hons.) with Research &bull; Semester III
    </p>
    <p style="color: #64748b; font-size: 0.8rem;">
      Amity University Noida &bull; Department of Computer Applications &bull; CSIT248: Advanced W
eb Technology
    </p>
  </footer>

  <!-- Script for Sandbox Real-time Computations -->
  <script>
    function switchCategory(cat, btn) {
      document.querySelectorAll('.sub-tab-btn').forEach(b => b.classList.remove('active'));
      document.querySelectorAll('.category-panel').forEach(p => p.style.display = 'none');
      btn.classList.add('active');
      document.getElementById('sandbox-' + cat).style.display = 'block';
    }

    function switchTab(tabId, btn) {
      document.querySelectorAll('.tab-panel').forEach(p => p.classList.remove('active'));
      document.querySelectorAll('.tab-btn').forEach(b => b.classList.remove('active'));
      document.getElementById('tab-' + tabId).classList.add('active');
      btn.classList.add('active');
    }

    // 1. Array Computations
    function computeArrayFunctions() {
      const raw = document.getElementById('array-input').value;
      const arr = raw.split(',').map(s => s.trim()).filter(s => s.length > 0);
      const container = document.getElementById('array-results');

      const count = arr.length;
      const unique = Array.from(new Set(arr));
```

```
  const reversed = [...arr].reverse();
  const sortedAsc = [...arr].sort();
  const sortedDesc = [...arr].sort().reverse();
  const first = arr[0] || 'N/A';
  const last = arr[arr.length - 1] || 'N/A';

  container.innerHTML = `
    <div class="output-card">
      <div class="func-name">count($arr)</div>
      <div class="func-desc">Number of elements in array</div>
      <div class="func-result">${count} elements</div>
    </div>
    <div class="output-card">
      <div class="func-name">array_unique($arr)</div>
      <div class="func-desc">Removes duplicate values</div>
      <div class="func-result">[${unique.join(', ')}]</div>
    </div>
    <div class="output-card">
      <div class="func-name">array_reverse($arr)</div>
      <div class="func-desc">Returns array in reversed order</div>
      <div class="func-result">[${reversed.join(', ')}]</div>
    </div>
    <div class="output-card">
      <div class="func-name">sort($arr)</div>
      <div class="func-desc">Sorts array ascending</div>
      <div class="func-result">[${sortedAsc.join(', ')}]</div>
    </div>
    <div class="output-card">
      <div class="func-name">rsort($arr)</div>
      <div class="func-desc">Sorts array descending</div>
      <div class="func-result">[${sortedDesc.join(', ')}]</div>
    </div>
    <div class="output-card">
      <div class="func-name">reset() / end()</div>
      <div class="func-desc">First and last elements</div>
      <div class="func-result">${first} &bull; ${last}</div>
    </div>
  `;
}

// 2. String Computations
function computeStringFunctions() {
  const s = document.getElementById('string-input').value;
  const container = document.getElementById('string-results');

  const len = s.length;
  const words = s.trim().length === 0 ? 0 : s.trim().split(/\s+/).length;
  const upper = s.toUpperCase();
  const lower = s.toLowerCase();
  const reversed = s.split('').reverse().join('');
  const sub = s.slice(0, 8);
  const replaced = s.replace(/PHP/gi, 'Modern Web');

  container.innerHTML = `
    <div class="output-card">
      <div class="func-name">strlen($str)</div>
      <div class="func-desc">Total character count</div>
      <div class="func-result">${len} chars</div>
    </div>
    <div class="output-card">
      <div class="func-name">str_word_count($str)</div>
      <div class="func-desc">Count words in string</div>
      <div class="func-result">${words} words</div>
    </div>
    <div class="output-card">
      <div class="func-name">strtoupper($str)</div>
```

```
          <div class="func-desc">Convert string to uppercase</div>
          <div class="func-result">${upper}</div>
        </div>
        <div class="output-card">
          <div class="func-name">strtolower($str)</div>
          <div class="func-desc">Convert string to lowercase</div>
          <div class="func-result">${lower}</div>
        </div>
        <div class="output-card">
          <div class="func-name">strrev($str)</div>
          <div class="func-desc">Reverse a string</div>
          <div class="func-result">${reversed}</div>
        </div>
        <div class="output-card">
          <div class="func-name">substr($str, 0, 8)</div>
          <div class="func-desc">Extract first 8 characters</div>
          <div class="func-result">"${sub}"</div>
        </div>
        <div class="output-card">
          <div class="func-name">str_replace("PHP", "Modern Web")</div>
          <div class="func-desc">Replace search term with replacement</div>
          <div class="func-result">"${replaced}"</div>
        </div>
      `;
    }

    // 3. Numeric Computations
    function computeNumericFunctions() {
      const x = parseFloat(document.getElementById('num-1').value) || 0;
      const y = parseFloat(document.getElementById('num-2').value) || 0;
      const z = parseFloat(document.getElementById('num-large').value) || 0;
      const container = document.getElementById('numeric-results');

      const absolute = Math.abs(x);
      const rounded = Math.round(x);
      const ceiling = Math.ceil(x);
      const floorVal = Math.floor(x);
      const squareRoot = Math.abs(y) >= 0 ? Math.sqrt(Math.abs(y)).toFixed(3) : 'NaN';
      const power = Math.pow(2, y);
      const formatted = new Intl.NumberFormat('en-IN', { minimumFractionDigits: 2, maximumFraction
Digits: 2 }).format(z);
      const randomVal = Math.floor(Math.random() * (999 - 100 + 1)) + 100;

      container.innerHTML = `
        <div class="output-card">
          <div class="func-name">abs(X)</div>
          <div class="func-desc">Absolute magnitude of ${x}</div>
          <div class="func-result">${absolute}</div>
        </div>
        <div class="output-card">
          <div class="func-name">round(X)</div>
          <div class="func-desc">Nearest integer</div>
          <div class="func-result">${rounded}</div>
        </div>
        <div class="output-card">
          <div class="func-name">ceil(X) / floor(X)</div>
          <div class="func-desc">Ceil vs Floor of ${x}</div>
          <div class="func-result">Ceil: ${ceiling} | Floor: ${floorVal}</div>
        </div>
        <div class="output-card">
          <div class="func-name">sqrt(|Y|)</div>
          <div class="func-desc">Square root of ${Math.abs(y)}</div>
          <div class="func-result">${squareRoot}</div>
        </div>
        <div class="output-card">
          <div class="func-name">pow(2, Y)</div>
```

```
          <div class="func-desc">Base 2 raised to power ${y}</div>
          <div class="func-result">${power}</div>
        </div>
        <div class="output-card">
          <div class="func-name">rand(100, 999)</div>
          <div class="func-desc">Random integer generator</div>
          <div class="func-result">${randomVal}</div>
        </div>
        <div class="output-card">
          <div class="func-name">number_format(Z, 2)</div>
          <div class="func-desc">Formatted with commas &amp;amp; decimals</div>
          <div class="func-result">${formatted}</div>
        </div>
      `;
    }

    function copyCode(elementId, btn) {
      const text = document.getElementById(elementId).innerText;
      navigator.clipboard.writeText(text).then(() => {
        const orig = btn.innerText;
        btn.innerText = "Copied! ✅";
        btn.style.background = "#10b981";
        setTimeout(() => {
          btn.innerText = orig;
          btn.style.background = "";
        }, 2000);
      });
    }

    window.addEventListener("DOMContentLoaded", () => {
      computeArrayFunctions();
      computeStringFunctions();
      computeNumericFunctions();
    });
  </script>

</body>

</html>
```

Supplementary Source File: functions_demo.php

```
<?php
/**
 * Experiment 21: Demonstration of Array, String, and Numeric Functions in PHP
 * Course: Advanced Web Technology (CSIT248) - Amity University Noida
 *
 * Objectives:
 *   Demonstrate core built-in PHP functions across three primary primitive/data structures:
 *   1. Array Functions (manipulation, searching, sorting, mapping, reduction)
 *   2. String Functions (length, casing, pattern matching, tokenizing, slicing)
 *   3. Numeric / Mathematical Functions (arithmetic, rounding, random generation, formatting)
 */

header('Content-Type: text/html; charset=utf-8');
?>
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Experiment 21 - PHP Built-in Functions Masterclass</title>
  <style>
    body { font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif; background: #0f172a; colo
r: #f8fafc; padding: 2rem; }
    h1, h2, h3 { color: #38bdf8; }
    .section { background: #1e293b; border: 1px solid #334155; border-radius: 10px; padding: 1.5re
m; margin-bottom: 2rem; }
```

```
    table { width: 100%; border-collapse: collapse; margin-top: 1rem; }
    th, td { padding: 0.65rem 0.85rem; border: 1px solid #334155; text-align: left; font-size: 0.9
rem; }
    th { background: #0f172a; color: #a5b4fc; }
    code { font-family: Consolas, monospace; color: #38bdf8; background: #0b1120; padding: 0.15rem
0.4rem; border-radius: 4px; }
    .result { font-weight: bold; color: #34d399; font-family: Consolas, monospace; }
  </style>
</head>
<body>

  <h1>Experiment 21: Array, String &amp;amp; Numeric Functions in PHP</h1>
  <p style="color: #94a3b8;">CSIT248 Advanced Web Technology &bull; Amity University Noida</p>

  <!-- ========================================
       SECTION 1: ARRAY FUNCTIONS
       ======================================== -->
  <div class="section">
    <h2>1. Array Functions in PHP</h2>
    <?php
      $fruits = ["Apple", "Banana", "Cherry", "Mango", "Banana"];
      $numbers = [45, 12, 89, 34, 67, 23];
      $student = ["name" => "Atul Anand", "roll" => "IT2023001", "branch" => "IT", "sem" => "VI"];
    ?>
    <table>
      <thead>
        <tr><th>Function</th><th>Input / Context</th><th>PHP Expression</th><th>Output Result</th>
</tr>
      </thead>
      <tbody>
        <tr>
          <td><code>count()</code></td>
          <td>["Apple", "Banana", "Cherry", "Mango", "Banana"]</td>
          <td><code>count($fruits)</code></td>
          <td class="result"><?php echo count($fruits); ?> elements</td>
        </tr>
        <tr>
          <td><code>array_unique()</code></td>
          <td>Remove duplicates from $fruits</td>
          <td><code>array_unique($fruits)</code></td>
          <td class="result"><?php echo implode(", ", array_unique($fruits)); ?></td>
        </tr>
        <tr>
          <td><code>in_array()</code></td>
          <td>Search for "Cherry" in $fruits</td>
          <td><code>in_array("Cherry", $fruits)</code></td>
          <td class="result"><?php echo in_array("Cherry", $fruits) ? 'TRUE (Found)' : 'FALSE'; ?>
</td>
        </tr>
        <tr>
          <td><code>array_reverse()</code></td>
          <td>Reverse list $numbers</td>
          <td><code>array_reverse($numbers)</code></td>
          <td class="result"><?php echo implode(", ", array_reverse($numbers)); ?></td>
        </tr>
        <tr>
          <td><code>sort()</code></td>
          <td>Sort ascending $numbers</td>
          <td><code>sort($numbers)</code></td>
          <td class="result"><?php
            $tempNums = $numbers;
            sort($tempNums);
            echo implode(", ", $tempNums);
          ?></td>
        </tr>
        <tr>
```

```
          <td><code>rsort()</code></td>
          <td>Sort descending $numbers</td>
          <td><code>rsort($numbers)</code></td>
          <td class="result"><?php
            $tempNums = $numbers;
            rsort($tempNums);
            echo implode(", ", $tempNums);
          ?></td>
        </tr>
        <tr>
          <td><code>array_keys()</code></td>
          <td>Associative student keys</td>
          <td><code>array_keys($student)</code></td>
          <td class="result"><?php echo implode(", ", array_keys($student)); ?></td>
        </tr>
        <tr>
          <td><code>array_merge()</code></td>
          <td>Merge two arrays</td>
          <td><code>array_merge(["A","B"], ["C","D"])</code></td>
          <td class="result"><?php echo implode(", ", array_merge(["A","B"], ["C","D"])); ?></td>
        </tr>
      </tbody>
    </table>
  </div>

  <!-- ========================================
       SECTION 2: STRING FUNCTIONS
       ======================================== -->
  <div class="section">
    <h2>2. String Functions in PHP</h2>
    <?php
      $sampleStr = "Advanced Web Technology in PHP";
      $paddedStr = "   Amity University Noida   ";
    ?>
    <table>
      <thead>
        <tr><th>Function</th><th>Input String</th><th>PHP Expression</th><th>Output Result</th></t
r>
      </thead>
      <tbody>
        <tr>
          <td><code>strlen()</code></td>
          <td>"<?php echo $sampleStr; ?>"</td>
          <td><code>strlen($sampleStr)</code></td>
          <td class="result"><?php echo strlen($sampleStr); ?> characters</td>
        </tr>
        <tr>
          <td><code>str_word_count()</code></td>
          <td>"<?php echo $sampleStr; ?>"</td>
          <td><code>str_word_count($sampleStr)</code></td>
          <td class="result"><?php echo str_word_count($sampleStr); ?> words</td>
        </tr>
        <tr>
          <td><code>strtoupper()</code></td>
          <td>Convert to uppercase</td>
          <td><code>strtoupper($sampleStr)</code></td>
          <td class="result"><?php echo strtoupper($sampleStr); ?></td>
        </tr>
        <tr>
          <td><code>strtolower()</code></td>
          <td>Convert to lowercase</td>
          <td><code>strtolower($sampleStr)</code></td>
          <td class="result"><?php echo strtolower($sampleStr); ?></td>
        </tr>
        <tr>
          <td><code>strpos()</code></td>
```

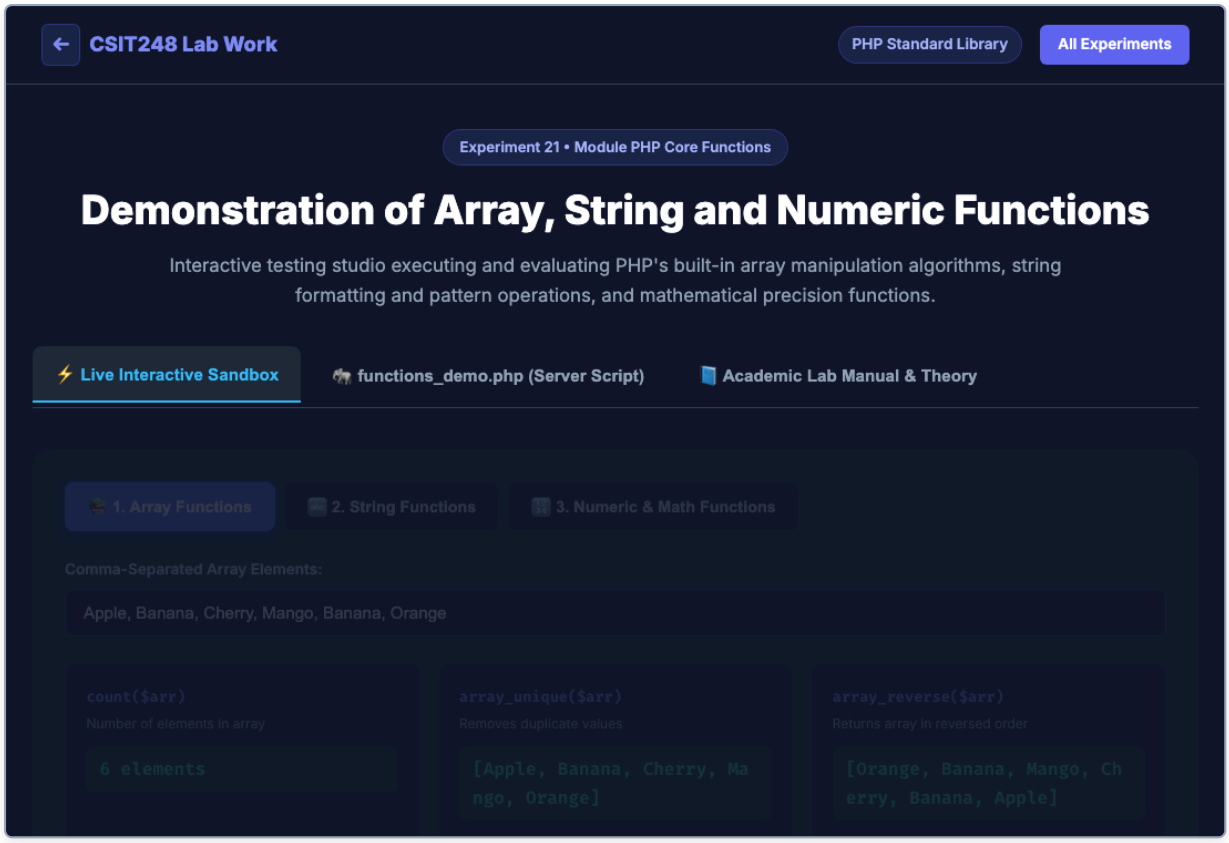
```
          <td>Find index of "PHP"</td>
          <td><code>strpos($sampleStr, "PHP")</code></td>
          <td class="result"><?php echo strpos($sampleStr, "PHP"); ?> (0-indexed position)</td>
        </tr>
        <tr>
          <td><code>str_replace()</code></td>
          <td>Replace "PHP" with "Modern Web"</td>
          <td><code>str_replace("PHP", "Modern Web", ...)</code></td>
          <td class="result"><?php echo str_replace("PHP", "Modern Web", $sampleStr); ?></td>
        </tr>
        <tr>
          <td><code>substr()</code></td>
          <td>Substring characters 0 to 8</td>
          <td><code>substr($sampleStr, 0, 8)</code></td>
          <td class="result">"<?php echo substr($sampleStr, 0, 8); ?>"</td>
        </tr>
        <tr>
          <td><code>strrev()</code></td>
          <td>Reverse string</td>
          <td><code>strrev("Amity")</code></td>
          <td class="result"><?php echo strrev("Amity"); ?></td>
        </tr>
        <tr>
          <td><code>trim()</code></td>
          <td>Strip surrounding whitespaces</td>
          <td><code>trim($paddedStr)</code></td>
          <td class="result">"<?php echo trim($paddedStr); ?>"</td>
        </tr>
        <tr>
          <td><code>explode()</code></td>
          <td>Split string into array by space</td>
          <td><code>explode(" ", $sampleStr)</code></td>
          <td class="result">[<?php echo implode(" | ", explode(" ", $sampleStr)); ?>]</td>
        </tr>
      </tbody>
    </table>
  </div>

  <!-- ========================================
       SECTION 3: NUMERIC / MATH FUNCTIONS
       ======================================== -->
  <div class="section">
    <h2>3. Numeric &amp;amp; Mathematical Functions in PHP</h2>
    <table>
      <thead>
        <tr><th>Function</th><th>Input Values</th><th>PHP Expression</th><th>Output Result</th></t
r>
      </thead>
      <tbody>
        <tr>
          <td><code>abs()</code></td>
          <td>-42.85</td>
          <td><code>abs(-42.85)</code></td>
          <td class="result"><?php echo abs(-42.85); ?></td>
        </tr>
        <tr>
          <td><code>round()</code></td>
          <td>3.14159265 to 3 decimals</td>
          <td><code>round(3.14159265, 3)</code></td>
          <td class="result"><?php echo round(3.14159265, 3); ?></td>
        </tr>
        <tr>
          <td><code>ceil()</code></td>
          <td>7.12 (ceiling integer)</td>
          <td><code>ceil(7.12)</code></td>
          <td class="result"><?php echo ceil(7.12); ?></td>
```

```
        </tr>
        <tr>
          <td><code>floor()</code></td>
          <td>7.89 (floor integer)</td>
          <td><code>floor(7.89)</code></td>
          <td class="result"><?php echo floor(7.89); ?></td>
        </tr>
        <tr>
          <td><code>sqrt()</code></td>
          <td>144</td>
          <td><code>sqrt(144)</code></td>
          <td class="result"><?php echo sqrt(144); ?></td>
        </tr>
        <tr>
          <td><code>pow()</code></td>
          <td>Base 2, Exponent 8</td>
          <td><code>pow(2, 8)</code></td>
          <td class="result"><?php echo pow(2, 8); ?></td>
        </tr>
        <tr>
          <td><code>max()</code> &amp;amp; <code>min()</code></td>
          <td>Values: 15, 82, 4, 99, 36</td>
          <td><code>max(...)</code> / <code>min(...)</code></td>
          <td class="result">Max = <?php echo max(15, 82, 4, 99, 36); ?> | Min = <?php echo min(1
5, 82, 4, 99, 36); ?></td>
        </tr>
        <tr>
          <td><code>rand()</code></td>
          <td>Random integer between 100 and 999</td>
          <td><code>rand(100, 999)</code></td>
          <td class="result"><?php echo rand(100, 999); ?></td>
        </tr>
        <tr>
          <td><code>number_format()</code></td>
          <td>1250000.758 to 2 decimals with commas</td>
          <td><code>number_format(1250000.758, 2)</code></td>
          <td class="result"><?php echo number_format(1250000.758, 2); ?></td>
        </tr>
        <tr>
          <td><code>is_numeric()</code></td>
          <td>Validate numeric string "542.10"</td>
          <td><code>is_numeric("542.10")</code></td>
          <td class="result"><?php echo is_numeric("542.10") ? 'TRUE (Valid Number)' : 'FALSE'; ?>
</td>
        </tr>
      </tbody>
    </table>
  </div>

</body>
</html>
```

Output :


CSIT248 Lab Work
PHP Standard Library
All Experiments
Experiment 21 • Module PHP Core Functions
Demonstration of Array, String and Numeric Functions
Interactive testing studio executing and evaluating PHP's built-in array manipulation algorithms, string formatting and pattern operations, and mathematical precision functions.
⚡ Live Interactive Sandbox
functions_demo.php (Server Script)
Academic Lab Manual & Theory
1. Array Functions
2. String Functions
3. Numeric & Math Functions
Comma-Separated Array Elements:
Apple, Banana, Cherry, Mango, Banana, Orange
count($arr)
Number of elements in array
6 elements
array_unique($arr)
Removes duplicate values
[Apple, Banana, Cherry, Mango, Orange]
array_reverse($arr)
Returns array in reversed order
[Orange, Banana, Mango, Cherry, Banana, Apple]

Experiment No.: 25

Title : Implementation of MySQL DML Commands (DELETE, ORDER BY, UPDATE).

Aim : To implement an interactive database manager in PHP executing atomic MySQL DML operations: DELETE to remove records, ORDER BY for multi-column sorting, and UPDATE with prepared statements.

Source Code :

```
<!DOCTYPE html>
<html lang="en">

<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <meta name="description"
    content="Experiment 22: Write a PHP Script to implement MySQL commands: a) DELETE b) ORDER BY
c) UPDATE - CSIT248 Advanced Web Technology">
  <title>Experiment 22 - MySQL DELETE, ORDER BY &amp;amp; UPDATE Operations</title>
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link
    href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&family=Fira+Cod
e:wght@400;500;600&display=swap"
    rel="stylesheet">
  <style>
    :root {
      --primary: #6366f1;
      --primary-hover: #4f46e5;
      --secondary: #06b6d4;
      --accent: #f43f5e;
      --success: #10b981;
      --warning: #f59e0b;
      --danger: #ef4444;
      --bg-dark: #0f172a;
      --card-bg: #1e293b;
      --card-border: #334155;
      --text-main: #f8fafc;
      --text-muted: #94a3b8;
    }

    * {
      box-sizing: border-box;
      margin: 0;
      padding: 0;
    }

    body {
      font-family: 'Inter', system-ui, -apple-system, sans-serif;
      background-color: var(--bg-dark);
      color: var(--text-main);
      line-height: 1.6;
      min-height: 100vh;
      display: flex;
      flex-direction: column;
    }

    /* Sticky Navigation */
    nav {
      position: sticky;
      top: 0;
      z-index: 100;
```

```
  background: rgba(15, 23, 42, 0.92);
  backdrop-filter: blur(12px);
  border-bottom: 1px solid var(--card-border);
  padding: 1rem 2rem;
  display: flex;
  justify-content: space-between;
  align-items: center;
}

.nav-brand {
  font-weight: 800;
  font-size: 1.15rem;
  color: #818cf8;
  text-decoration: none;
  display: flex;
  align-items: center;
  gap: 0.5rem;
}

.nav-brand span {
  background: rgba(99, 102, 241, 0.15);
  border: 1px solid rgba(99, 102, 241, 0.3);
  padding: 0.2rem 0.5rem;
  border-radius: 6px;
}

.nav-links {
  display: flex;
  align-items: center;
  gap: 1rem;
}

.nav-btn {
  background: var(--primary);
  color: #ffffff;
  padding: 0.45rem 1rem;
  border-radius: 6px;
  text-decoration: none;
  font-size: 0.85rem;
  font-weight: 600;
  transition: background 0.2s;
}

.nav-btn:hover {
  background: var(--primary-hover);
}

/* Main Container */
main {
  max-width: 1200px;
  width: 100%;
  margin: 0 auto;
  padding: 2.5rem 1.5rem;
  flex: 1;
}

/* Header */
.page-header {
  text-align: center;
  margin-bottom: 2rem;
}

.badge-pill {
  display: inline-block;
  background: rgba(99, 102, 241, 0.15);
  color: #a5b4fc;
```

```
  border: 1px solid rgba(99, 102, 241, 0.3);
  padding: 0.3rem 0.9rem;
  border-radius: 9999px;
  font-size: 0.825rem;
  font-weight: 600;
  margin-bottom: 0.75rem;
}

.page-title {
  font-size: 2.25rem;
  font-weight: 800;
  color: #ffffff;
  margin-bottom: 0.5rem;
}

.page-subtitle {
  color: var(--text-muted);
  font-size: 1.05rem;
  max-width: 820px;
  margin: 0 auto;
}

/* Tabs Nav */
.tabs-nav {
  display: flex;
  gap: 0.5rem;
  border-bottom: 1px solid var(--card-border);
  margin-bottom: 2rem;
  overflow-x: auto;
  padding-bottom: 0.25rem;
}

.tab-btn {
  background: transparent;
  border: none;
  color: var(--text-muted);
  padding: 0.75rem 1.25rem;
  font-size: 0.95rem;
  font-weight: 600;
  cursor: pointer;
  border-radius: 8px 8px 0 0;
  display: flex;
  align-items: center;
  gap: 0.5rem;
  transition: all 0.2s;
}

.tab-btn:hover {
  color: #ffffff;
  background: rgba(255, 255, 255, 0.05);
}

.tab-btn.active {
  color: #38bdf8;
  background: var(--card-bg);
  border-bottom: 2px solid #38bdf8;
}

.tab-panel {
  display: none;
}

.tab-panel.active {
  display: block;
  animation: fadeIn 0.3s ease-in-out;
}
```

```
@keyframes fadeIn {
  from { opacity: 0; transform: translateY(6px); }
  to { opacity: 1; transform: translateY(0); }
}

.card {
  background: var(--card-bg);
  border: 1px solid var(--card-border);
  border-radius: 16px;
  padding: 1.75rem;
  margin-bottom: 1.5rem;
  box-shadow: 0 10px 25px rgba(0, 0, 0, 0.3);
}

.card-title {
  font-size: 1.25rem;
  font-weight: 700;
  color: #38bdf8;
  margin-bottom: 1.25rem;
  display: flex;
  align-items: center;
  gap: 0.5rem;
  border-bottom: 1px solid var(--card-border);
  padding-bottom: 0.75rem;
}

/* Interactive Table */
.table-container {
  overflow-x: auto;
  margin-top: 1rem;
}

table {
  width: 100%;
  border-collapse: collapse;
  font-size: 0.9rem;
}

th, td {
  padding: 0.85rem 1rem;
  text-align: left;
  border-bottom: 1px solid #334155;
}

th {
  background: #0f172a;
  color: #38bdf8;
  font-weight: 600;
  user-select: none;
  cursor: pointer;
  transition: background 0.2s;
}

th:hover {
  background: #1e293b;
  color: #818cf8;
}

th.active-sort {
  color: #34d399;
}

.btn-action {
  padding: 0.35rem 0.65rem;
  border-radius: 6px;
```

```
  font-size: 0.8rem;
  font-weight: 600;
  cursor: pointer;
  border: none;
  transition: all 0.2s;
  margin-right: 0.35rem;
}

.btn-update {
  background: #6366f1;
  color: #ffffff;
}

.btn-update:hover {
  background: #4f46e5;
}

.btn-delete {
  background: rgba(239, 68, 68, 0.2);
  border: 1px solid var(--danger);
  color: #fca5a5;
}

.btn-delete:hover {
  background: #ef4444;
  color: #ffffff;
}

/* Modal / Edit Card */
.edit-box {
  background: #0f172a;
  border: 1px solid #38bdf8;
  border-radius: 12px;
  padding: 1.5rem;
  margin-bottom: 1.5rem;
  animation: fadeIn 0.3s;
}

.form-grid {
  display: grid;
  grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
  gap: 1rem;
  margin-bottom: 1rem;
}

label {
  display: block;
  font-size: 0.8rem;
  font-weight: 600;
  color: #cbd5e1;
  margin-bottom: 0.35rem;
}

input[type="text"],
input[type="number"] {
  width: 100%;
  background: #1e293b;
  border: 1px solid #475569;
  color: #ffffff;
  padding: 0.65rem 0.85rem;
  border-radius: 6px;
  font-size: 0.9rem;
  outline: none;
}

input:focus {
```

```
  border-color: #38bdf8;
}

/* Query Inspector */
.sql-inspector {
  background: #090d16;
  border: 1px solid #334155;
  border-radius: 10px;
  padding: 1.25rem;
  margin-top: 1.5rem;
  font-family: 'Fira Code', monospace;
  font-size: 0.85rem;
}

.sql-inspector .label {
  color: #a5b4fc;
  font-weight: 700;
  font-size: 0.8rem;
  text-transform: uppercase;
  margin-bottom: 0.5rem;
  display: block;
}

.sql-inspector code {
  color: #38bdf8;
  display: block;
  white-space: pre-wrap;
}

/* Code Viewers */
.code-container {
  background: #0f172a;
  border: 1px solid var(--card-border);
  border-radius: 12px;
  overflow: hidden;
  margin-bottom: 1.5rem;
}

.code-header {
  background: #1e293b;
  padding: 0.75rem 1.25rem;
  display: flex;
  justify-content: space-between;
  align-items: center;
  border-bottom: 1px solid var(--card-border);
}

.code-header span {
  font-family: 'Fira Code', monospace;
  font-size: 0.85rem;
  color: #38bdf8;
}

.btn-copy {
  background: #334155;
  color: #f8fafc;
  border: 1px solid #475569;
  padding: 0.35rem 0.75rem;
  border-radius: 6px;
  font-size: 0.8rem;
  cursor: pointer;
}

pre code {
  display: block;
  padding: 1.25rem;
```

```
      font-family: 'Fira Code', monospace;
      font-size: 0.875rem;
      line-height: 1.6;
      color: #e2e8f0;
      overflow-x: auto;
      white-space: pre;
    }

    footer {
      background: #0b1120;
      border-top: 1px solid var(--card-border);
      padding: 1.5rem 2rem;
      text-align: center;
      font-size: 0.85rem;
      color: var(--text-muted);
      margin-top: 3rem;
    }
  </style>
</head>

<body>

  <!-- Navigation -->
  <nav>
    <a href="../index.html" class="nav-brand">
      <span>&larr;</span> CSIT248 Lab Work
    </a>
    <div class="nav-links">
      <span style="font-size: 0.85rem; color: #a5b4fc; background: rgba(99, 102, 241, 0.15); borde
r: 1px solid rgba(99, 102, 241, 0.3); padding: 0.3rem 0.75rem; border-radius: 9999px; font-weight:
600;">
        MySQL DML Operations
      </span>
      <a href="../index.html" class="nav-btn">All Experiments</a>
    </div>
  </nav>

  <!-- Main Container -->
  <main>

    <header class="page-header">
      <span class="badge-pill">Experiment 22 &bull; Module PHP &amp; MySQL Data Manipulation</span
>
      <h1 class="page-title">Implementation of MySQL Commands: DELETE, ORDER BY, UPDATE</h1>
      <p class="page-subtitle">
        Interactive database workbench executing atomic records deletion, multi-column sorting (AS
C/DESC), and attribute updates with prepared queries and SQL statement visualization.
      </p>
    </header>

    <!-- Tabs Nav -->
    <div class="tabs-nav">
      <button class="tab-btn active" onclick="switchTab('workbench', this)">
        ⚡ Live Database Workbench
      </button>
      <button class="tab-btn" onclick="switchTab('php', this)">
        🐘 mysql_crud_demo.php (Server Script)
      </button>
      <button class="tab-btn" onclick="switchTab('sql', this)">
        🗄️ schema.sql (Table DDL &amp; Seeds)
      </button>
      <button class="tab-btn" onclick="switchTab('manual', this)">
        📘 Academic Lab Manual &amp; Theory
      </button>
    </div>
```

```
    <!-- TAB 1: Live Workbench -->
    <section id="tab-workbench" class="tab-panel active">
      <div class="card">

        <!-- Header & Controls -->
        <div style="display: flex; justify-content: space-between; align-items: center; margin-bot
tom: 1.25rem; border-bottom: 1px solid var(--card-border); padding-bottom: 0.75rem; flex-wrap: wra
p; gap: 0.75rem;">
          <h2 class="card-title" style="margin-bottom: 0; border: none; padding: 0;">
            📊 Live Table: employees (Click Column Headers to Test b) ORDER BY)
          </h2>
          <button type="button" class="btn-copy" onclick="resetEmployees()">🔄 Reset Sample Recor
ds</button>
        </div>

        <!-- Inline Update Editor (c) UPDATE -->
        <div id="update-container" class="edit-box" style="display: none;">
          <h3 style="color: #38bdf8; margin: 0 0 1rem 0; font-size: 1.1rem;">
            ✏️ Command c) UPDATE: Editing Employee <span id="edit-emp-name" style="color: #fff;">
</span> (<span id="edit-emp-id-badge" style="font-family: 'Fira Code', monospace; color: #a5b4f
c;"></span>)
          </h3>
          <form onsubmit="handleApplyUpdate(event)">
            <input type="hidden" id="edit-id">
            <div class="form-grid">
              <div>
                <label for="edit-dept">Department</label>
                <input type="text" id="edit-dept" required>
              </div>
              <div>
                <label for="edit-desig">Designation</label>
                <input type="text" id="edit-desig" required>
              </div>
              <div>
                <label for="edit-salary">Salary (₹)</label>
                <input type="number" id="edit-salary" step="1000" required>
              </div>
            </div>
            <div style="display: flex; gap: 0.5rem;">
              <button type="submit" class="btn-action btn-update" style="padding: 0.6rem 1.25re
m;">💾 Execute UPDATE Statement</button>
              <button type="button" class="btn-copy" onclick="cancelEdit()">Cancel</button>
            </div>
          </form>
        </div>

        <!-- Table Display -->
        <div class="table-container">
          <table>
            <thead>
              <tr>
                <th id="th-emp_id" onclick="sortByColumn('emp_id')">ID <span class="sort-icon">▲</
span></th>
                <th id="th-emp_name" onclick="sortByColumn('emp_name')">Employee Name <span class
="sort-icon"></span></th>
                <th id="th-department" onclick="sortByColumn('department')">Department <span class
="sort-icon"></span></th>
                <th id="th-designation" onclick="sortByColumn('designation')">Designation <span cl
ass="sort-icon"></span></th>
                <th id="th-salary" onclick="sortByColumn('salary')">Salary (₹) <span class="sort-i
con"></span></th>
                <th id="th-joined_date" onclick="sortByColumn('joined_date')">Joined Date <span cl
ass="sort-icon"></span></th>
                <th style="cursor: default;">Actions (UPDATE / DELETE)</th>
              </tr>
            </thead>
```

```
            <tbody id="employees-tbody">
              <!-- Rendered via JS -->
            </tbody>
          </table>
        </div>

        <!-- Real-time SQL Inspector -->
        <div class="sql-inspector">
          <span class="label">🔍 Executed MySQL Command Inspector:</span>
          <code id="sql-query-display">SELECT * FROM employees ORDER BY emp_id ASC;</code>
        </div>

      </div>
    </section>

    <!-- TAB 2: PHP Script View -->
    <section id="tab-php" class="tab-panel">
      <div class="card">
        <div class="code-container">
          <div class="code-header">
            <span>mysql_crud_demo.php (Complete PHP Implementation)</span>
            <button class="btn-copy" onclick="copyCode('code-crud', this)">📋 Copy PHP Script</bu
tton>
          </div>
          <pre><code id="code-crud">&lt;?php
/**
 * Experiment 22: PHP Script to Implement MySQL Commands:
 * a) DELETE  b) ORDER BY  c) UPDATE
 * Course: Advanced Web Technology (CSIT248) - Amity University Noida
 */

$pdo = new PDO("mysql:host=localhost;dbname=company_db;charset=utf8mb4", "root", "", [
    PDO::ATTR_ERRMODE => PDO::ERRMODE_EXCEPTION,
    PDO::ATTR_DEFAULT_FETCH_MODE => PDO::FETCH_ASSOC
]);

// ========================================
// a) DELETE Command Implementation
// ========================================
if (isset($_GET['action']) && $_GET['action'] === 'delete' && isset($_GET['id'])) {
    $delete_id = (int)$_GET['id'];
    $stmt = $pdo->prepare("DELETE FROM employees WHERE emp_id = :id");
    $stmt->execute([':id' => $delete_id]);
    echo "Employee #$delete_id successfully removed from database.";
}

// ========================================
// c) UPDATE Command Implementation
// ========================================
if ($_SERVER['REQUEST_METHOD'] === 'POST' && isset($_POST['action']) && $_POST['action'] === 'upda
te') {
    $id    = (int)$_POST['emp_id'];
    $desig = trim($_POST['designation']);
    $sal   = (float)$_POST['salary'];
    $dept  = trim($_POST['department']);

    $stmt = $pdo->prepare("UPDATE employees SET designation = :desig, salary = :sal, department =
:dept WHERE emp_id = :id");
    $stmt->execute([
        ':desig' => $desig,
        ':sal'   => $sal,
        ':dept'  => $dept,
        ':id'    => $id
    ]);
    echo "Employee #$id successfully updated.";
}
```

```
// ========================================
// b) ORDER BY Command Implementation
// ========================================
$allowedCols = ['emp_id', 'emp_name', 'department', 'designation', 'salary', 'joined_date'];
$sort = in_array($_GET['sort'] ?? '', $allowedCols) ? $_GET['sort'] : 'emp_id';
$order = (strtoupper($_GET['order'] ?? '') === 'DESC') ? 'DESC' : 'ASC';

$stmt = $pdo->query("SELECT * FROM employees ORDER BY $sort $order");
$employees = $stmt->fetchAll();
?&gt;</code></pre>
        </div>
      </div>
    </section>

    <!-- TAB 3: SQL Schema View -->
    <section id="tab-sql" class="tab-panel">
      <div class="card">
        <div class="code-container">
          <div class="code-header">
            <span>schema.sql (Table DDL &amp; Seed Data)</span>
            <button class="btn-copy" onclick="copyCode('code-schema', this)">📋 Copy SQL</button>
          </div>
          <pre><code id="code-schema">-- Experiment 22: Schema for DELETE, ORDER BY, UPDATE
CREATE DATABASE IF NOT EXISTS company_db;
USE company_db;

CREATE TABLE IF NOT EXISTS employees (
    emp_id INT AUTO_INCREMENT PRIMARY KEY,
    emp_name VARCHAR(100) NOT NULL,
    department VARCHAR(60) NOT NULL,
    designation VARCHAR(80) NOT NULL,
    salary DECIMAL(10, 2) NOT NULL,
    email VARCHAR(120) NOT NULL UNIQUE,
    joined_date DATE NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

INSERT INTO employees (emp_name, department, designation, salary, email, joined_date) VALUES
('Atul Anand', 'Information Tech', 'Full Stack Engineer', 95000.00, 'atul@company.com', '2023-06-1
5'),
('Priya Sharma', 'Product & Design', 'UI/UX Lead', 88000.00, 'priya@company.com', '2023-08-01'),
('Rohan Verma', 'DevOps & Cloud', 'Cloud Architect', 115000.00, 'rohan@company.com', '2022-11-2
0'),
('Ananya Sengupta', 'Data & AI', 'Machine Learning Eng', 105000.00, 'ananya@company.com', '2023-01
-10'),
('Vikramaditya Rao', 'Cybersecurity', 'Security Analyst', 82000.00, 'vikram@company.com', '2024-02
-18'),
('Sneha Kulkarni', 'Quality Assurance', 'Lead QA Specialist', 76000.00, 'sneha@company.com', '2023
-09-05');</code></pre>
        </div>
      </div>
    </section>

    <!-- TAB 4: Academic Manual -->
    <section id="tab-manual" class="tab-panel">
      <div class="card" style="line-height: 1.8;">
        <h2 style="color: #38bdf8; font-size: 1.5rem; margin-bottom: 1rem;">Experiment 22: Academi
c Lab Manual</h2>

        <h3 style="color: #818cf8; margin-top: 1.5rem; margin-bottom: 0.5rem;">1. Aim</h3>
        <p style="color: #cbd5e1;">
          To write and implement a PHP script that interacts with a MySQL database to execute the
standard SQL commands:
          <strong>a) DELETE</strong>, <strong>b) ORDER BY</strong>, and <strong>c) UPDATE</strong>
with parameter binding and exception handling.
        </p>
```

```
        <h3 style="color: #818cf8; margin-top: 1.5rem; margin-bottom: 0.5rem;">2. Command Syntax &
amp; Theoretical Notes</h3>
        <ul style="color: #cbd5e1; margin-left: 1.5rem;">
          <li><strong>a) DELETE:</strong> Removes specific tuples matching criteria:
            <br><code>DELETE FROM employees WHERE emp_id = ?;</code>
            <br><em>Security note:</em> Always use a strict <code>WHERE</code> clause and prepared
statement to prevent accidental table truncation.
          </li>
          <li style="margin-top: 0.75rem;"><strong>b) ORDER BY:</strong> Sorts the result set by o
ne or more columns in ascending (default) or descending order:
            <br><code>SELECT * FROM employees ORDER BY salary DESC;</code>
            <br><em>Security note:</em> Column names cannot be bound as PDO parameters; always san
itize through an explicit whitelist array.
          </li>
          <li style="margin-top: 0.75rem;"><strong>c) UPDATE:</strong> Modifies existing attribute
values in one or more records:
            <br><code>UPDATE employees SET designation = ?, salary = ? WHERE emp_id = ?;</code>
          </li>
        </ul>
      </div>
    </section>

  </main>

  <!-- Footer -->
  <footer>
    <p style="margin-bottom: 0.4rem;">
      Designed &amp;amp; Developed by <strong style="color: #818cf8;">Atul Anand</strong> &bull; BCA
(Hons.) with Research &bull; Semester III
    </p>
    <p style="color: #64748b; font-size: 0.8rem;">
      Amity University Noida &bull; Department of Computer Applications &bull; CSIT248: Advanced W
eb Technology
    </p>
  </footer>

  <!-- Script for Workbench Execution -->
  <script>
    const INITIAL_EMPLOYEES = [
      { emp_id: 1, emp_name: "Atul Anand", department: "Information Tech", designation: "Full Stac
k Engineer", salary: 95000, joined_date: "2023-06-15" },
      { emp_id: 2, emp_name: "Priya Sharma", department: "Product & Design", designation: "UI/UX L
ead", salary: 88000, joined_date: "2023-08-01" },
      { emp_id: 3, emp_name: "Rohan Verma", department: "DevOps & Cloud", designation: "Cloud Arch
itect", salary: 115000, joined_date: "2022-11-20" },
      { emp_id: 4, emp_name: "Ananya Sengupta", department: "Data & AI", designation: "Machine Lea
rning Eng", salary: 105000, joined_date: "2023-01-10" },
      { emp_id: 5, emp_name: "Vikramaditya Rao", department: "Cybersecurity", designation: "Securi
ty Analyst", salary: 82000, joined_date: "2024-02-18" },
      { emp_id: 6, emp_name: "Sneha Kulkarni", department: "Quality Assurance", designation: "Lead
QA Specialist", salary: 76000, joined_date: "2023-09-05" }
    ];

    let employees = JSON.parse(JSON.stringify(INITIAL_EMPLOYEES));
    let currentSort = 'emp_id';
    let currentOrder = 'ASC';

    function renderTable() {
      const tbody = document.getElementById("employees-tbody");
      tbody.innerHTML = "";

      if (employees.length === 0) {
        tbody.innerHTML = `<tr><td colspan="7" style="text-align: center; color: #94a3b8; padding:
2rem;">No records remaining. Click "Reset Sample Records" to restore data.</td></tr>`;
        return;
```

```
      }

      employees.forEach(emp => {
        const tr = document.createElement("tr");
        tr.innerHTML = `
          <td style="font-family: 'Fira Code', monospace; color: #a5b4fc;">#${emp.emp_id}</td>
          <td><strong style="color: #ffffff;">${escapeHtml(emp.emp_name)}</strong></td>
          <td>${escapeHtml(emp.department)}</td>
          <td><span style="background: rgba(99, 102, 241, 0.15); color: #a5b4fc; padding: 0.2rem
0.5rem; border-radius: 4px;">${escapeHtml(emp.designation)}</span></td>
          <td style="font-family: 'Fira Code', monospace; color: #34d399; font-weight: 700;">₹${ne
w Intl.NumberFormat('en-IN').format(emp.salary)}</td>
          <td style="font-family: 'Fira Code', monospace; color: #cbd5e1;">${emp.joined_date}</td>
          <td>
            <button type="button" class="btn-action btn-update" onclick="openEdit(${emp.emp_id})">
✏️ Edit (UPDATE)</button>
            <button type="button" class="btn-action btn-delete" onclick="handleDelete(${emp.emp_i
d})">🗑️ DELETE</button>
          </td>
        `;
        tbody.appendChild(tr);
      });

      updateSortIndicators();
    }

    // b) ORDER BY
    function sortByColumn(col) {
      if (currentSort === col) {
        currentOrder = (currentOrder === 'ASC') ? 'DESC' : 'ASC';
      } else {
        currentSort = col;
        currentOrder = 'ASC';
      }

      employees.sort((a, b) => {
        let valA = a[col];
        let valB = b[col];

        if (typeof valA === 'string') {
          return currentOrder === 'ASC' ? valA.localeCompare(valB) : valB.localeCompare(valA);
        } else {
          return currentOrder === 'ASC' ? valA - valB : valB - valA;
        }
      });

      document.getElementById("sql-query-display").textContent = `SELECT * FROM employees ORDER BY
${currentSort} ${currentOrder};`;
      renderTable();
    }

    function updateSortIndicators() {
      ['emp_id', 'emp_name', 'department', 'designation', 'salary', 'joined_date'].forEach(col =>
{
        const th = document.getElementById('th-' + col);
        if (!th) return;
        const icon = th.querySelector('.sort-icon');
        if (col === currentSort) {
          th.classList.add('active-sort');
          icon.textContent = currentOrder === 'ASC' ? ' ▲' : ' ▼';
        } else {
          th.classList.remove('active-sort');
          icon.textContent = '';
        }
      });
    }
```

```
    // a) DELETE
    function handleDelete(id) {
      const emp = employees.find(e => e.emp_id === id);
      if (!emp) return;

      if (confirm(`Execute MySQL DELETE statement for Employee #${id} (${emp.emp_name})?`)) {
        employees = employees.filter(e => e.emp_id !== id);
        document.getElementById("sql-query-display").textContent = `DELETE FROM employees WHERE em
p_id = ${id};\n-- Affected Rows: 1`;
        renderTable();
      }
    }

    // c) UPDATE
    function openEdit(id) {
      const emp = employees.find(e => e.emp_id === id);
      if (!emp) return;

      document.getElementById("edit-id").value = emp.emp_id;
      document.getElementById("edit-emp-name").textContent = emp.emp_name;
      document.getElementById("edit-emp-id-badge").textContent = `#${emp.emp_id}`;
      document.getElementById("edit-dept").value = emp.department;
      document.getElementById("edit-desig").value = emp.designation;
      document.getElementById("edit-salary").value = emp.salary;

      const editBox = document.getElementById("update-container");
      editBox.style.display = "block";
      editBox.scrollIntoView({ behavior: 'smooth' });
    }

    function cancelEdit() {
      document.getElementById("update-container").style.display = "none";
    }

    function handleApplyUpdate(e) {
      e.preventDefault();
      const id = parseInt(document.getElementById("edit-id").value);
      const emp = employees.find(e => e.emp_id === id);
      if (!emp) return;

      const newDept = document.getElementById("edit-dept").value.trim();
      const newDesig = document.getElementById("edit-desig").value.trim();
      const newSalary = parseFloat(document.getElementById("edit-salary").value);

      emp.department = newDept;
      emp.designation = newDesig;
      emp.salary = newSalary;

      document.getElementById("sql-query-display").textContent =
        `UPDATE employees\nSET designation = '${newDesig}', salary = ${newSalary}, department =
'${newDept}'\nWHERE emp_id = ${id};\n-- Rows matched: 1, Changed: 1`;

      cancelEdit();
      renderTable();
    }

    function resetEmployees() {
      employees = JSON.parse(JSON.stringify(INITIAL_EMPLOYEES));
      currentSort = 'emp_id';
      currentOrder = 'ASC';
      document.getElementById("sql-query-display").textContent = `SELECT * FROM employees ORDER BY
emp_id ASC;`;
      cancelEdit();
      renderTable();
    }
```

```
    function escapeHtml(str) {
      return str.replace(/&/g, "&amp;").replace(/</g, "&lt;").replace(/>/g, "&gt;").replace(/"/g,
"&quot;");
    }

    function switchTab(tabId, btn) {
      document.querySelectorAll('.tab-panel').forEach(p => p.classList.remove('active'));
      document.querySelectorAll('.tab-btn').forEach(b => b.classList.remove('active'));
      document.getElementById('tab-' + tabId).classList.add('active');
      btn.classList.add('active');
    }

    function copyCode(elementId, btn) {
      const text = document.getElementById(elementId).innerText;
      navigator.clipboard.writeText(text).then(() => {
        const orig = btn.innerText;
        btn.innerText = "Copied! ✅";
        btn.style.background = "#10b981";
        setTimeout(() => {
          btn.innerText = orig;
          btn.style.background = "";
        }, 2000);
      });
    }

    window.addEventListener("DOMContentLoaded", renderTable);
  </script>

</body>

</html>
```

Supplementary Source File: schema.sql

```
-- ================================================================================
-- Experiment 22: MySQL Commands Implementation Schema
-- Course: Advanced Web Technology (CSIT248) - Amity University Noida
-- Table: employees
-- Implements: a) DELETE  b) ORDER BY  c) UPDATE
-- ================================================================================

CREATE DATABASE IF NOT EXISTS company_db;
USE company_db;

CREATE TABLE IF NOT EXISTS employees (
    emp_id INT AUTO_INCREMENT PRIMARY KEY,
    emp_name VARCHAR(100) NOT NULL,
    department VARCHAR(60) NOT NULL,
    designation VARCHAR(80) NOT NULL,
    salary DECIMAL(10, 2) NOT NULL,
    email VARCHAR(120) NOT NULL UNIQUE,
    joined_date DATE NOT NULL
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

-- Seed Sample Records
INSERT INTO employees (emp_name, department, designation, salary, email, joined_date) VALUES
('Atul Anand', 'Information Tech', 'Full Stack Engineer', 95000.00, 'atul@company.com', '2023-06-1
5'),
('Priya Sharma', 'Product & Design', 'UI/UX Lead', 88000.00, 'priya@company.com', '2023-08-01'),
('Rohan Verma', 'DevOps & Cloud', 'Cloud Architect', 115000.00, 'rohan@company.com', '2022-11-2
0'),
('Ananya Sengupta', 'Data & AI', 'Machine Learning Eng', 105000.00, 'ananya@company.com', '2023-01
-10'),
('Vikramaditya Rao', 'Cybersecurity', 'Security Analyst', 82000.00, 'vikram@company.com', '2024-02
-18'),
```

```
('Sneha Kulkarni', 'Quality Assurance', 'Lead QA Specialist', 76000.00, 'sneha@company.com', '2023
-09-05');
```

Supplementary Source File: mysql_crud_demo.php

```php
<?php
/**
 * Experiment 22: PHP Script to Implement MySQL Commands:
 * a) DELETE
 * b) ORDER BY
 * c) UPDATE
 *
 * Course: Advanced Web Technology (CSIT248) - Amity University Noida
 */

// Database Connection Settings
$host = 'localhost';
$user = 'root';
$pass = '';
$db   = 'company_db';

$message = '';
$message_type = 'success';
$last_query = '';

try {
    $pdo = new PDO("mysql:host=$host;dbname=$db;charset=utf8mb4", $user, $pass, [
        PDO::ATTR_ERRMODE            => PDO::ERRMODE_EXCEPTION,
        PDO::ATTR_DEFAULT_FETCH_MODE => PDO::FETCH_ASSOC
    ]);

    // ========================================
    // 1. IMPLEMENTATION: a) DELETE COMMAND
    // ========================================
    if (isset($_GET['action']) && $_GET['action'] === 'delete' && isset($_GET['id'])) {
        $delete_id = (int)$_GET['id'];

        $delete_sql = "DELETE FROM employees WHERE emp_id = :id";
        $last_query = "DELETE FROM employees WHERE emp_id = $delete_id;";

        $stmt = $pdo->prepare($delete_sql);
        $stmt->execute([':id' => $delete_id]);

        if ($stmt->rowCount() > 0) {
            $message = "Record with Employee ID #$delete_id was successfully deleted.";
            $message_type = "success";
        } else {
            $message = "Record not found or already deleted.";
            $message_type = "error";
        }
    }

    // ========================================
    // 2. IMPLEMENTATION: c) UPDATE COMMAND
    // ========================================
    if ($_SERVER['REQUEST_METHOD'] === 'POST' && isset($_POST['action']) && $_POST['action'] ===
'update') {
        $emp_id      = (int)$_POST['emp_id'];
        $designation = trim($_POST['designation'] ?? '');
        $salary      = (float)$_POST['salary'];
        $department  = trim($_POST['department'] ?? '');

        if ($emp_id > 0 && !empty($designation) && $salary > 0) {
            $update_sql = "UPDATE employees SET designation = :desig, salary = :sal, department =
:dept WHERE emp_id = :id";
            $last_query = "UPDATE employees SET designation = '$designation', salary = $salary, de
partment = '$department' WHERE emp_id = $emp_id;";
```

```
            $stmt = $pdo->prepare($update_sql);
            $stmt->execute([
                ':desig' => $designation,
                ':sal'   => $salary,
                ':dept'  => $department,
                ':id'    => $emp_id
            ]);

            $message = "Employee #$emp_id successfully updated.";
            $message_type = "success";
        } else {
            $message = "Invalid input values for record update.";
            $message_type = "error";
        }
    }

    // ==========================================
    // 3. IMPLEMENTATION: b) ORDER BY COMMAND
    // ==========================================
    // Whitelist columns to prevent SQL Injection in ORDER BY clause
    $allowed_sorts = ['emp_id', 'emp_name', 'department', 'designation', 'salary', 'joined_date'];
    $allowed_orders = ['ASC', 'DESC'];

    $sort_by = in_array($_GET['sort'] ?? '', $allowed_sorts) ? $_GET['sort'] : 'emp_id';
    $order   = in_array(strtoupper($_GET['order'] ?? ''), $allowed_orders) ? strtoupper($_GET['ord
er']) : 'ASC';

    $next_order = ($order === 'ASC') ? 'DESC' : 'ASC';

    $select_sql = "SELECT * FROM employees ORDER BY $sort_by $order";
    if (empty($last_query)) {
        $last_query = $select_sql . ";";
    }

    $stmt_list = $pdo->query($select_sql);
    $employees = $stmt_list->fetchAll();

} catch (PDOException $e) {
    $message = "Database Exception: " . $e->getMessage();
    $message_type = "error";
    $employees = [];
}
?>
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Experiment 22 - DELETE, ORDER BY &amp;amp; UPDATE in MySQL</title>
  <link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;500;600;700;800&family=Fira+
Code:wght@400;600&display=swap" rel="stylesheet">
  <style>
    body { font-family: 'Inter', system-ui, sans-serif; background: #0f172a; color: #f8fafc; margi
n: 0; padding: 2rem; }
    .container { max-width: 1100px; margin: 0 auto; }
    .card { background: #1e293b; border: 1px solid #334155; border-radius: 12px; padding: 1.5rem;
margin-bottom: 1.5rem; }
    h1 { color: #fff; font-size: 1.85rem; margin-bottom: 0.5rem; }
    .alert { padding: 0.75rem 1rem; border-radius: 6px; margin-bottom: 1rem; font-size: 0.9rem; }
    .alert.success { background: rgba(16, 185, 129, 0.15); border: 1px solid #10b981; color: #6ee7
b7; }
    .alert.error { background: rgba(239, 68, 68, 0.15); border: 1px solid #ef4444; color: #fca5a5;
}
    table { width: 100%; border-collapse: collapse; font-size: 0.9rem; margin-top: 1rem; }
    th, td { padding: 0.75rem 1rem; border-bottom: 1px solid #334155; text-align: left; }
    th { background: #0f172a; color: #38bdf8; }
```

```
    th a { color: #38bdf8; text-decoration: none; display: flex; align-items: center; gap: 0.25re
m; }
    th a:hover { text-decoration: underline; }
    .btn { padding: 0.35rem 0.65rem; border-radius: 4px; font-size: 0.8rem; text-decoration: none;
cursor: pointer; border: none; font-weight: 600; }
    .btn-edit { background: #6366f1; color: #fff; }
    .btn-delete { background: #ef4444; color: #fff; }
    .sql-box { background: #000; border: 1px solid #334155; border-radius: 8px; padding: 1rem; fon
t-family: 'Fira Code', monospace; font-size: 0.85rem; color: #38bdf8; margin-top: 1rem; }
  </style>
</head>
<body>

<div class="container">
  <h1>Experiment 22: MySQL DELETE, ORDER BY &amp;amp; UPDATE</h1>
  <p style="color: #94a3b8; margin-bottom: 1.5rem;">CSIT248 Advanced Web Technology &bull; Amity U
niversity Noida</p>

  <?php if (!empty($message)): ?>
    <div class="alert <?php echo $message_type; ?>">
      <?php echo htmlspecialchars($message); ?>
    </div>
  <?php endif; ?>

  <div class="card">
    <h3 style="color: #38bdf8; margin: 0 0 0.5rem 0;">1. Command b) ORDER BY &bull; Click column h
eader to toggle ASC/DESC</h3>

    <table>
      <thead>
        <tr>
          <th><a href="?sort=emp_id&order=<?php echo $next_order; ?>">ID <?php echo ($sort_by ===
'emp_id') ? ($order === 'ASC' ? '▲' : '▼') : ''; ?></a></th>
          <th><a href="?sort=emp_name&order=<?php echo $next_order; ?>">Employee Name <?php echo
($sort_by === 'emp_name') ? ($order === 'ASC' ? '▲' : '▼') : ''; ?></a></th>
          <th><a href="?sort=department&order=<?php echo $next_order; ?>">Department <?php echo
($sort_by === 'department') ? ($order === 'ASC' ? '▲' : '▼') : ''; ?></a></th>
          <th><a href="?sort=designation&order=<?php echo $next_order; ?>">Designation <?php echo
($sort_by === 'designation') ? ($order === 'ASC' ? '▲' : '▼') : ''; ?></a></th>
          <th><a href="?sort=salary&order=<?php echo $next_order; ?>">Salary (₹) <?php echo ($sort
_by === 'salary') ? ($order === 'ASC' ? '▲' : '▼') : ''; ?></a></th>
          <th>Actions (UPDATE &amp;amp; DELETE)</th>
        </tr>
      </thead>
      <tbody>
        <?php if (!empty($employees)): ?>
          <?php foreach ($employees as $row): ?>
            <tr>
              <td style="font-family: 'Fira Code', monospace;">#<?php echo $row['emp_id']; ?></td>
              <td><strong><?php echo htmlspecialchars($row['emp_name']); ?></strong></td>
              <td><?php echo htmlspecialchars($row['department']); ?></td>
              <td><?php echo htmlspecialchars($row['designation']); ?></td>
              <td style="font-family: 'Fira Code', monospace; color: #34d399;">₹<?php echo number_
format($row['salary'], 2); ?></td>
              <td>
                <button type="button" class="btn btn-edit" onclick="populateEditForm(<?php echo ht
mlspecialchars(json_encode($row)); ?>)">Edit (UPDATE)</button>
                <a href="?action=delete&id=<?php echo $row['emp_id']; ?>&sort=<?php echo $sort_by;
?>&order=<?php echo $order; ?>"
                   class="btn btn-delete"
                   onclick="return confirm('Execute DELETE for Employee #<?php echo $row['emp_i
d']; ?>?')">DELETE</a>
              </td>
            </tr>
          <?php endforeach; ?>
        <?php else: ?>
```

```
          <tr><td colspan="6" style="text-align: center; color: #94a3b8; padding: 2rem;">No record
s found.</td></tr>
        <?php endif; ?>
      </tbody>
    </table>
  </div>

  <!-- Form for UPDATE -->
  <div class="card" id="edit-card" style="display: none;">
    <h3 style="color: #818cf8; margin: 0 0 1rem 0;">2. Command c) UPDATE Employee Record</h3>
    <form method="POST" action="mysql_crud_demo.php">
      <input type="hidden" name="action" value="update">
      <input type="hidden" name="emp_id" id="edit_emp_id">

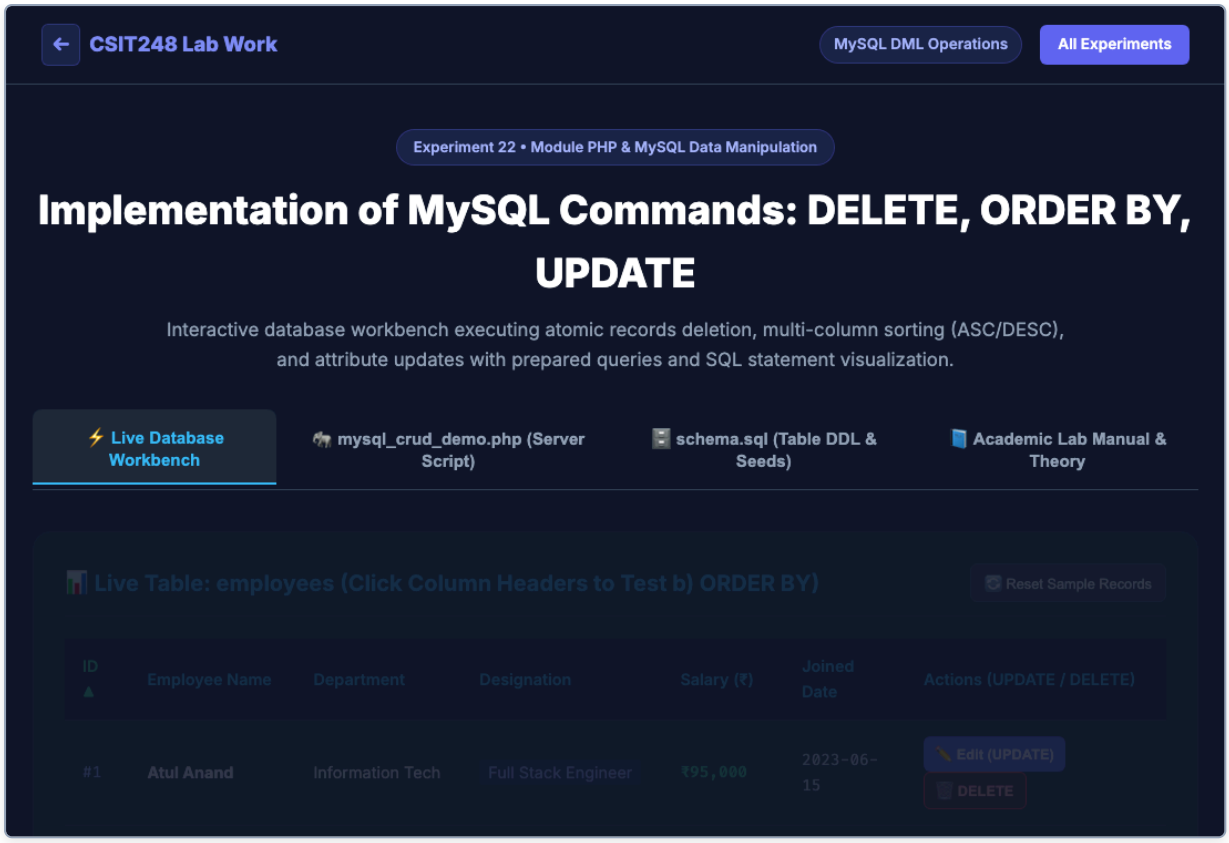
      <div style="display: grid; grid-template-columns: repeat(3, 1fr); gap: 1rem; margin-bottom:
1rem;">
        <div>
          <label style="display: block; font-size: 0.8rem; margin-bottom: 0.3rem;">Department</lab
el>
          <input type="text" name="department" id="edit_department" required style="width: 100%; b
ackground: #0f172a; border: 1px solid #334155; color: #fff; padding: 0.5rem; border-radius: 4px;">
        </div>
        <div>
          <label style="display: block; font-size: 0.8rem; margin-bottom: 0.3rem;">Designation</la
bel>
          <input type="text" name="designation" id="edit_designation" required style="width: 100%;
background: #0f172a; border: 1px solid #334155; color: #fff; padding: 0.5rem; border-radius: 4p
x;">
        </div>
        <div>
          <label style="display: block; font-size: 0.8rem; margin-bottom: 0.3rem;">Salary (₹)</lab
el>
          <input type="number" step="1000" name="salary" id="edit_salary" required style="width: 1
00%; background: #0f172a; border: 1px solid #334155; color: #fff; padding: 0.5rem; border-radius:
4px;">
        </div>
      </div>
      <button type="submit" class="btn btn-edit" style="padding: 0.6rem 1.25rem;">Execute UPDATE Q
uery</button>
    </form>
  </div>

  <div class="sql-box">
    <strong>Last Executed MySQL Command:</strong><br>
    <code><?php echo htmlspecialchars($last_query); ?></code>
  </div>
</div>

<script>
  function populateEditForm(emp) {
    document.getElementById('edit-card').style.display = 'block';
    document.getElementById('edit_emp_id').value = emp.emp_id;
    document.getElementById('edit_department').value = emp.department;
    document.getElementById('edit_designation').value = emp.designation;
    document.getElementById('edit_salary').value = emp.salary;
    document.getElementById('edit-card').scrollIntoView({ behavior: 'smooth' });
  }
</script>

</body>
</html>
```

Output :


CSIT248 Lab Work
MySQL DML Operations
All Experiments
Experiment 22 • Module PHP & MySQL Data Manipulation
Implementation of MySQL Commands: DELETE, ORDER BY, UPDATE
Interactive database workbench executing atomic records deletion, multi-column sorting (ASC/DESC), and attribute updates with prepared queries and SQL statement visualization.
⚡ Live Database Workbench
mysql_crud_demo.php (Server Script)
schema.sql (Table DDL & Seeds)
Academic Lab Manual & Theory
Live Table: employees (Click Column Headers to Test b) ORDER BY)
Reset Sample Records
ID ▲
Employee Name
Department
Designation
Salary (₹)
Joined Date
Actions (UPDATE / DELETE)
#1
Atul Anand
Information Tech
Full Stack Engineer
₹95,000
2023-06-15
Edit (UPDATE)
DELETE